

SERINGAL RIO BRANCO REDD+ PROJECT

SERINGAL RIO BRANCO

REDD+
P R O J E C T

ambipar
environment

 **BIOFÍLICA**

AMAZONIA
AGROINDÚSTRIA

Project title	Seringal Rio Branco REDD+ Project
Project ID	4834
Crediting period	15-March-2023 to 14-March-2053
Project lifetime	15-March-2023 to 14-March-2123; 100 years
(CCB) GHG accounting period	15-March-2023 to 14-March-2063; 40 years
Original date of issue	03-June-2024
Most recent date of issue	03-June-2024
Version	v.2
VCS Standard version	VCS Standard v4.7
CCB Standards version	CCB Standard v.3
Project location	Cruzeiro do Sul, Acre, Brazil
Project proponent(s)	Biofílica Ambipar Environmental Investments S.A. Amazônia Agroindústria EIRELI
Validation/verification body	Rina Brasil

History of CCB status	<i>Under validation</i>
Gold Level criteria	<i>Gold Level – Exceptional Benefits to Climate: project activities will provide significant support to communities and biodiversity in adapting to the impacts of climate change.</i> <i>Gold Level – Exceptional Benefits to Biodiversity: project activities will contribute to the conservation of 5 endangered species.</i>
Expected verification schedule	<i>The first VCS verification will take place with the VCS+CCB validation</i>
Prepared by	<i>Biofíllica Ambipar Environmental Investments S.A.</i>

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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or impact estimated by the end of project lifetime	Section reference
1) Expected benefits for the climate: the Seringal Rio Branco REDD+ project is expected to help mitigate climate change after its life cycle, based on the first baseline defined for the project, with total avoided emissions of 3,326,590 tCO₂eq. The avoided deforestation planned in the scenario with the Project is 5,487 hectares during the first 30 years of the Project.	Section 3
2) Expected Benefits for the Community: the benefits for the community located in the Project Zone and other stakeholders will be focused on supporting and promoting sustainable practices, along with coordinating efforts to educate people environmentally on issues related to hunting, fishing, and subsistence farming. Other aspects include optimizing non-timber forest management and collaborating with public and private entities to improve health conditions, sanitation, and infrastructure in the communities. Therefore, the aim is to influence social issues and living conditions in the communities surrounding the Project, reducing social vulnerability, adding value to adaptation to climate change, improving socio-economic conditions and promoting the well-being of families by improving economic and social well-being. With this, the project expects to potentially benefit 1,467 people, 168 of them women.	Section 4
3) Expected benefits for Biodiversity: the Seringal Rio Branco REDD+ project anticipates the maintenance of 147 species of fauna and 118 of flora confirmed in the Project Area, ensuring the protection and conservation of habitats and local biodiversity, including 17 species (5 species of flora and 12 of fauna) with some degree of threat according to the IUCN Red List. In addition, the Project Area is of national and international importance as a region of high biodiversity.	Section 5

1.2 Standardized Benefit Metrics

Category	Metric	Estimated by the end of project lifetime	Section reference
GHG emission reductions or carbon dioxide removals	Net estimated removals in the project area, measured against the without-project scenario	Not applicable, this project only works on reducing emissions.	N.A.
	Net estimated reductions in the project area, measured against the without-project scenario	3,326,590 tCO ₂ eq	Section 3
Forest ¹ cover	For REDD ² projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario	REDD+ APD: 5.487 hectares REDD+ AUD: to be defined	Section 3
	For ARR ³ projects: Estimated number of hectares of forest cover increased in the project area measured against the without-project scenario	Not applicable, this is a REDD+ project.	N.A.
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices are expected to occur as a result of project activities, measured against the without-project scenario	Not applicable, this is a REDD+ project.	N.A.
	Number of hectares of non-forest land in which improved land management practices are expected to occur as a result of project	Not applicable, this is a REDD+ project.	N.A.

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (VCS Program Definitions)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (VCS Program Definitions)

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (VCS Program Definitions)

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (VCS Program Definitions)

	activities, measured against the without-project scenario		
Training	Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities	Potentially 1467 people	Section 4
	Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities	Potentially 168 women	Section 4
Employment	Total number of people expected to be employed in project activities ⁵ , expressed as number of full-time employees ⁶	Data not available at time of validation.	Section 4
	Number of women expected to be employed as a result of project activities, expressed as number of full-time employees	Data not available at time of validation.	Section 4
Livelihoods	Total number of people expected to have improved livelihoods ⁷ or income generated as a result of project activities	Potentially 1467 people	Section 4
	Number of women expected to have improved livelihoods or income generated as a result of project activities	Potentially 168 women	Section 4
Health	Total number of people for whom health services are expected to improve as a result of project	Potentially 1467 people	Section 4

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. The Sustainable Livelihood Approach to Poverty Reduction. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

	activities, measured against the without-project scenario		
	Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	Potentially 168 women	Section 4
Education	Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	Potentially 1467 people	Section 4
	Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	Potentially 168 women and girls	Section 4
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Potentially 1467 people	Section 4
	Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Potentially 168 women	Section 4
Well-being	Total number of community members whose well-being ⁸ is expected to improve as a result of project activities	Potentially 1467 people	Section 4
	Number of women whose well-being is expected to improve as a result of project activities	Potentially 168 women	Section 4

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

Biodiversity conservation	Expected change in the number of hectares managed significantly better by the project for biodiversity conservation ⁹ , measured against the without-project scenario	29.616 hectares	Section 5
	Expected number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of project activities ¹¹ , measured against the without-project scenario	17 species, 5 of which are flora and 12 fauna, which have some degree of threat according to the IUCN Red List, will benefit from the project's activities.	Section 5

⁹ Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 PROJECT DETAILS

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

The Seringal Rio Branco REDD+ Project is a partnership between Biofílica Ambipar Environmental Investments S/A and Amazônia Agroindústria EIRELI and is located in the Alto Juruá region, between the municipalities of Cruzeiro do Sul and Porto Walter in the state of Acre in Brazil and has a rural property area of 29,616 hectares.

Of the property's 29,616 hectares, 29,611,0242 hectares have forest, more specifically Open Ombrophilous Forest, native vegetation characteristic of the western region of Acre. Given the scenario of surplus native vegetation on the property in the light of Brazilian environmental legislation and the scenario of agricultural expansion in the region, this is a project grouped into two instances, the first being the APD component, where the REDD+ project aims to act to contain planned deforestation on the farm, and a second future AUD instance, which will work to contain unplanned deforestation that threatens the property. Methodologically, the first APD instance applies VM0007, while the future second instance will apply VMM048.

Considering the scenario of the region before the implementation of the Project, it was identified that the main vectors of deforestation and degradation operating in the Reference Region during the historical period are: the creation of settlement projects with the opening of new branches, the conversion of areas for raising beef cattle, the opening of primary forest areas for the implantation of annual swiddens and the opening of roads for the selective collection of timber. These factors have contributed to the process of degradation and threats to the forest area, increasing environmental impacts in the region.

Thus, in order to promote forest conservation and mitigate climate change, the REDD+ project will reduce deforestation by improving asset surveillance, monitoring changes in land use and land cover using satellite images and strengthening the management of non-timber forest products through the sale of carbon credits. In addition to the climate benefits, the project also aims to generate socio-environmental benefits by monitoring biodiversity, strengthening local governance, articulating strategic partnerships for infrastructure and community development and training in sustainable economic development practices for the surrounding communities, while preserving the culture and traditions of the local people.

The development of activities combined with good project management, with continuous monitoring and evaluation of activities and results, will enable the project to achieve the objectives of reducing 3,326,590 tCO₂e in 30 years (REDD+ APD), improving the quality of life of potentially 1,467 people, including 168 women, and conserving 147 species of fauna and 118 species of flora, 17 of which are under some degree of threat according to the IUCN Red List.

2.1.2 Audit History (VCS, 4.1)

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	01-Jul-2024 – To be Defined	VCS and CCB	Rina Brasil	To be defined

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

Sectoral scope	14: Agriculture, forestry, and other land use (AFOLU)
AFOLU project category ¹²	Reducing Emissions from Deforestation and Forest Degradation (REDD)
Project activity type	Avoided Planned Deforestation (APD) and Avoided Unplanned Deforestation (AUD)

2.1.4 Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

The Seringal Rio Branco REDD+ Project is a grouped project with project activities within scope 14 for AFOLU projects, including two components:

- APD - Avoided Planned Deforestation: activities under this component are based on methodology VM0007 - REDD+ Methodology Framework v.1.7.
- AUD - Avoided Unplanned Deforestation: activities under this component are based on methodology VM0048 - Reducing Emissions from Deforestation and Forest Degradation v.1.0.

The project design assumes that the entire project boundary is well defined and that the proposed activities meet the scale of the project. All the information related to the eligibility of the grouped project is shown in section 2.1.6.1.

The Seringal Rio Branco REDD+ Project meets all the requirements established by the VCS program, that is, this grouped project is eligible within the VCS Standard scope (version 4.6), for AFOLU projects, as indicated below:

- REDD project activities take place on native ecosystem land, classified as forest, which has not been converted in the 10 years prior to the project start date.
- Project activities will not convert native ecosystems to generate GHG credits.
- The project activities constitute activities within the principles established for the AUD and APD components.

Through the initial commitment of the owner of the project area, the project guarantees that these eligibility conditions will be met through a rigorous document analysis process. In addition, the project ensures that the requirements are met through the analysis of geographic data in comparison with nationally and

¹² See Appendix 1 of the VCS Standard

scientifically recognized secondary data, comparing the area of the property with national remote sensing databases, such as the Rural Environmental Registry (CAR)¹³ and Mapbiomas¹⁴.

2.1.5 Transfer Project Eligibility (VCS, 3.23, Appendix 2)

Not applicable once REDD+ Seringal Rio Branco does not seek any transferring of project eligibility.

2.1.6 Project Design (VCS, 3.6)

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

2.1.6.1 Eligibility Criteria for Grouped Projects (VCS, 3.6; CCB, G1.14)

The Seringal Rio Branco REDD+ Project is a grouped project with multiple activities. As such, the set of eligibility criteria applied for the inclusion of new instances of project activities will comply with the VCS Standard (version 4.7).

According to the VCS standard (version 4.7, section 3.6.16), grouped projects must include one or more sets of eligibility criteria for the inclusion of new instances of project activities. Thus, for the inclusion of new instances of project activities, project developers must ensure the following set of eligibility criteria:

- 1) Comply with the applicability conditions defined in the methodology applied to the project.
 - The Seringal Rio Branco REDD+ Project was developed with multiple project activities (according to section 2.1.3 of the PDD).
 - The project developers guarantee that the project activities will be carried out in accordance with the applicability criteria established in methodology VM0007 for the first instance, and in accordance with VM0048 for the next instance already planned for the project.
- 2) Use the approach, technologies or measures specified in the project description.
 - As specified above, the Seringal Rio Branco REDD+ project is based on the grouped project approach and is divided into two different instances. The first instance is based on the VM0007 methodology, specifically aimed at conserving the area of the farm corresponding to its surplus Legal Reserve, where Brazilian legislation allows legal deforestation of the area after authorization by the competent agency, characterizing the use of the Avoided Planned Deforestation (APD) approach to contain planned deforestation of the area. While the next instance of the project will be based on the VM0048 methodology, focused on the Avoided Unplanned Deforestation approach, with the aim of containing unplanned deforestation that could occur due to external pressures on the area.

¹³ <http://www.car.ac.gov.br/#/>

¹⁴ <https://brasil.mapbiomas.org/>

- The developers of the Seringal Rio Branco REDD+ project undertake to assume that there will be no changes to the set of technologies or measures specified in the project description, used to estimate GHG emissions both in the baseline and in the leakage of the first and new instances of the project.
- 3) Apply the technologies and measures as specified in the project description.
 - The developers of the Seringal Rio Branco REDD+ project have technology and a technical team dedicated to the project with appropriate expertise for the area. The developers are responsible for the diagnostic studies aimed at the applicability of the methodologies, thus guaranteeing the eligibility of the project and the areas that will be included as new instances in the project.
 - 4) They are subject to the reference scenario determined in the project description for the specified project activity and geographical area.
 - The developers of the Seringal Rio Branco REDD+ Project guarantee that diagnostic studies will be carried out to determine the most plausible baseline scenario for each of the project instances, respecting the geographical area established in the project's grouped zone.
 - The project developers work on credit generation projects, with technology and field staff dedicated exclusively to the project. The developers have expertise and are responsible for analyzing and applying the diagnostics carried out to describe the most plausible reference scenario.
 - 5) The new instances of project activity have characteristics with regard to additionality that are consistent with the initial instances for the specified project activity and geographical area. For example, the new instances of project activity have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances or face the same investment, technological and/or other barriers as the initial instances.
 - The developers of the Seringal Rio Branco REDD+ Project, as already mentioned, work with the generation of carbon credits and have technology and field staff dedicated exclusively to the program. The developers therefore guarantee that all diagnostic studies will be carried out with the aim of describing in sufficient detail the financial, technical and/or other parameters to verify that the new instances of project activities are consistent with the first instances of project activities, as well as facing the same investment, technological and/or other barriers.

In addition, the grouped projects of the VCS Standard (version 4.6, section 3.6.17) allow the inclusion of new instances of project activities after the initial validation of the project. In this regard, the developers of the Seringal Rio Branco REDD+ Project guarantee that the following criteria will be met:

- 1) The new instance of activity programmed for the project will only take place within the grouped project zone of the Seringal Rio Branco REDD+ Project.
- 2) As mentioned above, the new project instance will fully comply with at least one complete set of eligibility criteria for inclusion of new instances of project activities. The developers of the Seringal Rio Branco REDD+ Project are aware that partial compliance with sets of eligibility criteria is insufficient.

- 3) New instances of project activities will be included in the monitoring report with sufficient technical, financial, geographical, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and allow sampling by the validation/verification body.
- 4) New project activities will be validated when they are checked against the applicable set of eligibility criteria.
- 5) New instances of project activity will have evidence of project property, in relation to each instance of project activity held by the Seringal Rio Branco REDD+ Project, from the respective start date of each instance (that is, the date on which the instance of project activity began to reduce or remove GHG emissions).
- 6) The inclusion of new instances of project activity will have a start date equal to or later than the start date of the Seringal Rio Branco REDD+ Project.
- 7) New instances of project activity will be eligible for credit from the date the instance begins until the end of the project period. In addition, the Seringal Rio Branco REDD+ Project is aware that when a new instance of project activity begins in a previous verification period, no credit can be claimed for GHG emission reductions or removals generated during a previous verification period (as described in section 3.4.4 of the VCS Standard v. 4.6) and new instances are eligible from the start of the next verification period.
- 8) New project activity cases will not leave a VCS project and will not enroll in another project in the sequence.

Finally, the Seringal Rio Branco REDD+ Project will validate the first instance of the project characterized with APD-type project activities, following the VM0007 methodology, and later, a new instance will be validated with AUD project activities, according to the VM0048 methodology. Both areas are already defined and characterized in the PDD, as are the project activities. It should be noted that the areas form a single farm, owned by the same landowner.

2.1.7 Project Proponent (VCS, 3.7; CCB, G1.1)

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Organization name	Biofílica Ambipar Environmental Investments S.A.
Contact person	Plinio Ribeiro, Soraya Pires and Leonardo Almeida

Title	CEO, Operations Director and Project Coordinator
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2.1.8 Other Entities Involved in the Project

Organization name	Vasta Insumos da Amazônia LTDA
Role in the project	Company responsible for operating sustainable forest management
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Title	Managing Partner
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Organization name	Soluções Técnicas e Serviços de Engenharia Ambiental LTDA - STA
Role in the project	Diagnostic developer
Contact person	Raniery Branco
Title	Managing Partner
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2.1.9 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

Amazônia Agroindústria EIRELI is the legitimate owner of the property where the Seringal Rio Branco REDD+ Project is being implemented and developed, as detailed in Section 2.5.5. In order to establish responsibility and rights over the project, as well as the percentage of carbon credits allocated to each party, a contract was signed between the proponents of the Seringal Rio Branco REDD+ Project.

2.1.10 Project Start Date (VCS, 3.8)

Project start date	March 15, 2023.
Justification	The project start date marks the beginning of project activities in the project area and in the case of REDD+ Seringal Rio Branco is defined by the installation of the project sign in the project area.

2.1.11 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Crediting period	30 years – the minimum for REDD+ projects are 20 years
Start date of first or fixed crediting period	March 15, 2023 to March 14, 2053
CCB benefits assessment period	March 15, 2023 to March 14, 2123

2.1.12 Differences in Assessment/Project Crediting Periods (CCB, G1.9)

The Project accreditation period is marked by the installation of the project sign in the project area, as mentioned in Section 2.1.10.

After official establishment of the partnership between Amazônia Agroindústria EIRELI and Biofílica Ambipar Environment Investments, the first investments are made to develop the technical studies to support the writing of the PDD, such as the determination of the baseline and the socio-economic and environmental diagnoses.

Subsequently, the activities related to the climate, community and biodiversity scope are developed, along with the monitoring of the attributes related to these scopes. At this point, the first commercialized carbon credits are collected, which come from the first verification of the project by the VCS certification in order to financially support the development of these activities and monitoring.

In this way, the assessment of changes related to climate, community and biodiversity benefits begins shortly after the start of the project's crediting period.

2.1.13 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
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15-March-2023 a 31-December-2023	266.699
1-January-2024 a 31-December-2024	272.750
1-January-2025 a 31-December-2025	278.801
1-January-2026 a 31-December-2026	284.852
1-January-2027 a 31-December-2027	290.903
1-January-2028 a 31-December-2028	296.954
1-January-2029 a 31-December-2029	303.005
1 January-2030 a 31-December-2030	309.056
1-January-2031 a 31-December-2031	315.107
1-January-2032 a 31-December-2032	321.158
1-January-2033 a 31-December-2033	55.201
1-January-2034 a 31-December-2034	49.892
1-January-2035 a 31-December-2035	44.583
1-January-2036 a 31-December-2036	39.274
1-January -2037 a 31-December-2037	33.965
1-January-2038 a 31-December-2038	28.656
1-January-2039 a 31-December-2039	23.347
1-January-2040 a 31-December-2040	18.038
1-January-2041 a 31-December-2041	12.729
1-January-2042 a 31-December-2042	7.420
1-January-2043 a 31-December-2043	7.420

<i>1-January-2044 a 31-December-2044</i>	7.420
<i>1-January-2045 a 31-December-2045</i>	7.420
<i>1-January-2046 a 31-December-2046</i>	7.420
<i>1-January-2047 a 31-December-2047</i>	7.420
<i>1-January-2048 a 31-December-2048</i>	7.420
<i>1-January-2049 a 31-December-2049</i>	7.420
<i>1-January-2050 a 31-December-2050</i>	7.420
<i>1-January-2051 a 31-December-2051</i>	7.420
<i>1-January-2052 a 31-December-2052</i>	7.420
<i>Total estimated ERRs during the first or fixed crediting period</i>	3.326.590
<i>Total number of years</i>	30
<i>Average annual ERRs</i>	110.886

2.1.14 Physical Parameters (CCB, G1.3)

The Seringal Rio Branco REDD+ Project, located on the property of the same name, covers an area of 29,616 hectares, within the municipal boundaries of Cruzeiro do Sul. In addition, the project's reference region (with an area of 739,065.43 ha) covers the municipal boundaries of Mâncio Lima, Rodrigues Alves, Cruzeiro do Sul and Porto Walter, as shown in figure 1 below.

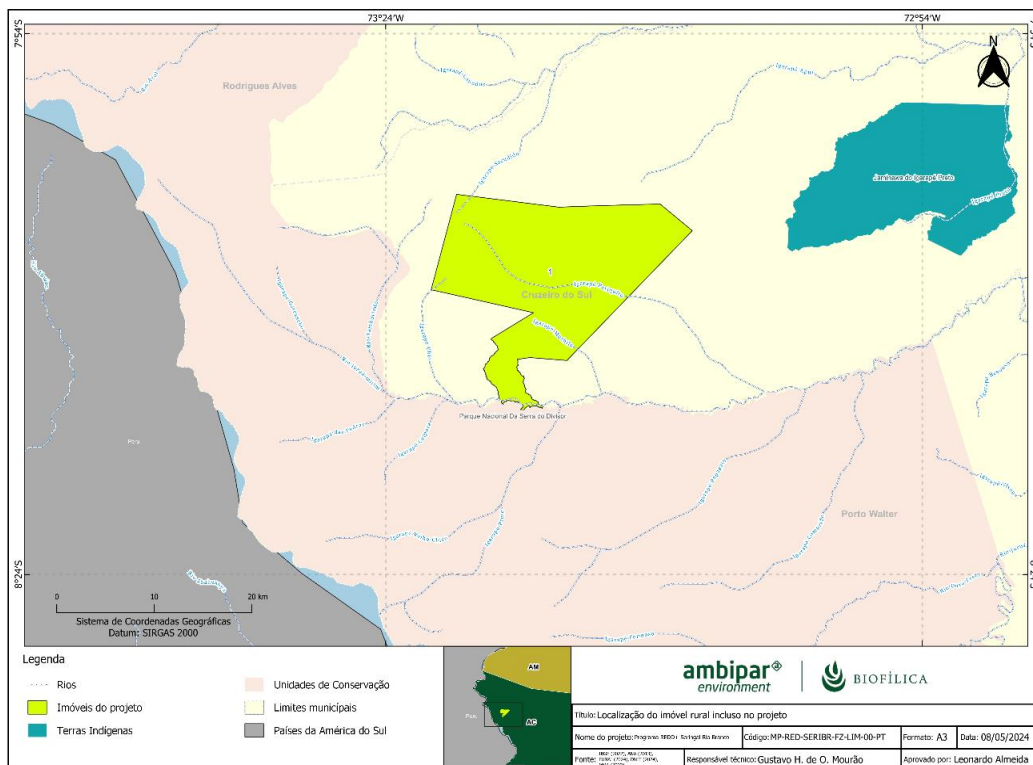


Figure 1: Location map of the Seringal Rio Branco REDD+ Project property

In order to characterize the physical parameters of the region and the project area, hydrographic, climatic, geological, geomorphological, pedological and mineral resources data were collected, which will be detailed below:

Climatic aspects:

The Köppen-Giger International Classification of climate types is based mainly on the amount and annual distribution of rainfall, as well as the average monthly and annual temperature and the average temperature of the coldest month. Based on this principle, the climate of the Amazon was classified by Köppen as type A, that is, a humid tropical climate, with the average temperature of the coldest month never falling below 18°C.

In the state of Acre, the climate is of the equatorial type (hot and humid), based on the Köppen classification, with two well-defined seasons in the year, characterizing the Af climate type (equatorial or humid tropical), with the occurrence of a short dry season and high levels of rainfall in the rainy season (SEMA, 2019)¹⁵. Similarly, according to the database provided by Alvares et al. (2013)¹⁶ in the reference

¹⁵ Environmental, geographical, and socio-economic context of the gregório river state forest complex and its surroundings. State of acre. Secretariat of state for the environment - sema. Invitation to tender no. xx/2020 - annex 3. 2019.

¹⁶ ALVARES, C. A.; STAPE, J. L.; SENTELHAS, P. C.; GONÇALVES, J. L. M.; SPAROVEK, G. Köppen's climate classification map for Brazil. Meteorologische Zeitschrift, v. 22, n. 6, p. 711-728, 2013.

region, as well as in the Seringal Rio Branco REDD+ Project area, there is a predominance of the Af climate type.

The Seringal Rio Branco REDD+ project area is in a zone where there is a predominance of an average temperature that shows little oscillation (25.29 °C to 25.35 °C), which represents greater viability for project installations that may require environments with constant annual weather variables, while taking into account seasonality throughout the year. For the reference region, the months of October to November are the hottest, while the months of May to August are the least hot.

To characterize rainfall in the reference region, a survey was carried out on the CHIRPS Product database, developed by the United States Geological Survey (USGS) and the Climate Hazards Group at the University of California, Santa Barbara (UCSB) (Funk et al., 2015)¹⁷. Based on this analysis, it was possible to observe that the average accumulated rainfall in the reference region of the Seringal Rio Branco REDD+ Project ranged from 70.73 mm to 271.27 mm, concluding that the rainiest months are from January to April, with the maximum occurring in March (Figure 2). Within the boundaries of the project area, figure 3 shows that rainfall varies on average from 173.87 mm to 179.06 mm, with an approximate variation of 6 mm.

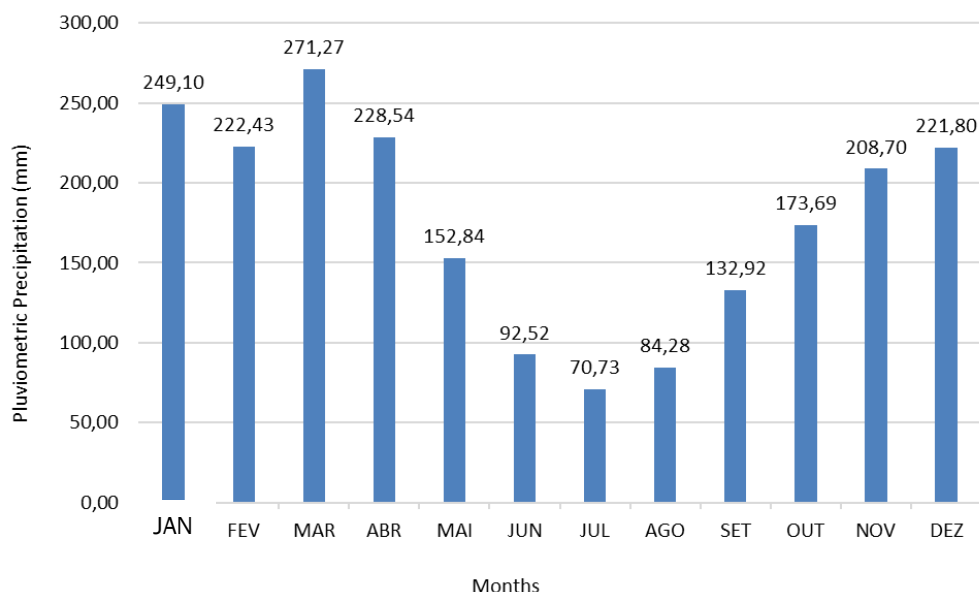


Figure 2: Average accumulated rainfall in the reference region of the Seringal Rio Branco REDD+ Project.
Source: CHIRPS/USGS (2023).

¹⁷ FUNK, C., PETERSON, P., LANDSFELD, M., PEDREROS, D., VERDIN, J., SHUKLA, S., HUSAK, G., ROWLAND, J., HARRISON, L., HOELL, A., MICHAELSEN, J., 2015. The climate hazards infrared precipitation with Stations a new environmental record for monitoring extremes. Scientific data 2, 1-21

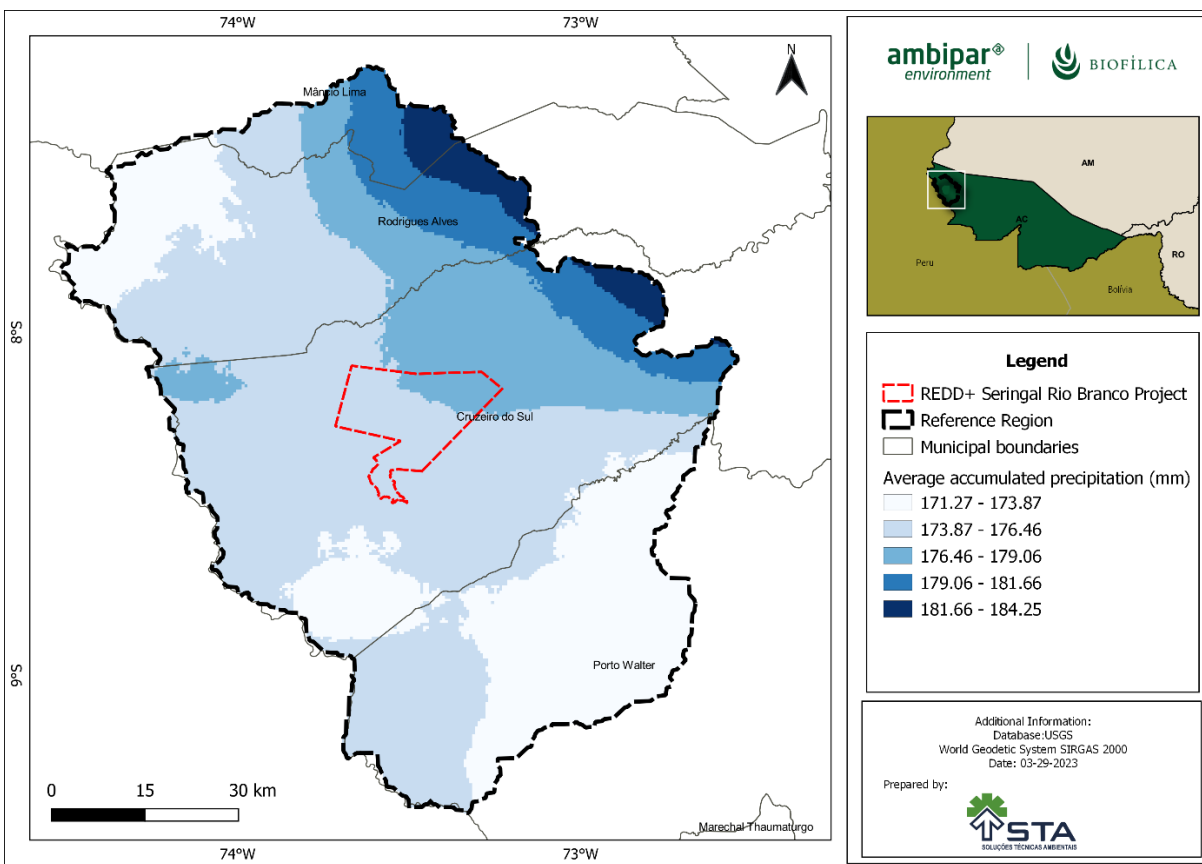


Figure 3: Spatialization of rainfall in the reference region of the Seringal Rio Branco REDD+ Project.

In order to observe the cumulative average wind speed, data from the WorldClim database version 2.1 was used for the period 1970 to 2000. Based on the results, it was observed that the lowest wind speeds were recorded in the months of April to August, with the lowest speed in April (2.27 km/h) and the months of November, December and January showed the highest speeds, with the maximum in December (3.37 km/h), that is, the accentuation of speeds occurs in the rainy season, especially in the months of November, December and January, which may be related to the formation of clouds and the occurrence of rainfall itself.

Hydrographic aspects:

The hydrographic network of the state of Acre makes up the great Amazonian hydrographic basin, consisting of the Juruá River and Purus River basins, tributaries of the right bank of the Solimões River. The state's main watercourses run in a southwest-northeast direction, practically parallel from east to west, as shown on the map in figure 4, and the main rivers that cut through the state have their sources in Peru, cross the state and flow into rivers in the state of Amazonas. These are tributaries of the right bank of the Solimões River, which, from Manaus, is called the Amazonas. The main watercourses are the Tarauacá, Purus, Gregório, Envira, Acre and Juruá rivers. They form the state's hydrographic network, divided into the Acre-Purus Basin and the Juruá Basin. The state's rivers are a very important means of transportation, so much so that most of Acre's cities and towns are built on the banks of rivers.

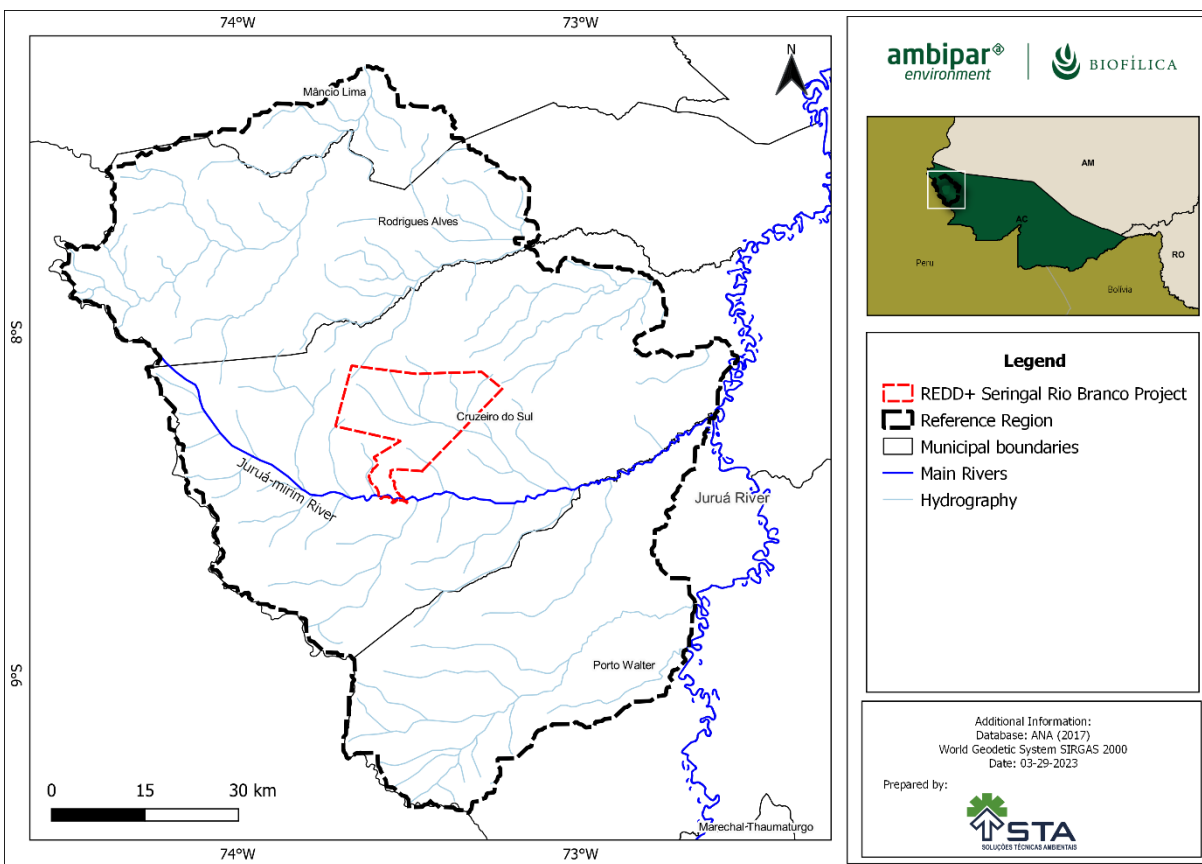


Figure 4: Hydrography in the reference region of the Seringal Rio Branco REDD+ Project.

Mineral Aspects:

In order to characterize the mineral resources in the project's reference region, a search was carried out on the Mining Geographic Information System (SIGMINE) website of the National Mining Agency (ANM). From the data collected, it was possible to see that there are no mineral resources with potential for exploitation within the Seringal Rio Branco REDD+ Project area. However, in the reference region to the north, the occurrence of some mineral resources with exploitation potential was identified, such as: Cassiterite (9805.61 ha); Amethyst (1709.97 ha); Diamond (11388.98 ha); Gold Ore (2664.46 ha).

Geological aspects:

To determine the geological aspects, information was gathered from the Environmental Information Database (BDiA) of the Brazilian Institute of Geography and Statistics (IBGE), where it was possible to verify the presence of the following lithostratigraphic units in the project's reference region: Structural Subprovincials; Indiscernible Cenozoic Coverage; Moa-Divisor; Solimões (Figure 5).

Considering the Seringal Rio Branco REDD+ project area, the predominant geological formation is the Solimões, which predominates in almost the entire state of Acre, extending across the border into Peruvian and Bolivian territory. This geological unit has the characteristic of being largely covered by sedimentation associated with the fluvial paleosystem of the Içá Formation, being exposed in the areas close to the valleys,

especially in those in the process of resuming erosion, in addition to presenting planar-parallel and cross tabular and channeled stratifications of small, medium and large amplitude and its sedimentation was related to the general subsidence of the basin, with submergence and uplift of varying intensity, resulting from the reactivation of old faults (Cavalcante, 2005)¹⁸.

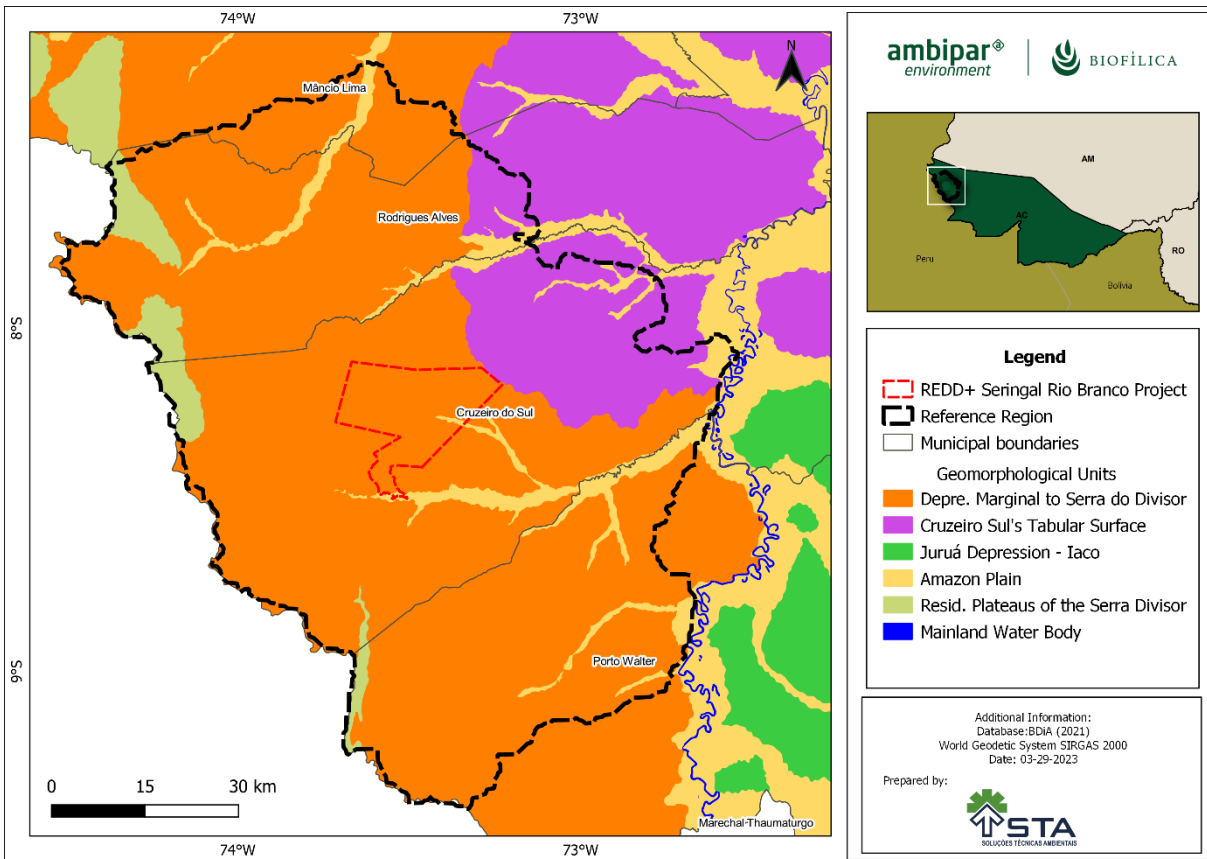


Figure 5: Geomorphological units of the reference region of the Seringal Rio Branco REDD+ Project. Source: BDIA/IBGE (2021).

Pedological aspects:

Acre's soils are of various orders, ranging from Latosols (well-developed soils) to Fluvial Neosols (young soils) (Do Amaral et al., 2021)¹⁹. In terms of soil composition, the reference region of the project area includes regions with: Red-yellow Argisol; Haplic Gleissol; Chromic Luvisol; Yellow Argisol; Argiluvial Plinthosol. The project area, on the other hand, predominantly has the occurrence of Red-Yellow Argisols, as shown in figure 6.

The Red-Yellow Argisols (PVA) have a clay accumulation horizon, B textural (Bt), with yellowish-red colors due to the presence of a mixture of the iron oxides hematite and goethite, characterized by being a deep

¹⁸ CAVALCANTE, L. M. Geological and geomorphological zoning between Feijó and Mâncio Lima - Acre. Rio Branco: Embrapa Acre, 2005. 24 p. il. color. (Embrapa Acre. Documentos, 99). 2005.

¹⁹ DO AMARAL, E. F.; LANI, J. L.; BARDALES, N. G.; DE ARAÚJO, E. A. Origin, formation, and diversity of the soils of Acre. Chapter 2. 2021.

and very deep soil, well-structured and with good drainage. In addition, the surface horizon A of the moderate and prominent type is predominant, with a medium/clayey texture, with the possibility of presenting medium/medium and medium/very clayey texture less frequently, in addition, they are soils whose natural fertility varies from low to very low, with a strongly acid reaction and low activity clays (Da Silva; Neto, 2021)²⁰.

This type of soil has the potential to be used to grow sugar cane, fruit, manioc, passion fruit and yams. However, in order to make good use of it, it needs to be fertilized and limed, given its medium to low natural fertility.

In this sense, due to their low and medium natural fertility in the Tabuleiros Costeiros, these soils need correctives and fertilizers in order to achieve good crop productivity, as well as the use of organic matter in the surface horizon, especially in sandy-textured soils.

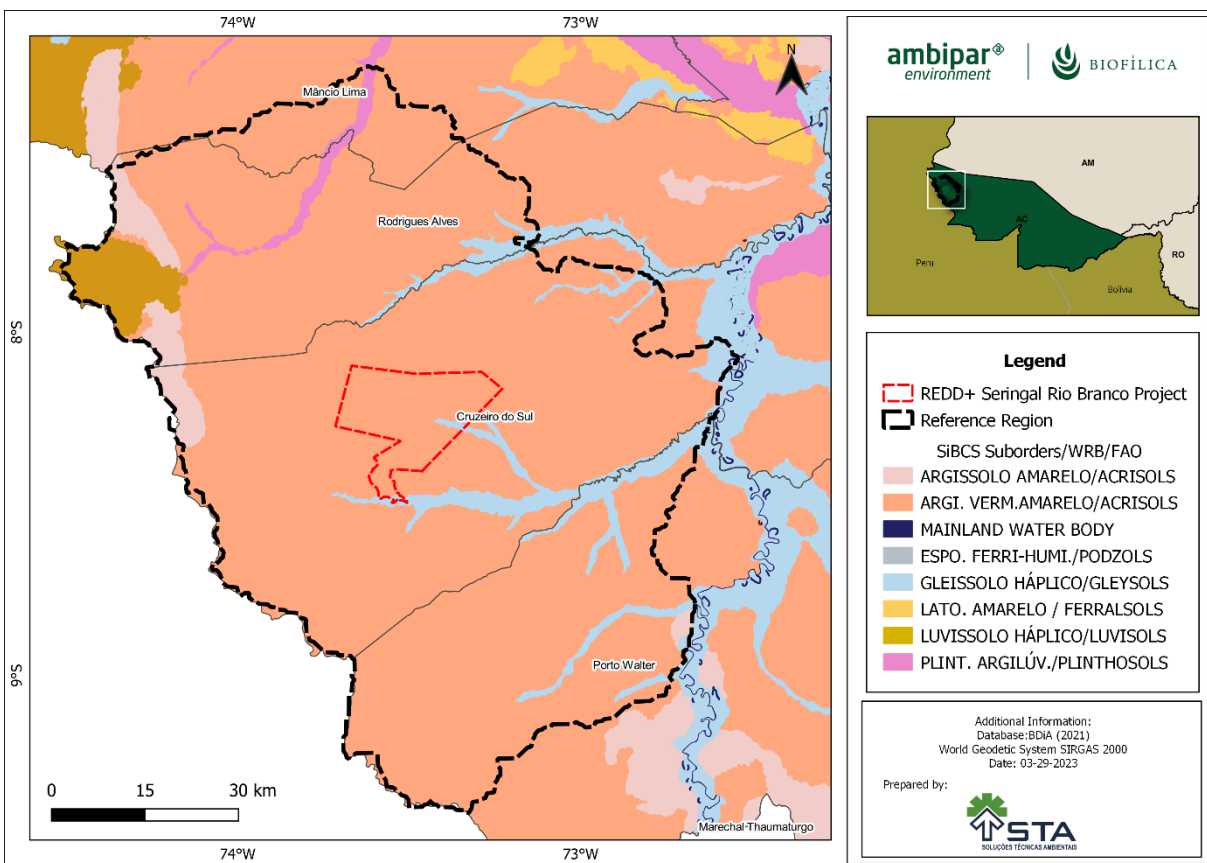


Figure 6: Pedological characterization in the reference region of the Seringal Rio Branco REDD+ Project.
Source: BDIA/IBGE (2021).

2.1.15 Social Parameters (VCS, 3.18; CCB, G1.3)

²⁰ DA SILVA, M. S. L.; NETO, M. B. de O. Red-Yellow Argisols. Brazilian Agricultural Research Corporation - EMBRAPA. 2021.

The occupation of the region known as the Juruá Valley, where the Seringal Rio Branco REDD+ Project is located, dates back to the ancestral occupation by indigenous populations. Divided into 49 tribes, these peoples, belonging to Pano-speaking ethnic groups such as the Ararauas, Catukinas and Curimas, inhabited the Juruá Valley, with the Náuas tribe on the banks of the Juruá River (CRUZEIRO DO SUL, 2023a)²¹. The municipality of Cruzeiro do Sul, initially inhabited by indigenous tribes, bore witness to this cultural heritage.

Porto Walter, which also has indigenous roots, is home to descendants of the Arara, Náuas, Amoaças, Kampas, Kulinas and Catianos nations. At the end of the 19th century, the arrival of migrants, mainly from northeastern Brazil, driven by the extraction of latex from the rubber tree, marked an ethnic transition in the region (PORTO WALTER, 2023)²². The formation of the rubber plantations attracted not only northeasterners, but also migrants from Peru.

The Seringal Rio Branco REDD+ Project area is located in the Juruá Valley Mesoregion, covering eight municipalities such as Cruzeiro do Sul, Tarauacá, Feijó, Mâncio Lima, Rodrigues Alves, Marechal Thaumaturgo, Porto Walter and Jordão, totaling 85,448 km² with a population density of 2.8 inhabitants/km² and an estimated population of 242,796. The Serra do Divisor National Park, present in the region, is an important environmental landmark (CRUZEIRO DO SUL, 2023a).

An analysis of the nearby riverside communities reveals that most of them were formed in the second half of the 20th century, between 1900 and 1950. The most recent community, Volta Grande, emerged in 1984. The Francisco Pimentel PDS settlement, established in 2005 and regularized by INCRA, represents a more recent initiative aimed at sustainable development (STA, 2023).

In general, the formation of the communities around the project is deeply rooted in the history of the local families. These families, who come from the rural region itself, were born in the communities where they live or in nearby localities in the municipalities of Cruzeiro do Sul and Porto Walter. The composition of these communities reveals a notable absence of significant migratory processes, highlighting the families' strong ties to the region.

Currently, the population of the group of nine communities around the project studied totals 783 inhabitants, distributed among 178 families. The Francisco Pimentel PDS, the largest among them with 171 families, has an estimated 684 residents, resulting in a total population in the project area of 1,467 people, distributed among 345 families.

During the field research, the communities provided data on gender distribution, revealing that the majority of the population is made up of men, accounting for 55%, while women account for 45%. This disparity contrasts with the state average, according to the 2010 Census, which indicates that 49.8% of the population of the state of Acre is made up of women, while men account for 50.2%. This analysis highlights the specific gender dynamics in the communities studied in relation to the state's demographic composition.

Table 1: Number of men and women in the communities. Source: Field research, 2023. Prepared by: STA, 2023.

²¹ CRUZEIRO DO SUL. Dados do município. Disponível em: <https://www.cruzeirodosul.ac.gov.br/municipio>. Acesso em 19/03/2023a.

²² PORTO WALTER. Dados do município. Disponível em: <https://www.portowalter.ac.gov.br/municipio>. Acesso em 19/03/2023.

Communities	Male	Female
Aldeota 2	37	37
Monte Cristo	97	56
Periquito	26	28
Prainha 1	63	62
Prainha 2	60	58
Primavera	75	40
Santo Antônio 1	21	22
São Luís	42	35
Volta Grande	12	12
Total	433	350

With regard to education, data shows that in the municipalities that make up the region of the communities studied, the school infrastructure is limited, resulting in low overall levels of schooling and high dropout rates, especially among young people. The table shows that in Cruzeiro do Sul, 71% of the population aged 10 or over has incomplete elementary education, while in Porto Walter this figure is 64%, compared to 70% in the capital, Rio Branco. The illiteracy rates for the population aged 15 and over are 18.7% in Cruzeiro do Sul, 34.9% in Porto Walter and 8.9% in Rio Branco.

Table 2: People aged 10 and over with no schooling and incomplete elementary education. Source: IBGE (2010). Prepared by: STA. 2023.

Color or Race	Cruzeiro do Sul	%	Porto Walter	%	Rio Branco	%
Yellow	357	1%	78	2%	2.881	2%
White	7.070	20%	1.093	23%	26.183	21%
Indigenous	539	2%	278	6%	164	0%
Brown	25.178	71%	3.028	64%	89.045	70%
Black	2.383	7%	282	6%	8.347	9%
Total	35.527	100%	4.759	100%	126.620	100%

In addition, the lack of daycare centers forces families to resort to private classes taught by community members for children up to 5 years old. In some localities, such as Aldeota 2, Santo Antônio, São Luís and Volta Grande, children have to travel to other communities to access education.

Access to health services in the surrounding communities is limited, characterized by a lack of infrastructure for basic care. Visits by community health workers are irregular, in some cases only occurring annually. In emergencies, families usually go to hospitals in Cruzeiro do Sul in search of medical assistance, as there are no medical facilities in the communities themselves.

In addition, the basic sanitation indicators in these communities are in line with the low levels observed in the surrounding municipalities. With regard to drinking water, the absence of a general distribution network is evident, with the river and streams being the main source of water for most families. According to the project's diagnosis, the lack of public sewage systems is notable, with the majority of houses (89%) having

less than half a bathroom or toilet, while 11% have none, as in the Santo Antônio 1 community. Inadequate sewage disposal, often dumped directly into nature or into the river, is an identified practice.

In addition, the problem of solid waste management is evident, since no community has a collection service. Families often burn waste on their plots (53%), throw it into the river or stream (24%), or leave it out in the open in local areas (24%). This situation contributes to the pollution of the Juruá-Mirim riverbed and streams, affecting the local environment.

In the communities studied, communication is challenging, especially the dependence on the radio to receive messages from family or friends in the city, since there is no internet access or cell phone network. Some communities rely on satellite dishes to obtain external information. In addition, the exchange of letters, carried by third parties along the river, is still a practice.

The lack of infrastructure also extends to the electricity supply, which is non-existent in the communities. To make up for this shortfall, the communities resort to motor generators or solar panels.

As for getting around, transportation options are limited to the waterways, using their own boats or those rented from third parties. This reality highlights the challenges faced by the communities, not only in terms of communication, but also in terms of access to essential services and interconnection with other locations.

The region's economic activities are mainly subsistence farming, combined with agro-extractive practices, small animal husbandry and even cattle breeding. Table 3 lists the means of obtaining family income, prioritizing the main productive activities that contribute to generating income. In addition to local activities, many families rely on social benefits, retirement or pensions as a source of livelihood.

Table 3: Main products sold in order of importance. Source: Field research, 2023. Prepared by: STA, 2023.

Communities	Classification of the main products sold			
	1st Product	2nd Product	1st Product	4th Product
Aldeota 2	Flour	Cattle	-	-
Monte Cristo	Flour		-	-
Periquito	Flour	Cattle	-	-
Prainha 1	Flour	Cattle	Piassava and titica brooms	Pig
Prainha 2	Flour	Cattle	-	-
Primavera	Flour		-	-
Santo Antônio 1	Flour		-	-
São Luís	Flour		-	-
Volta Grande	Flour	Cattle	-	-
PDS Francisco Pimentel	Flour	Corn/Rice	Pig	Açaí/Bacaba

The predominant family farming activity is the production of manioc to make flour. The agricultural production method adopted is known as itinerant agriculture, characterized by a few years of cultivation followed by long periods of resting the land. This method involves practices such as cutting, felling, and burning the land. Along with family farming, in the context of these communities, livestock farming, even on a small scale, plays a significant role. During the field research, it became clear that beef cattle breeding is an activity present in all the communities, contributing to the subsistence of some families. This practice

not only represents an additional source of income, but also highlights the diversification of economic activities in the communities, promoting the sustainability and resilience of their production practices.

2.1.16 Project Zone Map and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

Seringal Rio Branco, the rural property where the REDD+ project is located, is situated in the municipality of Cruzeiro do Sul, in the far west of the state of Acre, close to the nation of Peru (Figure 7). Its southern boundary is adjacent to the Juruá Mirim River and close to the Serra do Divisor National Park conservation unit, which reinforces the importance of the project for the extension of the protected area and for the conservation of local biodiversity (Figure 8). The property is located at the following geographical coordinates: longitude -73.246723 and latitude -8.120174, in the geographical coordinate system Datum SIRGAS 2000.

In addition, the property is located in the Amazon Biome. The Amazon Biome, covering more than 40% of Brazil's territory with an area of approximately 4,196,943 km², is mostly made up of a vast tropical rainforest. This biome stretches across several states, including Acre, Amapá, Amazonas, Pará, Roraima, parts of Maranhão, Mato Grosso, Rondônia and Tocantins, and is characterized by a diversity of ecosystems, such as dense dryland forests, seasonal forests, igapos, floodplains and savannas. Despite being the most preserved biome in Brazil, it has already lost around 16% of its forest cover, mainly due to threats such as deforestation, fires, mining, agropastoralism and biopiracy. These activities not only devastate the region, but also contribute to global climate change, considering the importance of the Amazon as an atmospheric "cooler" and the largest reservoir of biodiversity in the world.

Information on the location of the communities identified is described in detail in section 4.1, as well as the characterization and impacts of the project on these communities. Considering the project's other stakeholders, their characterization, location, and impacts can be found in section 2.3. The project area, which comprises the entire farm, is described in detail in section 3.1.13 of this PDD. All the areas with impacts on biodiversity are described in section 5.3 of the project, as well as a full characterization of the region's fauna and flora.

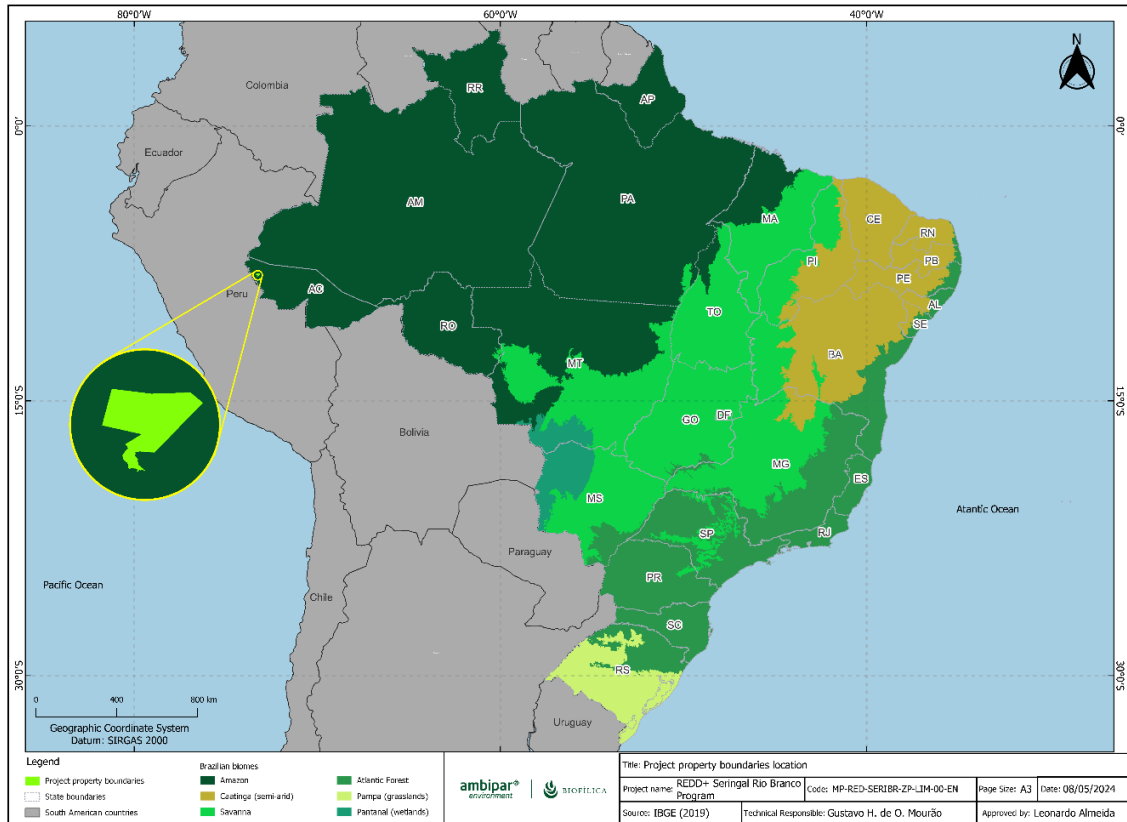


Figure 7: Location map of the Seringal Rio Branco REDD+ Project area.

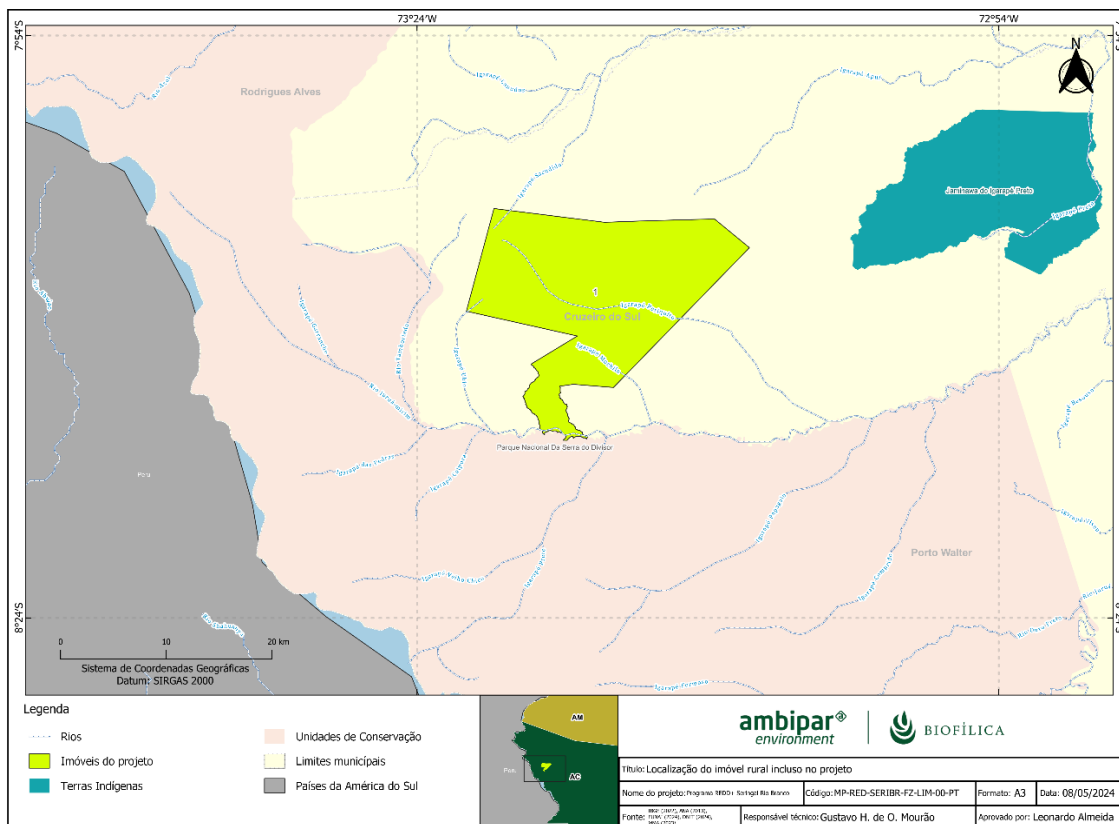


Figure 8: Location map of the Seringal Rio Branco REDD+ Project area.

2.1.17 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

A theory of change is a strategy that aims to change something, representing the "reasoning behind how and why a purpose or outcome should be achieved in a particular context". The use of results chains is key to addressing challenges, allowing project teams to articulate clear theories of change, focus efforts, prepare for effective performance monitoring and learn from experiences, identifying what works and what doesn't, as well as understanding the crucial conditions for success and practicing adaptive management.

In the context of designing the Seringal Rio Branco REDD+ Project, the results chain guided the application of the theory of change, representing a crucial step in the development of activities. This process helps developers clarify how their actions support the proposed strategic approach and will influence change in the problems identified.

The Seringal Rio Branco REDD+ Project aims to promote collaborative actions to generate net benefits for the climate, communities, and biodiversity. Its activities are mainly aimed at ensuring the conservation and protection of biodiversity and natural resources, reducing deforestation and greenhouse gas emissions, promoting local socio-economic development, social inclusion, stakeholder engagement and adaptive and assertive governance.

To ensure adequate funding to meet the objectives and maintain them throughout the life cycle of the Seringal Rio Branco REDD+ Project, financial resources will be generated, mainly through the marketing of REDD+ credits registered with the Verified Carbon Standard (VCS), linked to social development and the conservation of natural resources.

To this end, the following series of activities have been proposed for the development of the Project, after careful evaluation and consideration of the ideas and wealth of knowledge accumulated for the Project Zone. In addition, the activities defined are divided into scopes for better understanding as described below:

General Scope

- **Initial studies and articulations**

The start of the project included the first on-site visit to recognize the area, where a long-term partnership was defined with a view to environmental conservation and socio-economic development in the region, as well as meetings with technical partners to begin the project's specific diagnoses.

Diagnostics are initial studies that provide technical support for drawing up the project management plan. Among the studies carried out are the survey of the estimated forest carbon stock and the preparation of the deforestation baseline, which results in an understanding of the direct climate impacts; the Socioeconomic Diagnosis and Consultation with the communities, which deepened the studies already carried out in the area and resulted in an understanding of the direct social impacts; and the Environmental Diagnoses which, like the Socioeconomic Diagnosis, supported the construction of actions to stimulate economic development and the sustainable use of natural resources, as well as guaranteeing the conservation of the standing forest, underpinning the proposed activities and resulting in direct impacts on communities and biodiversity.

- **Implementation, monitoring, and evaluation of project activities**

As a fundamental part of obtaining net benefits from a REDD+ Project, it must be monitored within the framework of the activities, indicators and monitoring plan proposed in this document. For this to happen, proper project management is required, ensuring that the activities are carried out within a well-constructed and realistic schedule, that possible risks are mitigated and, furthermore, that there is continuous and adaptive improvement for the effective implementation of the activities proposed by the project.

In a nutshell, this activity will focus on monitoring and improving project management, adapting to the real scenario, checking for flaws and risks, and finally developing a space dedicated to managing, documenting, organizing, and making official the project's milestones, the changes that have taken place and the adaptations made throughout the credit generation period. Always associated with efficient and better-guided decision-making during the implementation and monitoring stages. This will lead to more effective adaptive management, promoting more assertive decisions that are aligned with the premises outlined in the PDD.

Thus, the "implementation, monitoring and evaluation of the activities carried out by the Project" activity is based on efficient management throughout the Project's life cycle, thereby achieving the expected positive impacts. In order to achieve the objectives, actions will be taken such as producing reports on the project's activities, evaluating indicators and results, defining the planning schedule and task priorities for all the years, tactical plans to be followed and implementing tools that can express the milestones and adaptive changes encountered for subsequent evaluation of the activities with the stakeholders.

If conducted properly, the actions will lead to a greater understanding of the impacts generated and opportunities for redirecting demands in priority levels, establishing procedures and protocols to be followed by the stakeholders. That said, constant alignment and common strategies between stakeholders are essential if this activity is to take place and demonstrate, in detail, the efforts of those involved in

implementing and monitoring the project. All this helps the project proponents to continuously improve the implementation of the proposed activities by identifying bottlenecks, inefficiencies, and failures.

- **Strengthening governance**

With a view to implementing the activities proposed by the Seringal Rio Branco REDD+ Project, the "Strengthening governance" activity has the scope to set up a working partnership to work on site, facilitating the day-to-day monitoring of the implementation of the activities proposed in this document. In this sense, through the implementation of a working partnership with a local partner, procedures will be structured to assist in the implementation of the project's activities, as well as in the monitoring of these activities, helping to ensure that the project is implemented in the best possible way, as well as helping the proponents to monitor and operationalize the demands and activities proposed.

That said, the consolidation of a partnership will strengthen the governance of the Seringal Rio Braco REDD+ Project, since it establishes the on-site presence of Project representatives for constant assistance, being a central pivot in understanding the local reality. In addition, it is hoped that the partners will be routinely present in the area, reflecting a greater power to contain deforestation, since it strengthens the relationship with the communities around the farm, preventing illegal entry and activities, as well as monitoring the effectiveness of the other activities proposed by the project.

Furthermore, this partnership will aim to define better strategies for engaging, working together and strengthening all stakeholders. With this in mind, in order to facilitate interaction with all the stakeholders and co-management with all the actors that make up this scenario, as well as mapping and resolving suggestions and complaints, the activity proposes the action of implementing and consolidating a communication channel that should dialog with what was proposed in the Project's communication plan. Strengthening the communication channel will allow stakeholders to collaborate on decisions and the implementation of project activities, thus helping to improve the performance of this partnership and the project in terms of climate, community, and biodiversity benefits.

Theme	Objectives	Expected benefits for climate, community and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Initial Studies	Carbon stock estimation	- Estimate the carbon stock for the Forest class and its reservoirs, through the Forest inventory in the Project area; - Generate the technical report.	- Generate knowledge about biomass and carbon stocks in the project area and its surroundings; - Enable reservoir monitoring and adjustments to stock projections over the project's lifetime; - Contribute to accounting for reduced emissions.	- Generation of inputs for long-term forest monitoring; - Identification of priority areas for stock conservation.

Theme	Objectives	Expected benefits for climate, community and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
	Determining the baseline	- Carrying out the study to determine the spatial limits of the project and determine the deforestation baseline; - Generation of technical report; - Modeling future deforestation	- Generation of knowledge about the dynamics of deforestation in the region; - Contribution to the accounting of reduced emissions; - Determination of the area most at risk for conducting field actions.	- Generation of inputs for long-term forest monitoring; - Generation of relevant data for use by the government in projecting future jurisdictional systems.
	Socio-economic diagnosis	- Drawing up a contextualization of the municipalities, communities and workers who will work on the project - Carrying out the socio-economic study; - Generating technical reports	- Generation of the updated socio-economic context of the region; - Providing input for the design of proposed activities in line with the local context; - Providing input for the work of other stakeholders.	- Improved socio-economic conditions; - Long-term prevention of deforestation in the Project area; - Ensuring the proper management of forest areas and the use of other natural resources.
	Biodiversity Diagnosis	- Drawing up a contextualization of local biodiversity aspects; - Carrying out the Biodiversity Study; - Generation of technical report.	- Updating biodiversity studies; - Providing input for the design of proposed on site biodiversity monitoring activities; - Providing input for the work of other stakeholders.	- Improved knowledge of local biodiversity; - Long-term prevention of deforestation in the project area; - Ensuring proper management of forest areas and the use of other natural resources.

Theme	Objectives	Expected benefits for climate, community and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
	Situational alignment and feedback to communities	- Informing stakeholders of the REDD+ Project; - Identifying, understanding, and prioritizing the problems encountered in the region's communities; - Conducting interviews and workshops with the communities directly and indirectly involved to design and present the Project's activities; - Generation of technical reports.	- Allowing adaptable management of the project to incorporate the needs and reality of the families; - Defining parameters to measure the benefits and impacts of the project on the communities; - Sharing information on REDD+ and promoting community involvement.	- Strengthening communication between stakeholders; - Improving the quality of life and socio-economic aspects of the communities; - Empowering communities with regard to their rights, duties and the importance of involvement in the project.
Implementation, monitoring, and evaluation of activities developed by the Project	The project aims to monitor the implementation of the proposed activities in order to mitigate risks and ensure continuous improvement of activities, increasing the quality of Project Management.	- Producing reports on the project's activities; - Evaluating the indicators and results of the actions implemented; - Defining the planning of activities for the following year; - Implementing milestones for evaluating the project's activities with stakeholders.	- Greater understanding of the impacts generated with stakeholders; - Opportunity to redirect project activities towards those of higher priority; - Refinement of data collection methods.	- Efficient project management and implementation; - Greater achievement of the project's expected impacts.

Theme	Objectives	Expected benefits for climate, community and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Strengthening Governance	By implementing an on-site partnership, the project aims to make it easier to monitor the implementation of the proposed activities on a day-to-day basis.	- Implementation of a working partnership to work on the project on-site; - Consolidation of procedures with the local partner to implement the project; - Implementation and consolidation of a communication channel with stakeholders.	- Implementation of project activities with stakeholders; - Stakeholders with access to project information; - Stakeholder collaboration and participation in decisions and implementation of project activities; - Mapping and resolution of stakeholder suggestions and complaints; - Creation of opportunities for sharing experiences.	- Strengthening governance; - Efficient management and implementation of the project; - Strengthening communication between stakeholders, helping the project to act on climate, community and biodiversity benefits.

Scope Climate

- **Improving asset surveillance within Seringal Rio Branco**

The activity of asset surveillance is closely linked to climate benefits, mainly aimed at mitigating and preventing deforestation by third parties in the Project Area, as well as reducing greenhouse gas emissions. Currently, the surveillance activities in the area are carried out by Seringal Rio Branco workers themselves, but without well-defined procedures and conduct. The proposal is to improve these processes, including training for the workers to take appropriate action in cases of illegal activities, as well as preventive measures against the unwanted entry of third parties. To this end, the project will seek to strengthen local partnerships, especially with inspection bodies. As part of these improvements, it is proposed to integrate the satellite image monitoring activity with asset surveillance, using unmanned aerial vehicles to expand monitoring in areas that are difficult to access. The satellite monitoring products will be used by the surveillance team to assess the areas detected, enabling more assertive action to be taken against illegal activities. In addition, the signposting of the project area with strategically installed signs on the banks of the river, the main access to the area, can help with asset surveillance, informing of the presence of the project and providing a channel of communication with the local community.

- **Monitoring deforestation via satellite images**

Remote monitoring is currently being carried out on the entire area of Seringal Rio Branco and its surroundings, by processing and analyzing satellite images due to human pressure. The assessment of human pressure focuses on identifying illegal logging, deforestation, forest degradation and possible fires.

The proponent of the project (Amazônia Agroindústria EIRELI) has signed a web interface called LandViewer. This interface is a direct market product for the public and different combinations of bands and images can be viewed in the areas of interest. This monitoring will result in bulletins with deforestation hotspots that will be forwarded to interested parties and, in this way, this remote monitoring will provide better support for the strategic patrol plan in the field. In addition, it is essential to focus on installing remote monitoring systems with forest fire alerts for properties, allowing for early detection and an agile response to fire outbreaks.

That said, this activity envisages maintaining and improving this remote monitoring of deforestation, developing reports that track and record deforested areas in the Project Zone, as well as strengthening and formalizing communication with surveillance actions.

These actions are directly related to containing deforestation and invasions, maintaining forest cover, controlling fires and, consequently, improving adaptations to climate change and, consequently, maintaining the benefits for the climate, community and biodiversity predicted by the scenario with the Project.

Theme	Objectives	Expected climate benefits		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Structuring an Asset Surveillance Program	The project aims to structure surveillance activities more broadly focused around the property, which will be guided by monitoring with satellite images.	- Map vulnerable invasion points to install surveillance posts; - Draw up procedures for interventions in the field; - Train and qualify security guards; - Develop communication strategies between the surveillance team and the remote monitoring team. - Implementation of signs to indicate the location of the Project.	- Greater understanding of the dynamics and agents of deforestation; - Conducting effective asset surveillance; - Re-establishing relations with the environmental inspection body and reporting illegal activities; - Identifying the presence of the Project in the area.	- Mitigation and prevention of deforestation; - Reduction of emissions from deforestation and forest degradation. - Knowledge and establishment of a communication channel by local communities.

Theme	Objectives	Expected climate benefits		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Monitoring deforestation via satellite images	The project aims to remotely monitor the entire area of the farm and its surroundings, by processing and analyzing satellite images to assess human pressure, and to produce bulletins informing of deforestation hotspots to strengthen surveillance actions.	- Monitoring and evaluating deforestation areas remotely; - Generating deforestation reports;	- Greater understanding of the dynamics of deforestation in order to conduct more effective surveillance; - Providing inputs for designing and improving interventions in the field;	- Mitigation and prevention of deforestation; - Reducing emissions from deforestation and forest degradation.

Social Scope

The socio-economic diagnosis of the Seringal Rio Branco REDD+ Project region revealed that there are 1,467 residents, 374 families and 10 communities in the Project Area. The activities were also defined by incorporating and addressing the main focal problems associated with these people's way of life, as well as the best opportunities for the Project to act. The project aims to reach as many people as possible over its 40-year duration, promoting and expanding community access to the activities provided for in the social scope of the Seringal Rio Branco REDD+ Project.

For this reason, all the activities of this nature that have been prescribed for the project will have a prior action that will focus on mapping and identifying potential stakeholders to join the activities as well as the main opportunities that are in line with the proper focus of each activity.

The details and specifics of this mapping applied to each social activity developed for the Project will be further detailed in the following activities:

- **Strengthening Non-Timber Forest Management in Seringal Rio Branco**

Gathering information on the uses of non-timber forest products revealed a significant variety of fruit species in the areas surrounding the communities, such as Açaí, Buriti, Bacaba, Patoá, Pupu da mata, among others. These areas are fundamental for meeting the basic needs of the local communities, being used to extract products for subsistence, especially for feeding families, and for the reserve of medicines and household utensils.

The use of these non-timber resources requires attention since, if not well controlled and recorded, it can have negative impacts on the forest and local biodiversity. These disorderly actions increase the risk of unsustainable extractions, favoring malicious practices related to illegal deforestation and forest degradation, contributing to an increase in greenhouse gas (GHG) emissions.

To deal with these challenges, monitoring community activities is essential. This monitoring aims to strengthen management of the territory, mitigating the risks of activities that could trigger deforestation and GHG emissions. In addition, the project seeks to map opportunities for improving the use of non-timber forest resources, promoting training and other actions in line with sustainable practices.

In this way, the initiative seeks not only to control the practices of extracting non-timber forest products, but also to contribute to the sustainable development of activities, promoting the appreciation of sustainable practices, generating income for stakeholders, strengthening territorial ties, keeping families in the territory, and preserving High Conservation Value Areas (HCVAs). This approach aims, over time, to promote socio-economic development in the Project Zone.

- **Promoting sustainable practices**

The communities located around Seringal Rio Branco rely on the forest as a vital source of natural resources for their subsistence, especially when it comes to food, extracting phytotherapeutic products and generating income for the families. The extraction of forest products, both non-timber and timber, guides a large part of the activities carried out by these families.

The diagnosis also identified the various economic activities carried out on the surrounding rural properties. The production of manioc flour stands out, accompanied by beef cattle, animal husbandry, the extraction of non-timber forest products (such as açaí and babaca), furniture making and forest management. In this context, the Seringal Rio Branco REDD+ Project proposes to extend its actions to include rural producers as well.

Given this scenario, the project proposes the activity of "Fostering Sustainable Practices", directing actions towards mapping the main development opportunities focused on sustainable practices. This will be done in line with the demands of communities and other stakeholders, with the aim of strengthening existing practices and those that have potential for implementation.

With this in mind, the project seeks to identify stakeholders with the potential to take part in actions to promote sustainable practices, implement partnerships for the development of activities, adding strength and financial resources.

In addition, the project will act on the educational basis of the communities, promoting training and other actions aligned with the mapped lines, which will contribute to sustaining these practices over time. Increasing awareness of the importance of maintaining natural resources and ecosystem services, along with the need to reconcile economic practices with sustainable development, will be a central focus. The promotion and articulation of environmental education actions play a crucial role in consolidating knowledge about the protection and conservation of the environment, in harmony with economic and subsistence activities, in order to avoid environmental damage and impacts.

- **Promoting environmental education for hunting and fishing**

Most of the communities close to the Project base their subsistence on hunting and fishing, considering these practices essential to guarantee an adequate diet. These activities are common to all the communities mapped in the Project Zone and are carried out in nearby areas and within the Project itself in order to preserve their food and subsistence culture.

Despite being considered fundamental to subsistence in the Project Zone, hunting and fishing are often associated with unsustainable and illegal practices, resulting in a decrease in local biodiversity.

It is important to note that the mapping of stakeholders and the formation of partnerships for the development of these actions will be carried out in line with the previously proposed activity of "Promoting Sustainable Practices". Thus, both activities aim to work with the same families and communities, seeking partnerships that meet the proposed objectives in an integrated manner.

As far as specific actions for this activity are concerned, they will be carefully planned to take into account the cultural elements associated with hunting and fishing, focusing on practices considered inappropriate, but respecting local habits and customs. Efforts will be directed more assertively towards the species that are culturally hunted and fished by the communities, concentrating on habits that are considered unsustainable.

However, the main difference in the commitment to sustainable practices is centered on the promotion of environmental education actions. This means addressing the root of the essential issues related to hunting and fishing, with the aim of establishing a more solid educational base, environmentally speaking, for the communities mapped by the Project. This approach seeks to transform the communities' relationship with traditional practices, making them more aware and aligned with long-term environmental preservation.

- **Strategic articulations for infrastructure and community development**

Riverside communities face significant challenges due to poor infrastructure and difficulties in accessing essential services. With regard to electricity, its absence in the communities requires the use of alternatives such as motor generators or solar panels to meet this basic need, which is often unfeasible given the financial conditions of each family. They also don't have access to the internet or other means of communication.

The lack of basic infrastructure, such as access to drinking water and sanitation, presents a significant challenge in isolated riverside communities. These populations often face difficulties in ensuring a regular supply of safe drinking water, thus increasing vulnerability to waterborne diseases. In addition, the lack of proper basic sanitation contributes to unsanitary conditions, becoming an obstacle to the sustainable development of these communities. The search for solutions that address these shortcomings becomes crucial to improving living conditions and promoting well-being in these isolated regions.

Transportation, which is vital for interaction between communities and access to urban resources, is restricted to waterways. Families use their own boats or those rented from third parties, facing a lack of land transportation options, which translates into isolation and prolonged periods without leisure or communication alternatives.

The challenge is exacerbated during periods when it is difficult to get around, in addition to the high costs, making it difficult to travel to deal with various issues in the city, such as access to education, health and banking services. This relative isolation has a direct impact on daily life, making it difficult for children to access higher levels of education.

Given this scenario, the project proposes actions that promote the engagement and coordination of governmental and non-governmental organizations to carry out actions aimed at improving infrastructure for these communities. This will be done in line with the demands of the communities and other interested parties, with the aim of strengthening existing practices and those that have potential for implementation.

With this in mind, the project seeks to identify interested parties with the potential to take part in actions to foment local needs, implementing partnerships for the development of activities, adding strength and financial resources.

Theme	Objectives	Expected benefits for the community		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Strengthening non-timber forest management in Seringal Rio Branco	The project aims to ensure that the NTFP extraction practices developed by community members around Seringal Rio Branco are maintained throughout the project's life cycle. To make this possible, the project will work to improve control over the NTFP extraction practices carried out by community members within the Project Zone and will contribute to better development of these activities, through training, safety and access procedures and protocols, as well as the search for partnerships that will boost the extraction of these products.	- Mapping stakeholders with the potential to take part in non-timber forest management actions within Seringal Rio Branco; - Mapping the main opportunities; - Delimiting and implementing procedures and protocols on non-timber forest management in Seringal Rio Branco; - Mapping and implementing partnerships for the development of activities; - Promoting training and other actions/interventions within the potential lines mapped by the project; - Monitoring stakeholders and their activities/actions developed within Seringal Rio Branco.	- Increased asset security; - Increased governance of Amazônia Agroindústria EIRELI - Maintenance and protection of HCVA 5; - Enhancement of sustainable practices; - Income generation for stakeholders involved in the actions; - Increased awareness of the importance of natural resources and ecosystem services.	- Improving the well-being of communities; - Mitigating and preventing deforestation; - Reducing emissions from deforestation and forest degradation.

Theme	Objectives	Expected benefits for the community		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Promoting sustainable practices	The activity aims to map out the main development opportunities, with a focus on improving sustainable practices applied to the livelihoods of potential stakeholders. The aim is to guarantee the sustainability of these families' subsistence activities, while encouraging new forms of production. This not only helps communities adapt to climate change, but also contributes to their sustainable development.	- Mapping stakeholders with the potential to take part in actions to promote sustainable practices; - Mapping the main development opportunities focused on sustainable practices; - Mapping and implementing partnerships for the development of the activities; - Training and other actions/interventions within the potential lines mapped by the project; - Promoting and coordinating environmental education actions in scenarios where there is synergy with the actions implemented.	- Development and enhancement of sustainable practices; - Income generation for the stakeholders involved in the actions; - Increased awareness of the importance of natural resources and ecosystem services; - Strengthening of territorial ties; - Permanence of families in the territory.	- Promoting socio-economic development in the Project Area; - Empowering community members; - Mitigating and preventing deforestation; - Reducing emissions from deforestation and forest degradation.
Promoting environmental education for hunting and fishing	Hunting and fishing are essential for the subsistence of families in the Project Area, but in the long term, these resources could become scarce and a major threat to local biodiversity. In order to reduce the damage caused by these activities, the project plans to combat inappropriate practices through environmental education, but without disregarding the cultural elements associated with the activity.	- Identification of the cultural elements associated with the hunting and fishing culture by the stakeholders; - Mapping of the unsustainable practices used for hunting and fishing; - Definition of environmental education solutions and actions that will be implemented by the project; - Mapping and implementation of partnerships for the development of the environmental education actions to be implemented.	- Improving the practices used for hunting and fishing; - Promoting and giving families access to environmental education; - Building values, knowledge, skills, attitudes and competencies aimed at conserving the environment for stakeholders.	- Reducing the impacts generated by hunting and fishing in the Project Zone; - Developing sustainability for activities linked to hunting and fishing; - Cultural preservation and guaranteeing better subsistence practices for families in the Project Zone.

Theme	Objectives	Expected benefits for the community		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Strategic articulations for infrastructure and community development	This activity aims to map development opportunities aligned with local demands, strengthening existing practices and boosting sustainable initiatives in partnership with stakeholders, governmental and non-governmental organizations, with the aim of promoting the insertion and maintenance of vital infrastructures in riverside communities.	- Mapping local conditions and their problems and deficits; - Identifying and involving stakeholders; - Strengthening Local Practices with Institutional Collaboration; - Implementing strategic partnerships with public authorities and other entities.	- Improvement in the practices used to get around; - Improvement in basic sanitation systems and drinking water; - Improvement in the basic health system; - Promotion of and access to communication; - Involvement and more active action by public authorities.	-Reducing the impacts of geographic isolation; - Improving the quality of life of riverside populations;

Scope Biodiversity

• Biodiversity Monitoring

In terms of biodiversity, the main objective of the Seringal Rio Branco REDD+ project is to promote activities to maintain biodiversity and forest cover in the project area. Located in the biogeographical subdivision of the Amazon region called the Inambari Endemism Center, the project area includes a great diversity of endemic fauna species, as well as a significant number of individuals of flora species endemic to the state of Acre and endangered species such as *Swietenia macrophylla* King (Mahogany) and *Cedrela odorata* L. (Cedar). The diagnostics carried out in the area showed the presence of 147 identified species of fauna, 12 of which were classified as vulnerable and 2 listed as species nearly threatened with extinction, while the floristic aspect confirmed the occurrence of 118 species in the project area, 5 of which were categorized with some degree of threat.

Considering the historical context of the region, the culture of predatory hunting and fishing is widespread among the local population and is used both for family subsistence and for craft, medicinal and commercial purposes. From this context, the project also aims to carry out activities aimed at raising awareness among the local community, seeking to promote understanding of the benefits of the project and, at the same time, the environmental damage linked to the irregular practice of hunting and predatory fishing. With this in mind, it is important to involve the local communities, given their fundamental role in the success of the project and the conservation of biodiversity in a sustainable way, thus contributing to the monitoring of the project.

In this way, the long-term aim of this activity is to generate a series of positive impacts on biodiversity, such as conservation of the species diagnosed, including those with some degree of threat, preservation of local

habitats, conservation of HCVA's, generation and dissemination of knowledge, dissemination of results to stakeholders, permanence of ecosystemic services, mapping of new areas of great relevance for conservation and maintenance of connectivity in the landscape.

Theme	Objectives	Expected benefits for biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Biodiversity Monitoring	The purpose of this activity is to monitor fauna and flora in the project area over time, contributing to the conservation and protection of local biodiversity during the course of the project. To this end, the activity seeks to engage stakeholders in this monitoring.	Organization and implementation of long-term biodiversity monitoring in the Project Area;	- Greater knowledge of regional biodiversity; - Obtaining data on the population dynamics and behavior of species, including endangered, endemic, invasive and the main species targeted by predatory hunting in the region; - Obtaining data on the main species of fauna and flora used by the population for subsistence and commercialization; - Local environmental engagement and awareness of biodiversity and its importance; - Greater clarification of the difficulties encountered in this activity, providing adjustments and control of changes in the implementation of the Project.	- Engagement and empowerment of stakeholders on local biodiversity; - Robust database on local biodiversity; - Balanced ecosystem for resident biodiversity; - Maintenance of threatened and endemic species on site. - Consolidated knowledge of invasive species for sound decision-making;

2.1.18 Sustainable Development Contributions (VCS, 3.17)

One of the objectives of the Seringal Rio Branco REDD+ Project is to promote sustainable development in the region, with the joint actions of all stakeholders, facilitated by Amazônia Agroindústria EIRELI, driving net benefits for the climate, biodiversity and local communities. Based on this support and the expected impacts, the project will contribute to the following UN Sustainable Development Goals:

Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture		
2.4	By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, help maintain ecosystems, strengthen adaptive capacity to climate change, extreme weather, droughts, floods, and other disasters, and progressively improve land and soil quality.	

The project combines the socio-economic demands of the communities with local opportunities for more resilient economic activities, through "Fostering sustainable practices". To this end, the project provides environmental education and community training, in association with different partners and extension workers, in order to promote knowledge about the importance of reconciling good production practices with the preservation and maintenance of natural resources. In addition to fostering perceptions about the importance and advantages of sustainable production systems, the project also works to raise awareness among extractivist families about the value of sustainable exploitation of non-timber forest resources for their survival, income generation, well-being and, consequently, the maintenance of the standing forest.

Goal 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

4.4 By 2030, substantially increase the number of young people and adults who have relevant skills, including technical and professional skills, for employment, decent work and entrepreneurship.

4.7 By 2030, ensure that all learners acquire the knowledge and skills necessary to promote sustainable development, including but not limited to education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and the contribution of culture to sustainable development.

The project enables and encourages access to education by providing technical courses and training in environmental and socio-economic areas, especially in resilient agricultural production practices, sustainable extraction of non-timber forest products, education on hunting, fishing, and preservation of endangered species. It also provides specific training for the workers of Seringal Rio Branco, especially those who belong to the asset patrol, with knowledge and guidelines on the appropriate measures to be taken in the event of illegal activities in the area. To this end, it relies on the support and collaboration of specialized partners in order to guarantee the effectiveness and engagement of stakeholders. These training activities promoted by the project enable the strengthening of local governance, the consolidation of a sense of belonging, access to information, better employment conditions and the diversification of income; mainly for extractivists and small rural producers, consequently leading to the maintenance of the forest and its resources.

4 QUALITY
EDUCATION



Goal 12: Ensure sustainable production and consumption patterns

12.2 By 2030, achieve sustainable management and efficient use of natural resources

12.8 By 2030, ensure that people everywhere have relevant information and awareness of sustainable development and lifestyles in harmony with nature.

12 RESPONSIBLE
CONSUMPTION
AND PRODUCTION



The project includes "Fostering sustainable practices" with actions aimed at identifying potential activities related to resilient subsistence agriculture, sustainable livestock farming and low impact extractivism; according to the demand and profile of local communities. In this sense, the project works to disseminate knowledge, instructions and experiences focused on the efficient use of natural resources and environmental preservation; focusing on sustainable business chains through greater integration between stakeholders, thus generating income, well-being and cultural identity for the communities fostered. The learning, engagement, and willingness of these families to improve productive and extractive practices increases the governance of the project and helps to maintain the forest cover and preserve its ecological aspects.

Goal 13: Take urgent action to combat climate change and its impacts

13.3 Improve education, raise awareness and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning.

13.b Promote mechanisms to build capacity for climate change-related planning and effective management in the least developed countries, including with a focus on women, youth, local and marginalized communities.

In general, all the activities carried out by the project are aimed at taking action to combat climate change and its impacts and, consequently, reducing environmental degradation in the project area. In this sense, in addition to engaging stakeholders and strengthening governance, the project also monitors deforestation via satellite and in the field. Asset surveillance within Seringal Rio Branco, armed with geospatial information, is more effective and assertive in preventing and combating illegal activities. For this to happen, the project works to improve the registration and dissemination of pressured and deforested sites in the project area, as well as investing in training workers to make appropriate decisions in cases where irregular activities are identified. As a result, the project has the potential to reduce greenhouse gas emissions, collaborating directly with Brazil's emissions reduction target. 3.326.590 tCO₂e of GHG emissions in 10 years.

13 CLIMATE ACTION



Goal 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

15.1 By 2030, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains, and drylands, in accordance with obligations under international agreements

15.2 By 2030, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally.

15.5 Take urgent and significant measures to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2030, protect and prevent the extinction of endangered species.

15.7 Take urgent action to end poaching and trafficking of protected flora and fauna species and address both the demand for and supply of illegal wildlife products.

15 LIFE ON LAND



15.a	Mobilize and significantly increase, from all sources, financial resources for the conservation and sustainable use of biodiversity and ecosystems.	
15.b	Mobilize significant resources from all sources and at all levels to finance sustainable non-timber forest management and provide adequate incentives to developing countries to promote sustainable non-timber forest management, including for conservation and reforestation	
15.c	Strengthen global support for efforts to combat poaching and trafficking in protected species, including by increasing the capacity of local communities to pursue sustainable livelihood opportunities.	
Considering the importance of Seringal Rio Branco for biodiversity, the project focuses its efforts on long-term monitoring of ecological indicators associated with the practices resulting from REDD+ conservation activities on populations and ecosystems, both inside and outside the boundaries of the property. In this way, the project aims to minimize habitat loss, landscape changes, overexploitation of species and climate change. To this end, it seeks to engage, involve, and raise awareness among all interested parties about the importance of the biodiversity of fauna and flora in providing ecosystem services, maintaining the connectivity of the landscape, controlling environmental degradation and limiting the overuse of natural resources. Environmental education talks on hunting and fishing, training on sustainable livelihoods and workshops and activities on local ecological aspects are aimed mainly at workers and community members from the surrounding area, including extractivists and riverside communities, who circulate through the property and use forest resources. This participatory involvement, access to knowledge and results, and knowledge of good production practices; promote the strengthening of local governance and the inhibition of irregular actions; ensuring positive impacts on biodiversity in the project area.		

2.1.19 Implementation Schedule (CCB, G1.9)

The schedule with the main dates and milestones for the development of the activities of the Seringal Rio Branco REDD+ Project has already been presented in section 2.1.17 (Project Activities and Theory of Change). The summary schedule of these activities is shown in table below:

Date	Milestone(s) in the project's development and implementation
1 to 1.5 years before validation and first verification	Activity planning meeting
	Articulation between institutions and identification of partnerships
	Consolidation of the Activity Schedule
	Carrying out socio-economic and environmental diagnoses
	Carbon stock estimation
	Determining the baseline and potential for generating credits
	Planning and workshops for project design

	Feedback and consultation with stakeholders
	Consolidation of the design, action plan and draft project concept document
	Review and translation of the project design document (PDD)
	Drawing up the monitoring report
Year of validation and first verification	Selection and contracting of the Validation/Verification body and the Credit Register platform
	Production of validation/verification audit follow-up bulletins
	Field audit follow-up
	Project and Credit Registration
Year 3 to year 30	Development and monitoring of socio-environmental management activities
	Monitoring deforestation and emissions
	Monitoring Biodiversity (fauna and flora) and High Conservation Value Areas (HCVA)
	Verification of credits (Selection and contracting of the verification body; Production of Monitoring Bulletins for verification; Monitoring of the field audit; Registration of credits)
	Conducting the credit marketing process

2.1.20 Risks to the Project (CCB, G1.10)

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Lack of interest from stakeholders in participating in project activities, especially surrounding communities, and public bodies	The lack of interest in participating in the project's activities has a direct impact on the benefits that would be generated at community level. Furthermore, a lack of interest on the part of public bodies could jeopardize potential partnerships for the development of project activities.	Strengthening and encouraging greater involvement of all parties involved in the design and decision-making processes in relation to the Project's activities, through the proposed actions that strengthen human capital through access to training, capacity building and technical assistance to

		encourage sustainable and adaptive production practices and models. Another extremely important measure is linked to the improvement and dissemination of communication tools among the actors involved, such as communication channels and grievance redress procedures (Section 2.3.11).
Market risk - Difficulty in marketing verified carbon credits	The difficulty in marketing the credits generated by the Seringal Rio Branco REDD+ Project could have a direct impact on the proposed activities, consequently impacting the three scopes of the project - climate, community, and biodiversity, since it would not be possible to invest in the activities.	Constant search for new funding opportunities, businesses, and activities, such as partnerships and donations for direct use in the Project's activities (not necessarily linked to the sale of credits). In addition, the aim is to consolidate and expand the network of business contacts in order to publicize the Project and strengthen the developers' business relationships, especially for the sale of Non-Timber Forest Products. To this end, Biofíllica Ambipar has a robust marketing and commercial sector responsible for developing materials to publicize the project and participating in national and international events related to the subject.
Failure to communicate with stakeholders	Failure to communicate with stakeholders can hinder the progress of the activities proposed by the project, resulting in a failure to implement them effectively and achieve the expected goals. There is also the possibility of establishing potential conflicts.	Facilitated availability of the available communication channels and greater publicity of the channel for receiving suggestions and complaints for conflict management.
Failure to manage financial resources	Difficulty in implementing the activities proposed by the Seringal Rio Branco REDD+ Project, in addition to the difficulty in allocating resources for	Biofíllica Ambipar has a financial team responsible for managing the project's resources.

conservation and sustainable development in the region.

2.1.21 Benefit Permanence (CCB, G1.11)

All the activities designed for the project and their short-, medium- and long-term positive impacts have been carefully designed and planned, and the need for the results to become self-sustaining in the long term has been taken into account in this planning.

That said, in order to maintain and enhance climate, community and biodiversity benefits during the 100-year period as well as beyond, a number of the Project's activities are focused on improving local capacities for governance, management, education, promotion, decision-making and increasing awareness and capacity for sustainable resource management for all stakeholders. These activities will have short-, medium- and long-term impacts and will therefore help to empower and guide all stakeholders in self-determining sustainable pathways to achieve climate, community, and biodiversity benefits far beyond the lifetime of the Project.

In this way, the strategies associated with each activity so that the benefits occur during the life cycle of the Project and afterwards are:

- Improved asset surveillance procedures:

By providing additional tools such as remote monitoring using high-resolution satellite images and offering training for the asset surveillance team, the project aims to increase the efficiency of asset surveillance operations. In this way, surveillance operations will have a major boost in the intelligence process related to territorial monitoring and management, which should have a direct impact on maintaining climate benefits beyond the life of the Project.

- Non-timber forest management in Seringal Rio Branco:

With knowledge of the main opportunities aimed at improving the use of non-timber forest products, as well as promoting training and other actions within the potential lines mapped, the project aims to increase stakeholder awareness of the importance of natural resources and ecosystem services, valuing sustainable practices, as well as strengthening territorial ties and, consequently, reducing the rural exodus. Thus, the project should guarantee the long-term maintenance of the benefits generated, based on the empowerment of stakeholders regarding the extraction of non-timber forest products, income generation and, consequently, the mitigation and prevention of deforestation and forest degradation.

- Promoting sustainable practices and environmental education related to hunting and fishing:

By mapping opportunities for development and improving sustainable practices, promoting training and other actions within the potential lines mapped, as well as developing environmental education actions related to hunting and fishing, the project aims in the long term to promote socio-economic development in the region, the empowerment of communities, the reduction of impacts generated by hunting and fishing and, consequently, knowledge about these issues, promoting mitigation and prevention of deforestation and unsustainable activities, as well as the consequent reduction of emissions and pressure on local biodiversity. Therefore, the skills, training and educational activities behind these activities will be learned by the stakeholders throughout the life of the project and will be maintained, providing lessons learned and a greater enrichment of learning beyond the life of the project.

- **Biodiversity Monitoring:**

The Project's main focus is the implementation of biodiversity monitoring in the Project Area throughout its life cycle, but going forward, it will seek to provide feedback and improve stakeholder engagement in this activity, seeking environmental involvement and awareness. In this way, it is hoped that the benefits associated with these actions will help, in some way, to change the stakeholders' view of local biodiversity, providing people with greater knowledge and thus reducing the impacts caused by the conflict between man and nature beyond the lifetime of the Project.

- **Promoting the insertion and maintenance of infrastructures:**

The project proposes strategic actions, such as mapping development opportunities in partnership with communities and public authorities, with the aim of strengthening existing practices and implementing infrastructure improvements. The identification of stakeholders and the implementation of partnerships add robustness and financial resources to the initiatives, consolidating an integrated and sustainable approach to the progress of riverside communities.

2.1.22 Financial Sustainability (CCB, G1.12)

The project's proponents have a solid partnership signed in 2022 with the aim of enabling investments in the conservation of the forest reserves of Amazônia Agroindústria EIRELI through the commercialization of environmental assets. The Seringal Rio Branco REDD+ Project will be an initiative that should make possible, in the medium and long term, the continuous investment of resources aimed solely at conservation and sustainable development in the region.

Considering the current premises of the carbon market and the potential for generating GHG Emissions Reductions, the financial flow of the Seringal Rio Branco REDD+ Project presents very attractive results. In this model, the proponents expect to recoup the investment made in the project when the GHG Emissions Reductions start to be sold.

Other information related to the financial analysis of the Seringal Rio Branco REDD+ Project and the financial health statements of the partner institutions (project proponents) is considered commercially sensitive information and was shared with the audit team on a confidential basis.

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project

Characterisation of the biotic and abiotic environment

The project area is characterised by Open Ombrophilous Forest, an ecosystem native to the western region of Acre, with a high diversity of endemic species. The climate in the region is equatorial (hot and humid), with two well-defined seasons, and is classified as Af (tropical humid) according to Köppen, with a short dry season and high rainfall rates during the rainy season. The Seringal Rio Branco REDD+ Project area is located in a region of endemism called the Inambari Endemism Centre, characterised by a high diversity of endemic species and close to the Serra do Divisor National Park, one of the Amazon's most biodiverse regions. The predominant vegetation is open ombrophilous forest, and the region's soils are typical of tropical forests, rich in organic matter but subject to degradation due to deforestation activities. More details on the physical environment of the project area can be found in section 2.1.14.

Extension of the BR-364 motorway

In 2003, the presidents of Brazil and Peru, respectively, decided to strengthen and consolidate bilateral relations between the two countries, with the aim of making progress in building a strategic alliance²³. At the time, important convergences were noted on the shared vision of co-operation and integration that guided bilateral relations, with the key elements being the complementation of economies, the promotion of trade, physical interconnection, investments in tourism, the fight against drug trafficking, the development of technical co-operation for the preservation of the environment, the protection of natural resources in pursuit of sustainable development and the fight against poverty.

The main integration proposal is the project to extend the BR-364 motorway between Cruzeiro do Sul in Brazil and Pucallpa in Peru. This extension project aims to further strengthen trade relations and integration between the border regions of the two countries, creating an important land link between the cities and contributing to the economic and social development of the region.

With the aim of promoting closer border trade relations between the State of Acre in Brazil and the Ucayali Region in Peru, the President of the Ucayali Region and the Governor of Acre signed an 'Act of Intent for the Development of the Pucallpa - Cruzeiro do Sul Trade and Integration Hub' during the Governor's visit to Pucallpa on 12 March 2004. This act established a more open dialogue for the implementation of joint actions to strengthen commercial, cultural, environmental, tourist and technical cooperation.

The extension of the BR-364 motorway is currently being analysed due to the socio-environmental impact it will have on the region. In addition, two other state roads in Acre are being planned by the government, broadening the consideration of road development in the region.²⁴

²³ FÓRUM BINACIONAL. Fórum Binacional de Integração e cooperação para o desenvolvimento sustentável da região Ucayali - Peru e estado do Acre - Brasil. Documento de Instalação do Fórum Binacional. Cruzeiro do Sul/Acre - Brasil, julho de 2006

²⁴ FÓRUM BINACIONAL. Fórum Binacional de Integração e cooperação para o desenvolvimento sustentável da região Ucayali - Peru e estado do Acre - Brasil. Documento de Instalação do Fórum Binacional. Cruzeiro do Sul/Acre - Brasil, julho de 2006.

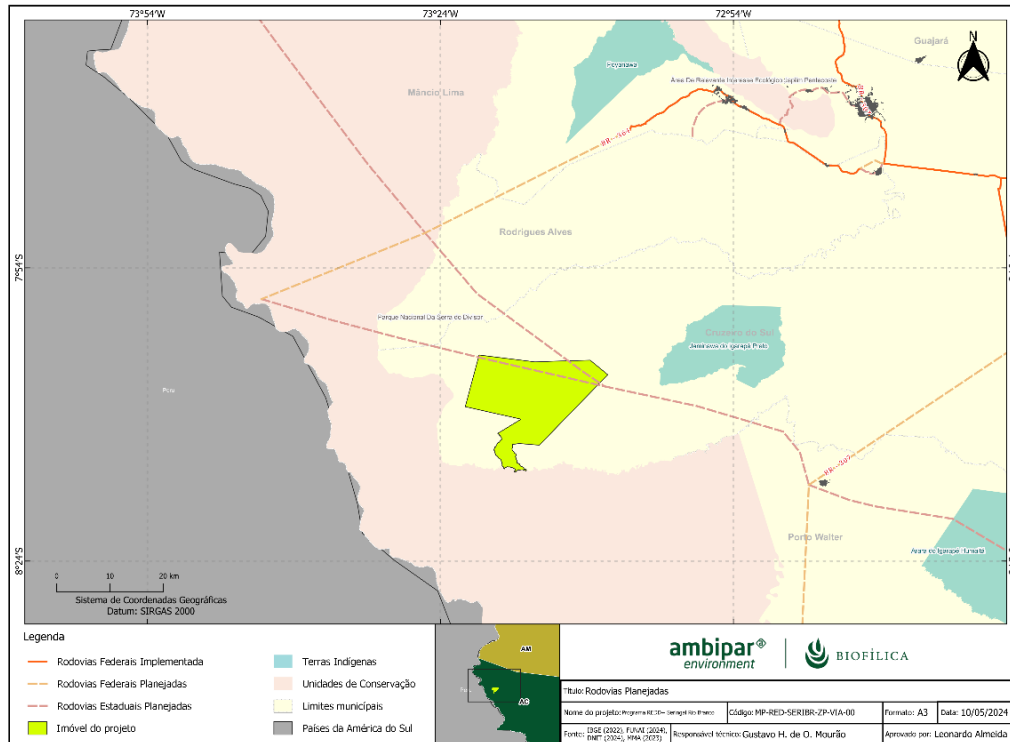


Figure 9 - Location map of planned federal and state roads in the Seringal Rio Branco REDD+ Project Zone

Oil exploration plan in the Project Zone

According to the National Petroleum Agency (ANP), the Acre Basin was entered into the 12th round of oil and gas bids by the National Petroleum Agency (ANP) in 2012. The so-called 'New Frontiers' areas, according to the ANP, aim to attract investment to regions that are still little known geologically or have technological barriers to overcome, enabling the emergence of new basins producing natural gas and conventional and unconventional oil resources (ANP, 2023).

Although there are currently no exploration activities, the Acre Basin plots included in the ANP's 12th bidding round are in the study, research and geological data survey phase. Should these plots be used for gas or oil exploration, there is a possibility that they will migrate to the Juruá Valley region, becoming a potential vector for deforestation in the future.

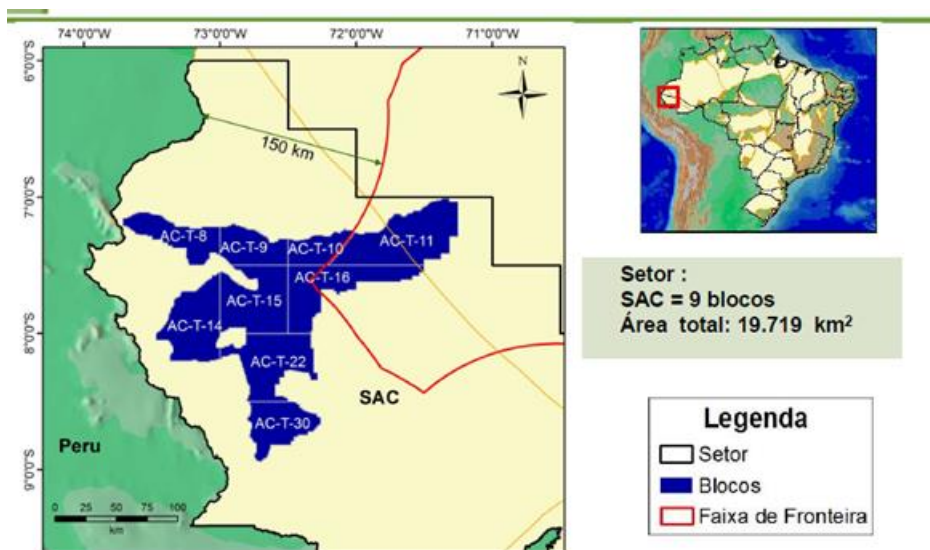


Figure 10 - Location of exploration blocks in the Acre Basin. Source: ANP (2023)

Access to the REDD+ area Seringal Rio Branco

Seringal Rio Branco is located on the left bank of the Juruá Mirim river, in the municipality of Cruzeiro do Sul. As it is located on the bank of a secondary river, travel is exclusively by river, with limitations during the Amazon summer period (dry season). Ferries and boats are used to transport people and goods in the region.



Figure 11 - Transport used by the local population for freight and goods. Source: Socio-economic diagnosis. Prepared by: STA (2023)

Acquisition of the area by the current owner and intentions of economic exploitation that do not include the standing forest

Since 2001, Amazônia Agroindústria has been carrying out farming practices in the Cruzeiro do Sul region. With three farms under its management in the same region - Seringal Valparaíso, Santa Maria II and Gama - dedicated to beef cattle production and agriculture, the company has a track record of successful operations. Managing a herd of 3,500 head of cattle, its aim is to expand cattle farming to meet demand from local meatpacking plants in the city of Cruzeiro do Sul and other cities in the state of Acre.

Given the context presented above, the owner's acquisition of the property was motivated by identifying the economic potential of the area and considering the feasibility of implementing livestock farming. Initially, this decision was influenced by the government's plans to expand infrastructure, such as the proposal to extend the BR-364 highway and other planned state roads, and which was maintained due to the viability of river transportation by barge or ferry during the river flood season. As a result, the plan to implement cattle ranching operations in compliance with Brazilian environmental legislation remained independent of the expansion of Acre's road network. The details of how land use would change from forest to cattle ranching in a future scenario without a project are described in section 3.1.4 using the VCS tool "VT0001 - Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities (T-ADD)", version 3.0.

2.2.2 Most-Likely Scenario Justification (CCB, G2.1)

The scenario most likely to occur in a scenario without the project is described according to the additionality tool for AFOLU VT0001 projects in section 3.1.4 and 3.1.5 in detail.

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

The fact that a REDD project aims to actively stop threats in order to protect the existing forest makes it very conducive to biodiversity conservation - the same interventions and land uses that protect carbon stocks also protect the habitat of plants and animals - (Pitman 2011). This goal is one of those that justifies the transformation of REDD projects into REDD+ projects (that is, an additional objective to be achieved by REDD projects).

In this sense, REDD+ projects are expected to have additional benefits (or positive impacts) for biodiversity conservation, especially considering the various ecosystem services that biodiversity provides (Karousakis, 2009). The benefits of REDD+ projects to biodiversity are widely discussed in the literature (Mant et al., 2014; McDermott et al., 2012). However, the efficiency of delivering biodiversity benefits will depend on how the REDD+ project is designed and implemented. The negative impacts of the Project must be well assessed and minimized in order to achieve the desired result overall, while the positive impacts of installing a REDD+ Project are explicit in the Project proposal. The reference region for the project's diagnostics, and more specifically the project area, has a high diversity of species, with the presence of species that present some degree of threat and regional endemism, as well as a stable and self-regenerating forest structure. These characteristics reveal the 'Exceptional Benefits' for Biodiversity and justify the need to conserve the ecosystem. In this way, the Seringal Rio Branco REDD+ Project area can be considered a 'key area' for biodiversity conservation (Langhammer et al., 2007), which reinforces the positive impact expected from the installation of the project. Additionally, the project can contribute to maintaining the local climate, minimizing global climate change, as well as maintaining habitat connectivity, especially considering a larger spatial scale, such as in the Reference Region. It also protects ecosystem services such as pollination, seed dispersal, soil protection, hydrological basins, etc.

In terms of protecting ecosystem services, in addition to endangered and endemic species, the protection of species useful to man stands out, such as those originating from extractive activities, both timber and non-timber species.

With the presence of the Project, there will be an increase in inspection and control of asset, especially within the Project area; but actions should also take place outside the Project area, for example in the leaking management area. Actions such as these are likely to have an impact on adjacent communities due to interests in the use of resources. For this reason, efficient management together with the adjacent communities should be considered, including the provision of training courses for the use of forest resources.

As described in Section 2.5.7, there are a number of laws, regulations and decrees (federal and state) that address issues related to the conservation of environmental and ecological heritage and respect for the rights of traditional peoples and communities. These regulations include, for example, Federal Laws 13.123 (2015) and 12.651 (2012). However, as described in the common practice scenario, these laws, in short, are not applied in practice. In other words, there is legal fragility and inefficiency regarding access to and use of land in Brazil and the protection of natural ecosystems and their resources, as well as the protection of endangered species of fauna and flora; reinforcing that the existence of such regulations does not guarantee their effective compliance and enforcement.

In addition, the analysis of the scenario without the Project demonstrates the existence of various barriers to the implementation of activities that could have positive impacts, such as those mentioned above. More details on the additionality of the Project for the community and biodiversity can be obtained and consulted in Sections 4.1.4 and 5.1.3.

2.2.4 Benefits to be used as Offsets (CCB, G2.2)

As described in Section 2.2.3, additionality for the community and biodiversity scope has been fully demonstrated. It should be noted that the benefits generated will not be used in other offset programs; the Seringal Rio Branco REDD+ Project only aims to produce offsets related to Reducing Emissions by Avoiding Deforestation, as described in Section 3 - Climate.

2.3 Safeguards and Stakeholder Engagement

2.3.1 Stakeholder Identification (VCS, 3.18, 3.19; CCB G1.5)

Most REDD+ projects are very dynamic and complex, and their impacts will not be the same for all stakeholder groups. The identification of the different stakeholder groups and sub-groups, their interests and interactions, and their likely reaction to project interventions or external pressures, are critical elements for an analysis of safeguards and possible impacts. Proper stakeholder identification and analysis allows for robust participatory consultation processes. A good stakeholder analysis is therefore essential.

The identification of stakeholders in the Seringal Rio Branco REDD+ Project was based on the Socio-Economic Diagnosis developed by STA (Soluções Técnicas Ambientais). The process of identifying rural communities took into account their relationship with the project area, identifying those who have some kind of direct or indirect use of the project area and who could be affected by the project. As a result, the stakeholders of the Seringal Rio Branco REDD+ Project were divided into two large groups: Institutional Stakeholders and Community Stakeholders.

For the Institutional Stakeholders group, the identification process began with brainstorming, listing all the institutions and/or groups that could have an influence on the project or be impacted by it.

Organizations working in the municipality of Cruzeiro do Sul were identified, including private organizations, non-governmental organizations and other relevant experiences in the area surrounding the Seringal Rio Branco REDD+ project. The following table shows the public and private organizations that carry out direct actions, both in terms of capacity building and training and in terms of strengthening community organizations and developing socio-productive projects:

Table 4: Public and private organizations that carry out direct actions with possible relationships with the Project.

Name	Nature	Scope	Area of activity
Federal Institute of Acre - Cruzeiro do Sul campus	Public	Federal	Teaching, research and extension
Federal University of Acre - Floresta campus	Public	Federal	Teaching, research and extension
National Rural Learning Service (SENAR)	Private	National	Promoting managerial training and technical assistance for rural areas.
SOS Amazonia	NGO	State	It is an organization that operates in Acre in projects to promote the conservation of forest cover and its environmental, social and economic services; the restoration of deforested areas, seeking to increase carbon stocks and the return of environmental services; the strengthening of forest conservation and biodiversity protection and the development of socio-biodiversity products. SOS Amazônia has already carried out actions in some of the communities surveyed.
Cooperative of Family Farmers and Solidarity Economy Producers of Nova Cintra (Coopercintra)	Cooperative	Regional	The cooperative is located on the right bank of the Juruá River, in the Nova Cintra community, in the rural area of Rodrigues Alves, a neighboring municipality to Cruzeiro do Sul. Its aim is to promote socio-biodiversity production chains.
Department of Agriculture and Supply and Department of Tourism and Entrepreneurship of the municipality of Cruzeiro do Sul	Other (Fair)	Municipality of Cruzeiro do Sul	Fish Market in Cruzeiro do Sul: In 2022, the Cruzeiro do Sul Fish Market was in its third edition. It took place in the parking lot of the Farmer's Market, where dozens of traders, fish farmers and fishermen made various types of fish available to consumers during the Holy Week period.

Name	Nature	Scope	Area of activity
			Flour Festival: The event promotes flour production and regional culture and identity.
Acre Environment Institute (IMAC)	Public body	State	The Acre Environment Institute (IMAC) operates as an autonomous body of the indirect administration, an autarchy linked to the Acre Environment Secretariat (SEMA). Its mission is to contribute to sustainable socio-environmental and economic development by carrying out environmental education, licensing, monitoring and inspection.
Environment Secretariat (SEMA)	Public body	State	
Acre Institute for Climate Change and Regulation of Environmental Services - IMC	Public body	State	The Institute is responsible for the technical coordination of the Environmental Services Incentive System (SISA), an instrument created by the Acre government that recognizes and encourages environmental conservation resulting from environmental products and services - such as forest preservation, good land use, forest product management, among others - provided by communities, companies and/or individuals throughout Acre.

For the Community Stakeholders group, communities were mapped using secondary data from the IBGE database. Six rural communities were identified along the Juruá Mirim River, some of which were located between the Serra do Divisor National Park and the project area. Also, through secondary searches, the Francisco Pimentel PDS settlement was mapped, which borders the project area.

A spatial analysis was also carried out using satellite images, delimiting a 20 km buffer from the edge of the project area in order to identify local populations, small communities or isolated properties in the area.

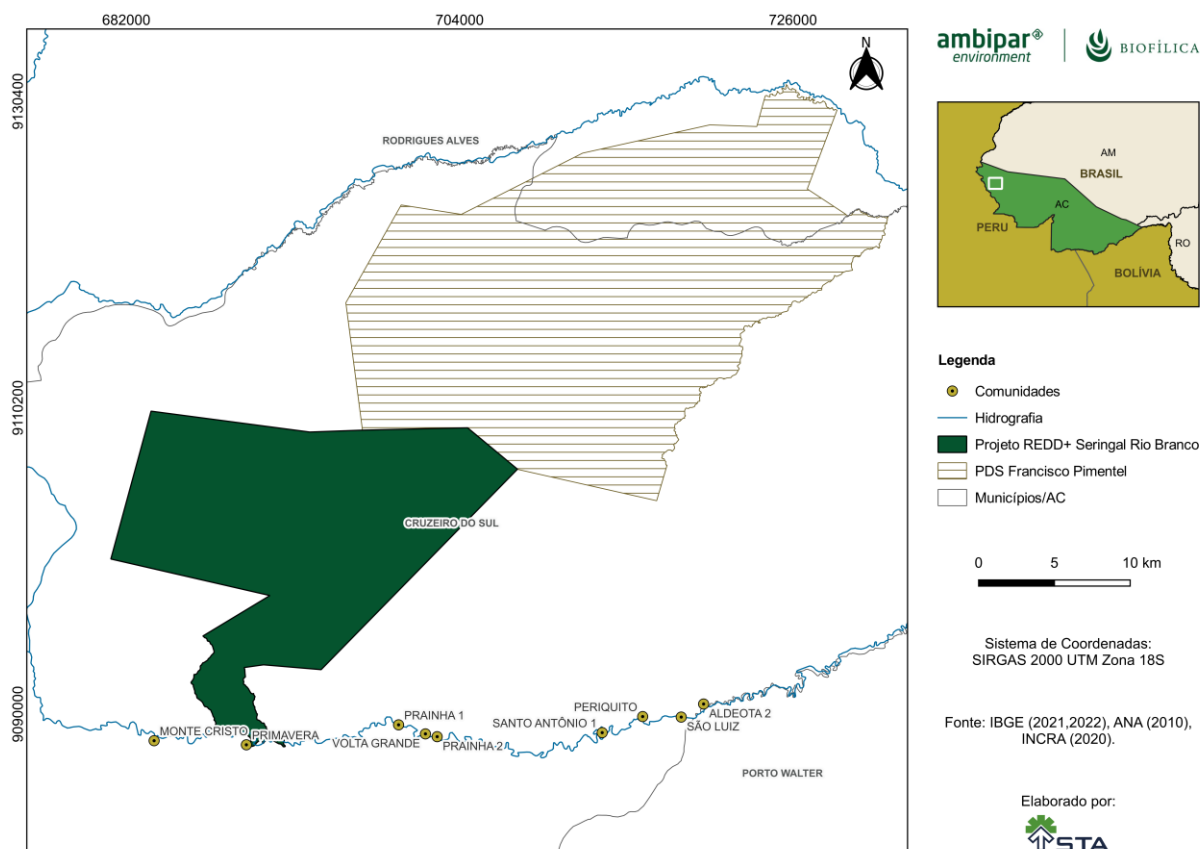


Figure 12. Communities identified around the Seringal Rio Branco REDD+ Project.Fonte: Source: Prepared by STA, 2023.

Based on this initial map, the team went into the field to carry out a pre-survey of community groups, in an activity called pre-field, which took place between February 8 and 11, 2023. The aim was to identify communities or population locations that could have some kind of relationship with the project, so the Juruá Mirim river was taken as a reference for the secondary data research. The communities were identified on the basis of two criteria: the first was geographical proximity to the project area and the second was direct or indirect use of the project area by the communities.

This will make it possible to analyze how these parties are impacted by the current situation and how the Seringal Rio Branco REDD+ Project will affect them. The following table shows all the communities identified in this 20 km buffer and their relationship with the project area.

Table 5: Communities identified within a 20 km buffer of the Project Area boundaries.

NO.	Communities	IBGE base	Identified in the pre-field	Relation to the project area?	Communities surveyed	Date of survey
1	Aldeota 1	No	02/11/2023	No	No	-
2	Aldeota 2	No	02/11/2023	Yes	Yes	02/18/2023
3	Avaí	Yes	Nonexistent	No	No	-
4	Boa Vista	Yes	02/08/2023	No	No	-
5	Cachoeira do Açaí	No	02/10/2023	No	No	-

6	Chico Mana	Yes	Nonexistent	No	-	-
7	Extrema	No	02/11/2023	No	No	-
8	Foz do Comprido	No	02/11/2023	No	No	-
9	Monte Cristo	No	02/08/2023	Yes	Yes	02/12/2023
10	Periquito	No	02/10/2023	Yes	Yes	02/21/2023
11	Prainha 1	No	02/09/2023	Yes	Yes	02/19/2023
12	Prainha 2	No	02/09/2023	Yes	Yes	02/20/2023
13	Primavera	No	02/08/2023	Yes	Yes	02/12/2023
14	Rio Branco	Yes	02/11/2023	No	No	-
15	Santo Antônio 1	Yes	02/10/2023	Yes	Yes	02/15/2023
16	Santo Antônio 2	No	02/10/2023	No	No	-
17	São Luís	Yes	02/10/2023	Yes	Yes	02/22/2023
18	Volta Grande	No	02/09/2023	Yes	Yes	02/14/2023
19	PDS Francisco Pimentel	Yes	-	Yes	Partially ²⁵	02/24/2023

Of the 18 communities located along the Juruá Mirim river during the pre-fieldwork, only 09 fit the criteria adopted for the in-depth DRP research, in addition to the Francisco Pimentel PDS, as shown in the table above. These ten communities will be monitored and followed up in section 4 of Community.

2.3.2 Stakeholder Descriptions (VCS, 3.18, 3.19; CCB, G1.6, G1.13)

Among the stakeholders identified, community groups and other interested parties with real relevance to the project's activities were listed, including a description of each of these stakeholders.

Table 6: Description of Stakeholders.

Stakeholder	Rights, interest, and overall relevance to the project
Communities	The communities located around the Seringal Rio Branco REDD+ Project that took part in the socio-economic study total ten communities, nine of which are riverside communities and one is an agrarian reform settlement. Within the productive structure of the surrounding communities, subsistence farming is the basis of productive-economic activities, combined with agro-extractive activities and the raising of small animals and even cattle. The relationship between these stakeholders and the project is unique and decisive because the pressure on forest areas, especially within the project area, from surrounding rural communities in search of wild animals for food, extractive activities in the area and the advance of cattle ranching, may increase more and more. In the long term, strengthening human skills and capacities in terms of economic productivity and sustainable use of resources can generate positive impacts for communities. The development of socio-productive activities that favor the generation of community income through sustainable activities. In this way, the Seringal Rio Branco REDD+ Project is an opportunity for synergy with the

²⁵ For the Francisco Pimentel PDS, data was collected using a simplified script with a key informant.

Stakeholder	Rights, interest, and overall relevance to the project
	communities with the aim of establishing sustainable consumption and cultivation practices that are aligned with environmental protection.
Federal Institute of Acre - Cruzeiro do Sul campus	The Federal Institute of Acre (IFAC), in the municipality of Cruzeiro do Sul, has the second largest IFAC campus, which is home to an agricultural unit that focuses on natural resources, environment and health and exact and earth sciences. The campus offers integrated technical, subsequent and higher education courses, has classrooms, laboratories, an auditorium, a library and a multi-sports gymnasium, and is a potential partner institution for the REDD+ Seringal Rio Branco Project.
Federal University of Acre - Floresta campus	The Federal University of Acre has a post-graduation program in Public Health that is developing the project "Food insecurity during pregnancy: childbirth care and maternal and neonatal outcomes". This project has studies being carried out in the municipality of Cruzeiro do Sul and aims to assess the factors associated with food insecurity during pregnancy in times of post-pandemic Covid-19 (UFAC, 2023) ²⁶ . The identification of comprehensive projects, such as this one involving public health as a whole, is important for envisioning future partnerships. The results of a study of this nature, disseminated in the communities of the Seringal Rio Branco REDD+ Project, could have an impact on people's lives, especially by talking about food insecurity and pregnancy.
Cooperative of Family Farmers and Solidarity Economy Producers of Nova Cintra (Coopercintra)	The Coopercintra cooperative has an Oil Extraction Plant with a solar and industrial dryer, breaker, kernel separation area, mechanical press, decantation tank and filter press, as well as transportation (ferry and tractor). The oil mill has a production capacity of 25 tons of murmuru butter per year. The cocoa chain is processed at the pre-processing unit located in the Novo Horizonte community, on the banks of the Juruá river, in the municipality of Guajará/AM. It has two barges for drying the kernels, a fermentation house and a warehouse. In the murmuru value chain, there are approximately 200 suppliers of raw materials distributed among the communities of Cruzeiro do Sul, Rodrigues Alves and Porto Walter, in the state of Acre, and also in Guajará and Ipixuna in Amazonas, all on the banks of the Juruá river. In addition to the murmuru chain, it works with wild cocoa, working with around 30 riverbank families in the Juruá. The cooperative's strong presence in the region demonstrates its enormous relevance to the future activities of the Seringal Rio Branco REDD+ Project.
SOS Amazonia	The NGO SOS Amazônia works in the project region to support community enterprises by investing in the structuring of processing centers, organizing

²⁶ UNIVERSIDADE FEDERAL DO ACRE - UFAC. Docentes realizam projeto sobre Insegurança Alimentar na Gestação. Disponível em: <https://www.ufac.br/floresta/noticias/2022/docentes-da-ufac-realizam-projeto-sobre-inseguranca-alimentar-na-gestacao>. Acesso em 26/05/2023.

Stakeholder	Rights, interest, and overall relevance to the project
	production processes and participatory management, and facilitating the expansion of actions in differentiated markets. In the development of socio-biodiversity products, it works with Buriti, Cacau, Murumuru, Seringueira, Tucumã, Andibora, Patauá and Copaíba. The NGO also works to develop socio-biodiversity value chains in the region, especially with the Native Rubber, Vegetable Oil and Wild Cocoa chains. As such, the Seringal Rio Branco REDD+ Project foresees numerous possibilities for synchronizing activities with this stakeholder.
National Rural Learning Service (SENAR)	SENAR carries out Rural Professional Training (FRP) and Social Promotion activities aimed at people in rural areas, contributing to their professionalization and improving their quality of life, as well as preparing them to exercise citizenship and the pursuit of sustainable development. The institution is based on developing the potential of the agricultural sector by continuously offering training in a wide range of activities linked to the rural sector, including property management. There are more than 250 face-to-face courses, all free of charge and certified. In addition to Rural Professional Training courses, SENAR-PR also works in Social Promotion, with activities aimed at improving the quality of life of the rural community. As part of its national Rural Professional Training (FPR) program, in June 2022 SENAR/AC concluded another class of the Rural Business course in the municipality of Cruzeiro do Sul. The course lasted five weeks and covered topics such as rural entrepreneurship, business management and strategic planning. Activities like these are very important focal points to be incorporated into the Seringal Rio Branco REDD+ Project.
Department of Agriculture and Supply and Department of Tourism and Entrepreneurship of the municipality of Cruzeiro do Sul	The municipality of Cruzeiro do Sul, through its secretariats, promotes various events that benefit local communities. The fish market is one of these events. It is organized by the Department of Agriculture and Supply, with the participation of the Department of Tourism and Entrepreneurship of the municipality of Cruzeiro do Sul. In its third edition, in 2022, the entrepreneur's fair was also set up with an exhibition of handicrafts, the sale of vegetables, fruit and other regional products. The Cruzeiro do Sul Flour Festival is another example of an event that celebrates production, culture, entrepreneurship, tourism and, above all, the identity of the municipality of Cruzeiro do Sul. In 2022 the festival was in its fifth edition, after two years without taking place due to the Covid 19 pandemic. All the municipal secretariats are involved in holding the festival, which welcomes around 30,000 visitors over the three days. Enabling the Seringal Rio Branco REDD+ Project community to take part in these events allows for the establishment of possible entrepreneurial actions and trade of local products.

Stakeholder	Rights, interest, and overall relevance to the project
Acre Environment Institute (IMAC)	One of IMAC's roles is to play a part in environmental education, making local communities aware of the importance of forest conservation and encouraging sustainable practices. This can help gain the support of local communities for the Seringal Rio Branco REDD+ Project and promote a culture of conservation. Furthermore, as part of its mission, the environmental agency seeks to promote sustainable development. Its participation in the Seringal Rio Branco REDD+ Project can help ensure that forest conservation activities are integrated into broader socio-economic development strategies, guaranteeing lasting benefits for local communities and the environment. This is why the institute plays an important role in monitoring and enforcing the law to prevent illegal activities, such as unauthorized deforestation or illegal hunting. This is crucial to ensuring the success of the Seringal Rio Branco REDD+ Project and the integrity of the protected forest areas.
Secretariat for the Environment (SEMA)	SEMA's main function is to promote integrated, shared and efficient environmental management, compatible with sustainable development, ensuring the preservation and conservation of the environment and improving quality of life. Partnerships between the Seringal Rio Branco REDD+ Project and SEMA can occur through technical and strategic collaboration, sharing of data and resources, coordination of efforts to implement forest conservation policies and engagement of local communities. SEMA can contribute significantly to ensuring compliance with environmental laws, promoting sustainable forest management practices and ensuring the integration of environmental conservation policies with socio-economic development. This collaboration is key to ensuring a comprehensive and integrated approach to forest conservation and sustainable development in Acre.
Acre Institute for Climate Change and Regulation of Environmental Services - IMC	Synergy with this stakeholder would enable technical collaborations for monitoring and evaluating projects, sharing relevant data and information, and coordinating efforts to implement integrated policies and programs. This collaboration strengthens the effectiveness and sustainability of REDD+ projects, integrating approaches to valuing environmental services and ensuring a holistic approach to forest conservation and combating climate change.

2.3.3 Stakeholder Access to Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

Communication between the Seringal Rio Branco REDD+ Project and its stakeholders will take place through three main channels: virtual, written and face-to-face/oral. The main objective is to ensure that all stakeholders have access to relevant documents and information relating to the project.

Virtual communication: the project design document and/or links to access it, important information for stakeholders, such as monitoring reports and other relevant documents, will be available virtually on the

Biofílica and Verra websites. News and updates about the project will also be posted on social media (LinkedIn and Instagram).

Written communication: a printed version of the summary of the project design document and versions of the respective monitoring reports will be made available at headquarters for consultation by all interested parties.

Face-to-face/oral communication: information, news and updates on the project will be passed on through meetings, events and other face-to-face encounters.

An additional means of communication that will stand out, especially given the scarcity of resources, is the transmission of information via radio. Using the local radio stations in both Cruzeiro do Sul and Rodrigues Alves, information will be disseminated effectively to the riverside communities. In addition, the public consultation event in Verra will be widely publicized to interested parties, who will have access to the draft Project Design Document.

2.3.4 Dissemination of Summary Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

The documents relating to the Seringal Rio Branco REDD+ Project will be made available and disseminated to all interested parties virtually on the official website of Biofílica Ambipar Environment and on the website of the Verra registration platform. In addition, a printed summary of the Project Design Document will be made available at the Seringal Rio Branco headquarters, as well as the monitoring reports, for free access and consultation by the communities and other interested parties.

As mentioned in the previous section, news and updates will be made publicly available through Biofílica Ambipar Environment's social networks, on its LinkedIn and Instagram accounts. In addition, all information and news will be reported orally through meetings, lectures and other face-to-face gatherings for communities, partners, proponents and other interested parties.

2.3.5 Informational Meetings with Stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

In order to implement appropriate social activities for the surrounding communities, a Participatory Rapid Rural Diagnosis (DRP) was carried out by the project proponents together with the company Soluções Técnicas Ambientais (STA), in which the REDD+ Project was presented in an effective and accessible way, as well as collecting the information needed to build the project. The approach used was through conversations, illustrations, drawings, diagrams, and observations made in the field. The research was carried out with the communities already described in Section 2.1.8.

During the same diagnosis, interviews were conducted with the only rural landowner in the region, using the same approach used to communicate with the communities, presenting the objectives and relevant information about the REDD+ Project. No interviews were conducted with workers, as Amazônia Agroindústria EIRELI was not carrying out any activities in the project area prior to the field research.

Finally, between July 11 and 16, 2023, as part of the project validation process, feedback was given to the communities and other local stakeholders on the results of the DRP, as well as the next steps for the project and validation of the proposed activities described in Section 2.3.7.

2.3.6 Risks from the Project and No Net Harm (VCS, 3.18, 3.19)

During the participatory diagnosis workshops in the rural communities, the technical team presented the steps of the study with the help of a banner, from the presentation of the team in the field, talking about what REDD projects are and the stages of a REDD project, to showing the community members the possible positive and negative impacts of the project.

It was also explained during the diagnosis that, with the project, there will be more control over the use of space in the project area, reinforcing the objective of REDD projects. And that there could be rules for use of the area, with restrictions on movement in the area and use of the forest, with a view to preventing deforestation.

Thus, according to the social impact assessment method, the following is an analysis of the social groups surrounding the project area, with an assessment of possible negative impacts, based on example scenarios constructed on the basis of the local reality and the possible risks identified.

Table 7: Analysis of possible risks to social groups that are related to the project area.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Family subsistence	Loss of access to essential subsistence products and extractive products	If there are unplanned restrictions on the use of the project area, there will be a loss of economic activity for families, causing a food and subsistence imbalance.	Activities to promote sustainable extractive and cultivation activities so that families can continue to make the best possible use of forest products and their area.
Hunting and fishing	Loss of access to subsistence hunting and fishing areas	A possible total restriction on families' access to fishing and hunting areas could have an impact on their diet and food security, generating greater dependence on purchased food.	Activities to promote sustainable fishing and animal husbandry practices for subsistence can meet the need for hunting and predatory fishing.

2.3.7 Community Costs, Risks, and Benefits (CCB, G3.2)

As mentioned above, a Participatory Rapid Rural Diagnosis (DRP) Workshop was held involving the communities selected for the Seringal Rio Branco REDD+ Project, as detailed in Sections 2.1.8 and 2.1.9. The activities carried out in conjunction with the communities made it possible to analyze the context of these communities through interviews.

Among the positive aspects of these communities, it is essential to highlight the production of non-timber forest products, especially açaí, buriti, lianas, straw, and medicinal plants, among other products. On the other hand, difficulties in getting around and accessing essential public services have been identified for some isolated communities. In addition, not all the communities have schools covering all levels of education, they don't have access to health centers, they don't have drinking water or basic sanitation and they lack effective technical assistance to improve the means of production.

The main threats identified in the context of the project include practices such as manioc farming, hunting and predatory fishing. These activities represent significant challenges for environmental conservation and sustainability in the region. In addition, other concerns, such as the lack of community engagement and the persistence of unsustainable production practices, are equally relevant. These factors highlight the importance of effective approaches to raising awareness, education, and the implementation of sustainable agricultural and extractive practices, with a view to mitigating the threats identified.

The results of the workshops were used as the basis for drawing up the activity plan presented in the Seringal Rio Branco REDD+ Project, as described in Section 2.1.11. The results were also fed back to the communities and the activities proposed by the project were validated. During this process, information materials were made available, such as folders, pamphlets, and a general presentation of the Project, containing information on the activities, scope, risks, impacts generated, communication channels and an overview of the climate context.

Throughout the project, more detailed information on costs, risks and potential benefits for communities will be provided at project meetings and consultations on the development of activities, as well as during follow-up meetings. In addition, it is important to emphasize from the outset and reinforce throughout the Project that participation is voluntary, and the decision to participate or not is not final and does not result in restrictions.

2.3.8 Information to Stakeholders on Validation and Verification Process (VCS, 3.18.6, 3.19; CCB, G3.3)

The potentially participating communities of the Project and other interested parties will be informed about the CCB validation and verification through the available means of communication (Section **Erro! Fonte de referência não encontrada.**) prior to the event.

Virtual channels such as Biofílica Ambipar Environment's website and newsletter, social media such as Instagram and LinkedIn, and communication channels such as radio broadcasts and the physical presence of communicators, will also be used to inform other stakeholders and the general public.

In addition, as mentioned in the previous sections, information about the project has already been disseminated during the Participatory Rapid Rural Diagnosis (DRP) and feedback of the results and presentation of the proposed activities to the stakeholders.

2.3.9 Site Visit Information and Opportunities to Communicate with Auditor (VCS, 3.18.6; CCB, G3.3)

The auditor's visit to the Project site will be informed to the communities, proponents, partners, and other stakeholders participating in the Seringal Rio Branco REDD+ Project by the available means of communication (Section 2.3.8) prior to the event.

In addition, the feedback on the results of the DRP and the validation of the activities proposed by the project with the stakeholders also mentioned the stages and next steps of the project, as well as the audit event.

Communication between communities and other stakeholders and the auditor, as well as the dissemination of information, will be facilitated through announcements on local radio stations, virtual channels, at the moment Biofílica's website and social networks such as Instagram and LinkedIn.

2.3.10 Stakeholder Consultations (VCS, 3.18; CCB, G3.4)

After coordination between Amazônia Agroindústria and Biofílica Ambipar Environmental, the REDD + Seringal Rio Branco Project moved on to the next step, which was to identify actors and partnerships to help develop technical studies for the implementation of the project, such as consultancies. The process of defining suppliers for the technical studies began in November 2022. With the technical partners defined, the diagnostics were started and completed in February 2023.

Initially, a kickoff workshop was held on January 27, 2023, between the proponents and partners, with the aim of forming a multidisciplinary team aligned with the demands and particularities of the Project, as well as defining the strategies for the different stages of the Project.

Pre-field activities were then carried out between February 8 and 11, 2023. The aim of this first activity was to establish initial contact with the communities and consolidate an agenda for carrying out the socio-economic diagnosis.

As explained above in section 2.3.1, communities were selected that had some connection with the Seringal Rio Branco, considering its importance to the families' subsistence, both primary and complementary.

Subsequently, during the month of February 2023, the socio-economic diagnosis was carried out with the communities initially selected for the project that had voluntarily agreed to take part in the study. The results obtained gave rise to the report developed by the STA company (See Section 2.1.6 and 4.1.1 for more details), which used the Participatory Rapid Rural Diagnosis (DRP) methodology, using on-site interviews, focus groups, diagrams, illustrations and participatory workshops to gather information. During the workshops, the participants were divided into working groups to discuss technical and descriptive points about the certification standards and their requirements in relation to fauna, flora, socio-economics, the physical environment, carbon inventory and determining the baseline. In these working groups, the strategic actions of the project and a preliminary assessment of the communities that would be selected to be directly involved in the first phase of the project were defined.

Throughout its execution, the work was conducted through a participatory and transparent process with the community members and other actors involved in the research, respecting the principles of clear, effective communication and transparency, as well as the principles of free and informed consent for the activities to be carried out.

For the research in each community, a pre-organized schedule was established to serve as a roadmap for the field approach, as shown in the table below. However, in some communities there was a need to adapt the itinerary due to local dynamics, in order to improve the progress of the research, according to the needs and reality of each community.

Table 8: Socio-economic survey approach schedule by community. Prepared by: STA, 2023.

Diagnostic Programming	Estimated time
Arrival in the community, initial contacts with local leaders to organize the space and dynamics for the diagnosis, preparation of the space reserved for the diagnosis and welcoming the residents.	30 minutes
Work opened with a presentation by the external team and speeches by local representatives.	15 minutes
Project presentation.	20 minutes
Presentation of the diagnosis, explanation of the dynamics of group and individual work, division of groups and start of work.	20 minutes
Research work in focus groups and individual interviews	2 hours
Finalization of the work, return to the collective space, openness to possible clarifications and information about the research	15 minutes
Closing	10 minutes

Thus, through these interventions in the field, the strengths and weaknesses of the stakeholders were listed, aligned and related to their potential and opportunities, resulting in the activities and actions proposed by the Project. The main results found in this research that were taken into account in the construction of the Project are described in Section 2.3.4.

In the photo below, the technician is using the identification of diseases by age group by means of a matrix, always supported by drawings, flowcharts, or maps for visual support.



Figure 13: Monte Cristo Community, 02/12/2023. Source: STA, 2023.

The information collected allowed us to understand local problems, providing input for the assertive construction of social activities, enabling a direct and indirect impact on the well-being of communities, whether by stimulating sustainable economic development through the promotion of sustainable practices, non-timber forest management or the promotion of environmental education for hunting and fishing, diversification of production, increased productivity, improving the communication process and the quality of life of residents through greater coordination with public authorities, with a view to improving infrastructure in the region. The activities and causal relationships between expected short, medium and long-term results are described in Section 2.1.11.

After the technical studies were completed, feedback was given to the communities and stakeholders from July 14 to 16, 2023. Before the activity took place, the STA team informed them in person and defined the days and times of the meetings, so that everyone was aware and consented to the activity on a voluntary and optional basis. During the meetings, participants were presented with the results of the socio-economic diagnosis of the region and the proposed activities that will be carried out by Amazônia Agroindústria EIRELI and Biofílica Ambipar Environment over the course of the project.

Four feedback meetings were held with local communities in the area surrounding the Seringal Rio Branco REDD+ project. It was possible to hold events by grouping some communities together, due to their proximity to each other. The Volta Grande, Monte Cristo, Prainha 2 and Periquito communities were therefore considered to be the hub areas for the meetings.

The Community Stakeholders involved in each feedback meeting and the details of the consultation process are listed below:

Stakeholder	Volta Grande Community
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Date of stakeholder consultation	14-July-2023
Stakeholder engagement process	The purpose of the meetings was to introduce the project and provide a space for participants to submit questions and suggestions related to the project's design. During these meetings, all participants had the opportunity to express their ideas and opinions with the aim of improving the proposed activities. In addition, the proposed communication channels were presented, intended to be structured and used throughout the project. The meetings were conducted by means of a presentation, orally conveying information about the Project with the aid of explanatory posters and pamphlets. The presentation covered the objectives of the meeting for stakeholders, identifying the Project proponents, clarifying the global climate scenario, discussing climate change, compliance with laws, statutes, and other regulatory frameworks, as well as detailing the Project's location, the surrounding area and the main drivers of deforestation and degradation in the region. Along with this comprehensive initial presentation, the material used emphasized crucial aspects such as the location of the area, the scope and type of the Project, the expected results, the summary and status of the main activities carried out. This summary detailed the planning, design and conception of the Project, the development, the validation and verification process, the follow-up to the audit process and, finally, the management and monitoring during implementation. In the process of communicating with stakeholders, the materials provided, such as pamphlets and explanatory letters, kept the communication channels open for possible suggestions, questions and complaints. To increase awareness and the availability of information to interested communities, the project document was left at the headquarters of Amazônia Agroindústria EIRELI.
Consultation outcome	Photos were taken, an attendance list was kept, and a voluntary questionnaire was administered to the participants to record their understanding of what was being said, as well as to give them a space to make criticisms and suggestions about the project and the dynamics followed during the feedback session. The documents were made available to the VBB team.



Figure 14: Volta Grande community, on 07/14/2023. Photo: STA, 2023.

Stakeholder input

The participants did not make any requests or suggestions for significant changes to the proposed Action Plan, but only demonstrated their commitment to monitoring the project's progress and their willingness to get involved in the future. In addition, all the participants received the project poster, pamphlet and printed presentation containing all the information covered to distribute in their respective communities.

Stakeholder	Primavera and Monte Cristo communities
Date of stakeholder consultation	15-July-2023
Stakeholder engagement process	During these meetings, all the participants had the opportunity to express their ideas and opinions with the aim of improving the proposed activities. In addition, the proposed communication channels were presented, which were intended to be structured and used throughout the project. The meetings were conducted by means of a presentation, orally conveying information about the Project with the aid of explanatory posters and pamphlets. The presentation covered the objectives of the meeting for stakeholders, identifying the Project proponents, clarifying the global climate scenario, discussing climate change, compliance with laws, statutes, and other regulatory frameworks, as

	well as detailing the Project's location, the surrounding area and the main drivers of deforestation and degradation in the region. Along with this comprehensive initial presentation, the material used emphasized crucial aspects such as the location of the area, the scope and type of the Project, the expected results, the summary and status of the main activities carried out. This summary detailed the planning, design and conception of the Project, the development, the validation and verification process, the follow-up to the audit process and, finally, the management and monitoring during implementation. In the process of communicating with stakeholders, the materials provided, such as pamphlets and explanatory letters, kept the communication channels open for possible suggestions, questions and complaints. To increase awareness and the availability of information to interested communities, the project document was left at the headquarters of Amazônia Agroindústria EIRELI.
Consultation outcome	Photos were taken, an attendance list was kept, and a voluntary questionnaire was administered to the participants to record their understanding of what was being said, as well as to give them a space to make criticisms and suggestions about the project and the dynamics followed during the feedback session. The documents were made available to the VBB team.  Figure 15. Primavera and Monte Cristo communities, on 07/15/2023. Photo: STA, 2023.



Figure 16. Primavera and Monte Cristo communities, on 07/15/2023. Photo: STA, 2023.

Stakeholder input

The participants did not make any requests or suggestions for significant changes to the proposed Action Plan, but only demonstrated their commitment to monitoring the project's progress and their willingness to get involved in the future. In addition, all the participants received the project poster, pamphlet and printed presentation containing all the information covered to distribute in their respective communities.

Stakeholder

Prainha 1 and Prainha 2 communities

Date of stakeholder consultation

15-July-2023

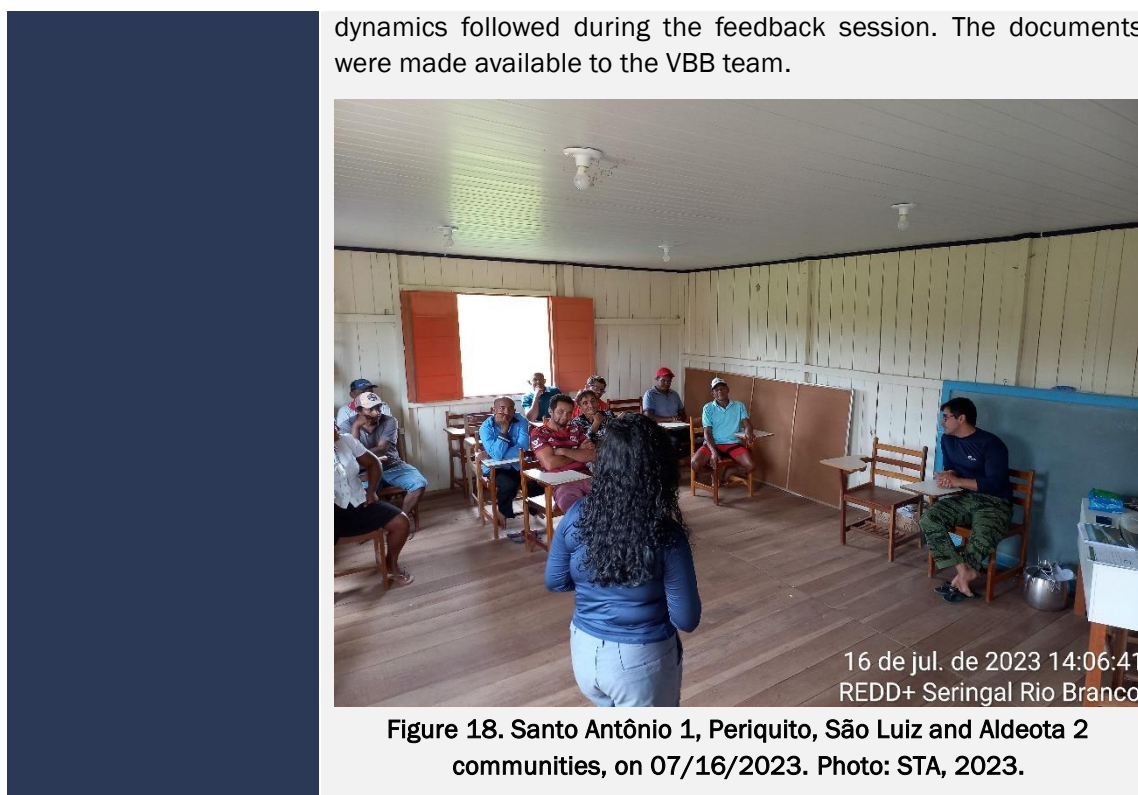
Stakeholder engagement process

The purpose of the meetings was to introduce the project and provide a space for participants to submit questions and suggestions related to the project's design. During these meetings, all participants had the opportunity to express their ideas and opinions with the aim of improving the proposed activities. In addition, the proposed communication channels were presented, intended to be structured and used throughout the project.

The meetings were conducted by means of a presentation, orally conveying information about the Project with the aid of explanatory posters and pamphlets. The presentation covered the objectives of the meeting for stakeholders, identifying the Project proponents, clarifying the global climate scenario, discussing climate change, compliance with laws, statutes and other regulatory frameworks, as

	well as detailing the Project's location, the surrounding area and the main drivers of deforestation and degradation in the region. Along with this comprehensive initial presentation, the material used emphasized crucial aspects such as the location of the area, the scope and type of the Project, the expected results, the summary, and status of the main activities carried out. This summary detailed the planning, design and conception of the Project, the development, the validation and verification process, the follow-up to the audit process and, finally, the management and monitoring during implementation. In the process of communicating with stakeholders, the materials provided, such as pamphlets and explanatory letters, kept the communication channels open for possible suggestions, questions and complaints. To increase awareness and the availability of information to interested communities, the project document was left at the headquarters of Amazônia Agroindústria EIRELI.
Consultation outcome	Photos were taken, an attendance list was kept, and a voluntary questionnaire was administered to the participants to record their understanding of what was being said, as well as to give them a space to make criticisms and suggestions about the project and the dynamics followed during the feedback session. The documents were made available to the VBB team.  15 de jul. de 2023 14:22:37 REDD+ Seringal Rio Branco Figure 17. Prainha 1 and Prainha 2 communities, on 07/15/2023. Photo: STA, 2023.
Stakeholder input	The participants did not make any requests or suggestions for significant changes to the proposed Action Plan, but only demonstrated their commitment to monitoring the project's

	progress and their willingness to get involved in the future. In addition, all the participants received the project poster, pamphlet and printed presentation containing all the information covered to distribute in their respective communities.
Stakeholder	Santo Antônio 1, Periquito, São Luiz and Aldeota 2 communities
Date of stakeholder consultation	16-July-2023
Stakeholder engagement process	The purpose of the meetings was to introduce the project and provide a space for participants to submit questions and suggestions related to the project's design. During these meetings, all participants had the opportunity to express their ideas and opinions with the aim of improving the proposed activities. In addition, the proposed communication channels were presented, intended to be structured and used throughout the project. The meetings were conducted by means of a presentation, orally conveying information about the Project with the aid of explanatory posters and pamphlets. The presentation covered the objectives of the meeting for stakeholders, identifying the Project proponents, clarifying the global climate scenario, discussing climate change, compliance with laws, statutes, and other regulatory frameworks, as well as detailing the Project's location, the surrounding area and the main drivers of deforestation and degradation in the region. Along with this comprehensive initial presentation, the material used emphasized crucial aspects such as the location of the area, the scope and type of the Project, the expected results, the summary, and status of the main activities carried out. This summary detailed the planning, design and conception of the Project, the development, the validation and verification process, the follow-up to the audit process and, finally, the management and monitoring during implementation. In the process of communicating with stakeholders, the materials provided, such as pamphlets and explanatory letters, kept the communication channels open for possible suggestions, questions and complaints. To increase awareness and the availability of information to interested communities, the project document was left at the headquarters of Amazônia Agroindústria EIRELI.
Consultation outcome	Photos were taken, an attendance list was kept, and a voluntary questionnaire was administered to the participants to record their understanding of what was being said, as well as to give them a space to make criticisms and suggestions about the project and the



Stakeholder input	The participants did not make any requests or suggestions for significant changes to the proposed Action Plan, but only demonstrated their commitment to monitoring the project's progress and their willingness to get involved in the future. In addition, all the participants received the project poster, pamphlet and printed presentation containing all the information covered to distribute in their respective communities.
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The consultation with the Institutional Stakeholders was carried out with one of the bodies identified in section 2.3.1.

Stakeholder	Acre Environment Institute - IMAC
Date of stakeholder consultation	22-February-2024
Stakeholder engagement process	Location: Cruzeiro do Sul - AC A meeting was held to present the REDD+ Juruá and REDD+ Seringal Rio Branco projects to the local IMAC body in Cruzeiro do Sul. During the meeting, the participation of Biofílica and its local partner, Amazonia Agroindustria, was highlighted. The initiative aimed to promote transparency and establish an open channel of

	communication between project proponents and local institutions, guaranteeing the effective participation of interested parties.
Consultation outcome	An attendance list was kept, and minutes drawn up. The documents were made available to the VBB team.
Stakeholder input	The participants made no requests or suggestions, only a commitment to follow the project's progress and a willingness to get involved in the future was demonstrated.

Stakeholder	Environment Secretariat (SEMA)
Date of stakeholder consultation	16-January-2024
Stakeholder engagement process	Location: Cruzeiro do Sul - AC A meeting was held to present the REDD+ Juruá and REDD+ Seringal Rio Branco projects, highlighting the participation of Bioflica and its local partner, Amazonia Agroindustria. During the meeting, points such as the request for support to produce monitoring reports on deforestation and fires in the Valparaíso and Rio Branco rubber plantations in Cruzeiro do Sul were also discussed, as well as updating the registration process for the Juruá REDD+ Project (Valparaíso Seringal) with the Institute for Climate Change and Regulation of Environmental Services. This initiative aimed to promote transparency and establish open communication between project proponents and local institutions, guaranteeing the effective participation of all interested parties.
Consultation outcome	An attendance list was kept, and minutes drawn up. The documents were made available to the VBB team.



Figure 19. Photographic record of the meeting at SEMA. Source: Entrepreneur Consultancy, January/2024.

Stakeholder input	The participants made no requests or suggestions, only a commitment to follow the project's progress and a willingness to get involved in the future.
Stakeholder	Acre Institute for Climate Change and Regulation of Environmental Services - IMC
Date of stakeholder consultation	19-January-2024
Stakeholder engagement process	Location: Rio Branco - AC A meeting was held to present the details of the Juruá REDD+ and Seringal Rio Branco REDD+ projects, highlighting the participation of Bioflica and its local partner, Amazonia Agroindustria. The main agenda was to discuss the IMC's new procedures for REDD+ projects and to exchange information and discuss the IMC's role in registering, following up and monitoring these projects at state level.
Consultation outcome	An attendance list was kept, and minutes drawn up. The documents were made available to the VBB team.



Figure 20. Photographic record of the meeting at IMAC. Source: Consulting Team, January 2024.

Stakeholder input

The President of the IMC gave a detailed explanation of the entire process of evolution of the IMC, where it can be seen that this Institute is prepared for the legal and institutional challenges that involve the services of actions of the State System of Incentives for Environmental Services of Acre (SISA). The meeting was very interesting and fruitful in view of the improvement of the methodologies used and the forecast of new regulatory frameworks, especially with regard to Bill 2.148/2015, which was approved on December 21, 2023, and is currently being processed in the Federal Senate with the expectation of approval and Presidential sanction in 2024.

There was a commitment to monitor the project's progress and a willingness to get involved in the future.

To the other stakeholders identified, the Federal Institute of Acre, the Federal University of Acre, the Cooperative of Family Farmers and Solidarity Economy Producers of Nova Cintra (Coopercintra), SOS Amazônia, the National Rural Learning Service (SENAR) and the Municipal Secretariats of Cruzeiro do Sul, will be sent letters presenting the project.

During the public consultation period, all institutional stakeholders will also be notified via letter, with all the details about the consultation and ways of participating.

Through the communication channels, until the next verification, the Project will communicate its development and implementation, including the results of the monitoring and the validation and verification processes of the VCS Program, making available all documents and information related to the Project, including the risks, costs and benefits that the project may bring to local stakeholders, the site visit of the validation/verification body and any relevant laws and regulations covering workers' rights in the host country.

2.3.11 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

From the communication plan created to continue communication and consultation between the project proponents and the community and other stakeholders, the following structure is followed:

Communication channels: communication is established through the emails of the project proponents (Amazônia Agroindústria and Biofilica), the website and telephone numbers. In order to improve this process, it is planned to set up an exclusive email for the project and provide contacts for the WhatsApp messaging application. In addition, a suggestion box will be installed at the project headquarters, both at the specific site of the project and at the Amazônia Agroindústria office in Cruzeiro do Sul. These tools will allow stakeholders to register doubts, suggestions, and complaints. In addition, interaction will be strengthened through meetings, talks, face-to-face events, and community visits.

Social mobilization strategy: social mobilizations will be carried out to hold meetings, activities, lectures, training sessions and other gatherings. Mobilizations will be carried out via mailing (e-mail), telephone, WhatsApp, radio and other necessary means of communication. At each event, it will be defined which parties will be mobilized depending on the topic and negotiations.

Communication procedures: through a defined structure, demands from interested parties will be received by the means of communication described, recorded on a standard form, analyzed, and forwarded to the areas responsible, if necessary, and resolved within a timeframe predetermined by the parties involved. Responsibility for responding depends on the complexity of the demand and may be carried out by the person responsible for project communication.

Conflict management: Conflict management will be based on the peaceful resolution of opposing interests directly linked to the Seringal Rio Branco REDD+ Project, seeking all possible means of negotiation, so that the decision-making process harmoniously serves all parties involved, taking into account the well-being of all. They will be carried out through dialogue using any of the means of communication mentioned above, prioritizing the means by which the parties feel most comfortable and with the shortest resolution time. The procedures and stages for resolving conflicts are described in detail in Section 2.3.12.

Through the communication channels, until the next verification, the Project will communicate its development and implementation, including the results of the monitoring and the validation and verification processes of the CCB & VCS Program, making available all documents and information related to the Project, including the risks, costs and benefits that the Project may bring to local stakeholders, the site visit of the validation/verification body and any relevant laws and regulations covering workers' rights in the host country.

The communication plan for the Seringal Rio Branco REDD+ Project may be adapted over the lifetime of the project as necessary, with a view to improving communication between stakeholders and responses to demands.

2.3.12 Stakeholder Consultation Channels (CCB, G3.5)

The project's activities are designed and implemented taking into account the wishes, characteristics and limitations of each community, as defined and verified during the DRP workshops and the feedback meeting.

As described in Section 2.3.4, workshops (DRP) and meetings have already been held between the communities and the project proponents. This communication and accessibility for discussion on the

progress of project activities between stakeholders and proponents will take place continuously throughout the duration of the project through the available communication channels, as described in Section 2.3.8. In addition, feedback was given to the communities, as described in Section 2.3.6, increasing the level of information sharing.

2.3.13 Stakeholder Participation in Decision-Making and Implementation (VCS, 3.18, 3.19; CCB, G3.6)

The decision-making and implementation processes, as well as the various activities linked to the project, are open to community participation. In addition, the Project will seek the equal participation of young people and women in the decision-making and implementation of Project activities. Involvement in the design, implementation, monitoring and evaluation takes place through the available communication channels (Section 2.3.8) and information meetings, in which all interested communities have the opportunity to participate.

2.3.14 Anti-Discrimination Assurance (VCS 3.19; CCB, G3.7)

The Code of Ethics and Conduct serves as a guideline that should guide the conduct of the company's members in relation to contact with the internal and external public, encouraging the dissemination and sharing of values and stimulating the improvement of behavior and attitudes.

Biofílica Ambipar Environmental, part of the Ambipar Group, has a code of ethics and conduct that represents all the Group's companies. The document serves as a primary instrument to guide the conduct of all parties involved in the adoption of good practices in relationships and in doing business, guiding attitudes of respect for diversity, not admitting attitudes of discrimination or prejudice of any kind (race, religion, age, gender, political conviction, origin, marital status, sexual orientation, or physical condition), as well as abusive attitudes, thus combating any form of prejudice.

The communication channels of the Seringal Rio Branco REDD+ Project (Section **Erro! Fonte de referência não encontrada.**), will be available to report any attitude that goes against the values described in the Code.

2.3.15 Feedback and Grievance Redress Procedure (VCS, 3.18.4; CCB, G3.8)

Development process

Amazônia Agroindústria EIRELI together with Biofílica Ambipar Environmental undertakes to establish a conflict resolution and feedback procedure that is accessible and adapted to the reality of the interested parties.

Conflict management within the scope of the Seringal Rio Branco REDD+ Project is based on the search for the peaceful resolution of differences of interest. The aim is to explore all possible means of negotiation, ensuring that decision-making harmonizes all parties involved and takes into account the general welfare. The process includes stages such as receiving the demand through any available communication channel, assessing the degree of seriousness/urgency of the demand, and verifying its veracity.

Subsequently, the issue is forwarded to the responsible department, where attempts are made at conciliation or

	negotiation. Management assesses the need for legal involvement, considering the possibility of mediation or the initiation of formal legal proceedings. The conflict is resolved in accordance with all legal parameters, ensuring that none of the parties concerned are harmed and always prioritizing the search for consensus between them.
Grievance redress procedure	Stages of conflict resolution When receiving complaints, doubts, suggestions, compliments, denunciations and requests related to the Seringal Rio Branco REDD+ Project, the proponents will consider the following stages, in order of priority: I) Negotiation or conciliation of less complex conflicts The process of feedback and redressing complaints will be carried out through the procedures and channels established in the Adaptive Management Plan, using the Project's email address and telephone with access to the WhatsApp messaging app to receive, listen to, respond to and try to resolve complaints received, as well as the suggestion box at the Seringal headquarters and at the company's headquarters in Cruzeiro do Sul, where questions, suggestions and complaints can be registered by interested parties on a specific form. Complaints and feedback can also be dealt with at meetings and other face-to-face events in the communities. Demands from interested parties will be received by the means of communication described, recorded on a standard form, analyzed, and forwarded to the areas responsible, if necessary, and resolved within a timeframe predetermined by the parties involved. Responsibility for responding depends on the complexity of the demand and may be carried out by the person responsible for the project communication or by the management of Amazônia Agroindústria EIRELI. The person responsible for the communication of the project will present information and documents with suggested solutions to the respective demand, in a way that is favorable to both parties. If the proposed measures are accepted, an agreement will be drawn up containing the following information: The purpose of the agreement and the deadline for its execution; - How to monitor the implementation of the measures and procedures applicable in the event of non-compliance with the agreement between the parties; - Formalization of the parties' acceptance by signing the document.

	- If there is no agreement between the parties, the demand will be recorded in the minutes of the meetings, so that new negotiation possibilities can be checked later. II) Mediation and arbitration of possible conflicts In cases where there is no solution through negotiation and conciliation, mediation will be attempted for more complex conflicts, in which a third party, who may or may not be the project proponents, is placed in a neutral and impartial manner, responsible for mediating conflicts and re-establishing communication between the parties involved, so that they themselves can reach a satisfactory solution. III) Legal proceedings Adoption of legal instruments to resolve highly complex conflicts, a judge or court with regular jurisdiction is expected to rule on the law regarding a person or property. We emphasize that the means of communication proposed for the project were presented and validated at the feedback event with the communities.
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2.3.16 Accessibility of the Feedback and Grievance Redress Procedure (VCS, 3.19; CCB, G3.8)

As mentioned in the previous topic, the Seringal Rio Branco REDD+ Project will adopt the reception of feedback, complaints, and denunciations in addition to conflict management and feedback procedures (Communication Plan Section 2.3.8 and Section 2.3.12), such as e-mail, suggestion boxes, WhatsApp applications and face-to-face communication at the Seringal headquarters and at the Amazônia Agroindústria EIRELI office. The proponents will ensure that the entire history of complaints, doubts, suggestions, compliments, denunciations and requests and their respective developments in procedures and decision-making are disclosed and accessible in a clear and transparent manner to all individuals, communities, and institutions, both in physical files (fill-in sheet) and virtual files, using the proposed communication channels.

It is worth emphasizing that in all channels for receiving feedback, complaints and denunciations, records may be anonymous, except in cases where individuals require identification.

2.3.17 Worker Training (VCS, 3.19; CCB, G3.9)

The qualification and empowerment of local actors is essential to guarantee the quality of the implementation of the actions proposed by the Project, as well as to ensure the permanence of the positive results and impacts in the long term. It is understood that, in order to ensure the effective implementation of the Seringal Rio Branco REDD+ Project, it is essential to work on generating local human capital, focusing mainly on the responsible management of natural resources. Thus, among the various actions proposed

by the project, part involves training and engaging local actors. The main proposals aimed at promoting the training of local actors, income generation and direct and indirect employment are described below.

- **Promoting sustainable agricultural practices:** the activity involves promoting training along the lines mapped out by the project, which may be related to the demands of the communities, as well as mapping stakeholders with the potential to join actions to promote sustainable practices and implementing partnerships ranging from financial backers to technical development. In addition, the activity promotes the improvement of practices developed by communities, promoting sustainable development, with environmental education actions in scenarios where there is synergy with the actions implemented;
- **Promoting environmental education for hunting and fishing:** the activity proposes seeking solutions and environmental education actions to be implemented in synergy with the cultural elements and practices used by community members, maintaining the source of income for families who practice hunting and fishing, improving the practice for sustainable development, and mapping and implementing partnerships for the development of environmental education actions, thus improving activities and practices;
- **Non-timber forest management:** the activity involves promoting training within the potential lines of extraction of non-timber products mapped by the Project, promoting sustainable development. In addition, the project proposes mapping the main opportunities and implementing partnerships for the development of the activities, increasing investment and improving the income of the families that depend on these activities;
- **Strengthening governance:** this involves expanding Amazônia Agroindústria EIRELI's action in the region, attracting investment and partnerships. By forming new partnerships, mainly for on-site work, the project should also stimulate the creation of indirect jobs and attract investment to the region;
- **Promoting actions that contribute to the insertion or maintenance of infrastructures:** the action involves improving the quality of life of communities by promoting the construction and maintenance of infrastructures through public incentives. It also strengthens partnerships with the municipality and various entities to raise standards in education, health, and leisure, while seeking to improve technological connectivity and facilitate river transportation, contributing to sustainable and inclusive progress.

In order to guarantee the efficiency and permanence of these actions listed above, the proponents must seek out the best techniques and procedures for conducting training for the people involved in the activities, always seeking to ensure the successful qualification of the team to work with the communities and fulfill the project's goals. These processes will follow all relevant laws and regulations relating to workers' rights. These processes will ensure that staff are engaged to meet the project's timeline and goals, with the aim of optimizing investments and avoiding the loss of human capital due to staff turnover. Other measures adopted to avoid the loss of acquired capacity will be the constant recording and reporting of procedures and the monitoring of the results obtained, since, in the event of staff turnover, procedures can be easily reproduced, mitigating impacts on the implementation of the project plan.

In addition, the project intends to seek partnerships with institutions capable of carrying out the proposed activities.

All activities are open to all residents of the communities surrounding the project. The participation of women, young people and marginalized people is encouraged by the proponents. In addition, the process of hiring workers involves all residents of the communities (Section 2.5.2).

2.3.18 Community Employment Opportunities (VCS, 3.19.13; CCB, G3.10)

The job opportunities offered by Amazônia Agroindústria EIRELI are equal to the surrounding communities, including management positions, if the requirements for the position are met. All job positions generated locally by the Project follow a Recruitment and Selection process, belonging to Amazônia Agroindústria, which carries out the entire process according to need and availability of vacancies.

No criteria of race, gender, sexual orientation, color, religion, age, ethnic origin, physical or mental disability or social class are adopted. All stages of the selection process, as well as the hiring of the professional, will be based on the criteria set out in the job description and a minimum qualification is desirable.

It should be noted that the project proponents already have teams made up mainly of local community members, which means that the project will only reinforce the actions already taken in this regard. Most of the people hired are local residents. In addition, they can be hired either through the interest of the community members themselves who come to Amazônia Agroindústria EIRELI, or through referrals from employees who already work on the farm.

2.3.19 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

There are no Amazônia Agroindústria workers in the project area. There are no productive activities in the project area.

However, within the scope of the socio-economic diagnosis of the Seringal Rio Branco REDD+ Project region and the demands of the project, there are plans for activities in the project area that may involve hiring workers. The first workers were recruited in late January and early February 2023, to support the field teams and the research itself within the project area. This is the first experience of dealing with workers.

Through the implementation of the actions proposed by the project, there will be a need to increase monitoring and surveillance actions, so it will be necessary to provide a specific team of asset security and monitoring, trained for the activities and with periodic training to meet the actions of the project.

In the table below, we present potential demands on workers related to project activities and proposals for training that can be carried out as part of the actions with workers in the Seringal Rio Branco REDD+ project.

Table 9: Potenciais demandas e propostas de ações formativas para futuros trabalhadores.

Potential demands on workers related to project activities	Training proposals
Access control and property security in the project areas.	Classification and training of workers for the surveillance function, training in security protocols and conduct in access control in the project area.
Registration of events that constitute threats to the project. This can be collected <i>on site</i> or remotely (drone).	Training in the registration and use of equipment related to the activity.
Registration of activities carried out within the project area.	Qualify the person responsible for correctly managing the data so that monitoring reports can be produced.

With regard to possible future risks associated with the project and mitigation measures that can be adopted if workers are involved, we present some suggestions in the table below:

Table 10: Possible risks x Mitigation procedures.

Possible risks	Suggested mitigation procedures
Avoid rumors about project activities.	There needs to be someone to mediate communication until trust is established.
Unequal working conditions for women	- Valuing women in different jobs with equal rights and salaries. - Creation of regular information campaigns on harassment, valuing women, gender relations, etc.
Workers with little or no information about the project	Regular lectures on the running of the farm and the responsibilities of each employee.
Workers with little or no information about their social and labor rights	- Create the Workers' Room as a kind of safe, non-judgmental place where workers can ask questions about their rights and duties and be well-informed and guided in all situations; - Hold regular information events on workers' rights.
Workers spending long periods in isolation, with no alternative leisure, transportation, radio, telephone or any kind of communication.	- Monitoring compliance with the time off regime in accordance with the CLT. - Implementation of leisure alternatives (such as: reading and television room, games and entertainment room), sports (such as: soccer field, sports equipment and collective games) and communication (such as: radio, satellite dish, internet, television channels).

In addition, Brazil has a series of rules that regulate formal work. The Regulatory Norms (NR) are complementary provisions to Chapter V (Occupational Safety and Medicine) of Title II of the Consolidation of Labor Laws (CLT), as amended by Law No. 6,514 of December 22, 1977. They consist of obligations, rights, and duties to be fulfilled by employers and workers with the aim of guaranteeing safe and healthy work, preventing the occurrence of illnesses and accidents at work.

With regard to the project's activities, the proponent is obliged to comply with all laws and regulations and, more specifically, to provide adequate protective equipment to the employees carrying out forest monitoring activities. In addition, the monitoring team is equipped with first aid kits.

The following are the main NR standards with possible links to the project:

- NR 4 - Specialized Services in Occupational Health and Safety (last update: MTPS Ordinance No. 510 of April 29, 2016).
- NR 5 - Internal Accident Prevention Commission (last updated: SIT Ordinance No. 247 of July 12, 2011).
- NR6 - Personal Protective Equipment - PPE.

- NR23 - Fire Protection.

- NR31 - Safety and Health at Work in Agriculture, Forest Livestock, Silviculture and Aquaculture.



Figure 21: Workers hired to support the research using personal protective equipment (PPE), 02/13/2023. Source: STA, 2023.

2.4 Management Capacity

2.4.1 Project Governance Structures (CCB, G4.1)

Project management will be the responsibility of Biofílica Ambipar Environment and Amazônia Agroindústria EIRELI. The obligations and commitments of the parties are described below:

Biophilic Responsibilities:

General coordination of socio-economic and environmental diagnoses (DSEA), baseline and carbon stock studies; construction of the PDD (Project Design Description); remote monitoring of forest cover and implementation/coordination of additional actions aimed at reducing/mitigating greenhouse gas (GHG) emissions; conducting validation/verification audits; dissemination of the Project; marketing of credits and co-management of the Project throughout its duration.

Responsibilities Amazônia Agroindústria EIRELI:

Investments needed to implement and validate the project (Capex); co-management of the project, as well as the development of all activities related to the environmental and social scopes, and infrastructure and logistical support for Biofílica and other professionals involved in the project. In addition, it must provide all the necessary support for the auditing processes, construction of promotional materials and other commercial processes.

During the development of the project, another organization (mentioned in Section 2.1.4) was involved in carrying out the diagnostic studies. Their responsibilities are described below:

Technical Solutions and Environmental Engineering Services - STA:

Development of the carbon stock estimation study, socio-economic diagnosis, and environmental studies, such as characterization of the physical environment and biodiversity of the region (flora and fauna) for the Seringal Rio Branco REDD+ Project.

In addition, as we have seen, the Seringal Rio Branco REDD+ Project is backed by human resources that have helped in its development and implementation.

2.4.2 Required Technical Skills (VCS, 3.19; CCB, G4.2)

The main technical skills required to implement the Seringal Rio Branco REDD+ Project are knowledge of the development and management of forest conservation projects in the Amazon biome, experience in the implementation, development, and assistance of programs for communities, implementation of effective land security and asset surveillance.

All the proponents involved in the project have the technical skills necessary for the successful completion of the Seringal Rio Branco REDD+ Project. Biofílica Ambipar Environment Investments is a Brazilian company that promotes the management of forest areas in the Amazon and Atlantic Forest biomes. The company has a specialized team and is a benchmark in the development of forest conservation projects, guaranteeing the quality and effectiveness of the REDD+ activities developed.

Amazônia Agroindústria is the legitimate owner and possessor of rural properties totaling approximately 54,903 ha (fifty-four thousand, nine hundred and three hectares) of forest, located in the state of Acre, in the municipalities of Cruzeiro do Sul and Porto Walter, in the Amazon biome, duly registered.

Amazônia Agroindústria maintains sustainable forest management activities on the properties, in its name, duly regularized and licensed with the competent environmental agencies and intends to maintain the use in line with the requirements for environmental conservation activities on the properties.

It is in Amazônia Agroindústria's interest to carry out projects aimed at the environmental conservation of the properties' forest areas, maintaining their ownership, developing environmental and social activities for the surrounding communities and the possibility of identifying new economic vocations for the forest area through the monetization of environmental services from the activities carried out on the property.

2.4.3 Management Team Experience (VCS, 3.19; CCB, G4.2)

Biofílica Ambipar Environment Investments

Plínio Ribeiro - Executive Director

Plínio Ribeiro has a degree in Business Administration from the Instituto de Ensino e Pesquisa - INSPER and a Master's degree in Public Administration and the Environment from Columbia University and the Earth Institute (USA). He has participated in various conservation projects on the lower Rio Negro, through the Institute of Ecological Research - IPÊ since 2005 and was one of the producers of Jean Michel Cousteau's documentary "Return to the Amazon". He has worked at Biofílica since 2008, where he has led projects, operations, and business management. He is currently the company's Executive Director and a shareholder.

Cláudio Padua - Scientific Director

Cláudio Pádua has a degree in Business Administration and Biology, a Master's in Latin American Studies and a PhD in Ecology from the University of Florida in Gainesville (USA). A retired professor from the University of Brasília, Pádua is currently rector of the School of Conservation and Sustainability and vice-president of the Institute for Ecological Research (IPÊ). He is also a Senior Associate Researcher at the Center for Environment and Conservation Studies at Columbia University (USA) and Director of International Conservation at the Wildlife Trust Alliance, as well as a consultant for the Brazilian Biodiversity Fund (FUNBIO) and WWF Brasil. Pádua represents Brazil on the International Advisory Group (IAG) of the G7 Pilot Program. In 2003, together with his wife, Suzana Pádua, he was named "Hero of the Planet" by Time magazine for his activities in favor of biodiversity conservation. Between 1997 and 2007, he won six conservation awards, three national and three international. Pádua has published two books and more than 30 articles in national and international scientific journals. Since 2008 he has led Biofílica's scientific involvement and production as Scientific Director and advisor.

Paula Conde - Financial and Administrative Analyst

Paula Conde has a degree in Business Administration from PUC São Luís and a postgraduate degree in Accounting and Financial Administration from FAAP. She has extensive experience, mostly in one of the largest media and education groups in Latin America - Editora Abril, where she worked with Financial Control and Reporting, Treasury, Accounting and Financial Reconciliation, Accounts Payable and Receivable and Royalties. At Biofílica, she is responsible for administrative and financial activities and logistical support for the team and projects.

Soraya Pires - Operational Director

Agronomist, graduated from Esalq/USP with a specialization in strategic management and finance from FGV, with 15 years' experience in agribusiness management and development and a career developed in large multinational companies in the sugar-energy sector (Adecoagro, BP and BP Bunge). She has been with Biofílica since August 2022, carrying out activities related to business development, operational management, financial modeling, strategic planning and AFOLU project management.

Caio Gallego - Operations Manager

Caio Gallego is a Forestry Engineer with a degree from the University of São Paulo (USP/ESALQ). He is a specialist in geoprocessing and remote sensing focused on environmental conservation, mapping and analysis of changes in land use. He has knowledge of sustainable forest management, environmental modeling and the use of alternative GIS for silviculture and agribusiness. He has advanced knowledge in the use of GIS software and analysis of changes in land use and land cover, such as ArcGIS, QuantumGIS and DinâmicaEGO. Caio Gallego has more than 8 years' experience working with the development and implementation of AFOLU projects for voluntary carbon markets. He has worked with project development, focusing mainly on the implementation of one of the first REDD+ projects within the VCS program in the Brazilian Amazon. His main expertise has always focused on the technical development of projects, having been responsible for more than 10 validation/verification processes. In more than 10 projects in which he was directly responsible for development, he worked on the application of socio-environmental safeguards, guaranteeing the generation of net positive impacts on the climate, the community and biodiversity, as well as ensuring the maintenance of exceptional criteria applicable to each of these scopes.

Laion Pazian - Commercial Manager

Laion Pazian has a degree in Economics from the University of São Paulo (USP/ESALQ) and an MBA in Commercial Management from the Getúlio Vargas Foundation. He manages Biofílica's carbon credits commercial team, key-accounts, commercial policy and strategy. He also monitors and directs carbon market intelligence analysis and is responsible for the area's pricing policy and planning.

Ricardo Cordeiro - Communications Coordinator

Ricardo Cordeiro is an advertising executive and on- and off-line art director with over 10 years' experience. He has worked in digital agencies, trade and live marketing. Experienced in UX, planning and digital strategies. Specialized in Digital Marketing and Web Project Management. At Biofílica he works as communications coordinator, responsible for digital marketing, branding and institutional communications.

Márcio Sales - Statistical Modeling Specialist

Márcio Sales is a statistician with a degree from the Federal University of Pará, a Master's degree in Geography from the University of California, Santa Barbara and a PhD in Production Ecology and Resource Conservation from Wageningen University in the Netherlands. He is a specialist in data analysis and conducts research into geostatistical modeling of processes distributed in space and time. He works at Biofílica producing projections of GHG emissions from future deforestation for project baselines and monitoring deforestation by satellite.

Leonardo Albiero Gomes de Almeida – Project Coordinator

Leonardo is a Forestry Engineer with a degree from São Paulo State University (UNESP-FCA) and an extension course at the Hochschule Weiheinstephan Triesdorf in Germany. Green Belt certified, he has experience in project management and process improvement in the forestry production chain. He has also worked with sustainable forest management, wood panel production and environmental licensing. At Biofílica Ambipar he coordinates 'REDD+ Seringal Rio Branco' (ID 4834), 'Renove ALM Brazil' (ID 4849), the 'Corridors for Life ARR Grouped Project' (ID 3727) and REDD+ Juruá (ID 3430), all of which are in the process of being validated or have already been validated by the VCS and CCB standards. He has also coordinated a feasibility study for REDD+ projects in Pará State Forests.

Shaxahmary de M. C. dos Santos – Project Coordinator

Shaxahmary dos Santos is a Forestry Engineer with a degree from the University of São Paulo (USP/ESALQ). She is currently a specialist in Project Management. In her last professional experiences, and during her undergraduate studies, she worked on natural resource conservation, climate change, payments for environmental services, geoprocessing, restoration, and forest hydrology. She has been with Biofílica since March 2021, working in management, more specifically in the development, implementation, and monitoring of AFOLU projects.

Amanda Rocha Fiallos - Project Analyst

Amanda Fiallos is a Forestry Engineer who graduated from the University of São Paulo (USP/ESALQ). She has experience in forest restoration and conservation projects, environmental education, geoprocessing, and remote sensing applied to the planning of planted forests. She has been working at Biofílica since May 2022, producing technical reports, and reporting on the activities of the AFOLU Projects, and evaluating reports and products delivered by suppliers regarding the socio-environmental diagnoses carried out.

Bianca de Oliveira - Project Analyst

Bianca de Oliveira is a Forestry Engineer who graduated from the Federal Rural University of Rio de Janeiro (UFRRJ) and has a postgraduate degree in Integrated Environmental Management from the State University of Rio de Janeiro (UERJ). She has experience in forest restoration and conservation projects, environmental licensing, and payments for environmental services. She has been working at Biofílica Ambipar since March 2023, producing technical reports, analyzing environmental diagnoses, and developing AFOLU projects.

Raphael Ramiro - FP&A Specialist

Raphael Ramiro is an administrator with a degree from São Paulo State University and a post-graduate degree in Corporate Finance from the Federal University of São Carlos. He has over twelve years' experience in analyzing and evaluating projects from an economic and financial point of view. He has been with Biofílica since May 2022, carrying out activities related to the Valuation and Financial Modeling of Carbon Projects - (AR, REDD+, ALM, Blue Carbon); Business Valuation (M&A); analysis of Financial Additionality - Carbon Projects; Executive Board / Board of Directors Presentations; Financial Performance Analysis / KPIs.

Rafael Costa - Legal Support

Rafael Costa holds a PhD in International Law from the University of São Paulo. Master's in political and Economic Law from Mackenzie Presbyterian University. He is a specialist in Public and Tax Law and has an MBA in Business Management from USP. He has 17 years' experience in litigation and advisory in the promotion of legal actions, as well as out-of-court dispute resolution in civil, corporate and tax matters. He has worked on international investigations into Compliance (Anti-Corruption) and has been teaching undergraduate and postgraduate law for 9 years.

Marco Antonio Martins - Geoprocessing Analyst

Marco Antonio Martins is a geographer with a degree from the University of São Paulo (FFLCH-USP) and a master's student in the areas of environmental planning and remote sensing, working on mapping priority areas for forest restoration on farms. During his degree, he worked on geoprocessing and remote sensing projects applied to forest monitoring, territorial planning, mapping coastal habitats and restored areas in conservation units. Since 2021, he has been working on AFOLU projects, developing geographic intelligence analysis for forest conservation projects, forest corridor restoration and low-carbon agriculture.

Franciane Almeida - Geoprocessing Analyst

Franciane Almeida is a Forestry Engineer who graduated from the Federal University of Lavras (UFLA). She has experience in geoprocessing and remote sensing applied to the areas of forest planning and harvesting, recovery of degraded areas and land regularization. She also has experience in AFOLU projects, mainly in monitoring the forest cover of REDD+ projects based on satellite images and available official deforestation databases, such as PRODES.

Amazônia Agroindústria EIRELI

James Cameli - Amazônia Agroindústria EIRELI

Location: Rio Branco, AC - Brazil

James Cameli, a successful businessman who is recognized throughout the western Brazilian Amazon. He owns a conglomerate of 5 companies, operating for over 30 years in the sectors of agriculture, telecommunications, construction, industrialization and processing of forest products and environmental services. He is an advocate of environmental conservation policies and projects and is convinced that the "standing forest" is one of the region's greatest assets and therefore represents a strategic asset for Brazil's social and economic development and its contribution to changing current global climate conditions.

Marc Maleika - Amazônia Agroindústria EIRELI

Location: Rio Branco, AC - Brazil

Marc Maleika holds a Master's degree in International Affairs with an emphasis on International Politics and Finance from Columbia University, New York and a Master's degree in Economics with an emphasis on Financial Economics from Maastricht University. He has research experience in the area of the voluntary carbon market, was the creator of a sustainability incubator, and has developed reports on natural disaster preparedness.

Scarlett Siqueira do Valle - Amazônia Agroindústria EIRELI (Legal Advisor)

Location: São Paulo, São Paulo, Brazil

Scarlett Siqueira do Valle is an independent lawyer (401449 OAB/SP) with a degree in Law from Centro Universitário de Brasília - UNICEUB and a PhD in Communication and Semiotics from Pontifícia Universidade Católica de São Paulo - PUC-SP. She has experience in the areas of Law for Startups; Digital Law; LGPD and Business Law. As a corporate lawyer, she has experience in environmental conservation projects.

Veriton Viana da Costa - Amazônia Agroindústria EIRELI (Technical Advisor and Local Support)

Location: Cruzeiro do Sul, AC - Brazil

Veriton Costa is an agronomist and agricultural technician who graduated from the Federal University of Acre (UFAC). He has experience in Financial Management, Traditional Population Registration, Physical and Chemical Soil Analysis, Geoprocessing (ArcGis), People Management, Project Preparation, Report Preparation (Power BI and Excel), Agricultural and Forestry Defense Inspection, Lectures and Courses. At Amazônia Agroindústria EIRELI he works as an Advisor and Local Support.

Gilberto do Carmo Lopes Siqueira - Amazônia Agroindústria EIRELI (Consultant)

Localização: Cruzeiro do Sul, AC - Brasil

Gilberto Siqueira is a civil engineer and agricultural technician who graduated from the Federal University of Mato Grosso do Sul (UFMS), specialising in wood and wood structures. He has experience in civil engineering and project management. He is currently a consultant at the Euvaldo Lodi Institute - Acre.

2.4.4 Project Management Partnerships and Team Development (VCS, 3.19; CCB, G4.2)

The Seringal Rio Branco REDD+ Project has all the necessary partnerships for the construction and implementation of forest asset conservation activities. Currently, the partner institutions mentioned in Section 2.1.8 are responsible for preparing the Socioeconomic and Environmental Diagnoses, Carbon

Stock and Baseline, which make up the Project Design Document, through service supply contracts. When other initiatives throughout the development of the Project require new technical expertise and partners, the proponents of the Rio Branco REDD+ Project will join forces with governmental, non-governmental and private sector organizations in order to enable the generation of net positive impacts on society and biodiversity.

2.4.5 Financial Health of Implementing Organization(s) (CCB, G4.3)

Biofílica Ambipar Environment Investments is a Brazilian company with 15 years of experience in the environmental services market in Brazil, through the generation and commercialization of carbon credits from nature-based solutions (NBS), with a diversified business line and investors who support the company's business.

Amazônia Agroindústria is the legitimate owner and possessor of rural properties totaling approximately 54,903 ha (fifty-four thousand, nine hundred and three hectares) of forest, located in the state of Acre, in the municipalities of Cruzeiro do Sul and Porto Walter, in the Amazon biome, duly registered.

Amazônia Agroindústria maintains sustainable forest management activities on the properties, in its name, duly regularized and licensed with the competent environmental agencies and intends to maintain the use in line with the requirements for environmental conservation activities on the properties.

It is in Amazônia Agroindústria's interest to carry out projects aimed at the environmental conservation of the properties' forest areas, maintaining their ownership, developing environmental and social activities for the surrounding communities and the possibility of identifying new economic vocations for the forest area through the monetization of environmental services from the activities carried out on the property.

2.4.6 Avoidance of Corruption and Other Unethical Behavior (VCS, 3.19; CCB, G4.3)

Biofílica Ambipar Environment Investments undergoes annual financial audits to ensure that its funds are allocated responsibly and free from corruption. The company's financial statements and minutes of assemblies are published on the JusBrasil website, the largest open legal community in Latin America. Biofílica Ambipar Environment Investments, part of the Ambipar Group, has a code of ethics and conduct that represents all the Group's companies. The document serves as a primary instrument to guide the conduct of all parties involved in the adoption of good practices in relationships and in doing business, guiding attitudes of respect for diversity, to combat any form of prejudice or harassment.

2.4.7 Commercially Sensitive Information (VCS, 3.5.2 – 3.5.4; CCB Rules, 3.5.13 – 3.5.14)

Some information required by the VCS and/or CCB standard is considered confidential or commercially sensitive and cannot be publicly disclosed by the project proponents. This information was fully provided to the audit team during the validation process attached to this document but was not included in the public version. Below is a list of the information that was made available:

- Land documents and legal status;
- Proponents' financial statements;
- Project financial performance spreadsheet (budget) and other related documents;

- Agreements and contracts signed between the parties involved;
- Inventories and other diagnostics.

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

Compliance with laws, statutes, and other significant regulatory bodies for the Seringal Rio Branco REDD+ Project is related to the Non-Timber Forest Product Management activity. In the state of Acre, the project's activities are being licensed by the Brazilian Institute for the Environment and Renewable Natural Resources (IBAMA), so federal legislation must be applied. Subordinate to federal legislation, state-level legislation applies.

Regarding REDD+ activities, there has been a history of initiatives despite the construction and negotiation of this concept through agreements and meetings at the United Nations Framework Convention on Climate Change (UNFCCC). In December 2015, Brazil's National REDD+ Strategy (ENREDD+) was established by MMA Ordinance No. 370, as a document that formalizes for Brazilian society and UNFCCC signatory countries how the Brazilian government has structured its efforts and intends to improve them by 2020, contributing to climate change mitigation by controlling deforestation and forest degradation, promoting forest recovery, and fostering sustainable development. In this context, in Brazil, Decree No. 10,144 (of 11/28/2019) established the National Commission for REDD+ (CONAREDD+) in order to coordinate, follow up, monitor and review the National Strategy for REDD+ and guide the development of requirements for access to payments for results of REDD+ policies and actions in the country. The following year, CONAREDD+'s internal regulations were published by means of an Ordinance (No. 544, of October 26, 2020).

At the same time, of a highly relevant nature, Bill No. 572/2020, which "Establishes the national system for reducing emissions from deforestation and degradation, conservation, sustainable forest management, maintenance and increase of forest carbon stocks (REDD+) and other measures", is currently under analysis. The text is currently before the Chamber of Deputies.

As far as the carbon market is concerned, there is a Bill (PL 528 of 2021) currently before the Chamber of Deputies which aims to establish the Brazilian Emissions Reduction Market (MBRE) and regulate the purchase and sale of carbon credits in the country arising from activities to Reduce Emissions from Deforestation and Forest Degradation, for example. The promotion of this voluntary carbon market is provided for in the Law that instituted the National Climate Change Policy (Law No. 12.187, of 12/29/2009).

After years of discussion and the stagnation of Bill 528 of 2021 in the National Congress, Decree 11.075 of 19/05/2022 was recently enacted, which deals with the implementation of a regulated carbon credit market in Brazil through the creation of the National System for Reducing Greenhouse Gas Emissions (Sinare) and establishes procedures for drawing up Sectoral Climate Change Mitigation Plans. In addition to these measures, the document also introduces new concepts relating to methane credit, recording the carbon footprint of processes and activities, native vegetation carbon, soil carbon and blue carbon.

The main relevant legislation and regulations at federal and state level are listed and detailed below. In addition, there is a brief analysis of the international climate agreements that have been driving the creation and development of REDD+ initiatives around the world.

Federal legislation

- Law No. 14.119, of 01/13/2021: Establishes the National Policy for Payment for Environmental Services; and amends Laws No. 8.212, of July 24, 1991, No. 8.629, of February 25, 1993, and No. 6.015, of December 31, 1973, to adapt them to the new policy. The project complies with the law because it is being carried out on a rural property that has been registered with the CAR, as required by Article 9, and because it is among the types of environmental services listed in Article 3.
- Law No. 12.727, of 12/17/2012: Provides for the protection of native vegetation; amends Law No. 12.651, of May 25, 2012, which provides for the protection of native vegetation; amends Laws Nos. 6.938, of August 31, 1981, 9.393, of December 19, 1996, and 11.428, of December 22, 2006; and repeals Laws 4.771, of September 15, 1965, and 7.754, of April 14, 1989, Provisional Measure 2.166-67, of August 24, 2001, item 22 of item II of art. 167 of Law 6.015, of December 31, 1973, and § 2 of art. 4 of Law 12.651, of May 25, 2012. The aim of the law in question is sustainable development and the establishment of general rules on the protection of native vegetation, Permanent Preservation Areas (PPA) and Legal Reserve areas (LR). The project complies with this law because it takes place on a property with a duly delimited and protected LR and PPA and because it has no activities that violate the limits of exploitation and management of these areas.
- Law No. 12.651, of May 25, 2012: Provides for the protection of native vegetation; amends Laws Nos. 6.938, of August 31, 1981, 9.393, of December 19, 1996, and 11.428, of December 22, 2006; repeals Laws Nos. 4.771, of September 15, 1965, and 7.754, of April 14, 1989, and Provisional Measure No. 2.166-67, of August 24, 2001; and makes other provisions. This law, known as the Forest Code, regulates how native vegetation can be exploited. The project complies with this law because it takes place on a property that complies with the forestry code's requirements regarding protected areas and exploitation limits. In addition, the only exploitation that the project contemplates is that of non-timber products, an activity that, according to art. 21, is free in a legal reserve area.
- Law No. 12.187, of 29/12/2009: Establishes the National Policy on Climate Change - NPCC and makes other provisions. The Project contributes to this National Policy because the activities carried out are in harmony with the objectives: the Project helps to reduce the impacts of anthropogenic actions on the climate system, in line with Art. 3, contributes to the reduction of anthropogenic greenhouse gas emissions and the conservation of environmental resources, in line with Art. 4, among others.
- Decree No. 11.075, of May 19, 2022: Establishes the procedures for drawing up Sectoral Climate Change Mitigation Plans, institutes the National System for Reducing Greenhouse Gas Emissions and amends Decree No. 11.003, of March 21, 2022.
- Decree No. 10.144, of 11/28/2019: Establishes the National Commission for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conserving Forest Carbon Stocks, Sustainable Forest Management, and Increasing Forest Carbon Stocks - REDD+.
- Decree No. 58.054, of 03/23/1966: Promulgates the Convention for the Protection of the Flora, Fauna and Scenic Beauties of the Countries of America.

- Decree No. 2.661, of 07/08/1998: Regulates the sole paragraph of art. 27 of Law No. 4.771, of September 15, 1965 (Forestry Code), by establishing precautionary rules regarding the use of fire in agropastoral and forestry practices, and makes other provisions.
- Decree No. 5.975, of 11/30/2006: Regulates articles 12, final part, 15, 16, 19, 20 and 21 of Law No. 4.771, of September 15, 1965, article 4, item III, of Law No. 6.938, of August 31, 1981, article 2 of Law No. 10.650, of April 16, 2003, amends and adds provisions to Decrees Nos. 6.514/08 and 3.420/00 and makes other provisions.
- CONAMA Resolution No. 16, of 12/07/1989: Institutes the Integrated Program for Environmental Assessment and Control in the Legal Amazon.
- CONAMA Resolution No. 378, of October 19, 2006: Defines projects that potentially cause national or regional environmental impact for the purposes of the provisions of item III, § 1, art. 19 of Law No. 4.771, of September 15, 1965, and makes other provisions.
- CONAMA Resolution No. 379, of 10/19/2006: Creates and regulates a data and information system on forest management within the National Environment System - SISNAMA.
- IBAMA Ordinance No. 218, of 05/04/1989: Provides for the felling and exploitation of native forests and native successor forest formations of the Atlantic Forest, and other provisions.
- IBAMA Ordinance No. 438, of August 9, 1989: Amends the wording of article 4 of Ordinance No. 218, of May 4, 1989.
- MMA Ordinance No. 103, of April 5, 2006: Provides for the implementation of the Document of Forest Origin - DOF and makes other provisions.
- MMA Ordinance No. 253, of August 18, 2006: Establishes, as of September 1, 2006, within the scope of the Brazilian Institute for the Environment and Renewable Natural Resources - IBAMA, the Document of Forest Origin - DOF to replace the Authorization for the Transport of Forest Products - ATPF.
- Ordinance No. 1896, of December 9, 2013: Amends Regulatory Standard no. 31.
- MMA Normative Instruction No. 1, of September 5, 1996: Provides for Compulsory Forest Replacement and the Integrated Forestry Plan.
- MMA Normative Instruction No. 07, of 04/27/1999: Provides for authorization for deforestation in the states of the Legal Amazon.
- MMA Normative Instruction No. 02, of 05/10/2001: Provides for the economic exploitation of forests on rural properties located in the Legal Amazon, including Legal Reserve areas and with the exception of permanent preservation areas established in current legislation, which will be carried out through sustainable forest management practices for multiple use.
- MMA Normative Instruction No. 06, of 12/15/2006: Provides for forest replenishment and the consumption of forest raw materials, and makes other provisions.
- IBAMA Normative Instruction No. 178, of 06/23/2008: Defines the guidelines and procedures for IBAMA's appraisal and approval of authorizations to suppress forests and other forms of native vegetation in areas larger than 2,000 hectares on rural properties located in the Legal Amazon and 1,000 hectares on rural properties located in other regions of the country.
- Regulatory Standard No. 31, of 03/03/2005: Approves the Regulatory Standard for Safety and Health at Work in Agriculture, Livestock, Silviculture, Forestry Exploration and Aquaculture.

State Legislation

- State Law no. 2.308, of October 22, 2010: "Creates the State System of Incentives for Environmental Services - SISA, the Incentives Program for Environmental Services - ISA Carbono and other Programs for Environmental Services and Ecosystem Products in the State of Acre and makes other provisions". Information on how the project relates to this law can be found in section 1.11.
- State Law No. 1.426 of 12/27/2001: "Provides for the preservation and conservation of the state's forests, establishes the State System of Protected Natural Areas, creates the State Forestry Council and the State Forestry Fund, and makes other provisions." The project complies with this law as it has a management plan for non-timber forest products and authorization to use forest raw materials from IBAMA and the Acre Environment Institute (IMAC).
- State Law No. 3.883, of December 17, 2021: Provides for the State System of Protected Natural Areas - SEANP, which includes state, municipal and federal conservation units, indigenous lands, legal reserves (LR) and permanent preservation areas (PPA). For LR and PPA, the SEANP aims to promote public policies and actions aimed at the sustainable use of natural resources, the formation of ecological corridors and other compatible measures. The project complies with this law as it includes activities aimed at the sustainable use of natural resources.
- State Law no. 1.548, of January 29, 2004: Amends art. 9 of Law no. 1.426, of December 27, 2001.
- State Law No. 2,836 of December 30, 2013: Amends Law No. 1,426 of December 27, 2001, which provides for the preservation and conservation of the state's forests, establishes the State System of Protected Natural Areas, creates the State Forestry Council and the State Forestry Fund, and makes other provisions.
- State Law No. 3,595 of December 20, 2019: Amends provisions of Laws No. 1,022 of January 21, 1992, Law No. 1,117 of January 26, 1994 and Law No. 1,426 of December 27, 2001.
- Complementary State Law No. 300, of July 9, 2015: Amends Complementary Law no. 247, of February 17, 2012, which provides for the administrative structure of the Executive Branch; Law no. 2,308, of October 22, 2010, which creates the State System of Incentives for Environmental Services -SISA, the Program of Incentives for Environmental Services - ISA Carbono and other Programs of Environmental Services and Ecosystem Products of the State of Acre; Law no. 1,426, of December 27, 2001, which provides for the preservation and conservation of the state's forests, establishes the State System of Protected Natural Areas, creates the State Forestry Council and the State Forestry Fund, and makes other provisions.
- State Law No. 1.460, of May 3, 2002. "Institutes the Program to Support Traditional Populations and Small Producers-PRÓ-FLORESTANIA and makes other provisions.
- State Law No. 1.904, of June 5, 2007. "Institutes the Ecological-Economic Zoning of the State of Acre - ZEE.
- State Law No. 1.963, of December 4, 2007. " Provides for Vegetal Sanitary Defense in the State of Acre.
- State Law No. 2.024, of October 20, 2008. " Creates the State Program of Incentive to Family Forestry and Agroforestry Production.
- Decree No. 503, of April 6, 1999. "Institutes the State Ecological-Economic Zoning Program for the State of Acre and makes other provisions.
- Decree No. 8.452, of August 14, 2003. "Establishes the structure and composition of the State Forestry Council and regulates the Forestry Fund.

- Decree No. 3.414, of September 12, 2008. "Provides for forest replenishment in the state of Acre due to the consumption of forest raw materials.
- Decree No. 3.415, of September 12, 2008. "Provides for the creation of the Acre State Commission for Environmental Risk Management.
- Decree No. 5.507, of July 15, 2010. " Provides for the State Council for Rural and Sustainable Forestry Development - CDRFS.
- Resolution/CEMACT No. 001 of June 22, 2005. "Approves the terms of IMAC/IBAMA Interinstitutional Ordinance No. 001 of June 4, 2005, which deals with the definition of minimum standards for the sustainable use of the cat's claw vine.

Municipal Legislation - Cruzeiro do Sul

- Municipal Law No. 453, of October 7, 2006: Provides for the participatory Master Plan of the municipality of Cruzeiro do Sul and makes other provisions"
- Municipal Law No. 457, of December 7, 2006: Provides for the Municipal Environmental Policy of the municipality of Cruzeiro do Sul, its purposes and mechanisms for formulation and application, establishing the Municipal Environmental System and altering the competencies of SEMMA and COMDEMA, and makes other provisions.

International agreements

- FCCC/CP/2005/Misc.1: Reducing emissions from deforestation in developing countries: approaches to stimulate action. Submission from Parties. COP 11, Montreal, 2005.
- FCCC/CP/2007/6/add.1: Report of the Conference of the Parties on its thirteenth session, held in Bali from 3 to 15 December 2007. Addendum. Part two: Action taken by the Conference of the Parties at its thirteenth session or "Action Bali Plan". COP 13, Bali, 2007.
- FCCC/CP/2009/Add.1: Report of the Conference of the Parties on its fifteenth session, held in Copenhagen from 7 to 19 December 2009. Addendum. Part Two: Action taken by the Conference of the Parties at its fifteenth session or "Copenhagen Accord". COP 15, Copenhagen, 2009).
- FCCC/CP/2010/7/Add.1: Report of the Conference of the Parties on its sixteenth session, held in Cancun from 29 November to 10 December 2010. Addendum. Part Two: Action taken by the Conference of the Parties at its sixteenth session or "Cancun Agreement". COP 16, Cancun, 2010).
- FCCC/CP/2011/9/Add. 1: Report of the Conference of the Parties on its seventeenth session, held in Durban from November 28 to December 11, 2011. Addendum. Part Two: Action taken by the Conference of the Parties at its seventeenth session, COP 17, Durban, 2011.
- FCCC/CP/2012/8/Add.1: Report of the Conference of the Parties on its eighteenth session, held in Doha from 26 November to 8 December 2012. Addendum. Part two: Action taken by the Conference of the Parties at its eighteenth session.
- FCCC/CP/2013/Add.1: Warsaw Framework for REDD-plus, held in Warsaw, Poland, from November 11 to 22, 2013, in particular the following decisions:
 - Decision9/CP.19: Work program on results-based finance to progress the full implementation of the activities referred to in decision 1/CP. 16, paragraph 70.
 - Decision10/CP.19: Coordination of support for the implementation of activities in relation to mitigation actions in the forest sector by developing countries, including institutional arrangements.

- Decision12/CP.19: The timing and the frequency of presentations of the summary of information on how all the safeguards referred to in decision1/CP.16, appendix I, are being addressed and respected.
- Decision13/CP.19: Guidelines and procedures for the technical assessment of submissions from Parties on proposed forest reference emission levels and/or forest reference levels.
- Decision14/CP.19: Modalities for measuring, reporting, and verifying.
- Decision15/CP.19: Addressing the drivers of deforestation and forest degradation.
- FCCC/CP/2015/Add.1: Report of the Conference of the Parties on its twenty-first session, held in Paris from 30 November to 13 December 2015. Addendum. Part two: Action taken by the Conference of the Parties at its twenty-first session.
- FCCC/CP/2015 Paris Agreement: Global, legally binding agreement that sets out a global framework to avoid dangerous climate change by limiting global warming to well below 2°C and pursuing efforts to limit it to 1.5°C. Entry into force on November 4, 2016.
- FCCC/CP/2016 Decisions adopted by the Conference of the Parties (COP): Especially decisions 1 (preparation into force of the Paris Agreement), 3 (Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts), 6 (National adaptation plans) and 7 (Long-term climate finance).
- FCCC/CP/2017, FCCC/CP/2018, FCCC/CP/2019 Decisions adopted by the COP: Especially decision 1 reporting on developments of the implementation of the Paris Agreement.
- Nationally Determined Contribution – Brazilian NDC submitted in September 2015 to the United Nations Framework Convention on Climate Change for mitigation, adaptation and means of implementation, in a manner consistent with the purpose of the contributions to achieve the ultimate objective of the Convention, under the terms of decision 1/CP.20, paragraph 9.
- CITES, of 03/03/1973: "Convention on International Trade in Endangered Species of Wild Fauna and Flora", signed in Washington D.C. on March 3, 1973, amended in Bonn on June 22, 1979.
- Article 6 of the Paris Agreement (2021): Decision 1/CP.21 mandated the SBSTA to operationalize the provisions of this Article through recommending a set of decisions to the COP serving as the meeting of the Parties to the Paris Agreement at its first session. At COP26, the Parties to the Paris Agreement at its third session (CMA 3) adopted three main decisions related to Article 6: decision 2 (on Article 6.2), decision 3 (on Article 6.4) and decision 4 (on Article 6.8).
- Glasgow Leaders' Declaration on Forests and Land Use (2021): Signatories (including Brazil) promise to reverse and end deforestation by 2030.
- Brazilian Nationally Determined Contribution (NDC): First Brazilian NDC submitted in September 2015 to the UN Framework Convention on Climate Change for mitigation, adaptation and means of implementation, in a manner consistent with the purpose of contributions to achieve the ultimate objective of the Convention, pursuant to Decision 1/CP.20, paragraph 9. The updated Brazilian NDC was presented at the COP26 on December 8th, 2022.achieved.

2.5.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18.2; CCB, G3.11)

It is guaranteed that all employees belonging to Amazônia Agroindústria EIRELI, Biofílica Ambipar Environmental, and the companies that provide services are legally hired in compliance with Brazilian labor

legislation (CLT), such as transparency in the conditions of working hours, payment of wages, vacations, requests and claims, obligations of the employer and employees, and guarantees of health and safety at work.

In addition, international agreements ratified by Brazil and issues relating to worker welfare are respected.

Every year, Biofilica's compliance with labor laws and regulations is audited, as it is an S.A. company. Its financial statements are published on the website of Jus Brasil, the largest open legal community in Latin America.

Procedures for informing workers of their rights will be strengthened and improved throughout the Project's life cycle, since the main objective of the "Strengthening governance" activity is to create actions and conditions to facilitate the operationalization and implementation of the activities proposed by the Seringal Rio Branco REDD+ Project. To this end, the Project intends to strengthen relations between stakeholders, above all by implementing a communication plan that guarantees transparency, assimilation of suggestions and conflict management (See Section 2.3.8 and Section 2.3.12). Thus, the Project has the intention and potential to create opportunities for the exchange of experiences between stakeholders and the dissemination of information of collective interest, including the rights and duties of the workers involved in and impacted by the Project.

Listed below are the relevant laws and regulations that protect workers' rights in Brazil, as well as the international agreements ratified by Brazil on labor issues.

Federal legislation and regulations:

- Decree-Law No. 5,452 of May 1, 1943: Approves the Consolidation of Labor Laws.
- Law No. 6.514, of December 22, 1977: Amends Chapter V of Title II of the Consolidation of Labor Laws, relating to occupational safety and medicine, and makes other provisions.

International agreements ratified by Brazil:

- International Labor Organization Convention No. 29 of 1930, ratified by Brazil on April 25, 1957: Provides for the abolition of forced labor.
- International Labor Organization Convention No. 87 of 1940: Provides for freedom of association.
- International Labor Organization Convention No. 97 of 1949, ratified by Brazil on June 18, 1965: Provides for migrant workers.
- International Labor Organization Convention No. 98 of 1949, ratified by Brazil on November 18, 1952: Provides for the right to associate and collective negotiation.
- International Labor Organization Convention No. 100 of 1951, ratified by Brazil on April 25, 1957: Provides for equal pay for men and women.
- International Labor Organization Convention No. 105, ratified by Brazil on June 18, 1965: Provides for the abolition of forced labor.
- International Labor Organization Convention No. 111 of 1958, ratified by Brazil on March 1, 1965: Provides for discrimination in matters of employment and occupation.
- International Labor Organization Convention No. 131 of 1970, ratified by Brazil on May 4, 1983: Provides for the fixing of minimum wages, especially in developing countries.

- International Labor Organization Convention No. 138 of 1973, ratified by Brazil on 06/28/2001: Provides for the minimum age for admission.
- International Labor Organization Convention No. 142 of 1975, ratified by Brazil on 11/24/1981: Provides for the development of human resources.
- International Labor Organization Convention No. 143 of 1975: Provides for immigration under abusive conditions and the promotion of equal opportunities for migrant workers.
- International Labor Organization Convention No. 155 of 1981, ratified by Brazil on May 18, 1992: Provides for the safety and health of workers.
- International Labor Organization Convention No. 169 of 1989, ratified by Brazil on 07/25/2002: Provides for indigenous and tribal rights.
- International Labor Organization Convention No. 182, ratified by Brazil on 02/02/2000: Provides for the prohibition of the worst forms of child labor and immediate action for their elimination.

2.5.3 Human Rights (VCS, 3.19)

Biofílica Ambipar has focused its efforts on building its own policy at institutional level to guarantee the protection of human rights. Partnerships and consultancies have been established and since then a work plan has been developed. A complete mapping of guidelines and procedures has already been established and the pillars of the Human Rights Protection policy have been defined.

Within these pillars, it is understood that the appropriation of the concepts of human rights, decent work and contemporary slavery and their main indicators is fundamental both as a starting point for drawing up a corporate human rights and decent work policy and so that the project can effectively identify the risks and adverse impacts on labor human rights, with an emphasis on the abolition of slave labor.

- **Human Rights**

Human rights are rights that belong to all people simply because they are human beings. As such, human rights are not granted by any state or system, and do not depend on elements such as gender, nationality, national or ethnic origin, race, religion, language, among others. These rights include basic issues such as the right to life, and rights aimed at guaranteeing a dignified life, such as the rights to health, food, freedom and work (UNHR)²⁷.

- **Decent Work**

Decent work, also known as dignified work, is part of the social dimension of human rights, related to their history and the historical process of guaranteeing these rights (COMPARATO, 2011)²⁸, known as second-dimension human rights, which were enshrined in national constitutional charters from the second decade of the 20th century in response to the increased exploitation of labor relations brought about by the industrial revolution and state abstention from regulating economic and labor relations.

²⁷ United Nations Human Rights (UNHR). What are human rights? Available in: <https://www.ohchr.org/en/what-are-human-rights>.

²⁸ Comparato, Fábio Konder. 2011. The historical affirmation of human rights. São Paulo, Saraiva, 589 p. Available in: https://disciplinas.usp.br/pluginfile.php/18490/mod_resource/content/1/CHY%20-%20Comparato%20-%20Introdu%C3%A7%C3%A3o.pdf.

From the perspective of the human right to work and work, decent work is understood as a vector through which human beings can acquire the full development of their identity, personality, potential and capacities. Through it, each individual can constitute themselves as an autonomous person, being able to access material well-being and enjoy the goods available to society, achieving the right to continuous improvement of living conditions and an adequate standard of living for themselves and their families (art. 25, UDHR)²⁹.

- **Contemporary slavery or work in conditions analogous to slavery**

Contemporary slavery, also known as work in conditions analogous to slavery, is the result of the permanence of a slave mentality and behavior that finds updated forms of exploitation of labor and subjugation of the working person.

In the contexts and situations in which slave labor is used, workers are treated as disposable labor, as a mere means of obtaining profit and wealth, as something that is denied basic and essential human rights. A crime with an economic bias, contemporary slave labor is motivated by the maximization of profit, to the detriment of life, dignity, and the irreducible and universal value of the human person.

In fact, with regard to contemporary slave labor, it is understood that situations of economic vulnerability, racial vulnerability, cultural and educational vulnerability, vulnerability to shocks and territorial vulnerability intersect and make poor, black, poorly educated, migrant people, victims of forced displacement, and people from remote, underdeveloped regions without access to services and infrastructure, potential victims of contemporary slave labor.

Adopting a precautionary approach to combating contemporary slave labor, such as the one adopted by the Seringal Rio Branco REDD+ Project, must include, in the risk mapping process, the vulnerability variables identified among workers, which may predispose them to victimization in the face of being subjected to working conditions that reduce them to those analogous to slavery.

Therefore, in order to prevent any risk to human rights that may be identified, the Human Rights Protection policy seeks to structure the following lines of action:

- drawing up guidelines for the construction and/or revision of a policy on human rights and decent work in the forest restoration chain;
- inputs for mapping, monitoring and managing risks of human and labor rights violations;
- inputs for the development of a responsive victim-centered protocol; and
- training in human rights and contemporary slave labor across the axes above.

Therefore, corporate and organizational responsibility to respect human rights must be consolidated in the existence of these points:

- i) political declaration of human rights;
- ii) human rights due diligence processes;

²⁹ United Nations Organization (UN). Universal Declaration of Human Rights of the UN. Available in: <https://www.un.org/sites/un2.un.org/files/2021/03/udhr.pdf>.

- iii) mechanisms for response and reparation in cases of human rights violations, the Guiding Principles became the conceptual and normative basis around which the project was developed.

Therefore, in addition to explaining the central concepts that are indispensable for full ownership of the project, such as human rights, decent work and contemporary slavery, each of the axes that make up the construction of Biofílica Ambipar's Human Rights Protection policy corresponds to a dimension of corporate and organizational responsibility, so that the projects, as a result, can be equipped with the conceptual, normative, procedural and instrumental bases for fulfilling this responsibility in its entirety.

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

The process of identifying and mapping risks and impacts carried out in the Socio-Economic Diagnosis made it possible to gather data and information on local issues that are considered important for human rights, the protection of indigenous populations and the protection of cultural heritage.

The occupation of the region, which today is known as the Juruá Valley, began with indigenous populations. The communities surrounding the project are located in a region historically occupied by indigenous populations. The municipality of Cruzeiro do Sul was inhabited by indigenous tribes. The people who inhabited the Juruá Valley, divided into 49 tribes, belonged to ethnic groups of Pano languages, and were divided into groups with their own particular names, such as Ararauas, Katukinas and Curimas. And along the banks of the Juruá River lived the Náuas tribe (CRUZEIRO DO SUL, 2023³⁰).

Within the area of influence of the project there are 3 indigenous lands, none of which are adjacent to the project area:

Table 11: Identification and mapping of indigenous populations.

Indigenous land	Language Family	Language	Population	Area (ha)	Year of Approval
Jaminawa do Igarapé Preto ³¹	Pano	Yamináwa	215	25.651,6	1998
Arara do Igarapé Umaitá ³²	Pano	Arara Shawanaua	658	87.571,7	2006
Campinas/Katukina ³³	Pano	Katukina do Acre	703	32.623,6	1993

³⁰ CRUZEIRO DO SUL. Dados do município. Available in: <https://www.cruzeirodosul.ac.gov.br/municipio>.

³¹ Terras Indígenas no Brasil - Terra Indígena Jaminawa do Igarapé Preto. Available in: <https://terrasindigenas.org.br/pt-br/terras-indigenas/3703#direitos>.

³² Terras Indígenas no Brasil - Terra Indígena Arara/Igarapé Humaitá. Available in: <https://terrasindigenas.org.br/pt-br/terras-indigenas/3598>.

³³ Terras Indígenas no Brasil - Terra Indígena Campinas/Katukina. Available in: <https://terrasindigenas.org.br/pt-br/terras-indigenas/3636#direitos>.

The Socio-Economic Diagnosis revealed that the majority of the population of the communities located around the project area are of black or brown ethnic origin. The table below shows the ethnic identification gathered during the research, based on the population's self-identification.

Table 12: Analysis and ethnic identification of communities.

Community	Classification	Identified ethnicity (%)			
		White	Black	Brown	Indigenous
Aldeota 2	Riverside population	-	20%	80%	-
Monte Cristo	Riverside population	25%	5%	70%	-
Periquito	Riverside population	-	10%	80%	10%
Prainha 1	Riverside population	-	10%	90%	-
Prainha 2	Riverside population	10%	20%	70%	-
Primavera	Riverside population	5%	80%	10%	5%
Santo Antônio 1	Riverside population	10%	10%	80%	-
São Luís	Riverside population	5%	15%	80%	-
Volta Grande	Riverside population	5%	95%	-	-
PDS Francisco Pimentel	Family farmers	No information.			

During the research in the Volta Grande community, while the team was applying the diagnostic tools inside the house of an old resident of the community, the focus group activity had to stop for a few moments because a family from another community brought a child who was ill to receive a prayer from the resident of the house. This practice of praying for healing, especially for children, is common in most rural communities in the Amazon and is a religious manifestation, called by some cabocla shamanism. It is a form of shamanism, which has its origins in the cultural convergence of blacks and indigenous people and coexists with the practices of Catholicism. Cabocla shamanism is characterized by the worship of the enchanted (MAUÉS, 1994)³⁴. Heraldó Maués observes that these, in turn, are entities that can be human beings, but a special class of them, "who have not died, but have been enchanted, and live in a world different from that of other men, underground or underwater cities, in the 'encante'". These beings can also be animals that inhabit natural spaces, such as snakes and fish.

According to a story given by community members during fieldwork, the "enchanted" cannot be seen by anyone because they are invisible, as if they inhabited another dimension. The presence of the enchanted is only perceived by the shamans, also called healers, who see and/or hear them.

According to reports, in the past, there were more healers in the communities, who carried out shamanism rituals, which are characterized by cults of possession, in which there is the incorporation of the entities

³⁴ MAUÉS, R. H. Medicinas populares e "pajelança cabocla" na Amazônia. Rio de Janeiro: Editora FIOCRUZ, 1994. Available in: <https://books.scielo.org/id/tjd4g/pdf/alves-9788575412763-06.pdf>.

worshipped, who use the healer's body to attend to the adepts. It is a ritual practice aimed above all at curing illnesses, whether physical or spiritual.

Still in the Volta Grande community, the owner of the house left the focus-group space, took some herbs she had in her kitchen, prepared them in the shape of a branch and began to pray over the child. She then went from the living room to her bedroom and continued praying. After witnessing this moment, the team began to ask about this practice in other communities. Reports say that there are still three "rezadores" (prayers) as they call them, who live in communities on the Juruá Mirim river and who pray to heal people, especially children.

At this lady's home in the Volta Grande community, which is close to the Prainha 1 community, the conversation was discreet and quiet. There is concern about protecting these people who pray.

Shamanism is not limited to religiosity or health; the elements that make it up and its relationship with the enchanted also guide the ways of acting and living in the space of the communities and permeate interpretations and meanings about it and the beings that shape it, both human and non-human. According to Maués (2002)³⁵, "their beliefs and practices are intimately interwoven with the socio-cultural system to which they belong", defining the cosmological system of these peoples and, furthermore, a system of knowledge of space as a whole.

Generally, in the backyards of families who practice some kind of prayer for healing, there is a diversity of medicinal plant species, which are always within reach, and this variety is much greater in these backyards than in the backyards of evangelical families, for example. Studies on religion have already indicated that the denial of traditional knowledge is a characteristic of the evangelical faith throughout the world, although there are isolated cases of people who manage to escape this reality. In the reports on knowledge of medicinal plants, the types were varied, but almost always repeated: "favaca, cravo defunto, manjirioba, capim santo, mastruz, agrião, copaíba, jatobá, quina quina, folha do abacate, folha do maracuja, graviola". In some cases, they mentioned diseases that were treated with some plant, such as malaria, flu, cough, high blood pressure, stomachache.

The person who performs the prayer for healing does not receive payment in cash, usually the family who has received the healing of a family member donates something, a gift, or makes a retribution to the prayer, which can be a hunt or another "forest product". This reality is reminiscent of Marcel Mauss' (1974)³⁶,) concept of the "gift"¹⁹, based on the triad of "giving", "receiving" and "giving back", which constitutes the genesis of a system of material, symbolic and mercantile exchanges, among others, such as the "donation of a hunt", "mutual aid" and the exchange of days of work.

In traditional communities, the inseparable link between the natural environment, the subject and the social environment, in which this relationship is embedded, is the basis for behavior and values. From this, narratives are formed that constitute a set of explanations of the world of each human group. These narratives transmit perspectives and knowledge, mediate human contact with the environment and

³⁵ MAUÉS, R. H. Catholicism and shamanism: a comparison between healing in the charismatic movement and rural Amazonian shamanism. *Ilha Revista de Antropologia*, v. 4, n. 2, p. 051-077, 2002. Available in: <https://periodicos.ufsc.br/index.php/ilha/article/view/15120>.

³⁶ MAUSS, M Essay on the gift. Form and reason of exchange in archaic societies. In: *Sociology and Anthropology*. v. II. São Paulo: EDUSP, 1974. Available in: https://ppgh.ufba.br/sites/ppgh.ufba.br/files/mauss_m_ensaio_sobre_a_dadiva.pdf.

underpin the way of living in and with the place. Edna Castro (1997)³⁷ observes that "the existence of biological resources is directly linked to an ancestral system of sustainable coexistence between humans and the environment, which is why these resources depend on the survival of this system".

This system also includes a series of behaviors and prohibitions based on knowledge that guarantees the safeguarding of the natural environment, the community's sources of subsistence and the reproduction of a cosmological system that encompasses this relationship. There are preservation areas scattered throughout the communities which are essential and strategic for the survival of the families. The ways of proceeding in these spaces are passed down from generation to generation. As in other regions of the Amazon, there is a belief in invisible beings, considered to be entities that communicate between the world of humans, gods, and nature (MAUÉS, 2002)³⁸, which form a parallel dimension to the dimension of human existence.

Regarding this system of local rules and traditions, there was a report about the appearance of an enchanted spirit called "caboquinho da mata", which is the enchanted spirit that protects hunting, as if it were the "owner of the game", so when a person hunts beyond what is "allowed by the spirit" or when they hunt too much with a dog, they get sick.

One local resident said that he had witnessed cases of hauntings in the woods, for example people who had fallen ill while hunting with a dog at night and had seen the dog get scared, as if it was being whipped in the woods. For the narrator, there is the conviction that the conduct was at odds with the local rules instituted by the enchanted man of the forest, in this case, the "caboquinho da mata", who in turn punishes people who "hunt a lot". Fever, headache, vomiting, and excessive sweating are some of the symptoms of this evil. In these cases, only a prayer would have the power to reverse the evil.

There was also an account of the "batedor", an invisible enchanted being that stays near the water. They told stories of some fishing spots where this enchanted creature is present and protects the places. The "batedor" would scare the person who disagreed with what was allowed. This whole cosmological conception of space and how to proceed in these spaces is passed down from generation to generation; it is knowledge of nature and the preservation of resources and sites that is fundamental to the families. However, during the field research, it wasn't possible to go hiking in the forest or along the watercourses to map these places reported by residents, as the diagnostic research focused on the center of the community, where the families' houses are located.

In order for the Seringal Rio Branco REDD+ Project to contribute to the protection of cultural heritage as part of its activities, integrated measures will be adopted. Firstly, it is crucial to establish an open and continuous dialog with the local communities, ensuring their participation from the start of the project and respecting their traditions and cultural practices. In addition, during the monitoring processes there will be continuous and detailed surveys to identify and value the cultural heritage present in the project area, including religious practices, traditional medicine, handicrafts, among other aspects.

³⁷ CASTRO, Edna Ramos. Territórios, biodiversidade e saberes de populações tradicionais. Paper nº 92 do Núcleo de Altos Estudos Amazônicos/ NAEA – 1997. Available in: <https://periodicos.ufpa.br/index.php/pnaea/article/download/11834/8202>.

³⁸ MAUÉS, R. H. Catolicismo e xamanismo: comparação entre a cura no movimento carismático e na pajelança rural amazônica. Ilha Revista de Antropologia, v. 4, n. 2, p. 051-077, 2002. Available in: <https://periodicos.ufsc.br/index.php/ilha/article/view/15120>.

Integrating these cultural practices into the project's plans and activities is essential and will promote the conservation and sustainable use of areas where these practices occur. It is equally important to recognize and respect the traditional knowledge of local communities, fostering mediation support to strengthen these practices and skills.

In addition, socio-environmental safeguard measures must be developed and implemented to protect the rights of local communities and avoid negative impacts on their cultural heritage. A participatory monitoring and evaluation system must be established, allowing communities to follow and evaluate the impact of the project on their cultural heritage.

Finally, it is essential to establish partnerships with local institutions, civil society organizations and experts in culture and cultural heritage to support the implementation of measures to protect and conserve cultural heritage. By adopting these measures in an integrated manner, the Seringal Rio Branco REDD+ Project will demonstrate its commitment to preserving and protecting the cultural heritage of local communities, promoting cultural diversity, and strengthening the socio-environmental sustainability of the region.

2.5.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

Amazônia Agroindústria EIRELI is the legitimate owner of the property where the Seringal Rio Branco REDD+ Project is being implemented and developed. The right of possession and use are demonstrated by the following documents, which were forwarded to the audit team:

- - Certificate of full content containing the description of the Registration;
- Rural Environmental Registry (CAR);
- Rural Property Registration Certificate (CCIR);
- Descriptive memorial and plan linked to the Land Management System (SIGEF)

Therefore, the demonstration of the right to use the Project area is respected in accordance with the criteria of section 3.7 of the VCS Standard v4.7:

- 1) Right of use resulting from or granted under statute, regulation or decree by a competent authority;
- 2) Right of use deriving from the law;
- 3) Right of use arising from a statutory, patrimonial, or contractual right over land, vegetation or a conservation or management process that generates reductions and/or removals of GHG emissions (where this right includes the right to use such reductions or removals and the project proponent has not been deprived of such right of use).

2.5.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

The Seringal Rio Branco REDD+ Project recognizes and respects all property rights, complying with significant statutory and regular requirements, as well as having the necessary approvals from local authorities. The Project recognizes and supports the rights to lands, territories, and resources, including the statutory and traditional rights of indigenous peoples, communities, and other stakeholders.

The proponents of the project act as mediators of potential conflicts, as well as maintaining a good relationship with the neighboring communities. The following aspects are described in detail:

- The company Amazônia Agroindústria EIRELI owns the rights to use and economically exploit the properties, as well as the right to access the natural resources on them, under the terms of the Brazilian Federal Constitution and the Civil Code, as it owns the properties where the Seringal Rio Branco REDD+ Project will be carried out.

2.5.7 Free, Prior and Informed Consent (VCS, 3.18; CCB, G5.2)

Project activities are limited to a single private property. There are no actions in the project that affect the property rights of third parties or neighboring properties. The owner involved in the project actions is the only one whose property is affected, and not only consents to the activities, with an agreement signed in a contract, but also actively participates in actions and negotiations related to the project.

Even so, if it is deemed necessary the Prior, Free and Informed Consent will be obtained throughout the life cycle of the project, always with an approach based on dialogue and consent between the parties involved. In addition, the project does not aim to carry out any activities on the private property of third parties, traditional and indigenous communities, or the government. With regard to social and biodiversity activities, guarantees that no activity will be carried out without the free, prior and informed consent of the parties involved.

Table 13: Description of process for obtaining consent.

Description of process for obtaining consent	All stakeholders who could be impacted in some way by the Seringal Rio Branco REDD+ Project were consulted. Workshops were held in the communities surrounding the project in order to pass on information about the project, as well as consultations on the opinions of the community members regarding the project, as described in Section 2.3.6. No activity related to the Project will lead to the relocation of activities that are important to the culture or livelihoods of the Property Rights Holders, nor will it aim to involuntarily relocate or remove them from their lands or territories, since there are no populations living within the project area
Outcome of FPIC process	The activities (participatory workshops in the communities; interviews with key players and research tools) and the methodology for consulting stakeholders were described in detail in Section 2.3.10. Throughout its execution, the work was conducted through a participatory and transparent process with the community and other actors involved in the research, respecting the principles of clear, effective communication and transparency, as well as the principles of free and informed consent for the activities to be carried out.

2.5.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

As mentioned above, no activity related to the Project will lead to the relocation of activities that are important to the culture or livelihoods of the Property Rights Holders, nor will it affect the property rights of any resident of the surrounding area.

There will be no relocations or involuntary removals from their lands or territories, since there are no populations living within the project area. As such, the benefit-sharing mechanism is not relevant.

2.5.9 Property Rights Protection (VCS, 3.18, 3.19; CCB, G5.3)

The implementation and development of the Seringal Rio Branco REDD+ Project must not lead to the involuntary removal or relocation of any party, and the activities that are important to the livelihoods of the communities residing in the Project Zone must be respected and supported by the Project. Thus, the Project proposes social activities that seek to promote sustainable practices and strengthen and improve existing economic activities, such as the exploitation of non-timber forest products. These initiatives seek to discourage the practice of illegal activities, but without disregarding the cultural and traditional aspects of the impacted communities.

In addition, the land regularization of the area is supported by Amazônia Agroindústria EIRELI and backed up by the responsible public institutions. As mentioned above, possible judicial inconsistencies are resolved following all the procedures established by the local jurisdiction.

2.5.10 Illegal Activity Identification (VCS, 3.19; CCB, G5.4)

Deforestation is the main illegal activity that can negatively impact the development of the Seringal Rio Branco REDD+ Project, as well as hunting and predatory exploitation of fauna and flora. The main causes of this illegal deforestation have been identified as family farmers who use agricultural crops based on the opening of swiddens through burning, and medium-sized rural livestock farmers.

The Project seeks to control and combat these illegal activities commonly found in the Project region through mitigating and preventive measures such as strengthening land and property inspection, as well as encouraging the involvement of other actors and stakeholders, social inclusion, and regional socio-economic development through the generation of economic alternatives to deforestation and discouraging predatory hunting and fishing.

By applying these measures, it is hoped to improve the well-being of the communities without creating a burden on the native forest and local biodiversity. Land surveillance aims to curb illegal deforestation practices, the extraction of plant species and the hunting and capture of wild animals by third parties. The mechanisms and procedures for land inspection and the mitigation and prevention of illegal activities are summarized in table below.

Table 14: Asset surveillance mechanisms and procedures in the Project Area.

Strengthening Asset Surveillance in the Project Area	
Purpose	Determine the inspection conditions on the land owned by Amazônia Agroindústria EIRELI and actions to mitigate or prevent illegal activities.

Strengthening Asset Surveillance in the Project Area	
General conditions	- Carry out regular patrols to ensure the protection of Amazônia Agroindústria EIRELI's land assets; - Prevent deforestation, forest fires or other acts of aggression against the environment; - Maintain good relations with communities and other stakeholders; - Promote environmental education in communities that practice predatory hunting and fishing and encourage the practice of sustainable activities; - Request support from the police and inspection authorities when necessary; Method of operation: - Deployment of a surveillance team to the scene of the incident to investigate the fact and the application of appropriate measures; - Activation of the legal sector for measures; - Registration of incidents involving trespassing, damage to property and illegal extraction of forest products; - Incidents involving aggression against the environment must be registered with the responsible bodies (IBAMA, Environmental Police, etc.); - In all situations involving land conflicts, it is necessary to avoid confrontation between the parties, respecting the laws in force in the country.
Specific conditions	<u>Monthly Surveillance Program</u> - When surveillance detects the occurrence of illegal activities, the team must take the appropriate measures and collect the geographical coordinates for internal recording;
Records	- Report of occurrences; - Photographic record of occurrences; - Monthly monitoring program; - Report on property security activities.
Surveillance Intelligence Strategies	- Monitoring using high-resolution satellite images, enabling the generation of monthly and annual reports on altered areas; - Acquisition of support equipment for the monitoring team;

2.5.11 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

The Seringal Rio Branco REDD+ Project does not have any ongoing or unresolved conflicts or disputes over land rights, territories, or resource.

2.5.12 Approvals (CCB, G5.7)

The project proponents achieved recognition and approval of the implementation of the Seringal Rio Branco REDD+ Project among stakeholders through meetings, talks and face-to-face meetings with the communities, partners, proponents, and authorities mentioned in Section 2.3.

The event to present the project and publicly consult the communities on the Action Plan took place between July 11, 2023, and July 16, 2023 and took place through broad community involvement, provided by effective face-to-face dissemination. During this feedback, participants were able to understand and collaborate on the design and development of the project, as described in Section 2.3.10.

In addition to the meetings and gatherings of community members and other stakeholders described above, the project will go through a public consultation event on the Verra registration platform for comments, suggestions, and clarification of doubts about the Seringal Rio Branco REDD+ Project. The importance of the engagement and collaboration of these stakeholders in this process was reinforced by sending formal invitations to those directly and indirectly involved in the forest conservation sector, such as community associations, non-governmental organizations (NGOs), educational institutions and private companies. These invitations were sent via mailing list, with information about the project and an invitation to take part in the public consultation.

In addition, letters were sent to the relevant local institutions in the state of Acre, including the State Public Prosecutor's Office and other government and federal agencies, containing information about the public consultation, the context of the project and the communication channels used.

It is worth mentioning that despite the progress made by Brazil's National REDD+ Strategy (ENREDD+), the creation of the State System of Incentives for Environmental Services (SISA), the Incentives Program for Environmental Services (ISA Carbono) and other Environmental Services and Ecosystem Products Programs in the state of Acre, demonstrated in section 2.5.1 - National and Local Laws, there are still no official national or jurisdictional REDD+ policies. However, the project's proponents are always on the lookout for new information, always present at federal and state government discussion forums in order to contribute to the formulation of these policies and regulations and are readily available to adapt the Project to the new rules officially established.

2.5.13 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB G5.9)

2.5.13.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes

☒ No

2.5.13.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

2.5.13.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

2.5.14 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.5.14.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

2.5.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.5.14.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”?

☐ Yes

☒ No

2.6 Additional Information Relevant to the Project

2.6.1 Leakage Management (VCS, 3.11, 3.15)

The Seringal Rio Branco REDD+ Project will define the leakage baseline when new properties are acquired by the proponent Amazônia Agroindústria EIRELI.

2.6.2 Further Information

There is no other information to be presented apart from what is already in the PDD.

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology (VCS, 3.1)

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF),	v1.6
Module	VMD0006	Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL).	v1.3
Module	VMD0001	Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools (CP-AB).	v1.1
Module	VMD0002	Estimation of carbon stocks in the dead wood pool (CP-D).	v1.0
Module	VMD0005	Estimation of carbon stocks in the long-term wood products pool (CP-W).	v1.1
Module	VMD0009	Estimation of emissions from activity shifting for avoiding planned deforestation/forest degradation and	V1.3

		avoiding planned wetland degradation (LK-ASP)	
Module	VMD0015	Methods for monitoring of greenhouse gas emissions and removals in REDD project activities (M-REDD).	v2.2
Module	VMD0013	Estimation of greenhouse gas emissions from biomass and peat burning (E- BPB)	v1.2
Module	VMD0017	Estimation of uncertainty for REDD+ project activities (X-UNC).	v2.2
Tool	VT0001	CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD).	v1
Tool	AFOLU Non-Permanence Risk Tool	Tool for AFOLU non-permanence risk analysis and buffer determination (T-BAR)	v4
Tool	-	CDM Tool for testing significance of GHG emissions in A/R CDM project activities (T-SIG).	v1

3.1.2 Applicability of Methodology (VCS, 3.1)

The methodology approved by the VCS, code VM0007, was used for the Seringal Rio Branco REDD+ Project and is applicable according to the applicability criteria specified:

ID/Title of reference	Condition of applicability	Justification for compliance
VM0007 - All REDD Activity Types	The land in the project area must qualify as forest (according to the definition used by the VCS) for at least 10 years prior to the project start date.	According to an analysis of satellite images, the Project Area has been classified as forest for more than 10 years.
VM0007 - All REDD Activity Types	The lands within the project area are peatlands or tidal wetlands and emissions from the soil carbon stock are considered significant	As shown in the following sections, there are no organic soils (or peatlands) in the Project Area. In addition, the soil carbon stock has been conservatively excluded.

ID/Title of reference	Condition of applicability	Justification for compliance
		Therefore, the project is not subject to the requirements of the VCS WRC.
VM0007 - All REDD Activity Types	The baseline for deforestation and forest degradation within the project area must fall into one or more of the following categories: - Unplanned deforestation (AUDD category of the VCS) - Planned deforestation/degradation (VCS category APD) - Degradation due to extraction of wood for fuel (production of firewood and charcoal) (VCS category AUDD)	The Seringal Rio Branco REDD+ Project falls into the category of avoided planned deforestation (VCS APD Category).
VM0007 - All REDD Activity Types	Activities to prevent leakage should not include: Agricultural land that is flooded to increase production (e.g. rice in wetlands); or, intensifying livestock production through the use of “feedlots”.	None of the Program's containment and leakage activities include actions to flood agricultural land or create enclosures.
VM0007 - Avoiding Planned Deforestation/ Degradation	Project activities to prevent planned deforestation can only be carried out in areas where there is legal permission to deforested.	According to current legislation in Brazil and in the state of Acre, the Seringal Rio Branco REDD+ Project area meets all applicable legal requirements and can be legally deforested.
VMD0001 (CP-AB)	This module is applicable to all forest types and age classes. The inclusion of the above-ground tree biomass set as part of the project boundary is mandatory, according to the REDD-MF framework module.	This module applies to all forest types and age classes. The inclusion of the above-ground tree biomass pool as part of the project boundary is mandatory according to the REDD-MF module;

ID/Title of reference	Condition of applicability	Justification for compliance
		thus the biomass of trees with DBH > 10cm will be considered. The above-ground non-woody biomass will be considered from the biomass of palm trees (with DBH > 10cm); as demonstrated, this biomass is significant, according to the T-SIG tool. Underground biomass (tree and non-wood) will also be considered in the carbon stock estimates, which is optional according to the requirements of the methodology; as shown, this biomass is significant according to the T-SIG tool.
VMD0002 (CP-D)	This module is applicable to all forest types and age classes. This module is applicable if the dead wood reservoir is included as part of the project.	The module is applicable to all forest types and age classes. This reservoir is included when it represents a significant component of forest biomass.
VMD0009 (LK-ASP)	The module is applicable for estimating leakage emissions due to the displacement of activities from legally authorized and documented forest land to be converted into non-forest land, including the displacement of activities into drained or degraded forest wetlands because of project implementation.	Mandatory module. The Seringal Rio Branco REDD+ Project met the applicability condition for this module because the baseline scenario is the conversion of legally authorized and documented forest land into non-forest land. Documented into non-forest land. This module is used to estimate the level of greenhouse gas emissions.
VDM0005 (CP-W)	This module is applicable to all cases where wood is harvested for conversion into wood products for commercial markets, for all forest types and age classes.	This module is applicable in all cases where timber may be harvested for conversion into wood products for commercial markets, for all forest types and age classes. This module is applicable in the baseline, since the

ID/Title of reference	Condition of applicability	Justification for compliance
		pool of wood products is included as part of the project boundary, according to the applicability criteria of the REDD-MF framework module, specifically.
VMD0013 (E-BPB)	This module applies to REDD project activities with emissions from biomass burning and REDD-WRC project activities with emissions from biomass and/or peat burning.	In the baseline scenario, fire is used to clear the land, resulting in greenhouse gas emissions. When used in the baseline, accounting should take place both in the baseline and in the project scenario, and both within the project area and in the leakage belt. When post-fire incidents occur in areas that coincide with deforested or degraded areas in the baseline, the module should be used to account for greenhouse gas emissions.
VMD0017 (X-UNC)	This module is mandatory when using the REDD-MF methodology.	Mandatory module. REDD+ Seringal Rio Branco is a REDD+ project.
VMD0015 (M-REDD)	Where selective logging is taking place in the case of the project. Emissions from logging can be omitted if it can be demonstrated that the emissions are de minimis using the T-SIG Tool. If emissions from logging are not omitted as de minimis, logging can only take place in forest management areas that hold and maintain a Forest Stewardship Council (FSC) certificate for the years in which selective logging takes place.	Emissions from logging have not been omitted, since commercial timber harvesting may take place in the project in the future, although there is no such activity today.

ID/Title of reference	Condition of applicability	Justification for compliance
	Logging operations can only carry out selective logging that maintains a land cover that meets the definition of forest within the project boundary. All trees cut for timber during logging operations must have a DBH greater than 30 cm. During logging operations, only the trunk of the felled tree may be removed. The top/crown of the tree must remain within the forest area. Logging practices may not include stacking and/or burning logging residues The volume of timber harvested must be measured and monitored.	
VMD0006 (BL-PL)	The module is applicable for estimating baseline emissions on forest land (usually privately or government-owned) that is legally authorized and documented to be converted into non-forest land.	The activity of preventing deforestation and planned degradation is applicable, given that it is legally permitted to convert forested areas to a deforested condition up to a limit of 20% of the total area of the property, as stipulated in article 12 of the Forest Code, Law 12.651 of 2012. No logging has been carried out within the project area to date.
VT0001	AFOLU activities that are the same as or similar to the proposed Project activities, whether registered as a VCS AFOLU Project or not, will not lead to a violation of any	The project's activities do not violate any applicable laws. They are all in line with government guidelines. The VM0007 methodology used by the Seringal Rio Branco REDD+

ID/Title of reference	Condition of applicability	Justification for compliance
	applicable law even if this law is not applied. The use of this tool to determine additionality requires that the baseline methodology provides a step-by-step approach that justifies the determination of the most plausible baseline scenario	Project provides a step-by-step approach to justify the determination of the most plausible baseline scenario.

3.1.3 Project Boundary (VCS, 3.12)

	Source	Gas	Included?	Justification/explanation
Baseline	Emissions from burning woody biomass	CO ₂	Included	Reductions in carbon stock due to burning are counted as a change in carbon stock
		CH ₄	Excluded	REDD-MF methodology says it is conservative to exclude No-CO ₂ gases
		N ₂ O	Excluded	REDD-MF methodology says it is conservative to exclude No-CO ₂ gases
	Emissions from combustion of fossil fuels	CO ₂	Excluded	The REDD-MF methodology says that it is conservative to exclude CO ₂ and that potential emissions of CH ₄ and N ₂ O are insignificant. In addition, the E-FFC module confirms this information.
		CH ₄	Excluded	
		N ₂ O	Excluded	
	Emissions from fertilizer use	CO ₂	Excluded	There is no use of fertilizers in the baseline. In addition, the REDD-MF methodology states that potential emissions are insignificant and/or the exclusion is considered conservative.
		CH ₄	Excluded	
		N ₂ O	Excluded	
Project	Emissions from burning woody biomass	CO ₂	Included	Justification similar to that given in the baseline
		CH ₄	Included	Should be included if fire occurs
		N ₂ O	Included	Should be included if fire occurs
	Emissions from fossil fuel combustion	CO ₂	Included	Justification similar to that presented in the baseline
		CH ₄	Included	
		N ₂ O	Excluded	

Source	Gas	Included?	Justification/explanation
Emissions from fertilizer use	CO ₂	Excluded	Justification similar to that presented in the baseline
	CH ₄	Excluded	
	N ₂ O	Excluded	

The Seringal Rio Branco REDD+ Program, located in the municipality of Cruzeiro do Sul, in the far west of the state of Acre, covers an extensive area of 29,613 hectares, comprising a single continuous rural property. This property is characterized by a unique land structure, consisting of just one parcel of forest within the Amazon biome. Although this land continuity simplifies some administrative processes, it does not replace the need to delimit the property following the official government bases for property management, and the use of tools such as the Rural Environmental Registry (CAR) and the Land Management System (SIGEF) is still mandatory in the context of environmental and land management of rural properties in Brazil. While the CAR is a mandatory tool aimed at the environmental regularization of rural properties, SIGEF plays a complementary role by integrating information from the rural registry administered by the National Institute for Colonization and Agrarian Reform (INCRA) with the rural registry maintained by the notary system.

A crucial aspect in understanding the interaction between the Rural Environmental Registry (CAR) and the Land Management System (SIGEF) is the possibility of a single property registered in the CAR covering several registrations in SIGEF. This feature reflects the complexity and diversity of rural land structures in Brazil, where a single property can be made up of several parcels of land, each with its own registration in the land registry.

The importance of this multifaceted relationship between CAR and SIGEF stands out mainly in environmental management and the regularization of rural properties. While CAR focuses on identifying and registering the environmental characteristics of a property, SIGEF guarantees the accuracy of information relating to the ownership and physical delimitation of these properties through their land registry. Thus, a rural property registered in the CAR, representing an environmental management unit, can be legally divided into several registrations in SIGEF, reflecting its land composition.

The CAR aims to map and regularize rural property areas, focusing especially on environmental aspects such as the delimitation of Permanent Preservation Areas (APPs), Legal Reserves (RL), and other areas of environmental interest. This regularization is fundamental for the sustainable management of natural resources and is a prerequisite for obtaining a permit for planned deforestation.

The boundaries of the Seringal Rio Branco farm were delimited using the Land Management System (SIGEF). However, it is important to note that it was necessary to obtain the property's CAR. This procedure was essential for the precise delimitation of the Permanent Preservation Areas (APP) and the Legal Reserve (RL), which are fundamental for the progress and environmental compliance of the project.

Project Area (VM0007)

The methodology applied to define the Project Area (PA) involves a detailed and rigorous approach, in line with VERRA standards. The definition of forests follows the UNFCCC (United Nations Framework Convention on Climate Change) classification, which considers forests to be areas of at least 0.05-1.0 ha with canopy cover (or equivalent density) of more than 10-30%, with trees with the potential to reach a minimum height of 2-5 meters at maturity in situ, excluding areas predominantly under agricultural or urban use. For the project, we only consider areas larger than one hectare to be forest.

The MapBiomass³⁹ is a collaborative initiative involving organizations from the science, technology and academic sectors, with the aim of mapping land cover and land use in Brazil. Using satellite images and advanced data processing algorithms, MapBiomass offers annual historical series of land use and land cover maps, from the 1985s to 2022. This tool provides valuable data for monitoring changes in the Brazilian landscape, including deforestation, forest regeneration, agricultural expansion, and urbanization.

The initiative produces data categorized into various land use and land cover classes, such as forests, agricultural areas, pasture areas, bodies of water, urban areas, among others, allowing for a detailed and temporal analysis of land transformations. The spatial resolution of these maps makes it possible to identify patterns and trends of change on a national, regional and local scale.

The methodology of MapBiomass Collection 8 focuses on the evolution of land use and land cover mapping in the Amazon, using Landsat images from 1985 to 2022. In this collection, classification was improved with the introduction of the Random Forest Classifier (RFC), optimized after analysis to select the best parameters. The images were filtered by sensor type, date range and cloud cover, using techniques such as Temporal Dark Outlier Mask (TDOM) and Band Quality Assessment (BQA). New features in the collection include a more refined integration of classification results by satellite route to produce annual land use and land cover maps. In addition, the Mapbiomas methodology uses additional techniques in the post-classification process to map specific areas, such as wetlands, using annual mosaics and stratified samples for better accuracy. Temporal and frequency filters are applied to ensure consistent transitions between classes over the years, with a final integration that included cross-cutting themes to further refine the annual maps. The accuracy of the Mapbiomas classifications is validated using a robust set of around 25,000 reference samples, ensuring the reliability and applicability of the data for environmental and conservation analysis. The overall accuracy achieved was 96.4%, with an allocation disagreement of 2.2% and a quantity disagreement of 1.4%, demonstrating the high accuracy and relevance of the results obtained.

The MapBiomass methodology identifies various land use and land cover classes, some of which can be classified as forests according to the FAO definition, as well as agricultural and pasture areas. Areas classified as forest are characterized by a significant densification of the tree canopy, generally forming a continuous canopy. Forests usually have a denser biomass with larger trees.

Only one stratum was needed to achieve the optimum accuracy of the project's estimates of net GHG emissions or removals. Thus, it was not necessary to use the VMD0016 (X-STR) module.

³⁹ <https://brasil.mapbiomas.org/o-projeto/>

MapBiomas for the year 2022 was used to select exclusively those areas classified as forest, excluding all other land use and land cover categories for the same year. Mapbiomas data was adopted because of its ability to discern the subtleties of vegetation cover, because it is proven to be accurate and is widely used in the literature (PEDREIRA; CRUZ, 2023⁴⁰; SANTOS, 2023⁴¹; FREITAS; CAVALCANTI; BRAZ, 2023⁴²).

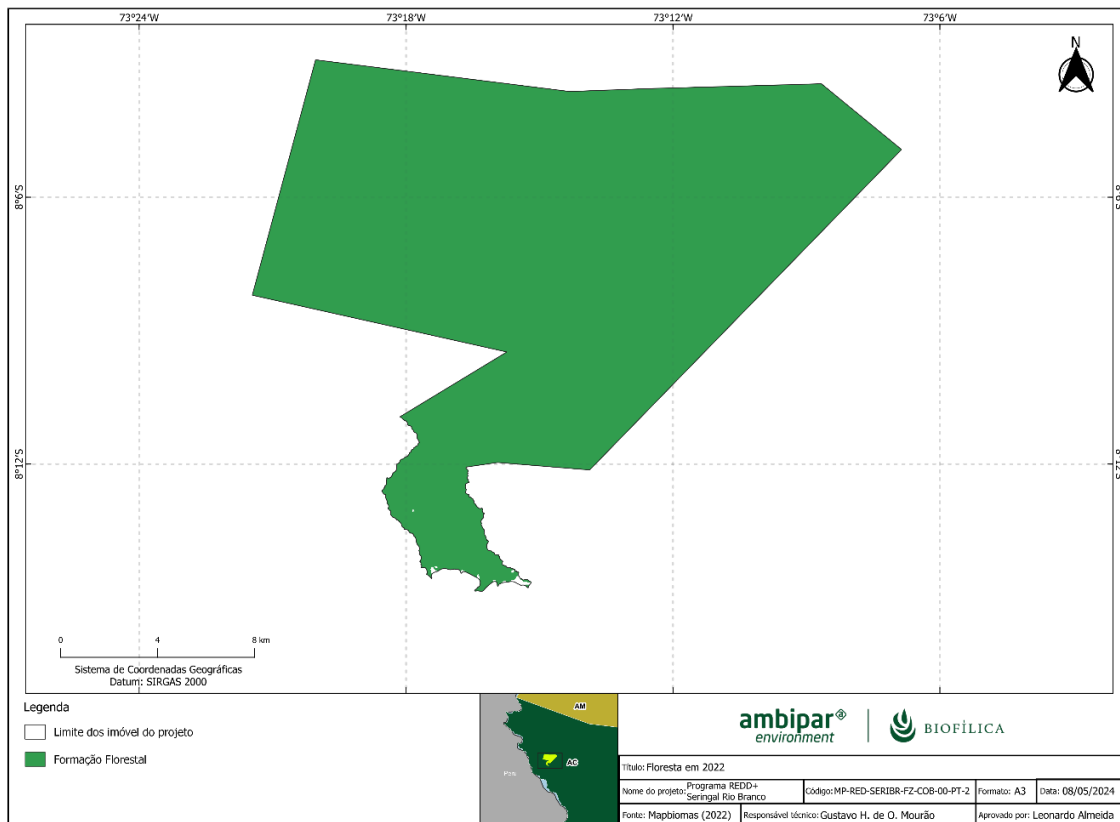


Figure 22: Areas classified as Forestry Formation by MapBiomas.

To improve the analysis, a visual inspection was carried out using high-resolution images from the mosaic of the Planet satellite array, which have a spatial resolution of five meters and are dated May 2023 (cloud-free image close to the project start date, March 15, 2023). These images are crucial for identifying recent deforestation that was not detected by automatic methods and for correcting misclassifications. During this review, a fragment was identified that had been wrongly classified as pasture, but which in fact belongs to the forest class, and was then correctly reclassified. In addition, due to the nature of raster images, the conversion process to vector can cause irregularities in the polygons, such as serrated edges. Therefore, the boundaries visually demarcated as non-forest were excluded to ensure a more precise delimitation of the forest area.

⁴⁰PEDREIRA¹, João Pedro das Neves Cardoso; CRUZ¹, Carla Bernadete Madureira. AVALIAÇÃO DA ACURÁCIA TEMÁTICA DO MAPBIOMAS PARA A CLASSIFICAÇÃO DE ÁREAS VERDES INTRAURBANAS. 2023.

⁴¹ SANTOS, Rosângela Leal et al. ANÁLISE DO USO E OCUPAÇÃO DO SOLO NA REGIÃO CACAUEIRA PÓS-VASSOURA DE BRUXA: USO APLICADO DO MAPBIOMAS. 2023.

⁴² FREITAS, Leonardo Cristiano da Silva; CAVALCANTI, Lucas Costa de Souza; BRAZ, Adalto Moreira. AVALIAÇÃO DA ACURÁCIA DAS CLASSES DE USO E COBERTURA DA TERRA DO MAPBIOMAS (COLEÇÃO 6) PARA O MUNICÍPIO DE CARPINA (PE). Caderno Prudentino de Geografia, v. 45, n. 2, p. 38-52, 2023.

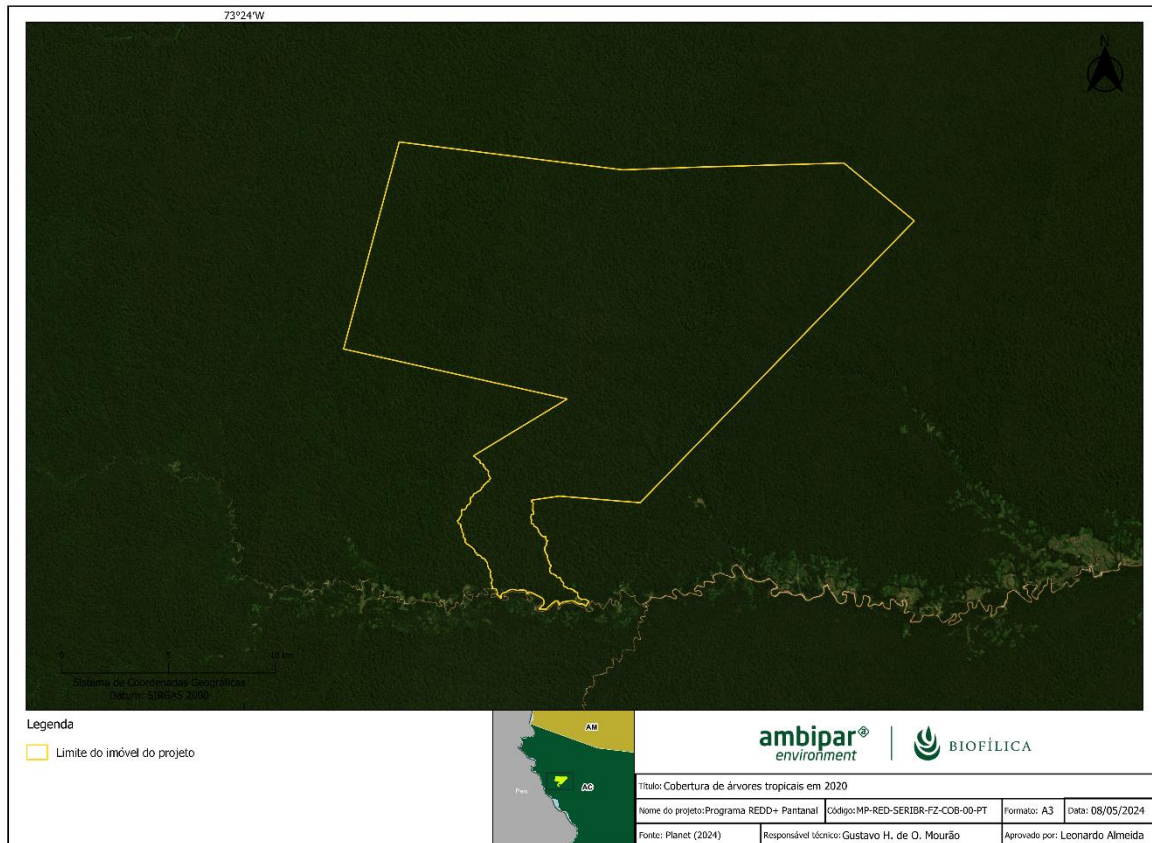


Figure 23: Delimitation of the property area.

APD projects require the stability of areas with tree cover. It is essential that these areas have remained as forest for at least a decade, with no deforestation occurring during this period. To verify this stability, we used the 2012 land cover maps from Mapabiomas to compare with the delimited areas of 2022. The analysis focused on forest classes and consisted of comparing the evolution of these classes over time. An intersection technique was used, keeping only areas where the classification remained unchanged between the two moments, consistently remaining as forest. Areas that met this stability criterion were then considered to be “stable forests”, qualifying for inclusion in the project.

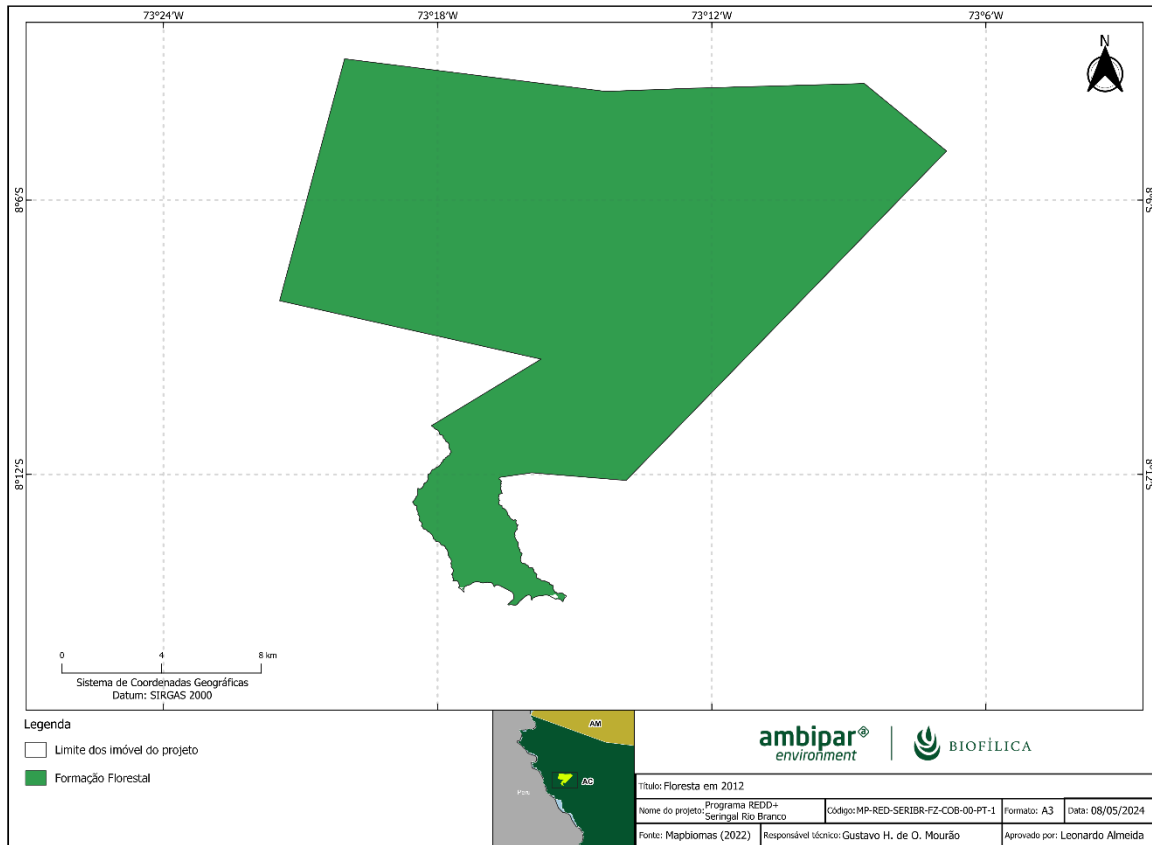


Figure 24: Areas classified as Forestry Formation by MapBiomas.

To ensure that forests can be cleared legally, following all Brazilian regulations, areas delimited as protected must be removed from the project. All the exclusions of forests that cannot be legally cleared will be detailed below.

The Legal Reserve is a protected area, required by law, within a rural property, intended for the preservation and restoration of native vegetation. These areas have significant restrictions on their use, with the aim of protecting natural resources, contributing to the conservation of biodiversity and helping to create balance. Due to these restrictions and their environmental conservation purpose, Legal Reserve areas are not suitable for APD projects.

Permanent Preservation Area (APP) is defined by Law 12.651/2012 as a protected area, which may or may not be covered by native vegetation, designed to preserve water resources, the landscape, geological stability and biodiversity. Its additional environmental functions are to facilitate the gene flow of fauna and flora, protect the soil and ensure the well-being of human populations. APPs are fundamental to environmental conservation, playing a crucial role in maintaining natural ecosystems and supporting life. It is forbidden to issue a deforestation permit for these areas.

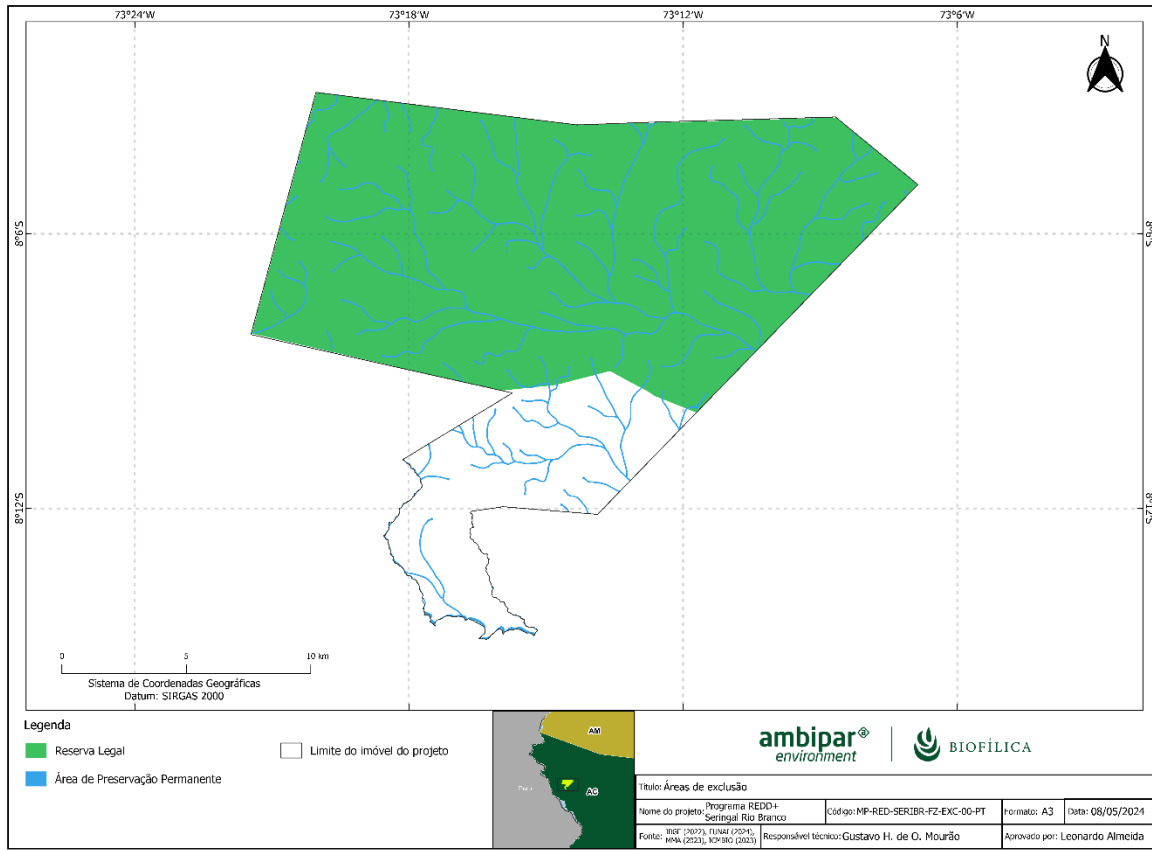


Figure 25: Legal Reserve and Permanent Preservation Areas.

According to the Brazilian Forest Code (Law No. 12.651/2012), deforestation of up to 20% is allowed on properties located in the Amazon biome. To this end, areas that cannot be deforested must be allocated to the Legal Reserve, and protected areas must go to Permanent Preservation Areas (APP). The surplus forest can be cleared up to a limit of 20% of the area of the rural property, ensuring that the use of natural resources occurs in a sustainable manner and in accordance with current legislation, balancing environmental conservation with economic development. The RL and APP of the property involved in the project were identified through the records in the Rural Environmental Registry (CAR). These areas were used as exclusion criteria in the process of delimiting the project area. Consequently, forests contained within the RL and APP were excluded from the project area.

Conservation Units (CUs) are protected areas established by government legislation, whether at federal, state or municipal level, with the main aim of conserving nature and guaranteeing the maintenance of ecosystem services. These areas are fundamental for preserving biodiversity, essential natural features, ecosystems and controlling vital ecological processes. Carrying out activities that significantly alter the natural environment, such as deforestation, even if planned and regulated, can be highly restrictive or even prohibited in PAs, especially in integral protection areas. In Brazil, PAs are mainly regulated by Law No. 9.985 of July 18, 2000, which established the National System of Nature Conservation Units (SNUC).

In the context of project development, it is crucial to check for any overlap with Conservation Units (CUs). This verification is fundamental because the presence of overlaps can lead to significant conflicts with

the bodies that manage these protected areas. In addition, overlaps with Conservation Units (UCs) can result in the necessary license not being granted for the vegetation suppression process, due to the strict regulations in these areas.

Therefore, before going ahead with the planning and implementation of DPA projects, it is essential to carry out a detailed analysis of geographical and cartographic information to identify any areas of overlap. This includes consulting official georeferencing databases. Detecting and resolving overlaps with PAs at an early stage in the project helps to avoid future legal conflicts and ensures that all proposed activities are in full compliance with current environmental legislation. This approach not only protects the integrity of the project but also ensures environmental conservation in accordance with the principles of sustainability and ecological responsibility.

The Serra do Divisor National Park, established by Decree No. 97.839 of June 16, 1989, covers areas in the municipalities of Mâncio Lima and Cruzeiro do Sul, in the state of Acre. This park was created with the aim of protecting the diversity of ecosystems present, conserving natural resources and facilitating controlled public use, as well as education and scientific research activities. Located in the far west of Acre, on the border with Peru, the park is a vital component for environmental preservation in the region.

Importantly, the relationship between the Serra do Divisor National Park and the Juruá Mirim River is of significant ecological importance. According to Article 2 of the park's creation decree, its eastern boundary runs along the Juruá Mirim River from its confluence with the Tamboriaco River, downstream to the mouth of a tributary. This route includes areas that are crucial for maintaining water quality and local biodiversity. It is vital for the ecological integrity of the region that these areas are protected, as they provide essential habitat for local fauna and flora and maintain the ecological corridors necessary for the migration and dispersal of species.

However, a potential delimitation conflict was identified related to the official government polygons that appear to overlap the boundaries of the project property located on the opposite bank of the Juruá Mirim River. This error in the polygons suggests that the conservation unit extends beyond the boundary of the river, which goes against what is stipulated in the decree, which defines the river as the natural boundary of the conservation unit on the opposite side of the property. This mistake in the government's georeferenced base needs to be corrected to ensure that the boundaries of the conservation unit are fully in line with what was officially decreed, thus respecting both the legislation and the owner's rights. Thus, the boundaries established in the decree were considered, and there is no longer any conflict between the boundaries of the property and the conservation unit.

After careful analysis and following all the legal regulations, an effective delimitation of protected areas was achieved, where the suppression of vegetation is prohibited. Respecting Brazilian environmental laws and the specific guidelines for the Brazilian Amazon, areas such as Legal Reserves, Permanent Preservation Areas and Conservation Units were clearly demarcated and all vegetation contained within them was excluded from the project.

At the end of the process of analysis and delimitation of the Seringal Rio Branco REDD+ project area, forest fragments of less than one hectare were excluded. This exclusion focuses on maintaining the

continuity of the forest, ensuring that only areas defined as forest by the most conservative definition of the UNFCCC (United Nations Framework Convention on Climate Change) are included in the project.

After the meticulous process of analysis and exclusion, the project area for the Seringal Rio Branco REDD+ Program was effectively delimited. This delimitation resulted in the selection of remaining forest plots, greater than one hectare, that strictly met the established criteria: all forest regions that had remained stable for a decade, without deforestation, and that were not located in Legal Reserves, Permanent Preservation Areas or Conservation Units. This careful process ensured that the project area encompassed only those regions that effectively aligned with the project's focus on regions with real potential to prevent planned deforestation, excluding sites that could not legally be deforested.

At the end of the process, 5487 hectares were delimited as the project area. These areas consist of forests that have been stable for at least ten years and can be legally cleared. The entire procedure strictly followed Verra's guidelines and the VM0007 methodology, ensuring that all the standards and requirements for validation and verification were met.

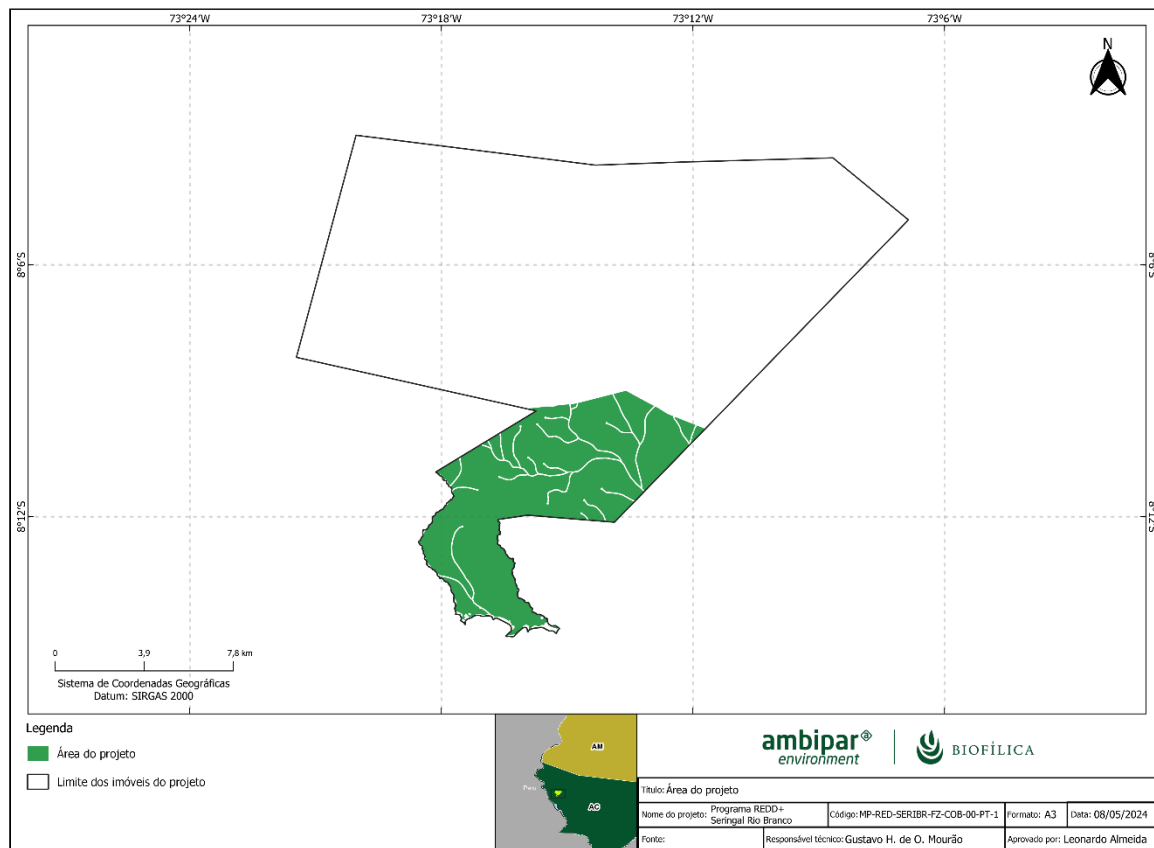


Figure 26: Delimitation of the Project Area.

The owner is committed to preventing any deforestation on his property. Considering that deforestation pressure in the region is expected to intensify, especially with the planning of BR 364, it has been decided that the part of the property that is not protected by the APD project will be allocated to another initiative.

This portion will be integrated into an AUD (Avoiding Unplanned Deforestation) project under Verra's VM0048 methodology, ensuring a comprehensive approach to forest conservation on the property.

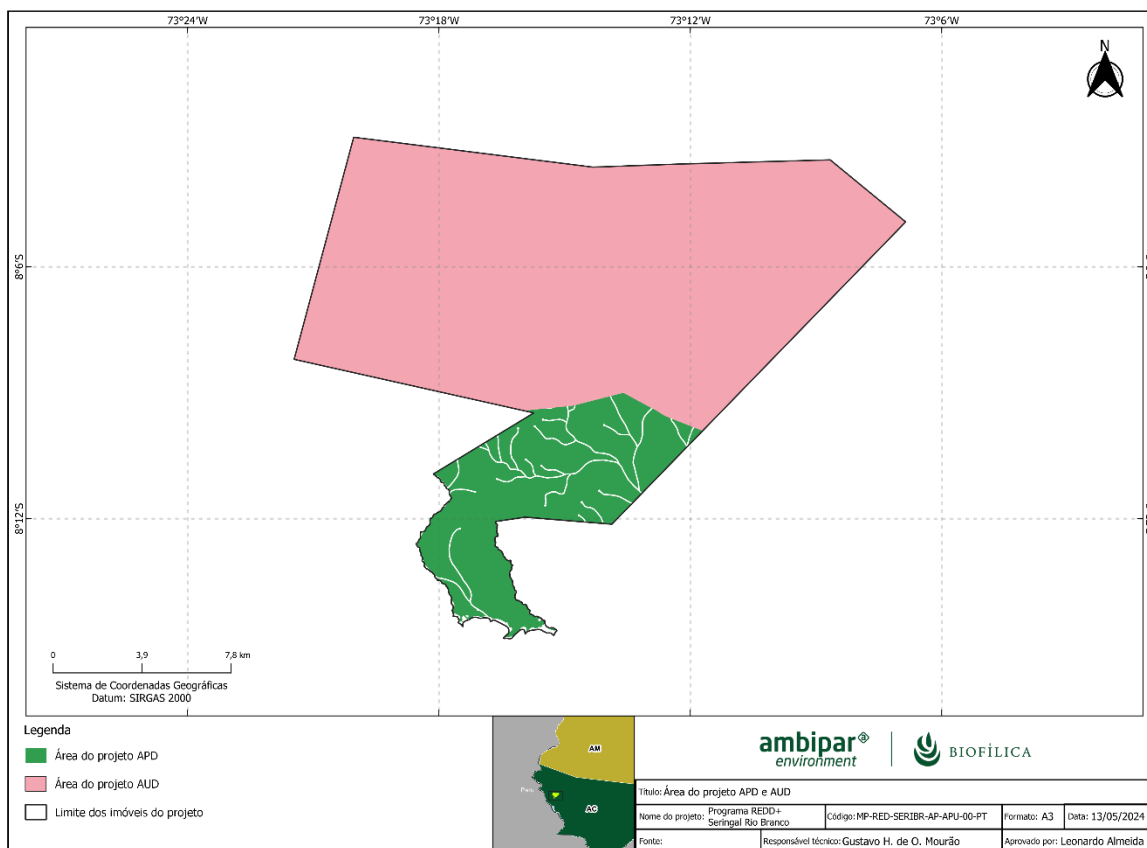


Figure 27: Area allocated to the APD and AUD Project.

3.1.4 Baseline Scenario (VCS, 3.13)

3.1.4.1 Stratification of the project area and identification of the planned deforestation agent in each stratum

Module VMD0006 v1.3 establishes that the Project Area be stratified based on the spatial distribution of carbon stocks, reflecting parameters related to forest productivity and activity parameters. Only one stratum was necessary to achieve the optimum accuracy of the project's estimates of net GHG emissions or removals, as mentioned in section 3.1.3.

The Seringal Rio Branco REDD+ Project, according to Module VMD0006 v1.3, has the scenario where the agent responsible for deforestation is an individual, an organization or an already defined corporation..

In the baseline scenario, the deforestation agent identified is the owner of Seringal Rio Branco, also a proponent of the Amazônia Agroindústria EIRELI Project. Since 2001, the company has been carrying out farming practices in the Cruzeiro do Sul region. With three farms under its management in the same

region - Seringal Valparaíso, Santa Maria II and Gama - dedicated to beef cattle production and agriculture, the company has a track record of successful operations. Managing a herd of 3,500 head of cattle, its aim is to expand cattle farming to meet demand from local meatpacking plants in the city of Cruzeiro do Sul and other cities in the state of Acre.

Considering a scenario without the Seringal Rio Branco REDD+ Project, the owner would choose to implement beef cattle ranching as the predominant activity in the area. This choice is due to the traditional nature of Brazilian cattle ranching, which originated in territorial expansion during the historical period known as “Colonial Brazil” (1530 to 1822). Today (2023-24), livestock farming still stands out as a strong economic activity capable of generating approximately 1.20 trillion reais with the presence of 230 million animals. In the scenario without the project, this monetary promise was seen as a favorable argument for deforestation and the entry of cattle into the Juruá Valley region, as the state of Acre has, since 2022, ranked 22nd in the generation of gross revenue from national cattle ranching with 2.50 billion reais and around 4.66 million head of cattle. The micro-region of Cruzeiro do Sul (AC), which includes the municipalities of Cruzeiro do Sul, Mâncio Lima, Marechal Thaumaturgo, Porto Walter and Rodrigues Alves, already has around 600,000 beef cattle (IBGE, 2022; MAPA, 2022). In addition, the beef market, whether processed or in natura, represents a high volume of exports for the country, with approximately 2.54 million tons of the 8.91 million produced (a third of the total), which makes it even more attractive to occupy new territories for this activity (IBGE, 2023). Despite the profitability, estimates show that around 70% of deforested areas in the Amazon are converted into pastures and, according to the Brazilian average of 0.97 U.A./ha from IBGE (2023), production is precarious and poorly technified, resulting in: low conservation of carbon in the soil, use of non-conservative agricultural practices, increasing greenhouse gas emissions and low efficiency in meeting the United Nations Sustainable Development Goal 2 (Zero Hunger and Sustainable Agriculture). In the state of Acre, in particular, approximately 81% of deforested areas are used to form cultivated pastures, the overwhelming majority of which are exotic species. Brizantão (*Brachiaria brizantha*), brachiaria (*Brachiaria decumbens*), humedícula (*Bracharia humedícula*) and colônia (*Panicum maximum*) pastures in particular are present in agricultural forage systems due to their easy adaptation (same climate of origin) and versatility (silage production). As a consequence, the loss of biodiversity is seen as a real danger to the Amazon ecosystem, leading to the interruption of the biophysical and biochemical processes of the local fauna and flora. In the Project Zone, the conversion of native forest into pasture is a common practice and is often the main reason for the sale of large rural properties.

At the same time, aware of the legal and environmental requirements, the owner drew up a Forest Exploration Plan for alternative land use (PEF). This plan, requested by IMAC (Acre's Environment Institute), is an essential document for the environmental licensing process for the area in question. This measure shows the intention to implement the activity in the project area if the Seringal Rio Branco REDD+ Project is not developed.

3.1.4.2 Aprovação do órgão ambiental competente

The environmental laws in force at the location of the Seringal Rio Branco REDD+ Project, i.e. in Brazil, in the state of Acre, in the municipality of Cruzeiro do Sul, were surveyed and analyzed in order to define

exactly where it is permissible to legally deforest. The laws that provide guidelines in this area are listed below:

Federal Law 12.651/2012 - Brazilian Forest Code:

The Brazilian Forest Code defines, among other factors, areas that must be protected and criteria for requests to suppress vegetation, these being:

Article 12 of this law stipulates that, outside the boundaries of the Legal Amazon, the minimum percentage of Legal Reserve that must be maintained by landowners is 20% of the total area of their property.

Article 4 of this law stipulates that a Permanent Preservation Area (APP) is considered to be the marginal strips of any natural perennial or intermittent watercourse, from the edge of the channel of the regular bed, with widths that vary according to the width of these watercourses, as stipulated by law.

Article 26 of this law also states that the suppression of native vegetation for alternative land use will depend on the property's registration in the CAR (Rural Environmental Registry) and prior authorization from the competent state agency of Sisnama (National Environment System).

Normative Ordinance No. 002, of June 25, 2004, which determines: Art. 1 Defining technical and administrative procedures for converting areas for land use, through the issuance of a Deforestation Authorization, Environmental License and Use of Raw Forest Material - AUMPF in the state of Acre.

As described in module VMD0006 v1.3, the possibility of deforestation in the project area and the intention to deforest, should government approval be required, can be demonstrated by means of a recent authorization from the relevant government department for the conversion of the forest to an alternative use, or by means of documentation proving the submission of an application for approval to the relevant government department for permission to clear and convert to an alternative use of the land. In Brazil, this documentation corresponds to an application for suppression and conversion of land to an alternative use, filed with the appropriate government department. The owner and proponent of the Amazônia Agroindústria EIRELI Project has a document filed with the environmental agency in force, IMAC (Acre's Environmental Institute), and has a Forest Exploration Plan for alternative land use aimed at implementing cattle ranching at the Seringal Rio Branco REDD+ Project site.

In general, deforestation permits are requested for areas smaller than 1,000 ha in Brazil. This common practice occurs because the procedures are simplified for this scale of deforestation. The implementation of agricultural projects larger than 1,000 ha depends on the preparation of an Environmental Impact Study (EIA) and the respective Environmental Impact Report (RIMA), to be submitted for approval to the competent state agency, in compliance with the federal requirements established in item XVII, art. .2 of CONAMA Resolution No. 01, of January 23, 1986, amended by CONAMA Resolution No. 11, of March 18, 1986. This common practice is also influenced by the short validity period of a deforestation permit and the high costs of suppressing operations, which also leads landowners to opt for suppressing small patches of forest each year, rather than large areas.

In this case, the owner chose to request initial management of 990.24 ha. The documents will be shared with the VVB team in order to demonstrate his intention to deforest.

Suitability of the project area for conversion to alternative non-forest use: This item is detailed in section 3.1.3, in the description of the delimitation of the boundaries of the Project Area.

3.1.4.3 Taxa relativa de desmatamento

In the event that the REDD+ Seringal Rio Branco project did not exist, and using Normative Ordinance No. 002 of June 25, 2004 to define legal deforestation, the owner would supply part of the forest allocated to the farm's boundaries. To this end, a third-party consultancy drew up an Alternative Land Use Forest Exploitation Plan that foresaw the removal of 5,487.10ha of Amazonian forest over 10 years, between 2024 and 2034. The area would be cleared by logging the large trees that would provide quality wood for civilian uses, followed by a controlled burn to clean up the remaining biomass. This type of practice is very common in the Brazilian Amazon, since burning forest biomass mineralizes the nutrients present in the vegetation and facilitates the planting of exotic grasses that will serve as fodder for livestock. This practice in the context of the planned deforestation of Seringal Rio Branco would occur with a relative deforestation rate of 548.71ha/year (fixed rate), as shown in table below:

Table 15: Framework of the native forest suppression project (clear cutting) in Seringal Rio Branco (AC).

Project: exploration plan (10 years)	Deforestation rate (ha/year)	Total accumulated area (ha)	Year of Start of Operation
1	548.71	548.31	2023
2	548.71	1,096.62	2024
3	548.71	1,646.13	2025
4	548.71	2,194.85	2026
5	548.71	2,743.56	2027
6	548.71	3,292.27	2028
7	548.71	3,840.98	2029
8	548.71	4,389.69	2030
9	548.71	4,938.40	2031
10	548.71	5,487.11	2032
Total	5,487.10	-	-

Then, in the hypothetical suppressed areas of 548.71ha (annually), as well as described in the Exploration Plan for advisory use, the economic activity would be livestock farming. The owner planned to produce approximately 500 animal units per year, which would be sold to local meatpacking plants in

the Juruá Valley mesoregion. In addition to the emissions generated by deforestation, this would result in the presence of methane released by the enteric fermentation of ruminant animals (via eructation), methane released by the decomposition of manure and nitrous oxide released by soluble solids (directly) deposited in the soil.

3.1.4.4 Probability of deforestation

As it is a private property, the probability of deforestation that makes up the first instance of the project is 100%, according to section 1.4 of module VMD0006 (page 12).

3.1.4.5 Risk of abandonment

Initially, five proxy areas located in the same municipality as the project area, Cruzeiro do Sul, were selected. This strategic choice confirms regional uniformity, as geographical proximity contributes significantly to the consistency of soil and vegetation factors. These elements are fundamental in analyzing the risk of abandonment in REDD projects, ensuring that the environmental conditions compared are as similar as possible.

The properties selected as proxy areas showed a high degree of similarity to the project area in terms of slope, pedology and elevation, as evidenced by the data collected. The analyses revealed that all the properties have an elevation of less than 500 meters and have more than 89% of their extension with a slope of less than 15%, indicating a predominantly flat topography, consistent in all the areas analyzed.

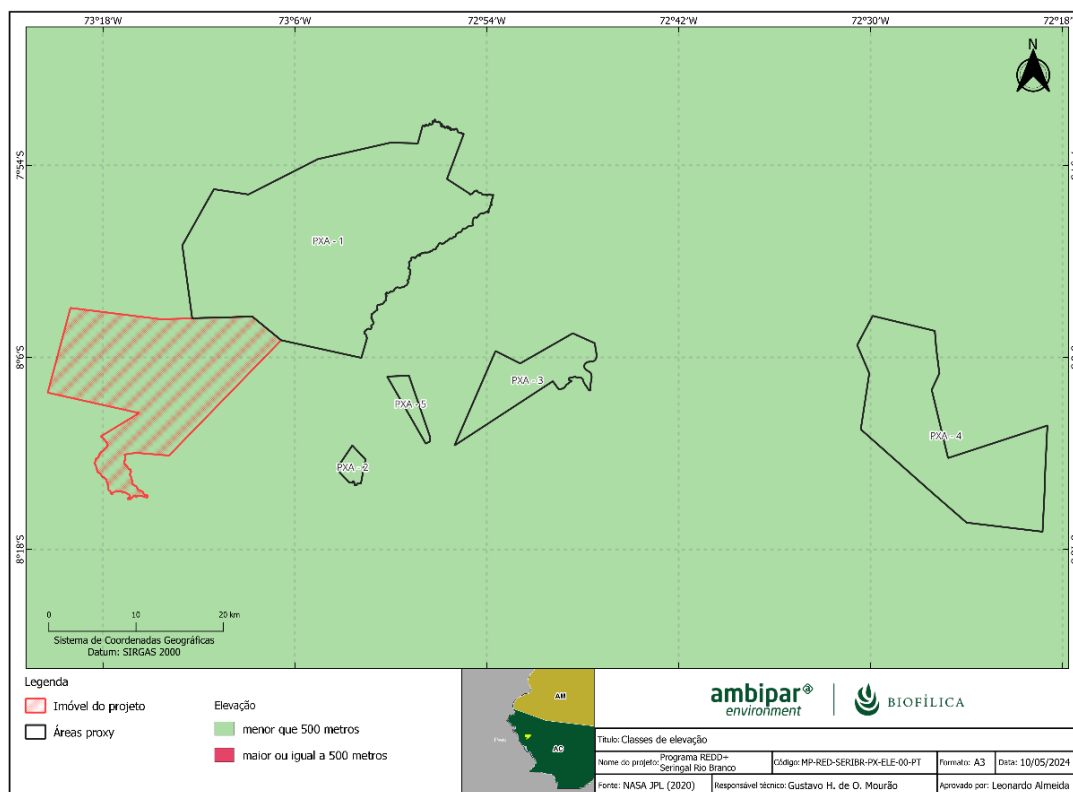


Figure 28: Proxy areas selected for risk analysis in relation to elevation.

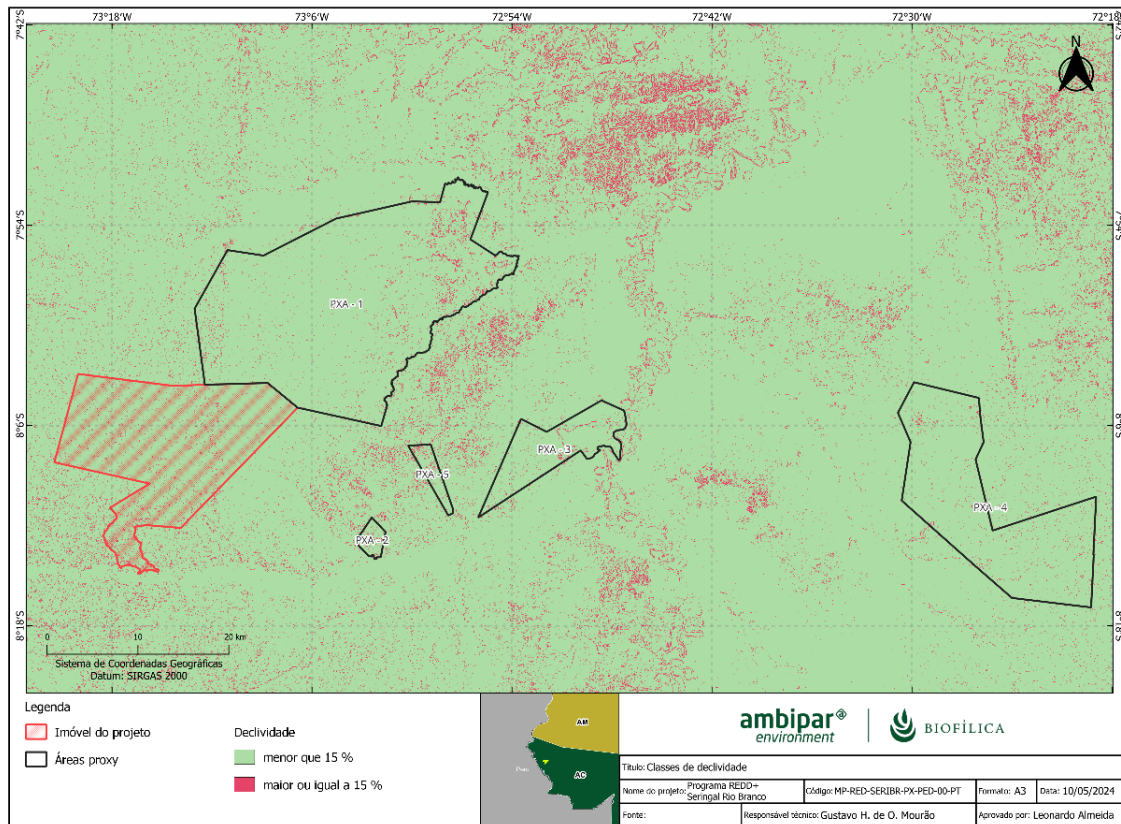


Figure 29: Proxy areas selected for risk analysis in relation to slop.

As far as pedology is concerned, all the properties, including the project area, are mostly made up of Argissolo, which accounts for 91% to 100% of the soil composition of the properties, with small variations of Gleissolo that account for a maximum of 9%. This uniformity in soil types is indicative of similar soil conditions, which can influence both the vegetation present and its conversion to different uses.

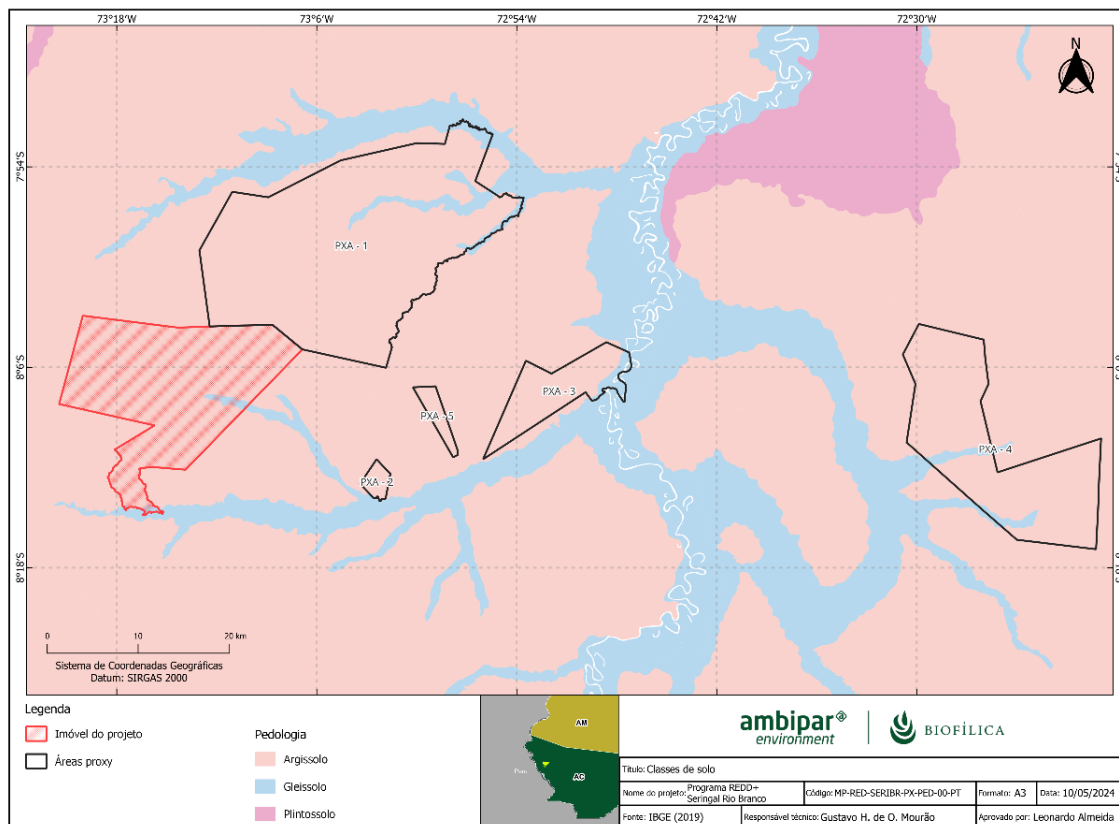


Figure 30: Proxy areas selected for risk analysis in relation to pedology.

With regard to phytophysiology, although there is a variation in the proportion between Open Ombrophilous Forest and Dense Ombrophilous Forest, it is essential to understand that both formations belong to the same humid forest ecological system. Open Ombrophilous Forest and Dense Ombrophilous Forest differ primarily in vegetation density, but share similar biological and climatic characteristics. The presence of 100% Open Ombrophilous Forest in the project area and in the other proxy properties, except for one, reinforces the consistency in the primary and secondary vegetation cover of the areas analyzed.

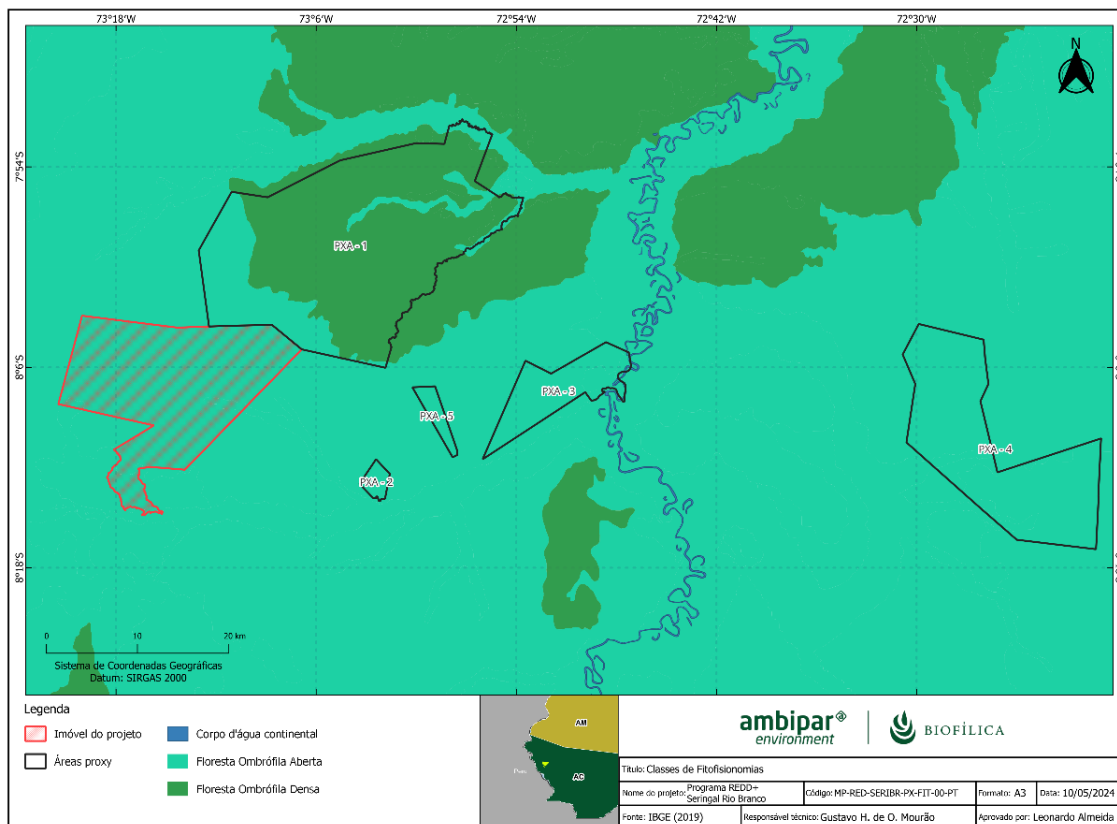


Figure 31: Proxy areas selected for risk analysis in relation to phytophysiology.

The methodology applied and the results obtained support the validity of the proxy areas selected to adequately represent the conditions of the project area. Below is a table detailing the percentage given to the area of each of the different classes investigated:

Table 16: Class categories for analysis of proxy areas.

Imóvel	Declividade		Solo		Fitofisionomia	
	< 15 %	>= 15	Argissolo	Gleissolo	Floresta Ombrófila Aberta	Floresta Ombrófila Densa
Área do Projeto	96%	4%	98%	2%	100%	0%
PXA - 1	97%	3%	94%	6%	25%	75%
PXA - 2	93%	7%	100%	0%	100%	0%
PXA - 3	96%	4%	91%	9%	100%	0%
PXA - 4	98%	2%	95%	5%	100%	0%
PXA - 5	89%	11%	100%	0%	100%	0%

After selecting the proxy areas, we began investigating the risk of the project being abandoned using MapBiomass data from 2012 to 2022. This analysis was crucial to understanding the dynamics of land

use and verifying the stability of agricultural activities in the region. It was observed that the areas that were used for grazing in 2012 remained in the same use until 2022, showing not only the persistence of farming activities, but also an increase in the extent of grazing areas over this period. This indicates that livestock activities in these areas are sustainable and not at risk of abandonment, reflecting a continuing trend of forest use and non-regeneration.

Although it is necessary to assess the risk of abandonment by March 2023, the MapBiomias data available only goes up to December 2022. This represents a limitation in the ability to directly monitor land use and vegetation cover at the beginning of 2023. However, based on the trends observed until 2022, it is not expected that land use practices will change abruptly in subsequent months, especially considering the established dynamics of agricultural expansion and the climatic context of the region, which favors the maintenance of these activities in the first wet months of the year. Therefore, even without MapBiomias raster data for 2023, the evidence up to the end of 2022 provides a solid basis for inferring the continuity of farming activities without risk of abandonment.

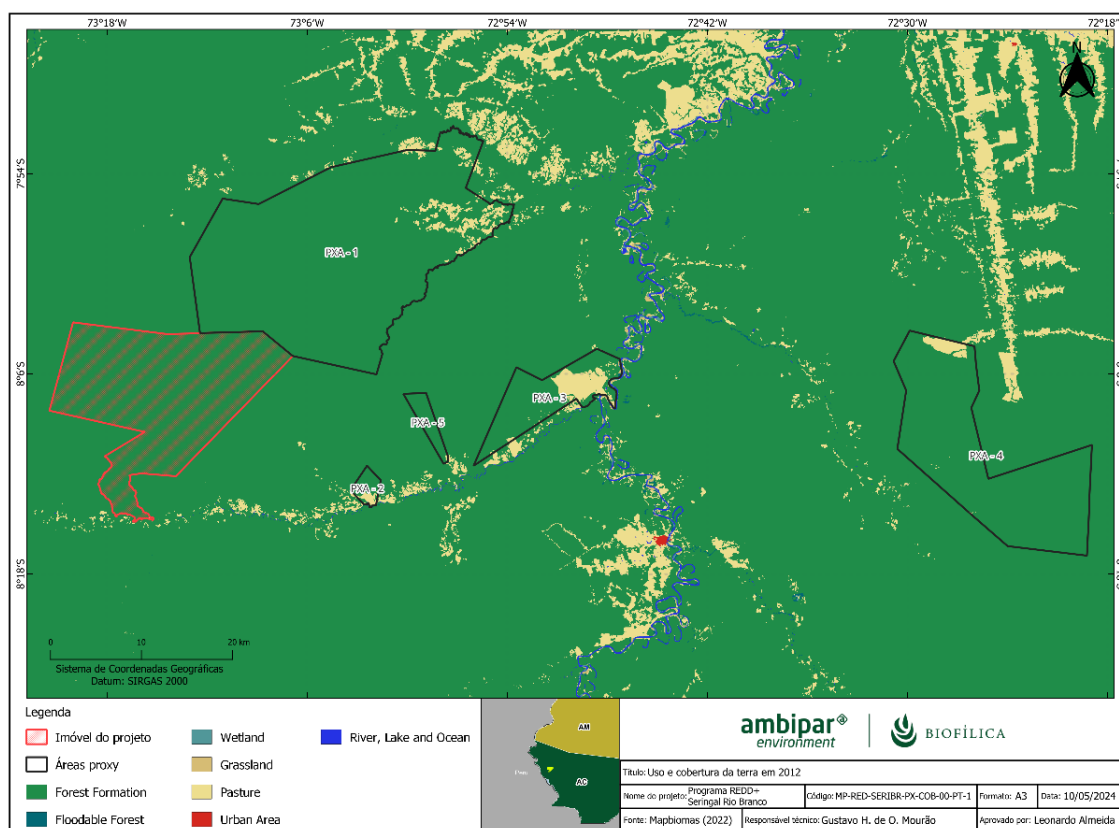


Figure 32: Land use and occupation data from MapBiomias.

3.1.4.6 Annual deforestation area ($AA_{planned,i,t}$)

The table below shows the estimated annual deforestation areas for the project area. The annual deforestation area was calculated by:

$$AA_{planned,i,t} = (A_{planned,i} \times D\%_{planned,i,t}) \times L - D_i$$

Where:

$AA_{planned,i,t}$: Baseline of planned annual deforestation area, for stratum i, in hectares (ha);

$A_{planned,i}$: Total area of deforestation planned for the baseline period, in hectares;

$D\%_{planned,i,t}$: Annual proportion of planned deforestation for stratum i, year t.

$L - D_i$: Probability of planned deforestation, equal to 100% for private properties;

Table 17: Baseline of the annual area of future deforestation in the project area (A (planned,i,t))

Project year t	Stratum i of the project area		Total	
	1- Floresta		annual	cumulative
	$ABSLPA_{i,t}$		$ABSLPA_t$	$ABSLPA$
	annual ha	cumulative ha	ha	ha
2023	549	549	549	549
2024	549	1.097	549	1.097
2025	549	1.646	549	1.646
2026	549	2.195	549	2.195
2027	549	2.744	549	2.744
2028	549	3.292	549	3.292
2029	549	3.841	549	3.841
2030	549	4.390	549	4.390
2031	549	4.938	549	4.938
2032	549	5.487	549	5.487
2033	0	5.487	0	5.487
2034	0	5.487	0	5.487
2035	0	5.487	0	5.487
2036	0	5.487	0	5.487
2037	0	5.487	0	5.487
2038	0	5.487	0	5.487
2039	0	5.487	0	5.487
2040	0	5.487	0	5.487
2041	0	5.487	0	5.487
2042	0	5.487	0	5.487
2043	0	5.487	0	5.487
2044	0	5.487	0	5.487
2045	0	5.487	0	5.487
2046	0	5.487	0	5.487
2047	0	5.487	0	5.487
2048	0	5.487	0	5.487
2049	0	5.487	0	5.487
2050	0	5.487	0	5.487
2051	0	5.487	0	5.487
2052	0	5.487	0	5.487

3.1.5 Additionality (VCS, 3.14)

3.1.5.1 Regulatory Surplus (VCS, 3.14)

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☐ No

The activities of the Seringal Rio Branco REDD+ Project are not required by any law, statute or regulation. It is worth stating that the proponent will follow all the rules and processes necessary to guarantee and prove the additionality of the activities proposed here.

3.1.5.2 Additionality Methods (VCS, 3.14)

Following the guidelines of Methodology VM0007 (item 7), the Seringal Rio Branco REDD+ Project uses the latest version of the tool referenced in the T-ADD to identify credible land use scenarios and determine the additionality of the project.

Thus, the tool “VT0001 - Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities”, version 3.0, dated February 1, 2012, was applied. The steps used, according to Tool VT0001, are described below:

STEP 1: Identification of alternative land use scenarios for the proposed AFOLU project activity

Sub-step 1a. Identify credible alternative land use scenarios for the proposed project activity VCS AFOLU

According to the application of the VT0001 tool, the following scenarios were identified for land use in the event of the Project not taking place::

1. Continuation of forestry on properties without financial incentives, such as the implementation of an AFOLU VCS-CCB Project

In this scenario, the current land use in the Project Area is assumed to remain unchanged, with forest cover remaining in the surplus area of the property, but without any financial incentive to maintain it.

In the regional context, although possible, this scenario is the most unlikely to be perpetuated. This is due to the direct costs associated with activities to protect these areas, as well as the opportunity cost related to the economic use of the properties, since the legislation allows deforestation. In addition, the

maintenance of rural properties requires the annual payment of fees and taxes, which implies the need to generate some economic revenue under penalty of financial losses or even the loss of the property to land reform programs.

These factors make it highly unlikely that land use will continue as it was before the project in the absence of financial incentives to sustain the surplus areas of the properties. Currently, the sources of revenue available to maintain these areas are limited, apart from the potential funding derived from carbon, or from converting this area to other land uses.

2. Carrying out REDD+ activities without registering as an AFOLU VCS-CCB Project

As indicated in scenario 1, maintaining forest cover without financial incentives is possible, but unlikely. In the second scenario, it is considered that there is no economic activity of any kind being implemented in the area, resulting in maintenance and surveillance costs, without any economic return, even with legal authorization to do so. In this scenario, due to the costs of conserving the forest areas, it is not considered to be financially sustainable in the long term, considering the territorial extension of the area, and much less attractive for the owner, without the revenue generated by carbon credits.

Thus, the additionality analysis shows that the project's activity, without revenue from carbon credits, is unlikely to occur and is not a common practice in the region. The project therefore depends on the economic viability provided by the sale of carbon credits to ensure forest conservation and the associated benefits, such as improved security, health, education and income diversification in local communities.

3. Legal deforestation by the landowner to expand livestock farming

Changing land use from forest to pasture within the property is the most likely scenario to occur, considering the predominance of livestock farming as a growing economic activity in the region. Analyses of secondary databases and field diagnostic surveys carried out by the Project show that many properties are already geared towards livestock activities, with the suppression of vegetation cover taking place over the last 10 years.

Livestock farming in Acre is consolidating its position as the state's main economic activity, with production expected to reach 120,000 tons of beef by 2024 (MAPA). This figure is highlighted by a study by the Business Forum for Innovation and Development in Acre, supported by the Foundation for the Support and Development of Teaching, Research and University Extension in Acre (Fundape), released in April 2024. According to the study, cattle farming accounts for 40% of Acre's Gross Production Value (GPV). In February 2024, the Ministry of Agriculture estimated Acre's agricultural GDP at R\$ 2.830 billion, of which R\$ 1.753 billion, or 61%, is attributed to cattle farming. IBGE data shows that, in 2022, the cattle herd in Acre totaled 4,635,381 animals, with the Lower Acre region concentrating 54% of this total. The area of pasture used by livestock in the state is 2.38 million hectares.

As mentioned in section 3.1.4, the company has a history of successful operations in the Cruzeiro do Sul region, with three farms under its management dedicated to beef cattle production and agriculture. The owner's goal is to expand cattle ranching to meet demand from local meatpacking plants in the city of

Cruzeiro do Sul and other cities in the state of Acre. In addition, the owner of the area has investments in livestock on other lands and a history of interest in expanding the business. In view of this, the scenario of expanding cattle farming on the property is highly likely without the presence of the project.

Sub-step 1b. - Consistency of credible land use scenarios with applicable laws and regulations

Scenarios 1 and 2 are in compliance with legal requirements, given that the scenarios do not take into account any type of economic activity, it is considered that the documentation of the property is the only aspect of greater relevance to the area and this is in compliance with legal requirements.

For scenario 3, deforestation and cattle ranching in the area are legally authorized under the Brazilian Forest Code, which allows 20% of the property's area to be used, with the remaining 80% being set aside for conservation as a Legal Reserve. The 20% area can be used for various purposes, subject to a prior license from the relevant environmental agency.

Thus, all three scenarios comply with local and national legislation and regulations.

Sub-step 1c. - Selection of the baseline scenario

Described in detail in Section 3.1 - Application of the Methodology, item 3.1.4 Baseline Scenario.

STEP 2: Investment analysis to determine that the proposed project activity is not the most economically or financially attractive of the land use scenarios identified

In development

STEP 3: Barrier analysis

VCS "VT0001 - Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities" requires either Investment Analysis (Step 2) or Barrier Analysis (Step 3). In this case, until the conclusion of this documentation, only the Investment Analysis, described in Step 2, was chosen.

STEP 2 - Barrier Analysis

This step aims to identify barriers and assess which of the identified scenarios would not be prevented from being realized by the barriers described below.

Sub-step 2a - Identification of barriers that would prevent the implementation of at least one of the alternative land use scenarios

According to the T-ADD tool, the barriers that the previously identified land use scenarios may face are as follows:

Investment barriers:

- Debt financing is not available for the land use scenarios.
- No access to international capital markets.
- Lack of access to credit.

Institutional barriers:

- Risk related to changes in government policies or laws.
- Lack of enforcement of forestry or land use legislation.

Barriers due to social conditions and land use practices:

- Widespread illegal practices (e.g. illegal grazing, extraction of non-timber products and logging).

Barriers related to land tenure, ownership, inheritance and property rights:

- Lack of adequate land tenure legislation and regulations to support security of tenure.
- Possibilities of high price risk due to fluctuations in the prices of products related to the project activity during the project period, in the absence of efficient markets.

STEP 4: Analysis of common practices

The analysis of common practices considered three policy mechanisms that encourage the conservation of vegetation cover in Acre: Environmental Reserve Quotas (CRAs), Ecological ICMS and the Environmental Services Incentive System (SISA), with a focus on the ISA Carboneo program.

CRAs are titles that represent areas with native vegetation cover and can be used to compensate for the lack of Legal Reserve (RL) on rural properties, in accordance with the Brazilian Forest Code (Law No. 12,651, of May 25, 2012). This mechanism, although provided for in the 2012 forest code, was only regulated in December 2018. Landowners with surplus RL can sell these titles to compensate for the RL deficit on other properties within the same biome. Each CRA represents one hectare of preserved native vegetation cover. However, CRAs have significant limitations when compared to the carbon market. CRA titles can only be traded within the same biome and are intended exclusively for environmental regularization, without taking into account the carbon stock in forests (Angelsen et al., 2012). Therefore, CRAs do not offer the same scope and financial benefits as carbon projects, which are aimed at any entity wishing to offset its greenhouse gas (GHG) emissions.

The Ecological ICMS is a tax mechanism that allows municipalities with environmental actions to access financial resources collected by the state through taxes. In Acre, the Ecological ICMS was created in 2019 and came into force in 2021 (Acre, Complementary Law No. 292, of December 30, 2019). It increases tax revenue for municipalities with a larger area of conservation units, but does not directly compensate landowners for the conservation of native vegetation. This mechanism has different regulations and

objectives from carbon projects, which allow landowners to directly access financial resources from the sale of carbon credits to implement conservation actions (Moura et al., 2016).⁴³.

SISA was established by State Law No. 2.308/2010 and provides incentives for a range of environmental services, including forest carbon (SISA, 2010)⁴⁴. The ISA Carbon Program, one of the most advanced within SISA, is carried out with financial resources from German Cooperation, through the REDD Early Movers Program (REM). SISA allows for integration with future incentive systems for environmental services at national, sub-national or international level. Although SISA, and specifically the ISA Carbon Program, offer a framework for conservation incentives, they still differ essentially from carbon projects. The ISA Carbon Program and its sub-programs allow carbon credits to be issued at both the state and individual project scale, but resources are often limited and dependent on international cooperation.

Although there are political mechanisms such as CRAs, Ecological ICMS and SISA in Acre, these do not offer the same benefits and purposes as REDD+ Projects. The CRAs are limited by the need for compensation within the same biome and do not account for carbon stocks. The Ecological ICMS benefits municipalities, but not landowners directly (Grinberg et al., 2016)⁴⁵. SISA and the ISA Carbon Program, although advanced, still depend on external resources and do not have the same financial flexibility as REDD+ Projects. From this analysis, it can be concluded that although these measures encourage environmental conservation, they do not match the financial benefits and scope that the Seringal Rio Branco REDD+ project can generate. The implementation of carbon projects such as this is essential for the long-term viability of forest conservation, as there are no similar practices, so we can say that the activity is additional to the REDD+ project.

3.1.6 Methodology Deviations (VCS, 3.20)

There are no methodological deviations to be presented when validating the Seringal Rio Branco REDD+ project.

3.2 Quantification of Estimated GHG Emission Reductions and Removals

3.2.1 Baseline Emissions (VCS, 3.15)

Estimating pre-deforestation carbon stocks

The pre-deforestation carbon stock was estimated using primary data from the forest inventory, considering only one stratum. According to the stock diagnosis studies, the definition of the stratum followed the typology of the IBGE classification system, identifying it only as Open Ombrophilous Forest for the Project area. Although there are many variables to describe vegetation communities, some do

43 Moura, E., Oliveira, A. D., & Sambuichi, R. H. (2016). Políticas públicas e REDD+ no Brasil: Acordos internacionais, projetos e questões de implementação. Ipea.

44 Sistema de Incentivos a Serviços Ambientais (SISA). (2010). Lei Estadual nº 2.308. Governo do Acre.

45 Grinberg, E., Rosa, M. L. P., & Lima, S. (2016). ICMS Ecológico: instrumento de conservação da biodiversidade e justiça socioambiental. Editora Annablume.

not differ at the lower levels of the IBGE (2012) hierarchical system. For this reason, it is common in the literature to find studies that describe and differentiate vegetation only at the higher levels, such as Open Forests and Dense Forests. Examples of this are the studies by Nogueira et al. (2008), which refer to the open forests of the southern Amazon, and the studies by Oliveira et al. (2012) and Scaranello et al. (2016), which deal with open and dense forests.

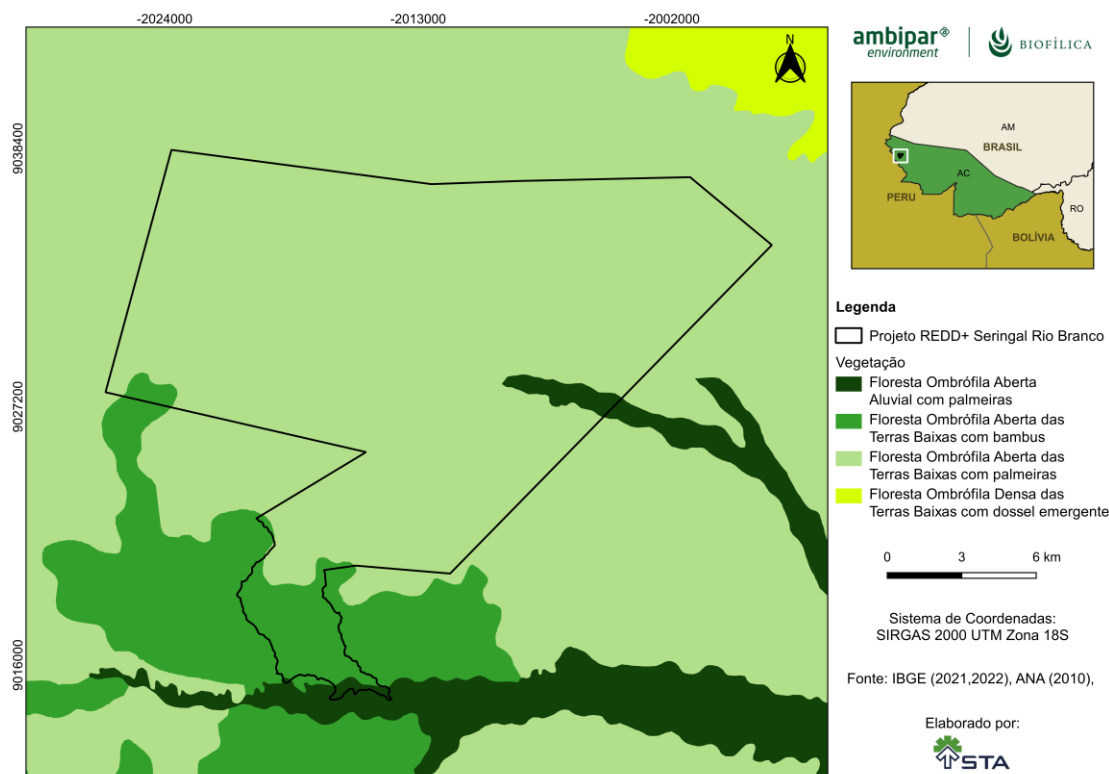


Figure 33: Vegetation cover in the Seringal Rio Branco REDD+ Project area, municipality of Cruzeiro do Sul, Acre.

The sampling design for allocating the inventory plots was defined considering the total forest area of the property, which is 29,611.0242 hectares. The inventory was carried out on the entire forest area with the future intention of implementing the AUD activity, using the VM0048 methodology. This area of 29,611.0242 ha was used as a benchmark for establishing the sampling process, thus forming the sample universe.

The aim of the sampling design was to guarantee a sampling error for estimating the average total biomass of less than 15% for the total area, considering a 95% confidence level. Preliminary analyses indicated conservative numbers of 26 plots measuring 100 x 100m in the forest area. The uncertainty found in the total biomass in the project area was 5.46% of the total number of individuals, which corroborates the lack of difference between the forest typologies found. The REDD+ Seringal Rio Branco Project used the fixed-area, permanent sample plot method. The location map of the allocated sample plots can be seen in Figure X.

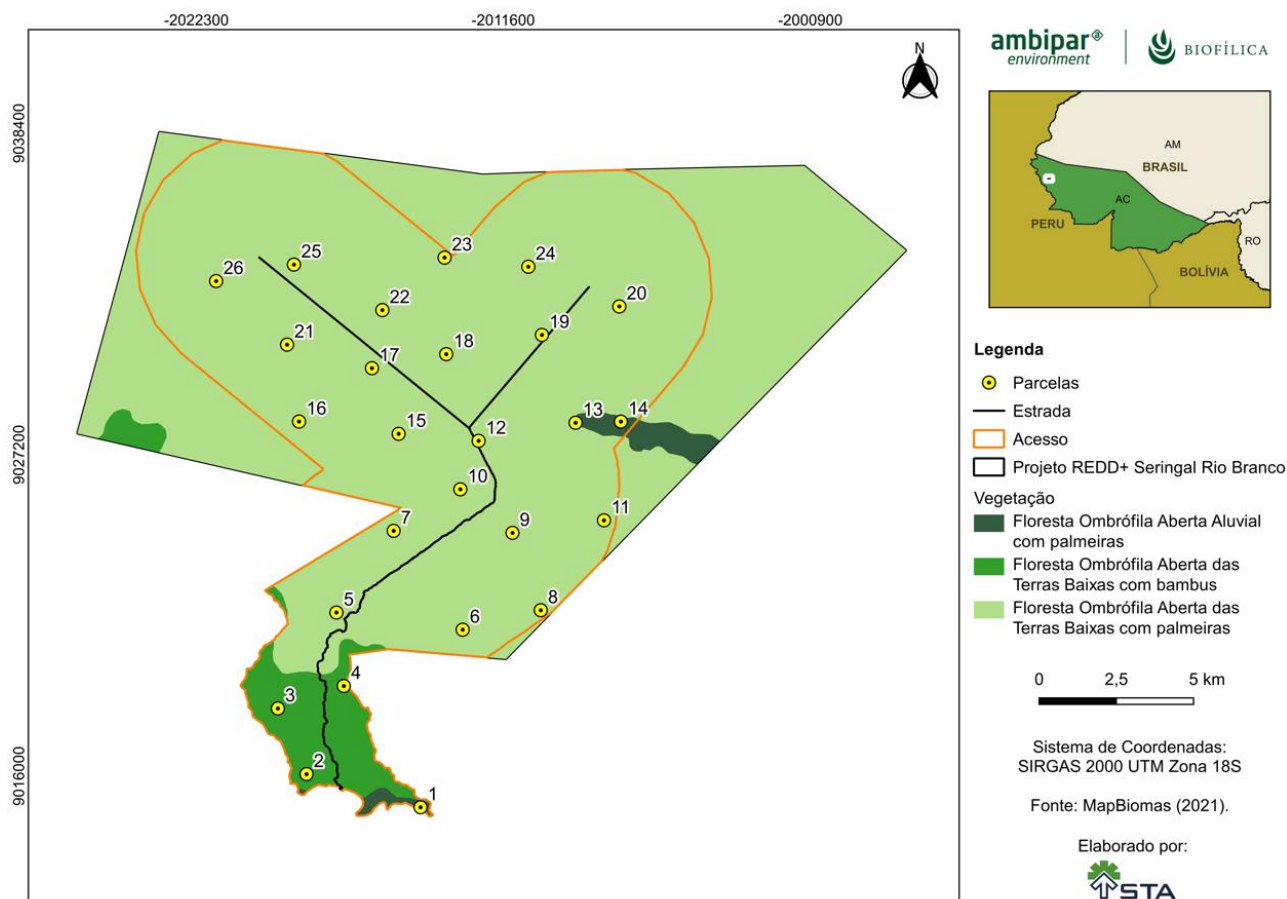


Figure 34: Location map of the sampling units (or plots) of the REDD+ Project Seringal Rio Branco, municipality of Cruzeiro do Sul, Acre.

3.2.1.1 Quantification of the Carbon Stock by Reservoir in the Baseline Scenario

A escolha da equação a ser utilizada no projeto REDD+ Seringal Rio Branco foi realizada em dois grupos vegetais: árvores (dicotiledôneas, “trees”) e palmeiras arborescentes (monocotiledôneas, “palm-tree”), uma vez que suas estruturas são diferentes. Para cada um desses grupos (árvores e palmeiras) foram testadas equações alométricas encontradas na literatura. As equações testadas foram inicialmente escolhidas em função dos dados inventariados no campo; ou seja, são equações que utilizam apenas o DAP como parâmetro para estimar a biomassa. Todos os detalhes sobre as análises feitas constam no diagnóstico de Estoque de Carbono disponibilizado para a auditoria.

3.2.1.1.1 Above Ground Living Biomass

After choosing the equations to be used to estimate the biomass of trees (Tree) (equation 10) and palms (Non-tree) (equation 11), the biomass of each tree and palm was estimated and the sum of the biomass of the plot was established according to the CP-AB Module.

$$C_{AB-tree}(i) = \sum_{i=n}^i \int_i^{\square} (X, Y \dots)_{(i)}$$

Equação 10

$$C_{AB-non-tree/palm}(i) = \sum_{i=n}^i \int_i^{\square}(X, Y \dots)_{(i)} \quad \text{Equação 11}$$

$$AGLB_{(total da parcela)} = \sum_{i=n}^i C_{AB}(\text{árvore } i) + \sum_{i=n}^i C_{AB-non-tree}(\text{palmeira } i) \quad \text{Equação 11}$$

Where:

$AGLB_{(plot total)}$ = live above-ground biomass (tree + non-tree) of plot i (Mg. ha⁻¹).

$\int_i^{\square}(X, Y \dots)$ = estimated biomass for tree or palm “i” from the chosen function;

$i = 1, 2, 3 \dots n$ trees or palms inventoried

$C_{AB-tree}(i)$ = Sum of the biomass of the trees in plot i (Mg);

$C_{AB-non-tree/palm}(i)$ = Sum of non-tree/palm biomass in plot i (Mg);

The averages and other statistical descriptors of $C_{AB-tree}$ e do $C_{AB-non-tree/palm}$ were calculated by stratum or forest typology, taking into account the number of plots inventoried (26 plots).

Remember that all the values were later transformed into MgC ha⁻¹ using the factor 0.47 and then transformed into MgCO₂e ha⁻¹ using the factor 44/12, as explained above.

3.2.1.1.2 Aboveground Dead Biomass

This reservoir was considered because it corresponds to 8.1% of the biomass of open forests in the Amazon, according to the Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change - Reference Report: Land Use Sector, Land Use Change and Forests, 2020.

As the project area is in a location that is difficult to access, the factor of 0.081 from the Brazilian Fourth Communication was used. This factor corresponds to the dead wood:above-ground biomass ratio found in Amazonian Open Ombrophilous Forests, originating from Nogueira et al. (2008).⁴⁶

$$C_{DW}(plot i) = AGLB_{plot total i} * R \quad \text{Equation 12}$$

Where:

$C_{DW}(plot i)$ = dead wood biomass of plot i (Mg. ha⁻¹);

R = ratio “above ground biomass : dead wood biomass” = 0.081 (ratio between above ground biomass and dead wood biomass);

$i = 1, 2, 3 \dots n$ inventoried plots;

⁴⁶ Nogueira, E.M.; Fearnside, P.M.; Nelson, B.W.; Barbosa, R.I.; Keizer, E.W.H. (2008). Estimates of forest biomass in the Brazilian Amazon: new allometric equations and adjustments to biomass from wood-volume inventories. *Forest Ecology and Management*, 256: 1853–1867.

$AGLB_{(plot\ total\ i)}$ = living above-ground biomass (trees + palms = tree + non-tree) of plot i; (Mg. ha-1).

The average and the other statistical descriptors of C_{DW} were calculated for the total area of the project considering the number of plots inventoried (26 plots).

Please note that all values were later transformed to MgC ha-1 using the factor 0.47 and then transformed to MgCO₂e ha-1 using the factor 44/12.

3.2.1.1.3 Below-ground biomass (roots)

According to the CP-AB Module, the root:shoot ratio (ratio of below-ground biomass to above-ground biomass) can be used; according to the IPCC (Table 4.4 of the 2006 IPCC Guidelines, Chapter 4) and the CP-AB Module itself, page 17), the root:shoot ratio in tropical forests is on average 24%. Therefore, the value of 0.24 was used for the root:shoot ratio, which is the average presented by the IPCC (2006). Below-ground biomass was calculated for each plot and not for each individual; the equation used for each plot was:

$$C_{BB\ (plot\ i)} = AGLB_{plot\ total\ i} * R \quad \text{Equation 12}$$

Where:

$C_{BB\ (plot\ i)}$ = below-ground biomass of plot i (Mg. ha-1);

R = “root:shoot” ratio = 0.24 (ratio of below-ground biomass to above-ground biomass);

i = 1, 2, 3 ... n plots inventoried;

$AGLB_{(total\ of\ plot\ i)}$ = living above-ground biomass (trees + palms = tree + non-tree) of plot i (Mg. ha-1);

The average and the other statistical descriptors of C_{BB} were calculated for the total area of the project, taking into account the number of plots inventoried (26 plots).

Please note that all values were later transformed to MgC ha-1 using the factor 0.47 and then transformed to MgCO₂e ha-1 using the factor 44/12.

3.2.1.1.4 Total Biomass

The total biomass of the plot is the result of the sum of the reservoirs analyzed:

$$\text{Total Biomass}_{plot\ i} = C_{AB-tree\ (plot\ i)} + C_{AB-non-tree/palm\ (plot\ i)} + C_{DW\ (plot\ i)} + C_{BB\ (plot\ i)} \quad \text{Equation 13}$$

Where:

$\text{Total Biomass}_{parcela\ i}$ = Total biomass of plot i (Mg ha-1);

$C_{AB-tree\ (plot\ i)}$ = Biomass of trees (with DBH > 10cm) (Mg ha-1);

$C_{AB-non-tree/palm (plot i)}$ = Biomass of non-trees/palms (with DBH > 10cm) (Mg ha⁻¹);

$C_{DW-(plot i)}$ = Biomass of dead wood (Mg ha⁻¹);

$C_{BB-(plot i)}$ = Understory/root biomass (Mg ha⁻¹);

The average and other statistical descriptors of total biomass were calculated for the total project area, taking into account the number of plots inventoried (26 plots).

Remember that all values were later transformed to MgC ha⁻¹ using the factor 0.47 and then transformed to MgCO₂e ha⁻¹ using the factor 44/12, as explained above.

For the trees in the Seringal Rio Branco REDD+ project, the Nogueira et al. (2008) equation was used in combination with the 1-Brown (1997) equation. The Nogueira et al. equation will be applied to trees with DBH < 95.0 cm, while the Brown equation will be applied to trees with DBH ≥ 95.0 cm.

The table below shows the intermediate data used in the emission calculations for estimating the carbon stock in the Seringal Rio Branco REDD+ Project areas.

Table 18: Statistical descriptors of Biomass, Carbon and CO₂e of open ombrophilous forests in the Seringal Rio Branco REDD+ Project area, municipality of Cruzeiro do Sul, Acre. Results obtained with 95% confidence; n=26.

Pool	Mean	Standard Deviation	Deviation Variation (%)	Standard Error of mean	Confidence interval =IC _{95%}	Uncertainty (%)
Biomass (Mg ha⁻¹)						
Tree	252,67	58,92	23,32	11,56	23,80	9,42
Palms (Non-Tree)	14,99	5,96	39,77	1,17	2,41	16,07
Dead Wood	21,68	4,77	22,02	0,94	1,93	8,90
Below Ground (Roots)	64,24	14,14	22,02	2,77	5,71	8,90
Total	353,59	77,85	22,02	15,77	31,45	8,90
Carbon (Mg ha⁻¹)						
Tree	118,76	27,69	23,32	5,43	11,19	9,42
Palms (Non-Tree)	7,05	2,80	39,77	0,55	1,13	16,07
Dead Wood	10,19	2,24	22,02	0,44	0,91	8,90
Below Ground (Roots)	30,19	6,65	22,02	1,30	2,69	8,90
Total	166,19	36,59	22,02	7,18	14,78	8,90

Pool	Mean	Standard Deviation	Deviation Variation (%)	Standard Error of mean	Confidence interval =IC _{95%}	Uncertainty (%)
Carbon equivalent – CO₂e (Mg ha⁻¹)						
Tree	435,44	101,54	23,32	19,91	41,02	9,42
Palms (Non-Tree)	25,84	10,28	39,77	2,02	4,15	16,07
Dead Wood	37,36	8,23	22,02	1,61	3,32	8,90
Below Ground (Roots)	110,71	24,37	22,02	4,78	9,85	8,90
Total	609,35	134,16	22,02	26,31	54,20	8,90

3.2.1.2 Estimating post-deforestation carbon stocks

The remaining carbon stock after deforestation (CP, post, i) can be calculated by applying the standard values adopted by the Ministry of the Environment (MMA) and presented in the Brazilian Fourth Communication⁴⁷ (Tabela 23, página 159).

The table below shows the three most likely types of use to be installed in the Amazon. The pasture values can be considered for two reasons: (1) the Socio-Economic Diagnosis report presented for the Seringal Rio Branco REDD+ Project area points to pastures for cattle breeding as the type of use most likely to occur according to the future perspective of the communities living in the surrounding area; and (2) the BL-UP module (page 31) advises that when faced with different land use possibilities, you can choose the one with the highest carbon stock, and this choice is considered conservative.

Table 19: Carbon stock by type of use after deforestation. Data source: The values in Mg C ha⁻¹ were extracted from the “Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference.

Post-deforestation land use	tC ha ⁻¹	tCO ₂ e ha ⁻¹
Pasture	10	36,67
Agriculture and pasture mosaic	7,57	27,76
Soybean planting	5,14	18,85

3.2.1.3 Change in Carbon Stock in All Pools in the Baseline Scenario

The change in carbon stocks in the project area was estimated by equation 13 of Module VMD0006:

⁴⁷https://www.gov.br/mcti/pt-br/acompanhe-o-mcti/sirene/publicacoes/comunicacoes-nacionais-do-brasil-a-unfccc/arquivos/4comunicacao/4_nat_com_brazil_web.pdf

$$\begin{aligned} \Delta C_{BSL,i,t} = & AA_{planned,i,t} \times (\Delta C_{AB_{tree,i}} - \Delta C_{WP,i} + \Delta C_{AB_{non\ tree,i}} + \Delta C_{LI,i}) + \\ & + \left(\sum_{t=10}^t A_{planned,i,t} \right) \times (\Delta C_{BB_{tree,i}} + \Delta C_{BB_{non\ tree,i}} + \Delta C_{DW,i}) \times \left(\frac{1}{10} \right) + \\ & + \left(\sum_{t=20}^t AA_{planned,i,t} \right) \times (C_{WP100,i} + \Delta C_{SOC,i}) \times \left(\frac{1}{20} \right) \end{aligned}$$

Where:

$\Delta C_{BSL,i,t}$: baseline estimate of the sum of changes in carbon stocks for all terrestrial reservoirs in stratum i and in year t, in tCO₂-e;

$AA_{planned,i,t}$: baseline estimate of the area of planned deforestation in the project area, in stratum i and year t, in hectares (ha);

$C_{WP100,i}$: estimate of the average carbon stock for the year in which deforestation occurred converted into forest products, and assumed to remain stored for 100 years, in tCO₂-e/ha;

$\Delta C_{R,i}$: baseline estimate of the change in the average carbon stock of the biomass of different terrestrial reservoirs in stratum i, in tCO₂-e/ha. The different pools are:

AB_{tree} : aboveground tree biomass; WP: forest products; WP : produtos florestais; $AB_{non\ tree}$: non-tree vegetation; LI : liteira; BB_{tree} : belowground tree biomass, DW : aboveground dead biomass and SOC: soil organic carbon.

The change in carbon stock is given by:

$$\Delta C_{R,i} = C_{R_{pre,i}} - C_{R_{post,i}}$$

Where:

$C_{R_{pre,i}}$ e $C_{R_{post,i}}$ are the baseline estimates of the pre- and post-deforestation stocks of reservoir R, stratum i. The following sections describe the steps for calculating these and the other components needed to obtain $\Delta C_{BSL,i,t}$.

Table 20: Change in net carbon stock in the baseline

Changes in carbon stock by initial forest class icl			Post-deforestation carbon stock changes			Total net change in the carbon stock of the project area	
ID _{icl} >	1	$\Delta C_{BSL,i,t}$	ID _{iz} >	1	$\Delta C_{BSL,i,t}$	$\Delta C_{BSL,i,t}$	
Name>	Florest	annual	Name>	Zone 1	annual	annual	cumulative
Project Year t	tCO ₂ -e	tCO ₂ -e	Project Year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2023	261.234	261.234	2023	0	0	261.234	261.234
2024	269.359	269.359	2024	2.012	2.012	267.347	528.581

Changes in carbon stock by initial forest class icl			Post-deforestation carbon stock changes			Total net change in the carbon stock of the project area	
ID _{icl} >	1	ΔC _{BSL,i,t}	ID _{iz} >	1	ΔC _{BSL,i,t}	ΔC _{BSL,i,t}	
Name>	Florest	annual	Name>	Zone 1	annual	annual	cumulative
Project Year t	tCO ₂ -e	tCO ₂ -e	Project Year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2025	277.484	277.484	2025	4.024	4.024	273.460	802.041
2026	285.609	285.609	2026	6.036	6.036	279.572	1.081.613
2027	293.733	293.733	2027	8.048	8.048	285.685	1.367.298
2028	301.858	301.858	2028	10.061	10.061	291.798	1.659.096
2029	309.983	309.983	2029	12.073	12.073	297.910	1.957.006
2030	318.108	318.108	2030	14.085	14.085	304.023	2.261.029
2031	326.233	326.233	2031	16.097	16.097	310.136	2.571.164
2032	334.357	334.357	2032	18.109	18.109	316.248	2.887.413
2033	73.123	73.123	2033	18.109	18.109	55.014	2.942.426
2034	64.998	64.998	2034	16.097	16.097	48.901	2.991.328
2035	56.873	56.873	2035	14.085	14.085	42.789	3.034.116
2036	48.749	48.749	2036	12.073	12.073	36.676	3.070.792
2037	40.624	40.624	2037	10.061	10.061	30.563	3.101.355
2038	32.499	32.499	2038	8.048	8.048	24.451	3.125.806
2039	24.374	24.374	2039	6.036	6.036	18.338	3.144.144
2040	16.250	16.250	2040	4.024	4.024	12.225	3.156.369
2041	8.125	8.125	2041	2.012	2.012	6.113	3.162.482
2042	0	0	2042	0	0	0	3.162.482
2043	0	0	2043	0	0	0	3.162.482
2044	0	0	2044	0	0	0	3.162.482
2045	0	0	2045	0	0	0	3.162.482
2046	0	0	2046	0	0	0	3.162.482
2047	0	0	2047	0	0	0	3.162.482
2048	0	0	2048	0	0	0	3.162.482
2049	0	0	2049	0	0	0	3.162.482
2050	0	0	2050	0	0	0	3.162.482
2051	0	0	2051	0	0	0	3.162.482
2052	0	0	2052	0	0	0	3.162.482

3.2.1.4 Non-CO₂ GHG emissions from biomass burning in the Baseline Scenario

Significant non-CO₂ greenhouse gas emissions occurring within project boundaries must be assessed; However, these emissions must also be significant, for which the T-SIG tool must be used.

Emissions are calculated through the sum of emissions caused by burning, the use of fossil fuels, the use of nitrogenous compounds (such as fertilizers) and livestock farming. Due to the practices that would occur in the baseline scenario of the REDD+ Seringal Rio Branco Project, the accounting of emissions

from the use of fossil fuels and nitrogenous compounds were removed, leaving GHG emissions due to the burning of biomass and livestock activities that would replace land use.

3.2.1.4.1 Non-CO2 GHG emissions from biomass burning in the Baseline Scenario

As described in section 3.1.4 of the Baseline Scenario, in the without-project scenario there would be deforestation followed by controlled burning of the forest. The burning of vegetation and/or peat results in emissions of CO₂, N₂O and CH₄. The quantification of the emission of these gases in the Baseline Scenario (and in the Project Scenario) is estimated using the module “VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB), v1.3” according to the equation below.

$$E_{biomassburn,i,t} = \sum_{g=1}^G \left((A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i}) \times 10^{-3} \right) \times GWP_g \quad (1)$$

Where:

$E_{biomassburn,i,t}$	= Greenhouse gas emissions due to biomass burning in stratum i in year t of each GHG (CO ₂ , CH ₄ , N ₂ O) (t CO ₂ e)
$A_{burn,i,t}$	= Area burnt for stratum i in year t (ha)
$B_{i,t}$	= Average aboveground biomass stock before burning stratum i , year (t d.m. ha ⁻¹)
$COMF_i$	= Combustion factor for stratum i (unitless)
$G_{g,i}$	= Emission factor for stratum i for gas g (kg t ⁻¹ d.m. burnt)
GWP_g	= Global warming potential for gas g (t CO ₂ /t gas g)
g	= 1, 2, 3 ... G greenhouse gases including carbon dioxide ¹ , methane and nitrous oxide (unitless)
i	= 1, 2, 3, ... M strata (unitless)
t	= 1, 2, 3, ... t^* time elapsed since the start of the project activity (years)

For the baseline calculations, the following data and parameters were used:

$A_{burn,i,t}$ – same area that would be legally deforested in the baseline scenario = 5487 ha

$B_{i,t}$ – stock presented in section 3.2.1.1 = 498,64 t.CO₂eq/ha

$COMF_i$ – biomass combustion factor in the single project stratum of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 0,57

$G_{CO_2,i}$ – carbon dioxide emission factor from biomass burning from 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 85,26 t.CO₂eq/ha

$G_{CH_4,i}$ – methane emission factor from biomass burning from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 7,81 t.CO₂eq/ha

GN_{20,i} – nitrous oxide emission factor from biomass burning 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 0,79 t.CO₂eq/ha

	Emissions of non-CO2 gasses from biomass burn at the baseline scenario		Total emissions of non-CO2 gasses from biomass burn at the baseline scenario	
ID _{icl} >	ID _{icl} =	1	annual	cummulative
Ano do Projeto t	ABSLPA _{icl} ,t (hectares)	EBBBSL _{totcl} ,t (tCO ₂ e ha ⁻¹)	EBBBSLPA _t (tCO ₂ e)	EBBBSLPA (tCO ₂ e)
2023	549	8,61	4.723	4.723
2024	549	8,61	4.723	9.445
2025	549	8,61	4.723	14.168
2026	549	8,61	4.723	18.890
2027	549	8,61	4.723	23.613
2028	549	8,61	4.723	28.335
2029	549	8,61	4.723	33.058
2030	549	8,61	4.723	37.780
2031	549	8,61	4.723	42.503
2032	549	8,61	4.723	47.225

3.2.1.4.2 Non-CO2 GHG emissions from livestock in the Baseline Scenario

The formalized Exploitation Plan, as well as the methodologies for quantifying emissions from the cattle farming system in the '2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 Chapter 10 and 11', made it possible to measure emissions from the possible baseline herd. To do this, livestock indices similar to those found in the IBGE Brazilian average (2023) were considered, using an average stocking rate of 0.97 A.U./ha (in the wet and dry seasons respectively). This made it possible to simulate the number of animals allocated to deforested areas each year. Furthermore, as mentioned above, the IPCC methodologies were used to quantify only emissions from enteric fermentation and manure (methane and nitrous oxide). In this way, table below allows us to identify the maximum number of animals emitting in the areas contained in the 10-year deforestation plan considered for the baseline.

Table 21: Livestock production baseline for the Seringal Rio Branco REDD+ project for the Exploitation Plan proposed to the owner.

Project Year t	Cumulative area of suppressed forest in the baseline scenario (hectares)	Cumulative number of animals in the area in the baseline scenario (animals)	Cumulative emissions avoided (t CO ₂ /year)
----------------	--	---	--

2023	548.31	532.25	741,98
2024	1,096.62	1,064.50	2.225,95
2025	1,646.13	1,596.75	4.451,89
2026	2,194.85	2,129.00	7.419,82
2027	2,743.56	2,661.25	11.129,73
2028	3,292.27	3,193.50	15.581,62
2029	3,840.98	3,725.75	20.775,50
2030	4,389.69	4,258.00	26.711,35
2031	4,938.40	4,790.25	33.389,19
2032	5,487.11	5,322.50	40.809,01
2033	5,487.11	5,322.50	48.228,83
2034	5,487.11	5,322.50	55.648,65
2035	5,487.11	5,322.50	63.068,47
2036	5,487.11	5,322.50	70.488,29
2037	5,487.11	5,322.50	77.908,11
2038	5,487.11	5,322.50	85.327,93
2039	5,487.11	5,322.50	92.747,75
2040	5,487.11	5,322.50	100.167,57
2041	5,487.11	5,322.50	107.587,39
2042	5,487.11	5,322.50	115.007,21
2043	5,487.11	5,322.50	122.427,03
2044	5,487.11	5,322.50	129.846,85
2045	5,487.11	5,322.50	137.266,67
2046	5,487.11	5,322.50	144.686,49
2047	5,487.11	5,322.50	152.106,31
2048	5,487.11	5,322.50	159.526,13
2049	5,487.11	5,322.50	166.945,95
2050	5,487.11	5,322.50	174.365,77
2051	5,487.11	5,322.50	181.785,59
2052	5,487.11	5,322.50	189.205,41

It was considered that the emissions from cattle ranching, for the baseline, extend over the entire time of the project, which has a period of 30 years. In this way, the total accumulated avoided by REDD+ Seringal Rio Branco, in the land use program with cattle ranching alone, is 222,457.05 t CO₂/ha.

Livestock emission estimates

Emission calculations for the assumed livestock in the project area (in the event of deforestation) were carried out according to the presence of animals in a nominal grazing system. The emission sources used were enteric fermentation and manure (see description in section 3.1.4 of Baseline Emissions). The

equations used were taken from the document '2006 IPCC Guidelines for National Greenhouse Gas Inventories', chapters 10 and 11. According to the VM0007 methodology, the insertion of new emissive components is legitimate as long as it meets the criteria: (1) calculation methodologies recognized by Verra and (2) demonstration in ex-ante and ex-post calculations.

The Tier 1 approach was used for all CO₂ sources, where only animal population data is needed to estimate emissions. The indices applied as components of the equations were also acquired from the IPCC itself, named “standard factors”. They are all described in the calculation procedure below.

Methane emissions via enteric fermentation in livestock:

As described in section 3.1.4 Baseline Scenario, the herd contained in the possible deforested area met the value of 0.97 A.U./ha. In this way, a (static) stocking rate of 532 U.A./year was considered for each deforestation cycle, starting with 532 animals in the first year and accumulating to 5,322 animals in the tenth year.

For methane emissions via enteric fermentation, equation 10.19 from chapter 10 (volume 4) of the '2006 IPCC Guidelines for National Greenhouse Gas Inventories' was used. The results were converted to t CO₂eq/ha using the Global Warm Potential (GWP) of 28.5 and adjusting the unit of measurement from Gg to Mg (dividing by 10³).

EQUATION 10.19 ENTERIC FERMENTATION EMISSIONS FROM A LIVESTOCK CATEGORY

$$Emissions = EF_{(T)} \bullet \left(\frac{N_{(T)}}{10^6} \right)$$

In which:

Emissions: methane emissions via enteric fermentation (Gg CH₄/year);

EF(T): emission factor defined for other cattle of 56 kg CH₄/year;

N(T): total number of animals.

Emission of waste deposited (on pasture) by the herd:

Methane emission:

It is understood that the decomposition of the solid organic matter contained in the animal waste will be broken down via fungi and bacteria present in the soil. Table 10.14 - Manure management methane emission factors by temperature for cattle, swine, and buffalo (kg CH₄ head⁻¹ yr⁻¹) from chapter 10, volume 4 of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories was used to calculate the emission factor. The value used was 1 kg CH₄/head.year, which was multiplied by the number of animals in the area.

Nitrous oxide emissions:

The equation '11.1 Direct N₂O emissions from Managed Soils (Tier 1)' from chapter 11, volume 4, of the '2006 IPCC Guidelines for National Greenhouse Gas Inventories' was used for nitrous oxide emissions present in the soluble fraction of manure (soluble solids, such as urea). The equation was adapted by removing the factors N₂O-NN inputs and N₂O-NOS due to the absence of nitrogen fertilization and organic soils in the project area. In this way, only N₂O-NPRP was quantified, which refers to the total direct nitrous oxide resulting from animal urine in the pasture. In addition, only cattle were considered as the type of animal present in the system, which led to the removal of the factors FPRP,SO and EF3PRP,SO (representing pigs and other animals).

EQUATION 11.1

DIRECT N₂O EMISSIONS FROM MANAGED SOILS (TIER 1)

$$N_2O_{Direct-N} = N_2O-N_{N\text{ inputs}} + N_2O-N_{OS} + N_2O-N_{PRP}$$

Where:

$$N_2O-N_{N\text{ inputs}} = \left[\left[(F_{SN} + F_{ON} + F_{CR} + F_{SOM}) \cdot EF_1 \right] + \left[(F_{SN} + F_{ON} + F_{CR} + F_{SOM})_{FR} \cdot EF_{1FR} \right] \right]$$

$$N_2O-N_{OS} = \left[\left(F_{OS,CG,Temp} \cdot EF_{2CG,Temp} \right) + \left(F_{OS,CG,Trop} \cdot EF_{2CG,Trop} \right) + \left(F_{OS,F,Temp,NR} \cdot EF_{2F,Temp,NR} \right) + \left(F_{OS,F,Temp,NP} \cdot EF_{2F,Temp,NP} \right) + \left(F_{OS,F,Trop} \cdot EF_{2F,Trop} \right) \right]$$

$$N_2O-N_{PRP} = \left[\left(F_{PRP,CPP} \cdot EF_{3PRP,CPP} \right) + \left(F_{PRP,SO} \cdot EF_{3PRP,SO} \right) \right]$$

Where:

N₂O-NN inputs: not included (value=0);

N₂O-NOS: not included (value=0);

N₂O-NPRP: annual direct emissions of N₂O-N from urine and manure inputs to grazed soils, kg N₂O-N/year;

FPRP,CPP: annual amount of urine deposited by animals on pasture and/or in the paddock, kg N/year;

EF3PRP,CPP: factor for N₂O emissions from urine deposited on pasture, strip and/or paddock, kg N₂O-N/kg N entered;

FPRP,SO: not included (value=0);

EF3PRP,SO: not included (value=0).

Following the IPCC Tier 1 approach, FPRP,CPP was calculated according to equation 11.5 'N in urine and dung deposited by grazing animals on pasture, range and paddock'. The fraction of total annual N excretion deposited on pasture (MS(T,PRP)) of 2% was adopted for beef cattle in accordance with the 'GHG Protocol Brazil' document for the state of Acre.

EQUATION 11.5
N IN URINE AND DUNG DEPOSITED BY GRAZING ANIMALS ON PASTURE, RANGE AND PADDOCK
(TIER 1)

$$F_{PRP} = \sum_T [N_{(T)} \bullet Nex_{(T)}] \bullet MS_{(T,PRP)}$$

Where:

FPRP, CPP: annual amount of urine deposited by animals on pasture and/or in the paddock, kg N/year;

N(T): total number of animals;

Nex(T): average annual N excretion of beef cattle, kg N/animal.year;

MS(T,PRP): fraction of total annual N excretion deposited on pasture.

To calculate the average annual N excretion of beef cattle (Nex(T)), still using the Tier 1 approach, equation 10.30 'Annual N excretion rate' from chapter 10, volume 4 of the document '2006 IPCC Guidelines for National Greenhouse Gas Inventories' was used. The standard factor for the rate of N excreted in kg N/1000 kg animal mass was taken from table 10.19 at 0.36 (Nrate(T)=0.36). In addition, the animal weight (TAM) was adjusted according to the standard animal unit of 450kg (TAM=450kg).

EQUATION 10.30
ANNUAL N EXCRETION RATES

$$Nex_{(T)} = N_{rate(T)} \bullet \frac{TAM}{1000} \bullet 365$$

Where:

Nex(T): average annual head N excretion from beef cattle, kg N/animal.year;

Nrate(T): rate of N excreted in beef cattle, kg N/1000 kg animal mass;

TAM: weight of beef cattle, adjusted to 450kg.

N₂O-N emissions were converted into N₂O emissions using the conversion equation provided in chapter 11, volume 4 of the '2006 IPCC Guidelines for National Greenhouse Gas Inventories', which has no numerical indication. In addition, the values were converted to t CO₂eq/year using the Global Warm Potential (GWP) of 273.

$$N_2O = N_2O-N \bullet 44/28$$

Where:

N₂O: is the emitted gas of interest for global warming;

N₂O-N: is the fraction of N that becomes N₂O in the atmosphere.

3.2.1.5 Baseline net GHG emissions

The net GHG emissions baseline for planned deforestation will be determined based on equation 1 of VMD0006 v.13:

$$\Delta C_{BSL,planned} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{BSL,i,t} + GHG_{BSL-E,i,t})$$

Where:

$\Delta C_{BSL,planned}$: Net GHG emissions in the baseline of planned deforestation up to year t^*

$\Delta C_{BSL,i,t}$: Net changes in the carbon stock in all reservoirs in stratum i of the baseline in year t ;

$GHG_{BSL-E,i,t}$: Greenhouse gas emissions as a result of deforestation activities within the project boundary in stratum i of the baseline in year t ;

i 1, 2, 3,...: M strata

t 1, 2, 3,...: t^* years elapsed since the projected start of project activity

Where:

$$GHG_{BSL-E,i,t} = E_{FC,i,t} + E_{BiomassBurn,i,t} + N_2O_{direct-N,i,t} + E_{livestock}$$

$E_{FC,i,t}$ - Net CO₂ equivalent emissions as a result of fossil fuel combustion within the project boundary in the single stratum of the project (i) at baseline in year t = null for REDD+ Seringal Rio Branco (disregarded)

$E_{BiomassBurn,i,t}$ - Methane and nitrous oxide emissions as a result of biomass burning within the project boundaries in the single project stratum (i) at baseline in year t = 8.61 t.CO₂eq/ha (section 3.2.1.4)

$N_2O_{direct-N,i,t}$ - Direct N₂O emissions as a result of nitrogen fertilizer application in alternative land use within the project boundaries in the single project stratum (i) at baseline in year t = null for REDD+ Seringal Rio Branco (disregarded)

$E_{livestock,i,t}$ - Net methane and nitrous oxide emissions as a result of livestock activities within the project boundaries in the single project stratum (i) at baseline in year t = 34.5 t CO₂eq/ha (section 3.2.1.5).

Table 22: Net greenhouse gas emissions in the baseline of planned deforestation.

Name> ID _{icl} >	ΔC_{BSLPA}		$GHG_{BSL-E,i,t}$		$\Delta C_{BSL,planned}$	
	annual	cumulative	annual	cumulative	annual	cumulative
Project Year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2023	261.234	261.234	5.465	5.465	266.699	266.699
2024	266.543	527.778	6.206	11.671	272.750	539.449

2025	271.852	799.630	6.948	18.619	278.801	818.250
2026	277.162	1.076.792	7.690	26.310	284.852	1.103.102
2027	282.471	1.359.262	8.432	34.742	290.903	1.394.005
2028	287.780	1.647.042	9.174	43.917	296.954	1.690.959
2029	293.089	1.940.131	9.916	53.833	303.005	1.993.964
2030	298.398	2.238.529	10.658	64.492	309.056	2.303.020
2031	303.707	2.542.235	11.400	75.892	315.107	2.618.127
2032	309.016	2.851.251	12.142	88.034	321.158	2.939.286
2033	47.782	2.899.033	7.420	95.454	55.201	2.994.487
2034	42.472	2.941.505	7.420	102.874	49.892	3.044.379
2035	37.163	2.978.669	7.420	110.294	44.583	3.088.962
2036	31.854	3.010.523	7.420	117.714	39.274	3.128.237
2037	26.545	3.037.068	7.420	125.133	33.965	3.162.202
2038	21.236	3.058.304	7.420	132.553	28.656	3.190.858
2039	15.927	3.074.232	7.420	139.973	23.347	3.214.205
2040	10.618	3.084.850	7.420	147.393	18.038	3.232.243
2041	5.309	3.090.159	7.420	154.813	12.729	3.244.971
2042	0	3.090.159	7.420	162.233	7.420	3.252.391
2043	0	3.090.159	7.420	169.652	7.420	3.259.811
2044	0	3.090.159	7.420	177.072	7.420	3.267.231
2045	0	3.090.159	7.420	184.492	7.420	3.274.651
2046	0	3.090.159	7.420	191.912	7.420	3.282.071
2047	0	3.090.159	7.420	199.332	7.420	3.289.490
2048	0	3.090.159	7.420	206.751	7.420	3.296.910
2049	0	3.090.159	7.420	214.171	7.420	3.304.330
2050	0	3.090.159	7.420	221.591	7.420	3.311.750
2051	0	3.090.159	7.420	229.011	7.420	3.319.170
2052	0	3.090.159	7.420	236.431	7.420	3.326.590

3.2.2 Project Emissions (VCS, 3.15)

In the scenario with the project, net emissions are calculated using equation 1 of the M-REDD module ($\Delta C_{WPS-REDD}$),

$$\Delta C_{WPS-REDD} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{P,DefPA,i,t} + \Delta C_{P,Deg,i,t} + \Delta C_{P,DistPA,i,t} + GHG_{P-E,i,t} - \Delta C_{P,Enh,i,t})$$

Equation 1 of the M-REDD

module

To achieve this goal, the first action is to verify land use change in the project area, using the monitoring method presented in section 3.3. Calculating these emissions requires monitoring the emissions resulting from different categories of disturbance, referred to as the “q” stratum in the M-REDD module. These are:

- 1) deforestation occurring in the project area ($\Delta CP, DefPA, i, t$);
- 2) degradation occurring in the project area ($\Delta CP, Deg, i, t$);
- 3) the natural disturbances occurring in the project area ($\Delta CP, DistPA, i, t$);
- 4) GHG emissions as a result of deforestation and degradation activities within the project area ($GHGP-E, i, t$);
- 5) of forest growth and sequestration during the project in areas projected to be deforested in the baseline ($\Delta CP, Enh, i, t$).

This last value ($\Delta CP, Enh, i, t$) can be conservatively set to zero, according to the M-REDD module. The other values will be broken down below for the entire project area, following the guidelines of the M-REDD module, in addition to the particularities for the calculations of the areas of planned deforestation, following the guidelines of the BL-PL module.

3.2.2.1 Change in Carbon Stock in all reservoirs in the Project Scenario

Monitoring of deforestation occurring in the project area ($\Delta CP, DefPA, i, t$)

The net change in carbon stock as a result of deforestation is equal to the deforested area multiplied by the emission per unit area; this formula is presented in equation 3 of the M-REDD module, as shown below:

$$\Delta C_{P, DefPA, i, t} = \sum_{u=1}^U (A_{DefPA, u, i, t} * \Delta C_{pools, P, Def, u, i, t})$$

Equation 3 of the M-REDD module

For this reason, it is important to analyze the land use change maps in order to define the ($A_{DefPA, u, i, t}$), which is the area of deforestation recorded in stratum i converted to land use u in the year, within the project area. In addition, the net changes in the carbon stock in all the reservoirs in the case of the land use project ($\Delta C_{pools, Def, u, i, t}$) are also required, which can be calculated using equation 5 of the M-REDD module, as shown below:

$$\Delta C_{pools, Def, i, t} = C_{BSL, i} - C_{P, post, i} - C_{WP, i}$$

Equation 5 of the M-REDD module

This equation takes into account the carbon stock (in CO₂e ha⁻¹) in all the reservoirs found in the baseline ($C_{BSL, i}$), originating from the forest inventory, the post-deforestation stock ($C_{P, post, i}$), considering the type of land use “u” (which is the type of use to which the area would be converted if it were deforested) and the carbon stock sequestered in wood products from logging ($C_{WP, i}$).

The remaining carbon stock after deforestation (CP, post, i) can be calculated by applying the standard values adopted by the Ministry of the Environment (MMA) and presented in the Brazilian Fourth Communication (Table 23, page 159). Table 17 shows the three most likely types of use to be installed in the Amazon. The values for pastures can be considered for two reasons: (1) extensive pastures for cattle breeding are the most likely type of use and (2) the BL-UP module (page 31) advises that when faced with different land use possibilities, you can choose the one with the highest carbon stock, which is considered conservative

Monitoring the degradation occurring in the project area ($\Delta CP, Deg, i, t$)

In general, methods using remote sensing technology are not able to make direct measurements of the degradation changes that may occur in the area, but they do have some ability to identify forest strata that have undergone a change in biomass, and this helps with monitoring. The calculation will be carried out using equation 7 of the M-REDD module, as shown below:

$$\Delta C_{P, Deg, i, t} = \Delta C_{P, DegW, i, t} + \Delta C_{P, SelLog, i, t}$$

Equation 7 of the M-REDD module

Equation 7 of the M-REDD module considers two types of change:

- 1) ($\Delta CP, DegW, i, t$) – the net change in carbon stock as a result of degradation through illegal logging of timber or fuelwood and charcoal in the project area in stratum i in the year; Mg CO2-e
- 2) ($\Delta CP, SelLog, i, t$) – the net change in carbon stock as a result of degradation originating from selective logging in FSC-certified forest management areas, considering stratum i in year t; ; Mg CO2-e. As there is no such possibility in the area, this value can be considered null.

To calculate $\Delta CP, DegW, i, t$ the first step is to determine whether there is potential for illegal activities related to the extraction of timber, firewood and charcoal. According to reports from the area, these activities do not seem to be significant in the Seringal Rio Branco REDD+ Project area. Therefore, this change can be assumed to be zero.

Ideally, throughout the existence of the project, these activities should be monitored every two years, using good remote sensing images, such as those from DETER, combined with field verification. Participatory Rural Diagnosis (PRA) can also be used, carried out every 2 or 3 years with the area's residents. If degradation is detected, it is necessary to identify the area that is potentially subject to degradation and monitor carbon changes using equation 8 of the M-REDD module.

$$\Delta C_{P, DegW, i, t} = A_{DegW, i} * \frac{C_{DegW, i, t}}{AP_i}$$

Equation 8 of the M-REDD module

Monitoring natural disturbances

Emissions from forest degradation caused by natural disturbances must be estimated. Natural disturbances arise from tectonic activities (such as earthquakes, landslides, volcanoes), extreme weather conditions (such as hurricanes), droughts or forest fires. These emissions can be omitted if they are considered insignificant using the T-SIG Tool. A risk assessment can also be carried out.

The net change in carbon stock as a result of natural disturbance is equal to the disturbed area multiplied by the emission per unit area.

When the disturbance event occurs ex post in the project area, the disturbed area should be defined and mapped as in ARRL,forest,t and ARRD,unplanned,hrp monitoring.

For planned deforestation, the sum of ADistPA,q,i,t will be equal to the area of overlap between the disturbed area and the sum of the planned deforestation area in the project area (D%planned,i,t * Aplanned,i), summed in the year in which the disturbance occurred.

The Net change in carbon stock as a result of natural disturbance in the project area in stratum i in the year ($\Delta C_{P,DistPA,i,t}$) is calculated using equation 20 of the same module, which considers all disturbance classes (or strata):

$$\Delta C_{P,DistPA,i,t} = \sum_{q=1}^Q (A_{DistPA,q,i,t} * \Delta C_{P,Dist,q,i,t})$$

Equation 20 of the M-REDD module

This equation 20 requires the area impacted by natural disturbance in each disturbance class or stratum ("q") ($A_{DistPA,q,i,t}$) and the net change in carbon stock as a result of these disturbance classes ($\Delta C_{P,Dist,q,i,t}$). Where fire occurs, $A_{DistPA,q,i,t}$ is similar to the burned area $A_{burnPA,q,i,t}$ and no other disturbance should be considered in that burned area.

These total changes resulting from each disturbance class ($\Delta C_{P,Dist,q,i,t}$) is calculated by equation 23 of the M-REDD module:

$$\Delta C_{P,Dist,q,i,t} = C_{BSL,i} - C_{P,Dist,q,i} - C_{WP,q,i}$$

Equation 23 of the M-REDD module

The CBSL i value corresponds to the carbon stock (in CO₂e ha⁻¹) in all the reservoirs found in the baseline, originating from the forest inventory. It is reduced by the values of CP, Dist, q, i and CWP Dist, q, i, which represent the stocks after the disturbance, CWP for timber products, which can be considered null as shown above, and CP, Dist for the carbon stock in the different reservoirs accounted for in the baseline (equation 24 of the M-REDD module can be used to calculate this). If there is burning, the amount of carbon stock after burning can be considered zero.

3.2.2.2 Non-CO₂ GHG emissions from biomass burning in the project scenario

As specified in VM0007 and module VMD0013, when emissions from biomass burning are accounted for in the baseline, accounting must also take place under the project scenario, both in the project area and in the leakage belt.

To enable quantification, the project will monitor the occurrence of fires in the project area and in the leakage belt using remote sensing. In this monitoring, hotspots will be detected using satellites with polar and geostationary orbits, which have different radiometric and temporal resolutions. The details of this monitoring method are presented in section 3.3.

If any heat sources are detected and validated within the project area, the emissions will be quantified by applying the VMD0013 module according to the equation below. Caso seja detectado e validado algum foco de calor dentro da área do projeto, as emissões serão quantificadas aplicando o módulo VMD0013 conforme a equação abaixo.

$$E_{biomassburn,i,t} = \sum_{g=1}^G \left((A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i}) \times 10^{-3} \right) \times GWP_g \quad (1)$$

Where:

$E_{biomassburn,i,t}$	= Greenhouse gas emissions due to biomass burning in stratum i in year t of each GHG (CO ₂ , CH ₄ , N ₂ O) (t CO ₂ e)
$A_{burn,i,t}$	= Area burnt for stratum i in year t (ha)
$B_{i,t}$	= Average aboveground biomass stock before burning stratum i , year t (d.m. ha ⁻¹)
$COMF_i$	= Combustion factor for stratum i (unitless)
$G_{g,i}$	= Emission factor for stratum i for gas g (kg t ⁻¹ d.m. burnt)
GWP_g	= Global warming potential for gas g (t CO ₂ /t gas g)
g	= 1, 2, 3 ... G greenhouse gases including carbon dioxide ¹ , methane and nitrous oxide (unitless)
i	= 1, 2, 3, ... M strata (unitless)
t	= 1, 2, 3, ... t^* time elapsed since the start of the project activity (years)

The following data and parameters will be used to calculate the project's emissions:

$A_{burn,i,t}$ – burnt area identified through project monitoring

$B_{i,t}$ – stock presented in section 3.2.1.1 = 498.64 t.CO₂eq/há

$COMF_i$ – biomass combustion factor in the single project stratum from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 0.57

$G_{CO_2,i}$ – carbon dioxide emission factor from biomass burning from 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 85.26 t.CO₂eq/ha

$G_{CH_4,i}$ – methane emission factor from biomass burning from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 7.81 t.CO₂eq/ha

GN_{20,i} – nitrous oxide emission factor from biomass burning from 2006 IPCC Guidelines for National Greenhouse Gas Inventories - Vol. 4 = 0.79 t.CO₂eq/ha

3.2.2.3 Non-CO₂ GHG emissions from cattle ranching in the Project scenario.

No emissions from cattle ranching are expected in the project scenario, since cattle ranching will not legally take place in the REDD+ APD project area.

3.2.3 Leakage Emissions (VCS 2.5, 3.2, 3.6, 3.15, 4.3)

To ensure efficient monitoring of leakage areas, it is essential to identify rural properties that belong to the same owner as the property included in the project area. To this end, we conducted a detailed analysis to identify other rural properties under the same ownership. This investigation resulted in the delimitation of two additional rural properties, both under the ownership of the same individual who owns the main project property.

To delimit the boundaries of the properties in the leakage areas, we used the database provided by the owner and information from the Rural Environmental Registry (CAR). Unlike the methodology used to define the project area, which requires forests to be stable for at least ten years before the start of the project and to remain so until the project start date, in the leakage areas it is necessary to ensure that all regions classified as forests are monitored at the time of the project start. This specific approach is adopted because it is a project requirement for effective leakage monitoring, ensuring that any transfer of deforestation impact outside the project area is properly tracked and documented.

The process for defining the leakage areas followed a careful methodology: initially, we identified the forest areas using the land use and occupation map from Mabiomas collection 8. We then proceeded to exclude areas that overlapped with Legal Reserves (LR) and Permanent Preservation Areas (PPA), to avoid overlapping with legally protected zones. In addition, we eliminated fragments of forest areas that were less than one hectare in size.

Of the 3 properties owned by the same landowner and which are not Seringal Rio Branco, leakage areas LK 1 and LK 2 are located in the municipality of Guajará, in the state of Amazonas, while area LK 3 is mainly in Cruzeiro do Sul, with a smaller part in Porto Walter, in the state of Acre. The LK3 property is being developed by the same proponent, Amazônia Agroindústria EIRELI, for the Juruá REDD+ Project, which is currently being validated and verified.

On the “LK 1” property, the total area is 1167 hectares, of which 84 hectares of forest have been identified for monitoring. In contrast, on the “LK 2” property, with a total area of 1947 hectares, no forest areas have been identified for leakage monitoring, as all the areas that could be legally cleared have already been converted, eliminating the possibility of monitoring remaining forest regions. On the other hand, the “LK 3” property covers a vast area of 25083 hectares, with 3845 hectares of forest that will be monitored due to the Juruá REDD+ Project.

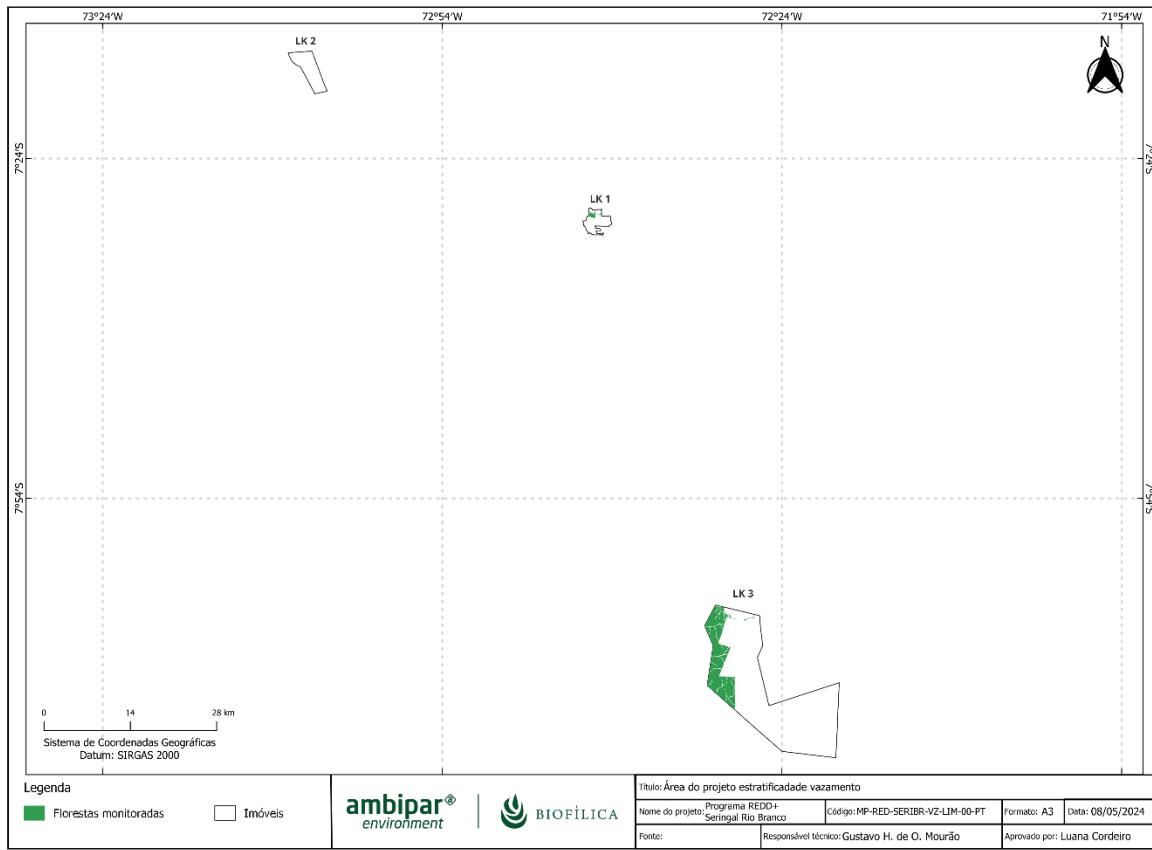


Figure 35: boundaries of the properties in the leakage areas.

Table 23: Properties included in the Leakage Belt area of the Seringal Rio Branco REDD+ Program.

ID	Property area	Monitoring area
LK 1	1167	84
LK 2	1947	0
LK 3	25083	3845
Total	28197	3929

Once these limits have been defined, the leakage emissions due to the change in activity of forest land that is legally authorized and documented to be converted into non-forest land can be estimated according to the criteria of VMD0009 v1.3.

However, since the deforestation agent has been identified for the Seringal Rio Branco REDD+ Project and, according to the survey carried out with the owners, the management plans of the other properties have not undergone significant changes as a result of the project, the leakage will not be considered in the baseline at this time, the leakage will be recalculated if the owner acquires new areas. Further details on monitoring the leakage are described in the Monitoring Plan in section 3.3.3.

3.2.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals (VCS, 3.15, 4.1)

Total net reduction in GHG emissions

The total net reductions in greenhouse gas emissions from the Seringal Rio Branco REDD+ Project activity are calculated based on equation 2 of VM0007:

$$NER_{REDD} = \Delta C_{BSL-REDD} - \Delta C_{WPS-REDD} - \Delta C_{LK-REDD}$$

Where:

NER_{REDD} : Total net GHG emission reductions from the REDD project activity up to year t^*

$C_{BSL-REDD}$: Net GHG emissions in the REDD baseline scenario up to year t^*

$C_{WPS-REDD}$: Net GHG emissions in the REDD project scenario up to year t^* - from Module M-REDD

$C_{LK-REDD}$: Net GHG emissions due to leakage from REDD project activity up to year t^*

Calculation of the AFOLU buffer account contribution

Following the guidelines indicated in VM0007 v1.6, for this project the buffer is equal to the net GHG emissions in the baseline minus the net emissions in the project case. Leakage emissions are not taken into account in the buffer calculations, and non-CO2 emissions from fossil fuels and fertilizer use have been conservatively excluded from the project scope.

To arrive at the buffer percentage, risk analysis was applied to the project. The risk analysis for the Seringal Rio Branco REDD+ Project was carried out using the AFOLU Non-Permanence Risk Tool, v4.2, resulting in a buffer of 12%, as indicated in the following table.

State the non-permanence risk rating (%)	12%
Has the non-permanence risk report been attached as either an appendix or a separate document?	☒ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO2e, the Long-term Average (LTA).	Not applicable
Has the LTA been updated based on monitored data, if applicable?	Not applicable
State, in tCO2e, the expected total GHG benefit to date.	

Is the number of GHG credits issued below the LTA?

Not applicable

Calculation of verified carbon units (VCUs)

To estimate the number of Verified Carbon Units (VCUs) generated from the implementation of the planned project activity (ex-ante), the following equation was applied:

$$VCU_t = Adjusted_NER_{REDD+} - Buffer_{Planned}$$

Where:

VCU_t: Number of verified carbon units in year t = t2 - t1 (VCU)

Adjusted_NER_{REDD+}: Total net GHG emission reductions from the REDD+ project activity adjusted for uncertainty (t CO₂e)

Buffer_{Total}: Total retention of the permanence risk buffer (t CO₂e)

Table 25: Estimated GHG Emission Reductions and Carbon Dioxide Removal

Vintage	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer allocation (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)
15-March-23 a 31-December-23	266.699	0	0	32.004	234.695
1-January-2024 a 31-December-2024	272.750	0	0	32.730	240.020
1-January-2025 a 31-December-2025	278.801	0	0	33.456	245.345
1-January-2026 a 31-December-2026	284.852	0	0	34.182	250.670
1-January-2027 a 31-	290.903	0	0	34.908	255.995

Vintage	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer allocation (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)
December-2027					
1-January-2028 a 31-December-2028	296.954	0	0	35.634	261.320
1-January-2029 a 31-December-2029	303.005	0	0	36.361	266.644
1-January-2030 a 31-December-2030	309.056	0	0	37.087	271.969
1-January-2031 a 31-December-2031	315.107	0	0	37.813	277.294
1-January-2032 a 31-December-2032	321.158	0	0	38.539	282.619
1-January-2033 a 31-December-2033	55.201	0	0	6.624	48.577
1-January-2034 a 31-December-2034	49.892	0	0	5.987	43.905
1-January-2035 a 31-December-2035	44.583	0	0	5.350	39.233

Vintage	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer allocation (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)
1-January-2036 a 31-December-2036	39.274	0	0	4.713	34.561
1-January-2037 a 31-December-2037	33.965	0	0	4.076	29.889
1-January-2038 a 31-December-2038	28.656	0	0	3.439	25.217
1-January-2039 a 31-December-2039	23.347	0	0	2.802	20.545
1-January-2038 a 31-December-2038	18.038	0	0	2.165	15.873
1-January-2040 a 31-December-2040	12.729	0	0	1.527	11.201
1-January-2041 a 31-December-2041	7.420	0	0	890	6.529
1-January-2042 a 31-December-2042	7.420	0	0	890	6.529
1-January-2043 a 31-	7.420	0	0	890	6.529

Vintage	Estimated baseline emissions (tCO2e)	Estimated project emissions (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated buffer allocation (tCO2e)	Estimated reduction VCU's (tCO2e)
December-2043					
1-January-2044 a 31-December-2044	7.420	0	0	890	6.529
1-January-2045 a 31-December-2045	7.420	0	0	890	6.529
1-January-2046 a 31-December-2046	7.420	0	0	890	6.529
1-January-2047 a 31-December-2047	7.420	0	0	890	6.529
1-January-2048 a 31-December-2048	7.420	0	0	890	6.529
1-January-2049 a 31-December-2049	7.420	0	0	890	6.529
1-Janeiro-2050 a 31-Dezembro-2050	7.420	0	0	890	6.529
1-Janury-2051 a 31-December-2051	7.420	0	0	890	6.529

Vintage	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer allocation (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)
1-January-2052 a 14-March-2052	7.420	0	0	890	6.529
Total	3.326.590	0	0	399.191	3.162.482

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation (VCS, 3.16)

Data / parameter	$\Delta C_{BSL,planned}$
Data unit	tCO ₂ e ha-1
Description	Net gas emissions in the baseline from planned deforestation up to year t. In other words, the Change in the Total Carbon Stock of the reservoirs evaluated, after planned deforestation.
Data source	Described in section 3.2.1
Value applied	Total: 3,326,590
Justification for the choice of data or description of the measurement methods and procedures applied	Required by the BL-PL, LK-ASP and M-REDD modules
Purpose of the data	Calculate the differences between before and after planned deforestation in the Project Area and the Leakage Belt.
Comments	n/a
Data / parameter	$A_{planned,i}$

Data unit	Ha
Description	Total area of planned deforestation during the fixed baseline period for stratum i
Data source	Described in section 3.1.4
Value applied	Total: 5,487
Justification for the choice of data or description of the measurement methods and procedures applied	Required by BL-PL module
Purpose of the data	Used in equation 5 of the BL-PL module
Comments	n/a

Data / parameter	$L-D_i$
Data unit	%
Description	Probability of deforestation in stratum i
Data source	Described in section 3.1.4
Value applied	100%
Justification for the choice of data or description of the measurement methods and procedures applied	Required by the BL-PL module
Purpose of the data	Used in equation 5 of the BL-PL module
Comments	n/a

Data / parameter	$C_{AB,tree}$
Data unit	$MgCO_2e\ ha^{-1}$
Description	Average carbon stock per hectare in the tree reservoir (DBH > 10 cm) in the project area.

Data source	Calculated using an allometric equation (Nogueira et al. 2008 and Brown 1997) on data measured in the field.
Value applied	435,44
Justification for the choice of data or description of the measurement methods and procedures applied	Above-ground biomass estimates were made using forest inventory data and allometric equations. Value required in the CP-AB module
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	$C_{BB_tree, i}$
Data unit	MgCO ₂ e ha ⁻¹
Description	Average carbon stock per hectare in the below-ground biomass carbon pool (trees and palms with DBH > 10 cm) in the Project area
Data source	Calculado por razão root:shoot
Value applied	110,71
Justification for the choice of data or description of the measurement methods and procedures applied	Estimates of below-ground biomass were made using forest inventory data and the root:shoot ratio (=0.22). Value required in the CP-AB module
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	$C_{AB, non-tree}$
Data unit	MgCO ₂ e ha ⁻¹

Description	Average carbon stock per hectare in the palm reservoir (DBH > 10 cm), in the project area
Data source	Calculated by allometric equation (Goodman et al. 2013) on data measured in the field
Value applied	25,84
Justification for the choice of data or description of the measurement methods and procedures applied	Above-ground biomass estimates were made using forest inventory data and allometric equations. Value required in the CP-AB module
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	$C_{DW, i}$
Data unit	MgCO ₂ e ha ⁻¹
Description	Average carbon stock per hectare in the deadwood carbon pool in the Project area
Data source	Calculated by dead wood:shoot ratio
Value applied	37,36
Justification for the choice of data or description of the measurement methods and procedures applied	Below-ground biomass estimates were made using forest inventory data and the dead wood:shoot ratio (=0.081). Value required in CP-D module
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	CF (conversion factor)
Data unit	Mg C

Description	Conversion factor from biomass (dry weight) to carbon
Data source	Standard value suggested by the CP-AB module
Value applied	0,47
Justification for the choice of data or description of the measurement methods and procedures applied	Value obtained for the forests of southern Amazonia Value required in the CP-AB module
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	44/12
Data unit	Mg CO2e
Description	Carbon mass conversion factor for CO2e
Data source	Scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU
Value applied	44/12
Justification for the choice of data or description of the measurement methods and procedures applied	IPCC default value. Value required in CP-AB module
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	Allometric Equation = $(\exp(-1.716 + (2,413 * \ln(DAP))))$
Data unit	Mg ha ⁻¹

Description	Equation for converting DBH to above-ground dry mass. This equation was applied to trees with DBH < 95.0cm.
Data source	Equation from Nogueira et al. (2008): Estimates of forest biomass in the Brazilian Amazon: New allometric equations and biomass adjustments of wood volume inventories. Forest Ecology and Management, v. 256, n. 11, p. 1853-1867, 2008.
Value applied	See field data table (Excel spreadsheet supplementary to this report)
Justification for the choice of data or description of the measurement methods and procedures applied	Equation developed for open forests in southern Amazonia. Equation required in the CP-AB module
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	Allometric Equation = $42,69 + (-12,8 * DAP) + (1,242 * DAP^2)$
Data unit	Mg ha ⁻¹
Description	Equation for converting DBH to above-ground dry mass. This equation was applied to trees with DBH > 95.0cm.
Data source	Equation from Brown S. (1997): Estimating biomass and biomass change of tropical forests: a primer. United Nations Food and Agriculture Organization (FAO), Rome. Forestry Paper, 134. Disponível em: https://www.fao.org/3/W4095E/w4095e00.htm#Contents Available at:19/01/2023
Value applied	See field data table (Excel spreadsheet supplementary to this report)
Justification for the choice of data or description of the measurement methods and procedures applied	Equation developed for Amazonian forests. Equation required in the CP-AB module

Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	Allometric equation $_{\text{palm}} = \exp(-3.3488 + (2.7483 \cdot \ln(\text{DAP})))$
Data unit	Mg ha ⁻¹
Description	Equation for converting DBH to above-ground dry mass. This equation was applied to palms with DBH > 10.0cm (Non-arboreal component).
Data source	Equation from Goodman et al (2013: Amazon palm biomass and allometry. Forest Ecology and Management, 310, 994-1004.
Value applied	See field data table (Excel spreadsheet supplementary to this report)
Justification for the choice of data or description of the measurement methods and procedures applied	Equation developed for Amazonian forest palms. Equation required in the CP-AB module
Purpose of the data	Calculation of baseline emissions
Comments	E value (stress factor) calculated via Table provided by Chave et al. (2014) See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project

Data / parameter	Root:shoot
Data unit	N/A
Description	Root:shoot ratio.
Data source	Average value shown for humid tropical forests with biomass > 125 Mg ha ⁻¹ , in Table 4.4. in IPCC GL AFOLU - The same Table is also shown in the CP-AB module

Value applied	0,24
Justification for the choice of data or description of the measurement methods and procedures applied	There is a lack of equations for open ombrophilous forests in southern Amazonia. Value required in the CP-AB module; which also guides the use of the IPCC GL AFOLU Table in the absence of equations.
Purpose of the data	Calculation of baseline emissions
Comments	Below-ground biomass was calculated for each plot and above-ground biomass is the sum of tree+non-tree, i.e. trees + palms. See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project.

Data / parameter	Dead wood ratio:shoot
Data unit	N/A
Description	Ratio of dead wood to live shoot, appropriate to the open ombrophilous forest type; note that, as defined here, the root to shoot ratio is applied as plot biomass.
Data source	Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change - Reference Report: Land Use Sector, Land Use Change and Forests, 2020.
Value applied	0,081
Justification for the choice of data or description of the measurement methods and procedures applied	Value accepted for VCS projects; Value required in the CP-D module.
Purpose of the data	Calculation of baseline emissions
Comments	See document: Carbon Stock Inventory of the Seringal Rio Branco REDD+ Project.

Data / parameter	<i>COMFi</i>
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Data unit	(dimensionless)
Description	Combustion factor for stratum i
Data source	IPCC (2006)
Value applied	0,5
Justification for the choice of data or description of the measurement methods and procedures applied	Value required in the E-BPB module Standard value for tropical rainforest given in Table 2.6 of IPCC (2006) - Volume 4, Chapter 2, Appendix 2). The combustion factor is a measure of the proportion of the fuel that is actually burned, which varies depending on the size and architecture of the fuel load (i.e. a smaller proportion of large, coarse fuels such as tree trunks will be burned compared to fine fuels such as grass leaves), the moisture content of the fuel and the type of fire (i.e. intensity and rate of spread).
Purpose of the data	Calculation of baseline emissions Project emissions
Comments	The default value should be updated whenever new guidelines are produced by the IPCC.

Data / parameter	Gg,i
Data unit	g kg ⁻¹ dry matter burned
Description	Emission factor for stratum i and gas g
Data source	IPCC (2006)
Value applied	Gas CH₄=6,8 Gas N₂O= 0,2
Justification for the choice of data or description of the measurement methods and procedures applied	Value required in the E-BPB module Default values in Table 2.5 of IPCC (2006) - Volume 4, Chapter 2, Appendix 2). Values considered conservative by the IPCC

Purpose of the data	Calculation of baseline emissions Project emissions
Comments	The default value should be updated whenever new guidelines are produced by the IPCC.

Data / parameter	<i>GWP g</i>
Data unit	t CO ₂ /t gas g
Description	Warming potential for gas g
Data source	Default values in Table 2.5 of IPCC (2007) - Fourth Assessment Report: Climate Change 2007.
Value applied	Gwp CH ₄ =25 Gwp N ₂ O= 298 Gwp CO ₂ = 1
Justification for the choice of data or description of the measurement methods and procedures applied	Value required in the E-BPB module Values considered conservative by the IPCC
Purpose of the data	Calculation of emissions from fires
Comments	The default value must be updated whenever new guidelines are produced by the IPCC.

Data / parameter	C _{POST} , planned,
Data unit	MgCO ₂ e ha ⁻¹
Description	Total carbon stock of the reservoirs evaluated, after planned deforestation
Data source	Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate

	Change - Reference Report: Land Use Sector, Land Use Change and Forests, 2020".
Value applied	Pasture: 36.67 Agriculture and pasture mosaic: 27.76 Soybean plantations: 18.85
Justification for the choice of data or description of the measurement methods and procedures applied	Required by BL-PL and M-REDD modules
Purpose of the data	To calculate the differences between before and after planned deforestation
Comments	

Data / parameter	A _{sp}
Data unit	Hectare (ha)
Description	Area and number of sample plots
Data source	Inventory
Value applied	26 plots of 1ha each
Justification for the choice of data or description of the measurement methods and procedures applied	
Purpose of the data	Indication of the sample size of the Project's stock.
Comments	During monitoring, this should be used in equations 2, 6 and 14 of the CP-AB module.

Data / parameter	A _i
Data unit	ha

Description	Area of forest stratum i (as the project called for a single stratum, this is the project area itself)
Data source	Geographical coordinates and SR maps
Value applied	5487
Justification for the choice of data or description of the measurement methods and procedures applied	Required by the M-REDD module
Purpose of the data	Calculation of baseline emissions
Comments	Must be monitored in baseline revisions

Data / parameter	$N_{rate(T)}$
Data unit	Kg N/1000 kg animal mass.day
Description	Standard value for the rate of N excreted
Data source	2006 IPCC Guidelines for National Greenhouse Gas Inventories (Chapter 10, volume 4).
Value applied	0.36
Justification for the choice of data or description of the measurement methods and procedures applied	Required for calculating the average excretion of N by beef cattle under a grazing system.
Purpose of the data	Used in equation 10.30 'Annual N excretion rates'
Comments	n/a

Data / parameter	TAM
Data unit	Kg
Description	Standard value for an animal unit in Brazil

Data source	-
Value applied	450kg
Justification for the choice of data or description of the measurement methods and procedures applied	Required for calculating average N excretion by beef cattle under a pasture system.
Purpose of the data	Used in equation 10.30 'Annual N excretion rates'
Comments	n/a

Data / parameter	$N_{(T)}$
Data unit	Total number of animals in the area increased per year
Description	Multiplication between average Brazilian stocking rate of 0.97 A.U./ha (IBGE, 2024) and area with potential for deforestation from the 2024 Exploitation Plan of 548.71ha
Data source	2006 IPCC Guidelines for National Greenhouse Gas Inventories (Chapter 11, volume 4).
Value applied	532 U.A./year
Justification for the choice of data or description of the measurement methods and procedures applied	Required for calculating the annual amount of urine deposited by beef cattle under a grazing system.
Purpose of the data	Used in equation 11.5 'N in urine and dung deposited by grazing animals on pasture, range and paddock'
Comments	n/a

Data / parameter	$MS_{(T, PRP)}$
Data unit	%
Description	Fraction of total annual N excretion deposited on pasture.
Data source	GHG Protocol methodology for agriculture (WRI, Brazil).

Value applied	2%
Justification for the choice of data or description of the measurement methods and procedures applied	Required for calculating the annual amount of urine deposited by beef cattle under a grazing system.
Purpose of the data	Used in equation 11.5 'N in urine and dung deposited by grazing animals on pasture, range and paddock'
Comments	n/a

Data / parameter	EF _{3PRP, CPP}
Data unit	kg N ₂ O-N/kg N inserted
Description	Factor for N ₂ O emissions from urine deposited on pasture, strip and/or paddock by beef cattle
Data source	2006 IPCC Guidelines for National Greenhouse Gas Inventories (Chapter 11, volume 4).
Value applied	0.02
Justification for the choice of data or description of the measurement methods and procedures applied	Required for calculating annual direct emissions of N ₂ O-N from urine and manure inputs to grazed soils
Purpose of the data	Used in equation 11.1 'Default emission factors to estimate direct N ₂ O emissions from managed soils'
Comments	n/a

Data / parameter	Manure methane emission factor
Data unit	kg CH ₄ / head.year
Description	Methane emission factor of beef cattle manure deposited on pasture.

Data source	2006 IPCC Guidelines for National Greenhouse Gas Inventories (Chapter 10, volume 4).
Value applied	1
Justification for the choice of data or description of the measurement methods and procedures applied	Required for calculating methane emissions from livestock in the project
Purpose of the data	Used to calculate methane directly. Specifically found in: 'Table 10.14 - Manure management methane emission factors by temperature for cattle, swine, and buffalo'
Comments	n/a

Data / parameter	EF _(T)
Data unit	kg CH ₄ /cabeça.ano
Description	Emission factor of methane produced by enteric fermentation of beef cattle under pasture production system
Data source	2006 IPCC Guidelines for National Greenhouse Gas Inventories (Chapter 10, volume 4).
Value applied	56
Justification for the choice of data or description of the measurement methods and procedures applied	Required for calculating methane emissions from livestock in the project
Purpose of the data	Required for equation 10.19 - Enteric Fermentation emissions from livestock category
Comments	n/a

3.3.2 Data and Parameters Monitored (VCS, 3.16)

Data / Parameter	A _{DefPA,i,u,t}
Data unit	ha
Description	Area of registered deforestation in the project area in stratum i converted to land use in year t
Data source	Calculated using remote sensing and data available from reliable sources
Description of measurement methods and procedures to be applied	Monitoring of the forest component through remote sensing using satellite images and data from reliable sources.
Frequency of monitoring/recording	Annual
Value applied	To be defined after monitoring
Monitoring equipment	Geotechnologies: remote sensing and geographic information systems
QA/QC procedures to be applied	In mapping changes in forest cover and defining land use classes, data obtained at medium spatial resolution (between 10m and 100m) will be used. Subsequently, for validation and refinement of the described mapping, data obtained by high-resolution sensors (up to 5m pixels) will be used. The minimum accuracy of the land use classification map and the land cover classification map is 80%.
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	N/A
Data / Parameter	Δ CWPS-REDD
Data unit	MgCO ₂ e

Description	Net GHG emissions in the REDD project scenario up to year t*
Data source	Required by VM0007 and M-REDD
Description of measurement methods and procedures to be applied	Net total GHG emissions in the project area
Frequency of monitoring/recording	Monitoring should occur at least every 10 years for baseline renewal.
Value applied	To be developed during the monitoring period
Monitoring equipment	To be developed during the monitoring period
QA/QC procedures to be applied	To be developed during the monitoring period
Purpose of data	Calculation of project emissions
Calculation method	Calculated based on VMD0015
Comments	N/A

Data / Parameter	Δ CLK-AS,planned
Data unit	tCO ₂ e
Description	Net greenhouse gas emissions due to leakage caused by the displacement of activities from projects that avoid planned deforestation
Data source	Module VMD0009 (LK-ASP)
Description of measurement methods and procedures to be applied	To be developed during the monitoring period

Frequency of monitoring/recording	Annual
Value applied	N/A
Monitoring equipment	To be developed during the monitoring period
QA/QC procedures to be applied	To be developed during the monitoring period
Purpose of data	Calculation of leakage
Calculation method	Calculated based on VMD0009 (LK-ASP)
Comments	N/A

Data / Parameter	Leakage Belt Forest Cover Benchmark Map
Data of unit	-
Description	Map showing the location of forested areas within the leakage belt at the beginning of the project's monitoring period.
Data source	Satellite images and GIS
Value applied	-
Purpose of data	Required by the M-REDD module
Calculation method	At least every 10 years before baseline renewal
Comments	The minimum accuracy of the map must be 90% for the classification of forest/non-forest in the remote sensing images. If the accuracy is below 90%, then the map is not acceptable for further analysis. More remote sensing data and field visits will be necessary to produce a product that achieves the 90% minimum mapping accuracy.

Data / Parameter	Aplanned, I, OP
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Data of unit	ha
Description	Total area of planned deforestation over the entire project lifetime for stratum i
Data source	Calculated using maps with GPS coordinates or remote sensing data
Value applied	0
Purpose of data	Required by the LK-ASP module
Calculation method	Used in equation 4 of the LK-ASP module
Comments	-

Data / Parameter	ARRL, forest, t
Data unit	ha
Description	Total area remaining forested in the reference region at time t
Data source	GPS coordinates or remote sensing data. Required by the M-REDD module
Description of measurement methods and procedures to be applied	To be developed during the monitoring period
Frequency of monitoring/recording	Must be monitored at least every 5 years or, if verification occurs less frequently than every 5 years, monitoring must be conducted before each verification event
Value applied	Not applied
Monitoring equipment	To be developed during the monitoring period
QA/QC procedures to be applied	

	To be developed during the monitoring period
Purpose of data	Calculation of baseline emissions
Calculation method	Not specified
Comments	N/A

Data / Parameter	Aplanned, i
Data unit	ha
Description	Total area of planned deforestation over the entire project lifetime for stratum i
Data source	Required by BL-PL and M-REDD modules
Description of measurement methods and procedures to be applied	Calculated using maps with GPS coordinates or remote sensing data. Ex ante, Aplanned, i must be determined as described in the BL-PL module
Frequency of monitoring/recording	Must be monitored at least every 5 years or, if verification occurs less frequently than every 5 years, monitoring must be conducted before each verification event
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	To be developed during the monitoring period
Purpose of data	Calculation of leakage emissions
Calculation method	N/A

Comments	N/A
Data / Parameter	$A_{DegW, i}$
Data unit	ha
Description	Area potentially impacted by the degradation process in stratum i
Data source	Ground verification and GIS. A PRA (Participatory Rural Appraisal) can be used. Module M-REDD
Description of measurement methods and procedures to be applied	This area can be measured by a buffer around all access points (access buffer), such as roads and rivers, or other previously cleared areas. The width of the buffer can be defined by the PRA
Frequency of monitoring/recording	Monitoring should occur whenever the probable existence of degradation is detected
Value applied	-
Monitoring equipment	To be developed during the monitoring period
QA/QC procedures to be applied	-
Purpose of data	Baseline calculations and project emissions
Calculation method	-
Comments	Ex ante, a quick survey can be conducted to determine the probable existence of degradation
Data / Parameter	AP_i
Data unit	ha
Description	Total area of sample plots of degradation in stratum i

Data source	Field verification and GIS or can be 3% of the buffer zone of AdegW, iExigido pelo módulo M-REDD
Description of measurement methods and procedures to be applied	The sampling plan should be designed using plots allocated along the buffer zone, sampling at least 3% of the buffer zone area.
Frequency of monitoring/recording	Must be monitored at least every 5 years or, if verification occurs less frequently than every 5 years, monitoring must be conducted before each verification event
Value applied	-
Monitoring equipment	To be developed during the monitoring period
QA/QC procedures to be applied	-
Purpose of data	Baseline calculations and project emissions
Calculation method	To be developed during the monitoring period
Comments	Ex ante, an estimate of the plot area must be made. This should be set to exactly 3% of the buffer zone Adeg, i
Data / Parameter	A _{DistPA, q, i, t}
Data unit	ha
Description	Area impacted by natural disturbances in stratum i, converted to natural disturbances in stratum q, in year t
Data source	Remote sensing images associated with field verification using coordinates (GPS)Exigido pelo módulo M-REDD
Description of measurement methods and procedures to be applied	The minimum monitoring unit must be equal to a minimum of 11 pixels if Landsat images are used or one hectare.

Frequency of monitoring/recording	Must be monitored at least every 5 years or, if verification occurs less frequently than every 5 years, monitoring must be conducted before each verification event
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Baseline calculations and project emissions
Calculation method	-
Comments	Ex ante estimates of emissions from natural disturbances should be based on the historical incidence of such events in the project region
Data / Parameter	$A_{burn, i, t}$
Data unit	ha
Description	Area impacted by natural disturbances in stratum i , in year t
Data source	Remote sensing images associated with field verification using coordinates (GPS). Module M-REDD
Description of measurement methods and procedures to be applied	The minimum monitoring unit must be equal to a minimum of 11 pixels if Landsat images are used or one hectare. This area should replace $A_{DistPA, i, t}$ in case of fire occurrence in the project area.
Frequency of monitoring/recording	Monitoring should occur with each verification made
Value applied	Calculated in the baseline and emissions of the project
Monitoring equipment	-

QA/QC procedures to be applied	To be developed during the monitoring period
Purpose of data	-
Calculation method	To be developed during the monitoring period
Comments	For ex-post calculation of emissions from wildfires in the project area
Data / Parameter	$C_{AB,tree,i}$
Data unit	t CO ₂ e ha ⁻¹
Description	Carbon stock in aboveground biomass in trees in stratum i
Data source	Required by the CP-AB module
Description of measurement methods and procedures to be applied	CP-AB module
Frequency of monitoring/recording	According to the CP-AB module
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	Developed according to the CP-AB module
Purpose of data	Calculation of baseline and project emissions
Calculation method	To be developed during the monitoring period

Comments	To be determined ex ante when accounting of GHGbiomassburn,i,t is included in the baseline scenario. To be monitored when accounting of GHGbiomassburn,i,t is included in the project scenario
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Data / Parameter	$C_{DW,i}$
Data unit	t CO ₂ e ha ⁻¹
Description	Carbon stock in aboveground biomass in trees in stratum i
Data source	Required by the CP-D module
Description of measurement methods and procedures to be applied	Módulo CP-D
Frequency of monitoring/recording	According to the CP-D module
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	Developed according to the CP-D module
Purpose of data	Calculation of baseline and project emissions
Calculation method	To be developed during the monitoring period
Comments	To be determined ex ante when accounting of $E_{biomassburn,i,t}$ is included in the baseline scenario. To be monitored when accounting of $E_{biomassburn,i,t}$ is included in the project scenario
Data / Parameter	$C_{LL,i}$
Data unit	t CO ₂ e ha ⁻¹

Description	Carbon stock in aboveground biomass in trees in stratum i
Data source	Required by the CP-L module
Description of measurement methods and procedures to be applied	CP-L module
Frequency of monitoring/recording	According to the CP-L module
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	Developed according to the CP-L module
Purpose of data	Calculation of baseline and project emissions
Calculation method	To be developed during the monitoring period
Comments	To be determined ex ante when accounting of $E_{biomassburn,i,t}$ is included in the baseline scenario. To be monitored when accounting of $E_{biomassburn,i,t}$ is included in the project scenario

Data / Parameter	$\Delta C_{BSL,i,t}$
Data unit	t CO ₂ e ha ⁻¹
Description	Summation of total changes in carbon stock in the project area due to unplanned deforestation in year t
Data source	Forest inventory Required by the BL-PL module
Description of measurement methods	Calculated through areas of forest loss detected in the Project Area and differences in carbon stocks considering ABSL,PA,t

and procedures to be applied	
Frequency of monitoring/recording	Monitoring should occur at least every 10 years for baseline renewal.
Value applied	-
Monitoring equipment	To be developed during the monitoring period
QA/QC procedures to be applied	To be developed during the monitoring period
Purpose of data	To be developed during the monitoring period
Calculation method	To be developed during the monitoring period
Comments	N/A
Data / Parameter	$\Delta C_{BSL\ planned}$
Data unit	t CO ₂ e ha ⁻¹
Description	Net GHG emissions in the baseline from planned deforestation up to year t
Data source	Forest inventory Required by the BL-PL module
Description of measurement methods and procedures to be applied	Calculations are made according to equations 1 and 13 of the BL-PL module, using $A_{planned, i}$
Frequency of monitoring/recording	Monitoring should occur at least every 10 years for baseline renewal.
Value applied	-

Monitoring equipment	To be developed during the monitoring period
QA/QC procedures to be applied	To be developed during the monitoring period
Purpose of data	To be developed during the monitoring period
Calculation method	To be developed during the monitoring period
Comments	N/A

Data / Parameter	$C_{DegW,I,t}$
Data unit	t CO ₂ e
Description	Carbon from the biomass of trees cut and removed through illegal logging, firewood, and charcoal extraction.
Data source	Calculated from the forest inventory Required by the M-REDD module
Description of measurement methods and procedures to be applied	Required by the M-REDD module
Frequency of monitoring/recording	Monitoring should occur at least every 5 years
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	To be developed during the monitoring period

Purpose of data	Quantification of carbon from the biomass of trees cut and removed through illegal logging, firewood, and charcoal extraction.
Calculation method	-
Comments	Ex-ante, an estimate of the likely degradation in the project case should be made. This estimate should be based on degradation rates in surrounding areas and the level of protection that will be in place (e.g., forest guards) in the project case.

Data / Parameter	$E_{BiomassBurn, i, t}$
Data unit	t CO ₂ e ha ⁻¹
Description	Emissions of gases other than CO ₂ due to biomass burning in stratum i in year t
Data source	Required by the E-BPB module
Description of measurement methods and procedures to be applied	Required by the E-BPB module
Frequency of monitoring/recording	Monitoring should occur whenever fires occur
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	To be developed during the monitoring period
Purpose of data	Quantification of gases emitted other than CO ₂ due to biomass burning in stratum i in year t

Calculation method	To be developed during the monitoring period
Comments	Ex-post calculation
Data / Parameter	Project Forest Cover Benchmark Map
Data unit	-
Description	Map showing the location of forested areas within the project area at the start of the project's crediting period.
Data source	Satellite images and GIS Used in Equation 3 of M-REDD to calculate $A_{DefPA,u,i,t}$, which is the area of deforestation recorded in project area stratum i converted to land use u in year t
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	At least every 5 years and at baseline renewal
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	The minimum map accuracy must be 90% for forest/non-forest classification in remote sensing images. If the accuracy is below 90%, then the map is not acceptable for further analysis. More remote sensing data and ground truth data will be needed to produce a product that achieves a minimum mapping accuracy of 90%.
Purpose of data	Monitoring of areas
Calculation method	-

Comments

Required by the M-REDD module

3.3.3 Monitoring Plan (VCS, 3.16, 3.20)

The Project will encompass continuous monitoring of the implementation of REDD+ activities in the Seringal Rio Branco property area.

Changes in carbon stock due to the conversion of forested areas to non-forested areas through planned deforestation will be monitored, as will changes in carbon stock due to forest fires and other catastrophic events. These changes will be deducted from the Project's baseline scenario in cases where they are significant. Two main activities to cover this monitoring consist of: 1) monitoring deforestation through satellite images; and, 2) improve surveillance within the property with the opportunity to verify in the field cases in which deforested areas were not detected by satellite and confirm areas that were detected.

In this way, monitoring of the Seringal Rio Branco REDD+ Project aims to monitor changes in carbon stock to estimate real changes in GHG emissions. These need to be evaluated to show an uncertainty of no more than 15% and to be significant for them to be considered. Therefore, not all changes in GHG emissions are considered.

In this sense, an essential activity in monitoring is to verify the uncertainty and significance of changes. For this, the project can use the guidance, respectively, from the X-UNC module and the CDM T-SIG tool. Additionally, monitoring for activities classified as ODA is guided by the BL-PL module, in addition to the monitoring module itself (M-REDD), which is mandatory in REDD-MF projects.

3.3.3.1 Monitoring deforestation using satellite images

Monthly, deforestation and degradation are monitored via satellite images, based on data produced from spectral bands and analysis by specialized technicians.

To monitor forest cover, specifically within the limits defined for the Seringal Rio Branco REDD+ Project as Project Area and Leakage Areas, data from the MapBiomas Alerta Project will be used, a system for validating and refining deforestation alerts through images high resolution. The current version is dedicated exclusively to deforestation in all Brazilian biomes.

Brazil is covered by more than a dozen deforestation warning systems, including DETER/INPE (Amazonia and Cerrado), SAD/IMAZON, GLAD/Univ. Maryland (all biomes). Together, these systems generate tens of thousands of alerts based on the analysis of satellite images of medium spatial resolution (20m to 60m). Currently, alerts generated by different providers are collected and used as a reference to locate deforestation alerts in daily high-resolution satellite images (3m).

Each alert is validated, refined and defined within a time window of occurrence. For each validated alert, a report is generated identifying images from before and after deforestation, making it possible to cross-reference areas with areas in the Rural Environmental Registry (CAR), National Registry of Conservation Units (CNUC), Indigenous Lands and other geographic limits (e.g. biomes, states, municipalities and river

basins) (MAPBIOMAS, 2023). This validation report is made from a visual inspection with high-resolution images from Planet (daily pass with 3.5 meters of spatial resolution) or images from Sentinel-2 (weekly pass, 10 meters of resolution).



Figura 36: MapBiomas page Alert consulted to monitor deforestation

The vegetation degradation analysis aims to monitor and map vegetation suppression that was not identified by the MapBiomas Alerta system and that did not present clear cuts and exposed soil. In this context, the type of suppression analyzed in the Seringal Rio Branco REDD+ Project corresponds mainly to management actions, when there is no burning and felling of a large continuous area, capable of being identified by Brazilian monitoring systems such as PRODES.

For this analysis, high spatial resolution satellite images of the Planet constellation are used. These images have approximately 5 meters of spatial resolution and are suitable for photointerpretation analyzes at a scale of approximately 1:15,000.

The methodology for identifying degradation consists of three main steps, namely:

- Selection of PLANET images for the monitoring period;
- Spatial overlap with the limits of the Project Area and Leakage Areas;
- Monitoring the area at a 1:15,000 scale, identifying pixels in the canopies and tree crowns that showed the color of exposed soil or characteristic of human activity.

This monitoring began in January 2023 with the aim of complementing the work of forest guards on patrol. The images are obtained through a free subscription plan on the Planet website (<https://www.planet.com/>) and are updated at the end of the first fortnight of each month and delivered at a spatial resolution of 3 to 5 meters (pixel width) and can be accessed via API through QGIS software.

With this resource, it is possible to have an overview of the progress of deforestation and the risk of invasions or incursions by illegal loggers both in the Project Area and nearby. Any opening of roads or

illegal actions that can be seen in the images are reported to the project management team and the necessary actions are defined, including strategies for future actions.

3.3.3.2 Fire monitoring using geographic information systems

So that quantification can be done, the project will monitor the occurrence of fires in the project area and in the leak areas through remote sensing. In this monitoring, hot spots will be detected using satellites with polar and geostationary orbits, which have different radiometric and temporal resolutions.

The system structured by the geoprocessing team collects data from the various sources shown in Table 24 and stores it in a database, where automatic microservices eliminate duplicate points and points outside the areas of interest for monitoring.

Table 26: Consulted data source.

Source	Satellite	Resolution
	NOAA - AVHRR (Advanced Very High Resolution Radiometer)	1km
	TERRA - MODIS (Moderate Resolution Imaging Spectroradiometer)	1km
	AQUA - MODIS (Moderate Resolution Imaging Spectroradiometer)	1km
	TERRA - MODIS (Moderate Resolution Imaging Spectroradiometer)	1km
	AQUA - MODIS (Moderate Resolution Imaging Spectroradiometer)	1km
	SUOMI NPP – VIIRS (Visible Infrared Imaging Radiometer Suite)	375m

Depending on the characteristics of the sensors embedded in satellites, a fire front may be represented by one or more heat points, in the same way that smaller fire fronts may be represented by a single point. This occurs because of the spatial resolution of the sensors.

On satellites embedded with MODIS sensors, the thermal image has a spatial resolution of 1km x 1km. Therefore, the heat point represents the center of the thermal anomaly pixel, which could mean one or more fires present within a 500m radius around the point.

Looking at Figure 37 below, we can see the influence of spatial resolution on fire detection, where sensors with higher spatial resolution will provide the most accurate location of the hotspot.

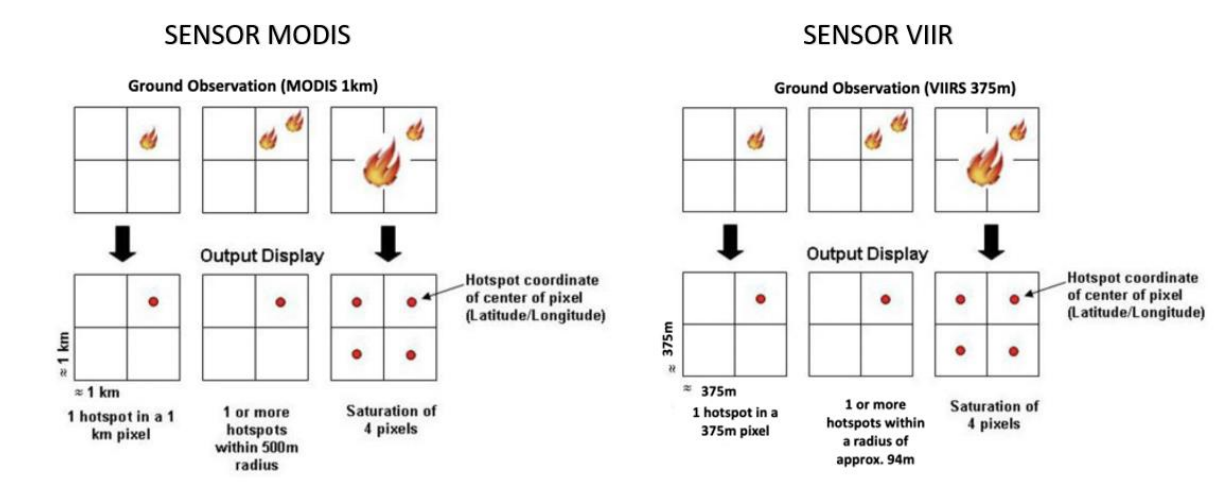


Figure 37: Detail of the influence of the spatial resolution of the sensors on fire detection.

Once the data has been collected and processed, the system checks whether there have been any hotspots in the project's areas of interest and, if there is an overlap, the system automatically sends an e-mail and/or SMS message to the project team members registered in the system, informing them of the total number of hotspots identified and their coordinates.

In order to increase the efficiency of fire prevention and firefighting in the project, the system monitors the following areas of interest:

- Boundary of the Seringal Rio Branco property with a 2km buffer around the property
- REDD+ project area with a 5km buffer around the project
- Leakage belt

3.3.3.3 Reassessment of baseline projections

The baselines will be reassessed over time in the Program, as directed by VM0007, because the agents, drivers and underlying causes of deforestation change dynamically. However, this update should not be done frequently or unexpectedly, as this can create serious uncertainties in the market. The baseline should be updated every 10 years in a planned manner. Two activities are necessary when updating the baseline:

1. Updating the information on changes in land use and land cover in the baseline;
2. Updating the carbon component of the baseline.

3.3.3.4 Quality control and assurance procedures

During the term of the Seringal Rio Branco REDD+ Project, all activities will be continuously monitored and evaluated by the proponent, which will allow for the development of a continuous feedback process in order to incorporate improvements in the procedures applied to all types of data to be monitored, consequently guaranteeing better quality for the Project.

The maps that show changes in carbon stock due to the conversion of forests into non-forest areas by unplanned and planned deforestation will undergo an analysis of the overall accuracy and the kappa index obtained from a confusion matrix, such as the matrix by Congalton and Green (1999), in which at least 100 points distributed randomly in relation to the area analyzed will be generated using a geographic information system. Validation will be carried out using high spatial resolution satellite images and/or data collected in the field. The minimum mapping accuracy, according to REDD-MF, for each class or category on the land use and cover map must be 90%. In addition to satellite images, field checks will be carried out in areas where conversion from forest to non-forest areas has been identified.

3.3.4 Dissemination of Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

The Climate Monitoring Plan and all its results will be made public on the VERRA registration page and on the Biofílica Ambipar website.

For interested parties who do not have access to the internet, a printed copy can be provided at strategic points (farm headquarters and partner offices). Information about the Program will also be disseminated via email and WhatsApp (with a pre-established list of stakeholder contacts), and on the project proponent's social media, such as Instagram and LinkedIn.

In this way, it is hoped that all the necessary efforts to provide full transparency and distribute the Program's key information and documents will be achieved.

3.4 Optional Criterion: Climate Change Adaptation Benefits

3.4.1 Regional Climate Change Scenarios (CCB, GL1.1)

The Seringal Rio Branco REDD+ Project is applying for Gold Level status under the CCB standard, based on providing significant support to communities and biodiversity in adapting to the impacts of climate change described below. Modeling and scenarios have been widely used in recent years to warn about the impacts of climate change in Latin America, despite their significant inaccuracy. For Brazil, great attention has been paid due to the number of anthropogenic effects in the Amazon and its importance and influence on the regional, national and global climate (DUBREUIL et al., 2012; ARRAUT et al., 2012; MARENGO, 2020).

The Amazon has great importance for maintaining local, regional and global rainfall through air masses, meteorological variables and the forest's hydrological cycle, and the importance of its “flying rivers” can be highlighted (ARRAUT et al., 2012; ZEMP et al., 2014; MARENGO, 2020; ARTAXO, 2020). Changes in land use, such as deforestation in the Amazon, impact not only the economy or biodiversity, but also the social context of many of the local populations (MARENGO et al., 2018). Eventos climáticos extremos ocorreram na região amazônica, como três grandes secas, nos anos de 2005, 2010 e 2016, e três grandes enchentes nos anos de 2009, 2012 e 2014, nos últimos 17 anos (MARENGO; OLIVEIRA, 2016; ZOU et al., 2015). Os cenários AR 5 para altas e baixas emissões no período do modelo do Painel Internacional de Mudanças Climáticas (IPCC) mostram aumentos de temperatura de 4 °C e redução de chuvas de aproximadamente até 40% na Amazônia (MARENGO, SOUZA JR; 2018).

In addition, for the current macro scale, it is possible to cite the IPCC document, WGII AR6, which has guided governments and stakeholders in defining the best scenarios for mitigation and adaptation to climate impacts, as well as its forecasts of changes in temperature, precipitation and extreme weather events (IPCC, 2022). According to the WGII AR6 scenario, temperatures will continue to rise until at least the middle of the century in all the scenarios (SSP1-1.9; SSP1-2.6; SSP2-4.5; SSP3-7.0 and SSP5-8.5) considered by the IPCC (IPCC, 2022).

The report also indicates that warming of 1.5 °C to 2 °C will be exceeded during the 21st century unless there are drastic reductions in CO₂ and other greenhouse gases in the coming decades. Variations of up to 40% are also indicated in scenarios of increased or decreased precipitation, varying according to location.

Warming in the Amazon region has been historically evidenced by various sources since 1950 (MARENGO et al., 2020; MARENGO, SOUZA JR, 2018). Variations in precipitation have also been evidenced over the last few years, in the northeast of the Amazon, the trend is for an increase in the occurrence of rainfall while in its southeastern portion the trend is for a reduction in the occurrence of rainfall based on observations from 1981 to 2017 analyzed from modeling and numbers of drought and flood events for the region (ESPINOZA et al., 2018).

In the world's largest rainforest, especially in the state of Pará, anthropogenic changes such as the construction of megaprojects, deforestation, arson, illegal mineral extraction and logging activities, among other factors, have led to a worsening of the situation already caused by climate change (SEMA PA, 2021; MOUTINHO et al., 2020). Considering the current scenario, the absence of the project would be limited in generating benefits for the climate, the community and biodiversity.

Considering the long-term dynamics of the Amazon rainforest balance, there is a synergy between the occurrence of three anthropogenic effects as drivers of change: (i) an increase in the local average temperature and consequent change in the precipitation regime, leading to severe droughts, (ii) increased regional deforestation increases the surface temperature in deforested areas by 1°C to 3°C degrees and reduces the cycling of water by vegetation and, (iii) the combination of increased surface temperature, extreme droughts and continuous burning driven by entropy transforms the once. fire-resilient forest into a biome more vulnerable to fires (NOBRE, 2020). These factors together result in the point of no return of the forest to its initial state, i.e. the “tipping point” as described below (MARENGO, SOUZA JR; 2018; ARTAXO et al., 2021).

That said, there will be many changes in the climate over the coming decades that will have an impact on the dynamics of the Amazon ecosystem. This scenario, combined with the scenario without the Project and its activities, will tend to induce and boost illegal practices, such as commercial logging for timber companies, sawmills and charcoal factories; associated with the conversion of land for subsistence agriculture under conventional slash and burn practices, agricultural production of grains, raising beef cattle on extensive pastures and the possibility of setting up and maintaining infrastructure and logistics projects. Thus, the environmental degradation and worsening of climate threats mentioned above would be leveraged by increased deforestation pressure in the reference region, gradually advancing to the limits of the project area, something that already occurs in the Region without the REDD+ Project.

3.4.2 Climate Change Impacts (CCB, GL1.2)

Amazonian populations in Brazil are expected to be more vulnerable to the impacts of climate change when compared to people elsewhere in the country. The high dependence on the consumption of freshwater fish, fruit and other forest extraction products as a source of food and income leaves these populations with an uncertain future due to future climate change, especially with regard to changes in temperature and rainfall patterns, as well as the length and occurrence of the dry and rainy seasons.

Based on the socio-economic diagnosis carried out in the project area and the surrounding region, it was possible to identify the community's perception of climate change. This information was recorded during the survey, which focused on obtaining reports from families on issues relating to the quality and availability of water, the quality of the soil for agriculture and the availability of fish and other game. The following table shows the community's perceptions of climate change and the possible impacts that these changes could have on the well-being of the communities.

Table 27: Perceptions and impacts of climate change. Source: Socio-economic Diagnosis REDD+ Project Seringal Rio Branco

Axis	Perception of climate change	Possible impacts on community well-being
Temperature	• High temperatures; • Rivers and streams dry up for longer; • Increase in respiratory diseases; • Allergies to heat exposure.	• They spend less time in the fields because of the heat; • Decrease in production, reducing family income; • Skin cancer and respiratory diseases are more common, damaging health in general.
Water	• Change in the rainfall regime, decreasing the flow of water for agriculture; • Rivers and streams are running dry for longer - decreasing their level; • Loss of biodiversity;	• Damages the seasonal production of agricultural crops; • Damages the quality and quantity of native fruits, impacting on food and income generation; • Restriction and availability of drinking water for families; • Decrease in fish, impacting on families' diets.
Fauna	• Disappearance of wild animals; • Decrease in fish species; • Low reproduction of wild animals	• Restriction in the diet of families that consume animals (game) and fish.

It is important to note that the municipalities in the project region have an economy based on agricultural activities, especially rice, beans, corn and manioc, as well as livestock and the extraction of timber products. These facts, combined with the low human development index and reduced per capita income in these locations, make them more vulnerable to climate impacts (IBGE, 2022). Even so, the quality of

human health in the Amazon region is already lower than in other regions of the country, however, with the intensification of climate change there is a greater propensity for the emergence of endemic diseases.

The communities located in the Seringal Rio Branco REDD+ project area are classified as riverside communities and rural settlements, which are all dependent, to a greater or lesser extent, on productive activities related to agriculture. The risk to food sovereignty and the risk of damage to the health and housing of these communities, negatively impacted by the rains, increase depending on the physical and biological aspects of their location.

That said, a large part of the communities in the Project Zone could be negatively impacted by extreme weather events, such as droughts, which would impact their food sovereignty by reducing the flow of rivers that supply fish and even causing losses in their subsistence crops. In addition to the subsistence base of these families, we can also mention economic activities such as agriculture, livestock farming and extractive activities (açaí and piassava) that could be highly impacted by rising temperatures. This can occur not only due to an increase in the intensity of the temperature and/or a decrease in rainfall, but also due to a change in the periods of the dry and rainy seasons, affecting production, harvesting and marketing (MARENGO et al., 2018).

With regard to biodiversity, it is necessary to highlight the impacts of climate change on the Amazon Rainforest, which contains a large part of the world's biodiversity and is home to more than 12% of all flowering plants (Gentry, 1982). According to the study by Sala et al. (2000) on changes in biodiversity by the year 2100, it was identified that, for tropical biomes, the main agents affecting biodiversity are land use and climate change. Thus, threats to the existence of the Amazon rainforest indicate serious threats to biodiversity. These impacts can be exemplified by the savannization of the forest and its consequent change in carbon stocks, local and/or regional extinction of animal and plant species, alteration of the hydrological cycle, among other factors that contribute to environmental degradation caused by climate change.

Consequently, modifications to the forest component have an impact on species that have suffered or are suffering fragmentation in their environments and may have a decrease in their population size due to increased inbreeding, which generates a loss of genetic variability that results in a decrease in adaptability and reproductive capacity.

3.4.3 Measures Needed and Designed for Adaptation (CCB, GL1.3)

The area and especially the Project zone are vulnerable to the potential impacts of climate change, which represents an additional challenge for the proponents of the Seringal Rio Branco REDD+ Project, since climate change predicted in the global and local context will have impacts over time on the functioning of ecosystem services, the livelihoods and health of communities and the perpetuation of the rich biodiversity found. In the Project area, the aim will be to strengthen community (human) and environmental (non-human) resilience by implementing adaptation options through a variety of Project activities (see details in section 2.1.11).

In the social sphere, in general terms, the Project's benefits seek to reduce the vulnerability of local communities to climate change through income diversification, environmental education and better sustainable practices associated with the livelihoods of these families.

With regard to the Project's expected adaptations to biodiversity, the main one is to ensure, through continuous monitoring and adaptive management of the Project, the long-term conservation of wildlife associated with curbing unplanned deforestation and, consequently, maintaining natural habitats and reducing the loss of landscape connectivity.

4 COMMUNITY

4.1 Without-Project Community Scenario

4.1.1 Descriptions of Communities at Project Start (CCB, CM1.1)

The occupation of the region known as the Juruá Valley, where the Seringal Rio Branco REDD+ Project is located, dates back to the ancestral occupation by indigenous populations. Divided into 49 tribes, these peoples, belonging to Pano-speaking ethnic groups such as the Ararauas, Catukinas and Curimas, inhabited the Juruá Valley, with the Náuas tribe on the banks of the Juruá River (CRUZEIRO DO SUL, 2023). The municipality of Cruzeiro do Sul, initially inhabited by indigenous tribes, bore witness to this cultural heritage.

Porto Walter, also with indigenous roots, is home to descendants of the Arara, Náuas, Amoaças, Kampas, Kulinas and Catianos nations. At the end of the 19th century, the arrival of migrants, mainly from northeastern Brazil, driven by the extraction of latex from the rubber tree, marked an ethnic transition in the region (PORTO WALTER, 2023). The formation of the rubber plantations attracted not only northeasterners, but also migrants from Peru.

The Seringal Rio Branco REDD+ Project area is located in the Juruá Valley Mesoregion, covering eight municipalities such as Cruzeiro do Sul, Tarauacá, Feijó, Mâncio Lima, Rodrigues Alves, Marechal Thaumaturgo, Porto Walter and Jordão, totaling 85,448 km² with a population density of 2.8 inhabitants/km² and an estimated population of 242,796. The Serra do Divisor National Park, present in the region, is an important environmental landmark (CRUZEIRO DO SUL, 2023).

The analysis of the nearby riverside communities reveals that most of them were formed in the second half of the 20th century, between 1900 and 1950. The most recent community, Volta Grande, emerged in 1984. The Francisco Pimentel PDS settlement, set up in 2005 and regularized by INCRA, represents a more recent initiative aimed at sustainable development.

In general, the formation of the communities around the project is deeply rooted in the history of the local families. These families, who come from the rural region itself, were born in the communities where they live or in nearby localities in the municipalities of Cruzeiro do Sul and Porto Walter. The composition of these communities reveals a notable absence of significant migratory processes, highlighting the families' strong ties to the region.

- **Population characteristics of the communities**

Currently, the population of the group of ten communities around the project studied totals 783 inhabitants, distributed among 178 families. The PDS Francisco Pimentel, the largest among them, with 171 families, has an estimated 684 residents, resulting in a total population in the project area of 1,467 people, distributed among 345 families.

Table 28: Number of families and residents per community. Source: Field research, 2023. Elaboration: STA, 2023.

Communities	No. of families	No. of residents
Aldeota 2	16	74
Monte Cristo	33	153
Periquito	13	54
Prainha 1	31	125
Prainha 2	19	118
Primavera	23	115
Santo Antônio 1	11	43
São Luís	23	77
Volta Grande	5	24
PDS Francisco Pimentel ⁴⁸	171	Not informed
Total	345	783

According to data from CIGA (2015), within the Francisco Pimentel settlement project there are at least 10 rural communities or localities of residents. The population living in these communities is practically isolated, with no means of land transportation, leaving only the streams as a way of getting around for a period of the year. Table below shows the communities with the number of families and an estimate of the number of residents, based on an average of four people per family, based on references from IBGE (2010), which estimate an average of 3.2 people per family for the state of Acre in 2012, opting for this study to round up to four people.

CIGA's data shows that in 2015 there were 120 families and these families added up to a population of 480 people living in the 10 communities of the PDS.

Table 29: Number of families and residents per community in the Francisco Pimentel PDS⁴⁹. Source: CIGA, 2014. Elaboration: STA, 2023.

Communities	No. of families	Estimated number of residents
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⁴⁸ Data was collected using a simplified script with a key informant.

⁴⁹ The communities estimated the number of residents during the diagnostic survey.

Foz do Apuí	10	40
Zé do Eduardo	4	16
Santa Maria I	16	64
Santa Maria II	8	32
Cicero	26	104
Oliveira	18	72
Zé Tuta	8	32
Chico Preto	15	60
São João I	7	28
São João II	8	32
Total	120	480

Calculating the same average for the current data, according to the key informant, on the number of families in the PDS, which totals 171 families, there would be a total of 684 people living in the PDS. Adding these totals up, we would have a total population of the area surrounding the project of 1,467 people, distributed among 345 families.

In general, the communities have historically been made up of local families who were born in the surrounding rural region, in the communities where they live or in nearby communities in the municipalities of Cruzeiro do Sul and Porto Walter. There has been little or no migration process in the formation of these communities. With the exception of Santo Antônio 1, which is made up of some migrant families from outside the region, from the municipality of Ipixuna, in the state of Amazonas.

- **Age of the population in the communities**

According to the data collected during the research, adult men stand out as the most significant population group in the area, totaling 252 individuals in the nine communities analyzed. Children are the second largest group, accounting for 210 people, followed by women, with a total of 168, and adolescents, with 116 individuals. The elderly have the smallest contingent, totaling just 37 people.

When analyzing the distribution of the population by age group and sex in the communities surveyed, it can be seen that the population is concentrated in three age groups, ranging from zero to 35 years old. The zero to 15 age group represents 37% of the total local population, while the 15 to 25 age group covers 24% of the population. The 26 to 35 age group accounts for 17% of the total, and people over 65 make up 4% of the population. This demographic profile highlights the region as characterized by a significant majority of young people and people of working age.

Table 30: Population by age group and sex of the communities surveyed. Source: Field research, 2023. Elaboration: STA, 2023. *PDS Francisco Pimentel não consta nessa quantificação.⁵⁰

⁵⁰ Nota: As comunidades estimaram as informações sobre o quantitativo de moradores por sexo e faixa etária durante a pesquisa de diagnóstico.

Age group	Men	%	Women	%	Total	%
Up to 15 years	139	32%	148	42%	287	37%
From 15 to 25 years old	107	25%	79	23%	186	24%
From 26 to 35 years old	80	18%	51	15%	131	17%
From 36 to 45 years old	47	11%	38	11%	85	11%
From 46 to 55 years old	33	8%	16	5%	49	6%
From 56 to 65 years old	5	1%	7	2%	12	2%
Over 65	22	5%	11	3%	33	4%
Total	433	100%	350	100%	783	100%

- Ethnicity of the communities' population**

The majority of the population of the communities located around the project area is of black or brown ethnic origin, according to the self-declared identification of the individuals, as detailed in table below during the survey.

Table 31: Percentage of each ethnic group in the population of the communities surveyed. Prepared by: STA, 2023.

Community	Classification	Identified ethnicity (%)			
		White	Black	Brown	Indigenous
Aldeota 2	Riverside population	-	20%	80%	-
Monte Cristo	Riverside population	25%	5%	70%	-
Periquito	Riverside population	-	10%	80%	10%
Prainha 1	Riverside population	-	10%	90%	-
Prainha 2	Riverside population	10%	20%	70%	-
Primavera	Riverside population	5%	80%	10%	5%
Santo Antônio 1	Riverside population	10%	10%	80%	-
São Luís	Riverside population	5%	15%	80%	-
Volta Grande	Riverside population	5%	95%	-	-
PDS Francisco Pimentel ⁵¹	Family farmers	No information.			

⁵¹ Although the information was not collected, the majority of the population of the Francisco Pimentel PDS is identified as Brown.

According to the IBGE, brown is one of the five ethnic groups that make up the Brazilian population, along with white, black, yellow, and indigenous. The black category refers to African ancestry, regardless of territory or social construction, and stands out for its dark-skinned phenotype. The brown category, on the other hand, includes people with diverse ethnic backgrounds, resulting from a mixture of whites, blacks, and indigenous people. This miscegenation includes descendants of different combinations, such as blacks with whites, blacks with indigenous people, Indians with whites, as well as other direct or indirect interracial interactions. Due to the history of racism and discrimination in Brazil, many people from the black/negro ethnic group, even if they didn't have dark skin, self-identified as brown, reflecting the complexity of miscegenation in Brazilian society.

- **Presence of women in the communities**

During the field research, the communities passed on information to the team about the number of men and women in the communities, considering the inhabitant population. The data shows that the majority of the population is made up of men, 55%, compared to 45% of women. According to the 2010 Census, 49.8% of the population of the state of Acre is made up of women, compared to 50.2% of men.

The survey showed that most communities have similar proportions of men and women. According to data from the PNAD Contínua (Continuous National Household Sample Survey) for 2021, the number of women in Brazil is higher than the number of men. The Brazilian population is made up of 48.9% men and 51.1% women. Reversing the national average, the Monte Cristo and Primavera communities have a higher percentage of men, 63% and 65% respectively.

Table 32: Number of men and women in the communities. Source: Field research, 2023. Elaboration: STA, 2023.

Communities	Male	Female
Aldeota 2	37	37
Monte Cristo	97	56
Periquito	26	28
Prainha 1	63	62
Prainha 2	60	58
Primavera	75	40
Santo Antônio 1	21	22
São Luís	42	35
Volta Grande	12	12
Total	433	350

The following table shows municipal data for the population by sex for the year 2010, which is data available from the Brazilian Census system. Data was collected for the two reference municipalities for the research around the project area, which are the municipalities of Cruzeiro do Sul and Porto Walter.

Table 33: Population by sex of the municipalities Cruzeiro do Sul and Porto Walter for the year 2010. Source: ACRE (2017). Elaboration: STA, 2023.

Municipality	Total	Male	%	Female	%
Cruzeiro do Sul	78.507	39.220	49,96	39.287	50,04
Porto Walter	9.176	4.794	52,24	4.382	47,76

- **Data on femicide**

The topic of violence against women, and especially femicide, was a sensitive one to address during the research in the communities, mainly because of the insecurity in dealing with the topic and the fact that the women were not alone during the research activities. Only general data on the state of ACRE was collected, due to the lack of information from the communities studied.

According to data on public safety in Brazil, combined with data from other organizations that compute information on violence in the country, it is possible to draw a picture of violence against women and femicide at national and state level. Data for municipalities in the state of Acre was not accessed. The publication of the Brazilian Yearbook of Public Safety gathers data from the State Secretariats of Public Safety and/or Social Defense, the Observatory of Criminal Analysis, the Coordination of Statistical Information and Criminal Analysis - COINE/RN, the Institute of Public Safety/RJ (ISP), the Brazilian Institute of Geography and Statistics (IBGE) and the Brazilian Forum of Public Safety. The publication of the 2022 Yearbook presented rates per 100,000/women of homicides against women and femicide by state.

In Acre, in 2021, 29 cases of homicide against women were recorded and of these, 13 cases were recorded as femicide, representing 45% of the total number of homicides. In Acre, the proportion of femicides in relation to homicides of women increased when comparing the years 2020 (38.7%) and 2021 (44.8%), representing a high number, above the national average, which was 33.9% for 2020 and 34.6% for 2021.

Table 34: Homicide of women and femicide - Brazil and the Legal Amazon - 2020-2021. Source: Brazilian Public Safety Yearbook, 2022.

Brazil and Federative Units	Homicide (Female Victims)					Feminicide					Proportion of femicides in relation to homicides of women	
	NS. Absolute		Rate ⁽¹⁾		Variation (%) 2020	NS. Absolute		Rate ⁵²		Variation (%)	In percent (%)	
	2020	2021	2020	2021		2021	2020	2020	2020		2020	2021
BRAZIL	3999	3878	3,7	3,6	-38,0	1354	1341	1,3	1,2	-1,7	33,9	34,6
Acre	31	29	6,9	6,4	-7,8	12	13	2,7	2,9	6,8	38,7	44,8
Amapá	20	23	4,6	5,2	12,9	9	4	2,1	0,9	-56,4	45	17,4

⁵² Rate per 100,000/women; (2) Update of information published in the Brazilian Yearbook of Public Safety, year 15, 2021.

Amazonas	68	110	3,2	5,2	59,3	16	23	0,8	1,1	41,6	23,5	20,9
Maranhão	125	134	3,5	3,7	6,5	65	58	1,8	1,6	-11,3	52	43,3
Mato Grosso	104	85	6,0	48,0	-19,3	62	43	3,6	2,4	-31,5	59,6	50,6
Pará	181	183	4,2	4,2	0,0	67	64	1,5	1,5	-5,5	37	35
Rondônia	56	49	6,3	5,5	-13,5	14	16	1,6	1,8	13	25	32,7
Roraima	16	26	5,3	8,3	57,5	9	5	3	1,6	-46,1	56,3	19,2
Tocantins	39	38	4,9	4,8	0,7	10	21	1,3	2,6	107,6	25,6	55,3

- **Aspects of education**

The general data from this diagnosis shows the lack of school infrastructure in the municipalities that make up the region where the communities studied are located. In addition, poor educational quality prevails, reflected in high dropout rates. This scenario is exacerbated by the lack of an adequate structure for young people in rural communities to remain in school and complete their studies.

Table below details the educational situation in the reference municipalities, including the capital, Rio Branco, in the 2010 Census, classifying people aged 10 and over with no education and/or incomplete elementary education, by color or race. In Cruzeiro do Sul, 71% of the population in this condition was brown, a similar percentage to the proportion observed in the capital, Rio Branco, which reached 70%. Porto Walter shows a slight discrepancy, with 64% of the brown population in this condition. These figures reflect the urgency of interventions in education in these localities to reverse the precarious situation..

Table 35: People aged 10 and over with no education and incomplete elementary education. Source: IBGE (2010). Elaboration: STA, 2023.

Color or Race	Cruzeiro do Sul	%	Porto Walter	%	Rio Branco	%
Yellow	357	1%	78	2%	2.881	2%
White	7.070	20%	1.093	23%	26.183	21%
Indigenous	539	2%	278	6%	164	0%
Brown	25.178	71%	3.028	64%	89.045	70%
Black	2.383	7%	282	6%	8.347	9%
Total	35.527	100%	4.759	100%	126.620	100%

As there is no daycare provision for the children, the families reported that children up to the age of 5 receive private tuition in their homes, given by community residents or family members. School-age children from the Aldeota 2 community go to the Aldeota 1 community school to attend classes. The Santo Antônio, São Luís and Volta Grande communities also don't have schools, so school-age children go to other nearby communities to receive lessons, such as Santo Antônio 2, Periquito and Prainha 2, respectively.

The Periquito and Prainha 2 communities have elementary and high schools. The Monte Cristo, Prainha 1 and Primavera communities have elementary schools.

Table 36: Education infrastructure and services available in the communities. Source: Field research, 2023.
Elaboration: STA, 2023.

Infrastructure and services items / Communities	Daycare	Public elementary school	Private Elementary School	Public High School	Literacy course for young people and adults	Other (uses another community's structure)
-	-	-	-	-	No	They go to school in the Aldeota 1 community
Sim	-	Yes	-	No	-	-
Sim	-	Yes	-	Yes	-	-
Sim	-	Yes	-	-	-	-
Sim	-	Yes	-	Yes	-	-
Primavera	-	Yes	-	-	-	-
Santo Antônio 1	-	-	-	-	-	They go to school in the Santo Antônio 2 community
São Luís	-	-	-	-	-	They go to school in the Periquito community
Volta Grande	-	-	-	-	-	They attend the Prainha 2 community school

With regard to school services, such as transportation, school meals and internet access, it was found that no community has school internet. With regard to school meals, the families reported that school meals are available for the students attending the schools in the communities and that the food is of good quality and quantity. With regard to school transportation, the communities reported that they have a boatman who picks up the students at their homes and drops them off after school.

- **Health aspects**

Access to health services in the surrounding communities is precarious. There is practically no basic health care infrastructure available. And community health agent visits, which take place in practically every community, are infrequent, sometimes occurring only once a year.

The Cruzeiro do Sul city hall pays for an emergency "ambulância" (boat) to transport the sick people to the city in urgent cases. The emergency boats are available in the Aldeota 2 and Prainha 2 communities.

The residents of PDS Francisco Pimentel receive visits from the health team from the municipality of Rodrigues Alves. The following table systematizes the information per community about the available infrastructure and other services.

**Table 37: Infraestrutura e serviços disponíveis de saúde nas comunidades⁵³. Source: Field research, 2022.
Elaboration: STA, 2023.**

Communities	Health center	Basic Health Unit	Visit by a Community Health Agent (ACS)	ACS visit frequency	Other infrastructure / alternative
Aldeota 2	-	-	Yes	1 time per year	-
Monte Cristo	-	-	Yes	2 years without a visit	They have an emergency boat, which functions as an ambulance
Periquito	-	-	Yes	1 time per year	-
Prainha 1	-	-	Yes	1 time per year	Microscopist (for malaria examination)
Prainha 2	-	-	Yes	1 time per year	They have an emergency boat, which functions as an ambulance
Primavera	-	-	Yes	1 time per year	-
Santo Antônio 1	-	-	-	-	-
São Luís	-	-	Yes	1 time per year	-
Volta Grande	-	-	-	-	-

In cases where more serious health situations occur, such as accidents with fractures, heart attacks, cancer, diabetes, snake bites, the need for specialized consultations, among others, families seek care in hospitals in nearby cities, such as Cruzeiro do Sul or Porto Walter.

With regard to the most frequent illnesses in the communities by age group, the table below lists the illnesses that most affect children, young people, adult women, elderly women, adult men and elderly men. It can be seen that children and young people are most affected by viruses, mycoses, diarrhea and respiratory infections in general, such as the flu, but there are also cases of malaria and reports of animal bites (such as snakes and stingrays).

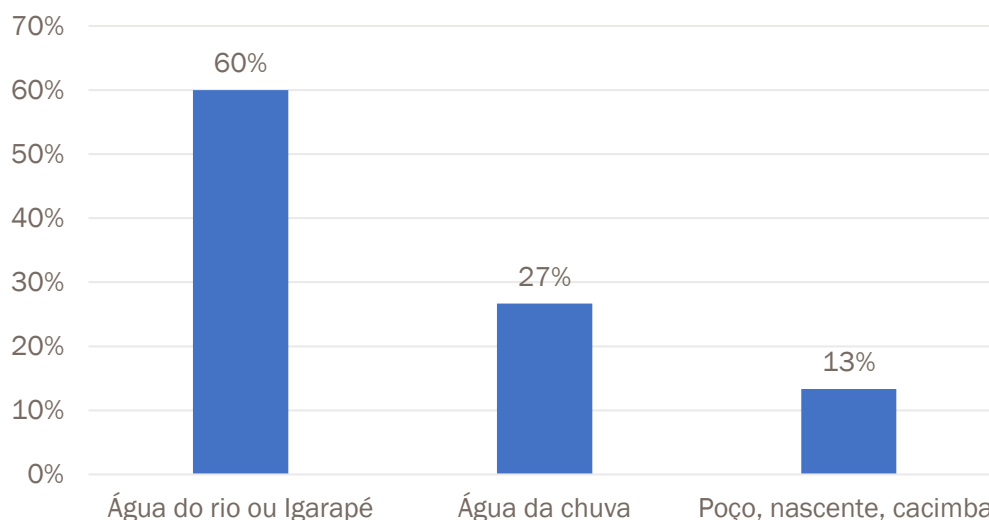
Malaria is common in all communities and groups, affecting the population as a whole. Dengue fever is also frequent and affects all communities and groups.

- **Aspects of basic sanitation**

The communities' basic sanitation data also reflects the reality of the region's municipalities, with low levels of assistance and access to services.

⁵³ Social data was collected from key informants.

Regarding the origin of drinking water and its quality. Regarding access to drinking water, the communities reported that no community has a general distribution network (water tank) and that the source of water for family consumption comes largely from the river and streams. 60% of the communities use river or stream water for consumption on a daily basis. 27% use rainwater for consumption and 13% use water for consumption from cisterns, wells, or springs, as shown in Graph X.



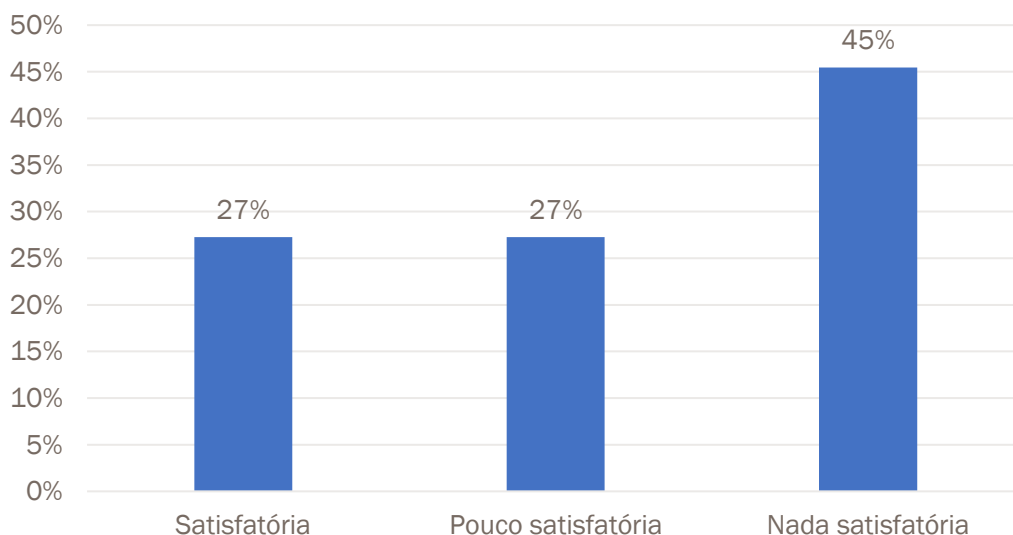
Graph 1: Percentage of the origin of water for household consumption. Source: Field research, 2023. Elaboration: STA, 2023.

Table 38: Origin of water for consumption in the communities. Source: Field research, 2023. Elaboration: STA, 2023.

Communities	Forms of access to drinking water in the community
Aldeota 2	River water and rainwater
Monte Cristo	River water and Cistern
Periquito	River water
Prainha 1	River water and rainwater
Prainha 2	River water and rainwater
Primavera	River water and Cistern
Santo Antônio 1	River water and rainwater
São Luís	River and stream water
Volta Grande	Stream (using a water pump)

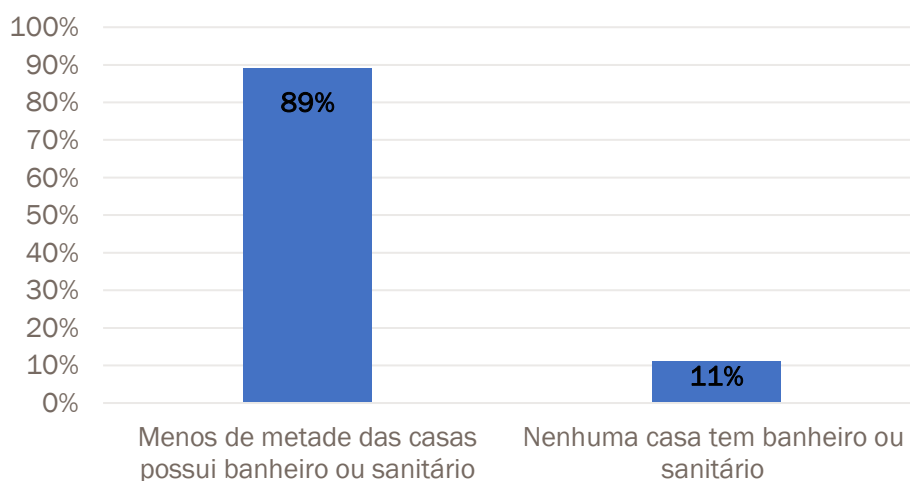
With regard to the quality of drinking water, families rated whether or not the water was satisfactory for consumption. In Graph X below, 45% of the communities say that the quality of the water is not at all satisfactory; 27% classify it as not very satisfactory and 27% as satisfactory. Cross-referencing the health data with the list of the main diseases affecting children and young people, there is a high frequency of

diseases linked to diarrhea and worms, which may be directly associated with the quality of the drinking water.



Graph 2: Quality of drinking water for household consumption. Source: Field research, 2023.
Elaboration: STA, 2023.

With regard to sanitary sewage in the communities, there is no public sewage system identified. The survey identified that for the majority of the surrounding communities, 89%, less than half of the houses have a bathroom or toilet. And 11% have nothing, neither a bathroom nor a toilet, as is the case in the Santo Antônio 1 community.

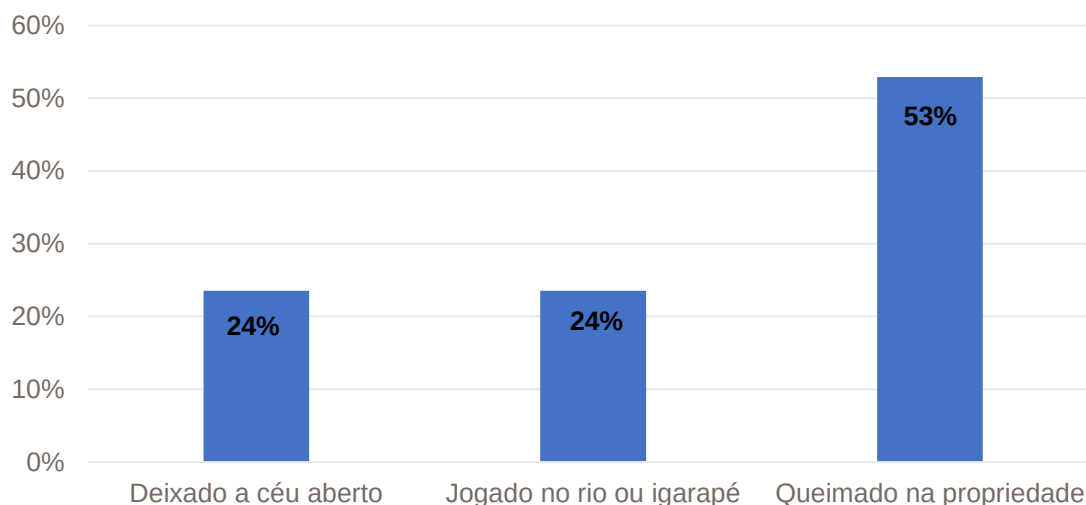


Graph 3: Existence of a bathroom or toilet in the communities Source: Field research, 2023.
Elaboration: STA, 2023.

Broadening the analysis, when asked where the sewage from the (existing) bathrooms or toilets is discharged, the communities reported that the sewage is discharged directly into nature or into the river.

With regard to the disposal of household waste in the communities, it can be seen that no community has a solid waste collection service. As such, families burn it on their plots or leave it out in the open. However, it was found that the Juruá-Mirim riverbed and the streams also receive waste that is discarded by residents, or even by boats traveling through the region.

Families burn it on their plots (53%), throw it in the river or stream (24%) or leave it in the open in some area of the community (24%).



Graph 4: Ways of disposing of waste in the communities. Source: Field research, 2023. Elaboration: STA, 2023.

- **Infrastructure and communication aspects**

As for the infrastructure and communication in the communities, it is notable that they all communicate via radio, receiving messages from relatives or friends in the city. However, there is no internet access or cell phone network. Some communities rely on satellite dishes to receive external information. In addition, the exchange of letters, carried by third parties along the river, is a practice still used for communication.

As far as building infrastructure is concerned, there is a significant variety in the communities. During the field research, it was observed that all the communities have some kind of infrastructure available. Using these spaces, such as community headquarters or churches, group meetings were held for the diagnosis.

In the Francisco Pimentel PDS, there are two public schools with school internet access, four Catholic churches and one Evangelical church. In addition, the settlement has only one shop to serve the families. In terms of access to electricity, the communities do not have a power grid, but rely on motor generators or solar panels. The predominant means of transportation is by river, using their own boats or those rented from third parties to travel between communities, the municipal headquarters and other locations.

- **Cultural aspects**

With regard to the social, cultural and historical heritage of the rural communities surrounding the Project Area, it is common to observe religious manifestations. The practice of praying for healing, especially for children, is widespread in most rural communities in the Amazon. Called cabocla or rural pajelança by some authors, this manifestation, a form of shamanism, has its roots in the cultural convergence between blacks and indigenous people, coexisting with Catholic practices and characterized by the cult of the enchanted.

The religious influence in the region is noticeable in the expansion of evangelical churches, which are moving into rural communities, including indigenous lands. This movement, which has been going on for decades, turns shaman leaders into evangelical pastors, often repressing other religious practices. Most of the communities have become evangelical in the last two decades due to an evangelization movement in the Juruá Valley.

Old traditions, such as the novenary, have been replaced in evangelical communities, showing a change in religious identity. In families that practice prayers for healing, the backyards have a diversity of medicinal plants, reflecting the integration between religious practices and traditional knowledge. In addition to religious practices, the communities preserve socio-cultural manifestations, such as religious and sporting activities, which were highlighted during the diagnostic survey and in the focus groups. These elements form a significant part of the communities' socio-cultural heritage.

Table 39: Religious events and festivities that communities hold or participate in. Source: Field research, 2023. Elaboration: STA, 2023.

Communities	Event name
Aldeota 2	Religious services
Monte Cristo	Religious services
Periquito	Soccer tournament / Religious services in other communities
Prainha 1	Soccer tournament / Religious services
Prainha 2	Soccer tournament / Religious services
Primavera	End of the year at church / Soccer tournaments / Religious services
Santo Antônio 1	Religious services
São Luís	No information on events.
Volta Grande	Christmas and New Year's Eve

Within the productive structure of the surrounding communities, subsistence farming stands out as the basis of productive-economic activities. This practice is complemented by agro-extractive activities, small animal husbandry and, in some cases, even cattle breeding.

The means of obtaining family income are varied, as shown in table below, which prioritizes the main productive activities contributing to income generation. In addition to the activities carried out in the area, many families rely on social benefits and/or retirement/pensions as an additional source of income.

Table 40: Activities that are sources of income for families. Source: Field research, 2023. Elaboration: STA, 2023.

Communities	Means of obtaining family income in the communities				
	1st Activity	2nd Activity	3rd Activity	4th Activity	5th Activity
Aldeota 2	Agriculture	Social Benefits	Retirement /Pension	Civil servant	Livestock
Monte Cristo	Agriculture	Social Benefits	Civil servant	Retirement /Pension	-
Periquito	Agriculture	Social Benefits	Civil servant	Livestock	-
Prainha 1	Agriculture	Social Benefits	Civil servant	Extractivism	Livestock (pig farming)
Prainha 2	Agriculture	Social Benefits	Civil servant	-	-
Primavera	Agriculture	Social Benefits	Civil servant	Retirement /Pension	-
Santo Antônio 1	Agriculture	Social Benefits	Civil servant	-	-
São Luís	Agriculture	Social Benefits	Retirement /Pension	-	-
Volta Grande	Agriculture	Social Benefits	Livestock (pig farming)	Civil servant	-
PDS Francisco Pimentel	Agriculture	-	-	-	-

As for family income, the majority of families receive one minimum wage per month, supplementing this fixed income with social benefits, pensions or retirements, especially Bolsa Família and Emergency Aid.

In the context of paid activities, the Monte Cristo community has women working as lunch ladies at the local school, as well as people working as boatmen. In Primavera, there are pensioners and civil servants, while in Volta Grande, most of the household groups earn a minimum wage. In Prainha 1, the Bolsa Família is considered a fixed income.

In family farming, manioc is the main product, mainly used to produce flour. The agricultural practice is characterized as itinerant farming, with a few years of cultivation followed by long periods of resting the land. The production method includes manual techniques, using tools such as sickles, axes and machetes.

Annual production is planned, with planting beginning in September and harvesting the following year. The Prainha 1 community also excels at producing and selling beans. In Aldeota 2, families often obtain advances in goods or money, and pay later with the flour produced, including items such as rice, beans, milk, sugar, coffee, preserves and oil. These practices reveal the economic dynamics and importance of family farming in these communities.

Table 41: Main products traded in order of importance. Source: Field research, 2023. Elaboration: STA, 2023

Communities	Classification of the main products sold			
	1st Product	2nd Product	3rd Product	4th Product
Aldeota 2	Flour	Cattle	-	-
Monte Cristo	Flour		-	-
Periquito	Flour	Cattle	-	-
Prainha 1	Flour	Cattle	Piassava and titica brooms	Pig
Prainha 2	Flour	Cattle	-	-
Primavera	Flour		-	-
Santo Antônio 1	Flour		-	-
São Luís	Flour		-	-
Volta Grande	Flour	Cattle	-	-
PDS Francisco Pimentel	Flour	Corn/Rice	Pig	Açaí/Bacaba

4.1.2 Interactions between Communities and Community Groups (VCS, 3.19; CCB, CM1.1)

The interaction between communities and community groups is good, mainly because of the geographical proximity between them. In this way, it is possible to see that students attend schools in communities to which they do not belong, depending on the availability of educational institutions in each community.

The communities also have a relationship with Seringal Rio Branco as they use the resources of the forest reserves, both for timber and non-timber extraction and for hunting and fishing. Section 4.1.3 describes the relationship between the communities and the Project Area in detail.

The interactions described will be strengthened and possible negative impacts, especially on relations between the communities and the Project Area, should be mitigated by the activities of the Seringal Rio Branco REDD+ Project.

4.1.3 High Conservation Values (CCB, CM1.2)

The concept of High Conservation Values (HCV) was developed by the Forest Stewardship Council (FSC, 1996)⁸¹ for the certification of timber products from responsible forest management, according to standardized Principles and Criteria that reconcile environmental and ecological safeguards with social benefits and economic viability (FSC, 2015)⁵⁴.

Every forest has some environmental and social value. The values that forests contain can include, among others, the presence of rare species, recreational areas, or resources collected by local people.

⁵⁴ Forest Stewardship Council (FSC). International Generic Indicators. FSC International Center, Policy and Standards Unit, Charles-de-Gaulle-Str. 5. Bonn, Germany

When these values are considered to be exceptional or of critical importance, the forest area can be defined as a High Conservation Value Forest (HCVF). The key to the concept of High Conservation Value Forests is the identification of High Conservation Value attributes: it is these values that are important and need to be protected. High Conservation Value Forests are simply forest areas where these values are found or, more precisely, the forest area that must be managed appropriately so that the identified values are maintained or enhanced.

In summary, a High Conservation Value Forest is the forest area required to maintain or increase a High Conservation Value attribute. By identifying the key values and ensuring that they are maintained or enhanced, it is possible to make rational management decisions that are consistent with protecting the important environmental and social values of a forest area.

Of the six criteria listed by the FSC, one is directly related to the forest and surrounding communities:

HCVA5: Areas and resources that are fundamental to maintaining the basic needs of local communities (subsistence, food, health, water, etc.);

In all the communities, forest reserve areas were identified, water reservoirs that are fundamental to supplying the basic needs of the local communities, such as subsistence, mainly linked to the use of water, feeding families and reserve of medicines, and building materials.

High Conservation Value	Essential areas and resources to maintain the basic needs of local communities (HCVA 5)
Qualifying Attribute	Areas used by the surrounding communities to extract medicinal plants, fruit, wood and water, as well as essential areas for subsistence hunting and fishing.
Focal Area	The areas used by the communities, such as forest reserves, streams and the Juruá Mirim River, require actions and initiatives to ensure that the structure of the environment is maintained and that forest resources are not depleted. The Seringal Rio Branco REDD+ Project has the potential to contribute to maintaining this attribute of high conservation value, whether by curbing deforestation and degradation of these areas by strengthening asset surveillance and remote monitoring in the Project Area or mainly through Social Scope activities, with actions to strengthen and organize non-timber forest management, encourage sustainable agricultural practices and promote environmental education, with direct and indirect impacts on the conservation of these areas in the Project Zone.

4.1.4 Without-Project Scenario: Community (CCB, CM1.3)

Current socio-economic indicators characterize a region with low welfare conditions for communities and few productive economic alternatives, causing families to seek better living conditions. As a result, some scenarios could lead to further deforestation in the region.

Economic activities: Family farming in the region tends to persist in the same pattern, with the traditional practice of slash-and-burn for the cultivation of manioc standing out, followed by the production of flour, the main source of income for families. This method gradually depletes the land, resulting in the future exhaustion of cultivable areas. This process, in turn, leads to expansion into the surrounding forest areas, increasing the areas of shrubbery and decreasing the areas of native forest. Finally, many communities practice hunting and fishing for subsistence and report that hunting, in particular, is mostly practiced by local people, which contributes to the loss of local biodiversity.

Low income: with the lack of alternative productive economic activities in the region, as well as informal work, many families find themselves in a situation of economic and food vulnerability. In addition, their income prospects diminish as they depend directly on the middlemen (Regatões) both for the marketing and channeling of family farming products and for the purchase of external inputs such as coffee, sugar, oils and fuel. The persistence of this dependency is attributed to the remote location of the communities, making access difficult, especially during the summer, when river levels are lower, making it difficult for larger boats to navigate. In addition, the high cost of local transportation, notably due to the high price of fuel for residents' boats, contributes to maintaining this low-income scenario.

Education levels: The difficult access to the communities by river limits education to elementary school, especially for children and young people, due to the lack of schools offering more advanced levels. The absence of daycare and private classes for children under five contributes to the low level of education in the region. Limitations in public policies for high school education and school transportation make it difficult to change this scenario. With the lack of qualifications and knowledge of new production techniques, income generation in the communities remains unchanged, with the exploitation of forest resources. The shortage of qualified labor restricts formal employment opportunities for young people in rural activities.

Lack of infrastructure: riverside communities face significant challenges due to their remote location, resulting in poor transportation, sanitation, energy, and telecommunication infrastructures. Geographical isolation and the lack of public policies contribute to this reality. This situation is reflected in various obstacles, such as the delay in families moving to more developed municipalities in search of better public services. In addition, the lack of basic sanitation leads to the use of water from rivers and streams in unsatisfactory conditions. Communication with the communities is affected by the lack of internet signal, cell phones and electricity, making it not instantaneous. Faced with this reality, these communities seek unsustainable alternatives to ensure their survival.

In view of this, it can be seen that the socio-economic conditions and infrastructure of the region can encourage illegal activities, such as predatory logging and the extraction of non-timber forest products, as well as illegal hunting and fishing, leading to a series of negative impacts on the ecological processes

of the forest and the depletion of natural resources of interest (ASNER et al., 2009)⁵⁵. It is also confirmed that agricultural activities with traditional slash-and-burn techniques, other agricultural crops, monocultures and logging and non-timber exploitation are the basis of subsistence and may represent the greatest potential for increased deforestation processes.

This factor sets up a future scenario in which the depletion of cultivable areas due to the repeated use of fire and swidden techniques requires new areas to be opened up. In addition, with the selective cutting of timber forest species, the communities' wood reserves could disappear. As a result, in the medium and long term, the pressure on the forest areas of Seringal Rio Branco could increase.

Given this situation, we can foresee two possible scenarios for deforestation in the Project Reference Region. Scenario 1 represents the continuation of the status quo (business as usual), without the REDD+ Project, leading to increasing pressure on forest resources and a consequent increase in deforestation. Scenario 2 shows actions aimed at socio-economic development based on the REDD+ Project, which could mitigate the impacts on forest resources and prevent deforestation in the region.

Actions within the scope of the Seringal Rio Branco REDD+ Project that stimulate the increase and improvement of income, especially with regard to "Promoting sustainable practices" (Section **Erro! Fonte de referência não encontrada.**) linked to the potential lines mapped by the project, are essential for achieving the objectives of reducing emissions from deforestation and degradation, enabling families to remain in rural areas and increasing the supply of properly produced food.

There is a need to stimulate the search for actions that can contribute to the development of public policies aimed at education, improving income conditions and infrastructure, as well as access to goods and services. Educating the rural population is essential for optimizing knowledge about the forest and its management, as well as guaranteeing better income and employment conditions. In addition, education is an important tool for the population to participate more in political and decision-making spaces regarding natural resources.

Another important measure for the success of these actions is "Strengthening Governance" and "Implementing, monitoring and evaluating the activities carried out by the Project" (Section **Erro! Fonte de referência não encontrada.**), promoting a more consolidated relationship with stakeholders, by strengthening and consolidating a local partnership to act in loco, and by monitoring the effectiveness of the activities proposed and implemented by the Project, contributing to the management of risks associated with the activities and driving the improvement of socio-economic aspects, guaranteeing access to a positive scenario and the smooth running of the Project.

Therefore, it can be concluded that the most likely scenario for the communities in the absence of the Project would be the continuation of the chain of events that lead to deforestation, such as low levels of income, little diversification of production combined with low productivity and unsustainable economic

⁵⁵ ASNER, G. P.; RUDEL, T. K.; AIDE, T. M.; DEFRIES, R.; EMERSON, R. A Contemporary Assessment of Change in Humid Tropical Forests. *Conservation Biology*, 23(6), 1386-1395. 2009. doi:10.1111/j.1523-1739.2009.01333.x

activities, difficulty in accessing basic infrastructure for a good quality of life, difficulty in accessing public policies and public services, among others.

The work of the Seringal Rio Branco REDD+ Project plays a crucial role in mitigating the challenges faced by the communities. Without the implementation of the project, the problems would persist, including limited access to public policies, the continuation of low-tech itinerant agriculture, characterized by low profitability, productivity and sustainability. In addition, the absence of sustainable economic practices, both in production processes and in hunting and fishing activities, would be a constant reality. Difficulty of access and precarious traffic would remain insurmountable obstacles, highlighting the importance of the Seringal Rio Branco REDD+ Project in the search for effective and sustainable solutions for these communities.

In this scenario, considering no significant improvement in public management models, the tendency would be for the rate of deforestation to remain the same or to increase, and with this the socio-economic context shown above would remain stagnant or worsen due to population growth and increased pressure from hidden causes of deforestation. In the event of a catastrophic scenario, it is possible that the situation of the communities surrounding Seringal Rio Branco will deteriorate even further:

- a) Social: continuity of levels of education, health, access to public policies that guarantee goods and services, communication, infrastructure such as better traffic on rivers and streams;
- b) Economic: stagnation and decrease in family income, agriculture and alternatives to promote diversification and verticalization of production, outlets for production, as well as the lack of sustainable production;
- c) Environmental: forest degradation, increased plundering of existing natural resources, as well as an increase in illegal hunting and fishing;

This scenario could lead to a rural exodus, that is, residents moving to the cities, where there is a risk of marginalization due to the poor conditions for absorbing labor in the region.

In the scenario with the presence of the Seringal Rio Branco REDD+ Project, the communities can be seen with increased levels of socio-economic conditions, reaching levels of development from their production to access to public policies that guarantee the continuity of families in the communities, avoiding rural exodus.

4.2 Net Positive Community Impacts

4.2.1 Expected Community Impacts (CCB, CM2.1)

The project's impacts were assessed based on the theory of change analysis and causal relationships, detailed in Section 2.1.11. Benefits, costs and risks were identified for all the communities in the initial diagnosis: Aldeota 2, Monte Cristo, Periquito, Prainha 1, Prainha 2, Primavera, Santo Antônio 1, São Luís, Volta Grande and PDS Francisco Pimentel.

In terms of direct impacts, the project offers opportunities such as capacity building for producers, access to training in sustainable practices, technical assistance services, the creation of spaces for participation, income generation, environmental awareness, environmental education, facilitating movement, community empowerment, communication with new markets and increased knowledge of agricultural production and REDD+.

The indirect impacts resulting from the proposed actions include empowerment in resource management, access to market information, increased self-esteem, greater access to public policies, conscious use of natural resources, keeping the rural community in the countryside, mitigating climate risks, increasing food availability and dialog with public agents.

As for costs, no significant costs are expected for community groups, only the investment of producers' time is considered a cost.

The potential risks for communities are related to the lack of stakeholder engagement, population growth and the consequent reduction in the supply of natural resources. The project aims to mitigate risks such as land invasions and deforestation through asset surveillance activities.

No additional negative impact is expected, as participation is voluntary, and there are no restrictions on land use for the established communities. The communities not served by the Project are also not restricted, avoiding negative impacts, as they will also not suffer any kind of restriction on land use, or be restrained from changing their way of life.

Community group	Aldeota 2, Monte Cristo, Periquito, Prainha 1, Prainha 2, Primavera, Santo Antônio 1, São Luis, Volta Grande and PDS Francisco Pimentel
Impact(s)	1. Producers trained in better production techniques; 2. Access to training on sustainable practices for the use of natural resources; 3. Access to technical assistance and rural extension services tailored to the reality of each community; 4. Creation of new spaces for participation, generating opportunities for direct communication with other stakeholders; 5. Income generation and diversification; 6. Environmental awareness and the permanence of families on their land; 7. Environmental education within the scope of hunting and fishing; 8. Facilitating movement and access to services and public goods;

	9. Integration with new markets; 10. Access to training on sustainable practices for the use of natural resources..
Type of benefit/cost/risk	<u>Expected benefits:</u> - Direct and indirect impacts on the community <u>Costs:</u> - Human and physical resources related to capacity building, training, and technical assistance. <u>Risks:</u> - Lack of engagement and interest from public institutions; - Lack of community engagement and the continuation of unsustainable agricultural and subsistence practices.
Change in well-being	- Income improvement; - Food security; - Technical improvement of economic activities - Improved local governance; - Territorial belonging.

4.2.2 Negative Community Impact Mitigation (VCS, 3.19; CCB, CM2.2)

As previously highlighted (Section 4.2.1), the Seringal Rio Branco REDD+ Project has no adverse impacts on the well-being of local communities. However, some potential risks have been identified, such as the lack of interest from other stakeholders, the possibility of population decline due to rural exodus and the lack of community engagement.

To mitigate these risks, it is possible to adopt measures such as strengthening the involvement of all stakeholders in decisions related to project activities, especially in meetings, and improving existing communication tools. With regard to rural exodus, mitigating strategies include the active participation of community members in discussions and decisions about project activities, as well as their participation in training and capacity building, with the aim of improving the well-being of the population.

In order to preserve and improve the High Conservation Value Area (HCVA), activities are proposed that involve the sustainable use of the area by the community for the extraction of non-timber forest products, the promotion of sustainable practices and environmental education related to hunting and fishing (Section 2.1.11). In addition, the implementation of actions such as improving asset surveillance, remote monitoring and encouraging sustainable practices contribute to the preservation of forest cover, which is crucial to the livelihoods of the surrounding communities.

4.2.3 Net Positive Community Well-Being (VCS, 3.19; CCB, CM2.3, GL1.4)

The Seringal Rio Branco REDD+ Project aims to promote sustainable socio-economic development in the communities surrounding the Project area, focusing on training and capacity building in sustainable practices. In the scenario without the Project, characterized by low income and lack of access to public policies and services, families resort to unsustainable and unplanned economic activities to supplement their income.

The Seringal Rio Branco REDD+ Project proposes activities aimed at the socio-economic and sustainable development of the communities, improving their well-being and quality of life, as well as promoting links with public and private partnerships to encourage investments and other actions aimed at the development of the communities.

The project aims to create opportunities for communities by causing the following net positive impacts:

- Improving production systems related to all the lines mapped out by the project, making them sustainable, implementing partnerships and increasing family income;
- Increasing community engagement through their participation in project activities, as well as strengthening human skills, knowledge and capacities related to economic activities;
- Increasing levels of knowledge about sustainable practices, both in terms of the extraction of timber and non-timber forest resources, as well as hunting and fishing activities, promoting the protection and conservation of forest cover and biodiversity, which are the families' means of subsistence and income;
- Permanence of families in the communities;
- Efficient communication, strengthening partnerships and integration with markets.

The main problems to be tackled in this context are:

- Low access to public policies, related to goods and services and infrastructure;
- Unsustainable economic activities, with low diversification and productivity;
- Difficulty of mobility and access;
- As a project, the aim is to influence social issues and the living conditions of the communities surrounding the project area, in order to reduce social vulnerability and rural exodus, providing families with an improved quality of life and income stability that will enable them to obtain the goods and services that promote economic and social well-being.

4.2.4 High Conservation Values Protected (CCB, CM2.4)

So far, during the preliminary assessment conducted with the DSEA studies (socio-economic and environmental diagnosis), no impacts were identified on the attributes of high conservation value related

to social issues (HCVA 5). However, should these be identified at some point in the future, measures should be taken to ensure that there are no net negative impacts on the attributes.

In order to guarantee that no HCVA related to the well-being of the communities will be negatively affected, the activities proposed by the Seringal Rio Branco REDD+ Project incorporate measures aimed at protecting and conserving forest areas used by the communities as a means of subsistence, food, health, as well as cultural and historical aspects. Activities related to promoting sustainable practices, non-timber forest management, environmental education related to hunting and fishing activities, the promotion of public incentives aimed at improving infrastructure, as well as activities related to monitoring forest cover and asset surveillance, promote the protection of HCVAs, preventing the loss of natural resources and illegal activities in these areas.

4.3 Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.1)

In the context of the Seringal Rio Branco REDD+ Project, no adverse impacts on other stakeholders are expected, which makes their occurrence unlikely. On the other hand, there are positive impacts that could generate benefits for various stakeholders, such as:

- All local communities and other residents in the project region, regardless of their participation in the activities, will benefit from the positive impacts related to the conservation and protection of forest cover and biodiversity.
- The sustainable development promoted by the project will bring benefits to all communities and other stakeholders, providing opportunities that will improve quality of life and well-being.
- In addition to the direct benefits of the project's activities, all stakeholders in the region will benefit from improved access to public policies.

It is important to note that, as indicated above, the likelihood of adverse impacts resulting from these activities is low, with the following being potential but unlikely:

- Lack of engagement of communities and other stakeholders in project activities and other initiatives.
- Possible failures in the communication of the project's actions, which could result in conflicts during the implementation and conduction of the activities.

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

As mentioned above, the occurrence of adverse impacts between the various actors involved in this project is not anticipated. In order to mitigate possible negative effects, participatory strategies are being adopted both in the design of activities and in decisions regarding the most appropriate time and structure for interaction. This includes the collaborative drafting of the agenda in order to minimize overlapping activities, a practice that is already underway. In addition, a communication plan has been

drawn up which incorporates procedures for resolving conflicts. If these measures do not prove effective, it is recommended to adapt the ways in which conflicts are communicated and dealt with.

4.3.3 Net Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

As explained and detailed in Section 4.3.1, no other adverse effects on the well-being of different groups of local actors are anticipated, since the project does not restrict access to natural resources in the area in question for any of the agents originally dependent on these resources. The actions planned in relation to the surrounding communities are based primarily on collaboration with government agencies and other local institutions, with the aim of promoting improvements in living conditions, expanding access to public policies and improving practices already in place. The scope of the activities outlined and proposed for this project is aimed exclusively at impacts that foster the inclusion and well-being of the communities and other parties involved.

4.4 Community Impact Monitoring

4.4.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

Monitoring the project's impact on communities and other stakeholders is a crucial management tool, making it possible to effectively evaluate the effectiveness of activities in relation to the established objectives. In this context, it is proposed to implement a monitoring system based on the stipulated goals to define the indicators to be collected. This system should incorporate verification tools and follow specific procedures for analyzing and evaluating results, in order to identify, when necessary, the measures needed to improve the desired progress.

Monitoring the benefits to communities has four components:

- Non-timber forest management, with monitoring of the number of families practicing this activity, as well as initiatives such as courses, training, procedures and established partnerships that seek to improve non-timber forest management practices in the communities in order to guarantee subsistence and income in the long term;
- Promoting sustainable practices, by monitoring initiatives such as courses, training and partnerships that aim to develop and improve sustainable practices in the communities, as well as promoting adaptive production systems that minimize the impacts of climate change;
- Promotion of environmental education for hunting and fishing, with the monitoring of initiatives, such as actions and strategic partnerships established with a focus on environmental education in the communities, with the aim of raising awareness among stakeholders in relation to the main species culturally hunted and fished, mapping unsustainable practices and seeking environmental education solutions to be implemented to guarantee the cultural habits of the communities while seeking a mechanism for maintaining forest resources susceptible to depletion and climate change;
- Articulations with organizations for the insertion or maintenance of infrastructures: constant monitoring of initiatives will be carried out, including actions and strategic partnerships focused on the

implementation and maintenance of crucial infrastructures in these remote communities. The central objective is to ensure not only the effective implementation, but also the long-term sustainability of these improvements. During this process, there will be a special emphasis on sensitizing stakeholders to the specific needs of these communities, seeking to actively involve them in the development and maintenance of infrastructure.

As such, the Community Impact Monitoring Plan basically covers quantitative indicators related to both the processes and products achieved in the short term and the medium and long term results. It is planned to refine this initial monitoring plan at a later date, through evaluation and validation by stakeholders.

Finally, the plan aims to preserve the High Conservation Value Areas (HCVA) through socio-environmental activities (detailed in Section 2.1.11) aimed at curbing deforestation and degradation in these areas. It also seeks to organize and improve traditional economic and subsistence activities, such as non-timber forest management and other sustainable practices .

The Community Impact Monitoring Plan includes process indicators and part of the results indicators relating to the activities carried out for the project. Whenever necessary, this initial monitoring plan will be revisited and supplemented, always with the assessment and validation of the stakeholders.

a) Data to be collected:

Data / Parameter	Number of partnerships contacted
Data unit	Number
Description	The purpose of this parameter is to measure the number of negotiations, approaches and partnerships that the project has made and/or tried to make throughout its life cycle, with the aim of contributing to the development and improvement of actions and activities linked to the project's social activities.
Data source	Reports (e.g. follow-up report on project activities that have been implemented), contracts, memos, emails, meeting minutes and/or other documents that corroborate as evidence that a partnership has been established and built up.
Description of the measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the project's accreditation period. In this way, the realization of partnerships linked to the Project's social activities will be monitored and accounted for
Monitoring/recording frequency	At each verification event
Value applied	To be accounted for after the start of the project
Monitoring equipment	Not applicable
QA/QC procedures to be	The systematized information on the partnerships established for social

applied	activities will be validated among the proponents, allowing for greater reliability and quality of the data. In addition, the project will undergo continuous evaluation of the information generated, by identifying improvements in the collection and recording processes and, when appropriate, incorporating adjustments into the project's strategic planning.
Purpose of the data	Demonstrate that the project is dedicating itself to and expanding its social scope through its partnerships
Calculation method	Not applicable
Comments	Not applicable

Data / Parameter	Number of trainings
Data unit	Number
Description	The purpose of this parameter is to measure the number of courses, qualifications and trainings carried out in the social activities proposed by the Project
Data source	Reports (e.g. follow-up reports on project activities that have been implemented), participants' attendance lists, contracts, photos, publicity materials, among other documents
Description of the measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the project's accreditation period. In this way, the training courses carried out as part of the Project's Social Scope activities will be monitored and accounted for
Monitoring/recording frequency	Annual
Value applied	To be accounted for after Project registration
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information from the qualifications, courses and training proposed by the social activities will be validated among the proponents, allowing for greater reliability and quality of the data. In addition, the project will continually evaluate the information generated, identifying improvements in the collection and recording processes and, when appropriate, incorporating adjustments into the project's strategic planning.
Purpose of the data	Not applicable
Calculation method	Not applicable
Comments	Not applicable

Data / Parameter	Number of people benefited
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Data unit	Number
Description	This data will count all the people or groups of people who have benefited from the Project, through the planned actions and activities related to the Social Scope, over the years of implementation and monitoring.
Data source	Reports (e.g. activity monitoring reports), interviews, feedback on results and/or consultations, attendance lists, presentations, and other documents that corroborate evidence that a person, whether a community member or not, has benefited from the project.
Description of the measurement methods and procedures to be applied	All the documents produced will be stored in digital files throughout the project's accreditation period. In this way, information on the people benefiting from the Project's Social Scope activities will be monitored and accounted for.
Monitoring/recording frequency	Annual
Value applied	To be accounted for after Project registration
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information on the people benefiting from the social activities will be validated among the proponents, allowing for greater reliability and quality of the data. In addition, the project will undergo continuous evaluation of the information generated, by identifying improvements in the collection and recording processes and, when appropriate, incorporating adjustments into the project's strategic planning.
Purpose of the data	Not applicable
Calculation method	Not applicable
Comments	Not applicable

Data / Parameter	Community engagement and adherence to the project
Data unit	Number
Description	The purpose of this parameter is to measure the adherence and involvement of community members in the project's proposals and activities, as a direct impact of the actions foreseen in the Project's Social Scope.
Data source	Reports (e.g. activity monitoring reports), interviews, feedback on results and/or consultations, attendance lists, presentations, and other documents that corroborate evidence that a person, whether a community member or not, has benefited from the project.
Description of the measurement methods and	All the documents produced will be stored in digital files throughout the project's accreditation period. In this way, the information on the increase

procedures to be applied	in the social organization of these community members, based on the development of the actions of the Social Scope of the Project, will be monitored and accounted for.
Monitoring/recording frequency	At each verification event
Value applied	To be accounted for after the start of the project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information on the increase in the social organization of these community members will be validated among the proponents, allowing for greater reliability and quality of the data. In addition, the project will undergo continuous evaluation of the information generated, identifying improvements in the collection and recording processes and, when appropriate, incorporating adjustments into the project's strategic planning.
Purpose of the data	Not applicable
Calculation method	Not applicable
Comments	Not applicable

Data / Parameter	Infrastructure, goods and services implemented
Data unit	Number
Description	The purpose of this parameter is to measure the number of improvements and investments made in infrastructure, the acquisition of goods and the provision of services to the community, which have been made possible by the activities provided for in the Project's social scope.
Data source	Reports, contracts, projects, invoices, photographs, meeting minutes and/or other documents that corroborate as evidence that there have been improvements and investments in infrastructure, goods and services for the communities involved in the project.
Description of the measurement methods and procedures to be applied	All the documents produced will be stored in digital files throughout the project's accreditation period. In this way, information on improvements and investments in infrastructure, goods and services for the communities will be monitored and accounted for.
Monitoring/recording frequency	At each verification event
Value applied	To be accounted for after the start of the project
Monitoring equipment	Not applicable
QA/QC procedures to be	The systematized information on improvements and investments in

applied	infrastructure, goods and services for the communities will be validated among the proponents, allowing for greater reliability and quality of the data. In addition, the project will undergo continuous evaluation of the information generated, identifying improvements in the collection and recording processes and, when appropriate, incorporating adjustments into the project's strategic planning.
Purpose of the data	Not applicable
Calculation method	Not applicable
Comments	Not applicable

b) Summary of the data collection procedure

Data will be collected during and after the activities with the stakeholders, as well as whenever a dialog is established with a possible local partner responsible for carrying out some activity on site. This information will be systematized and presented in the project's social activity reports.

c) Quality control and assurance procedures

The data collected and shown in the reports will be presented and validated during meetings between the proponents and potential partners, as well as in meetings with stakeholders, with the aim of presenting the results achieved by the project, receiving suggestions and criticisms for improving the actions.

d) Data archiving

All the data and reports produced by the Seringal Rio Branco REDD+ Project will be stored by Biofílica Ambipar Environment Investments in digital files throughout the project's life cycle. Original (physical) reports, minutes of meetings and field records produced during the execution of social activities will be digitized and subsequently stored by Amazônia Agroindústria EIRELI, as well as by a possible local partner who will act in locu. Biofílica Ambipar Environment Investments will receive and keep a copy of these documents in digital format for the duration of the project.

e) Organization and responsibilities of the parties involved in the above

All monitoring activities are the responsibility of Biofílica Ambipar Environment Investments, Amazônia Agroindústria EIRELI and a possible local contracted partner.

4.4.2 Monitoring Plan Dissemination (CCB, CM4.3)

The monitoring plan, as well as the results obtained from monitoring the Seringal Rio Branco REDD+ Project, will be made available to the public via a page on the official website of Biofílica Ambipar Environment Investments. The summarized documents relating to the monitoring plan and results, as well as relevant information, will be made available to communities and interested parties through meetings, lectures and by physical means on the premises of Amazônia Agroindústria EIRELI.

5 BIODIVERSITY

5.1 Without-Project Biodiversity Scenario

5.1.1 Existing Conditions (VCS, 3.19; CCB, B1.1)

The Seringal Rio Branco REDD+ Project is located in the southwestern region of the Amazon, where the area of endemism known as the Inambari Endemism Center is found, characterized by a high diversity of endemic species. In addition, the region is located close to the Serra do Divisor National Park, which is considered to be one of the richest regions in terms of biodiversity in the Amazon, giving the region a high level of endemism (Juen & De Marco, 2012; Pedroza et al. 2020).

This region, as well as the whole Amazon, is of great importance on the national and international scenario due to its biological relevance. Because of this, several studies have looked at future scenarios for the Amazon, where it is estimated that, by 2050, the loss of forests in the Amazon will reach its most severe scenario, being between 42 and 76% (depending on the scenario adopted), with evidence for severe population losses of rare species in more than 80% of their individuals (Ter Steege et al., 2015). In this sense, deforestation is one of the main problems from the point of view of local biodiversity conservation. In October 2023, around 290 square kilometers of deforestation were detected in the Legal Amazon, 8% of which was located in the state of Acre alone (Fonseca et al. 2023). This deforestation has presented a different scenario in recent years, where areas that were once designated for family farming have been converted to livestock farming, registering an exponential increase in cattle herds in recent years, with approximately 4 million head of cattle in 2021, an increase of 48% in five years (IBGE, 2019).

With a view to the sustainable use of forests and its preservation, Acre has a historical context with the composition of barriers against non-forest occupation in its territory. Where 46% of the state's area is occupied by protected areas, made up of conservation units and indigenous lands (ACRE, 2009). The area of influence of the project has a protected area percentage of 37.37%, containing 3 conservation units (PARNA Serra do Divisor, RESEX Riozinho da Liberdade and Area of Relevant Ecological Interest Japiim Pentecoste), as well as 4 indigenous lands.

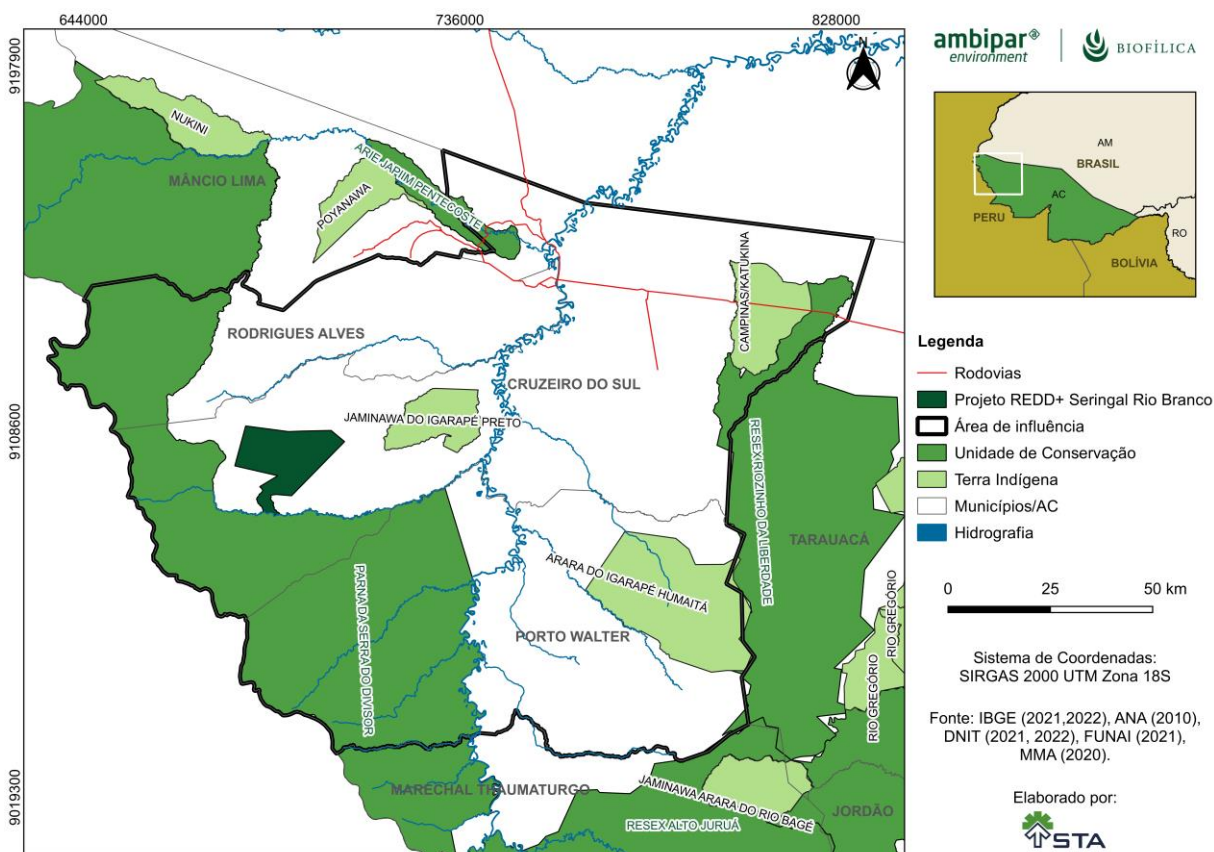


Figure 38: Map of the Seringal Rio Branco REDD+ Project Area in the context of the protected areas (conservation units, indigenous lands and settlement projects) in the project's area of influence.

Flora

The classification of vegetation in the project area was based on IBGE data, where three types of vegetation were observed: Alluvial Open Ombrophilous Forest with palm trees (Aap/FAP-Alluvial - 1.525% of the project area), Lowland Open Ombrophilous Forest with bamboos (Abb/FABD - 5.430% of the project area) and Lowland Open Ombrophilous Forest with palm trees (Abp/FAP - 93.040% of the project area). The phytosociological survey was carried out on eight plots of 1 hectare each chosen at random from the 34 plots established for the carbon stock inventory in the project area. 2 plots were allocated to the FAP-Alluvial (plots 1 and 13), 2 plots to the FABD (plots 3 and 4) and 5 plots to the FAP (7, 18, 20 and 26); the greater number of plots in the FAP is justified by the predominance of this typology in the project area (Table 2 shows that this vegetation corresponds to 93% of the project area). These plots were divided into 5 transects measuring 20 x 100m, which were used to calculate the frequency of species. All the trees and palms with a DBH > 10 cm were counted, their circumference measured at a height of 1.30 m ("CBH", in centimeters) and marked with an aluminum plate for later checking; subsequently, the CBH was transformed into diameter (DBH = CBH/π, expressed in meters).

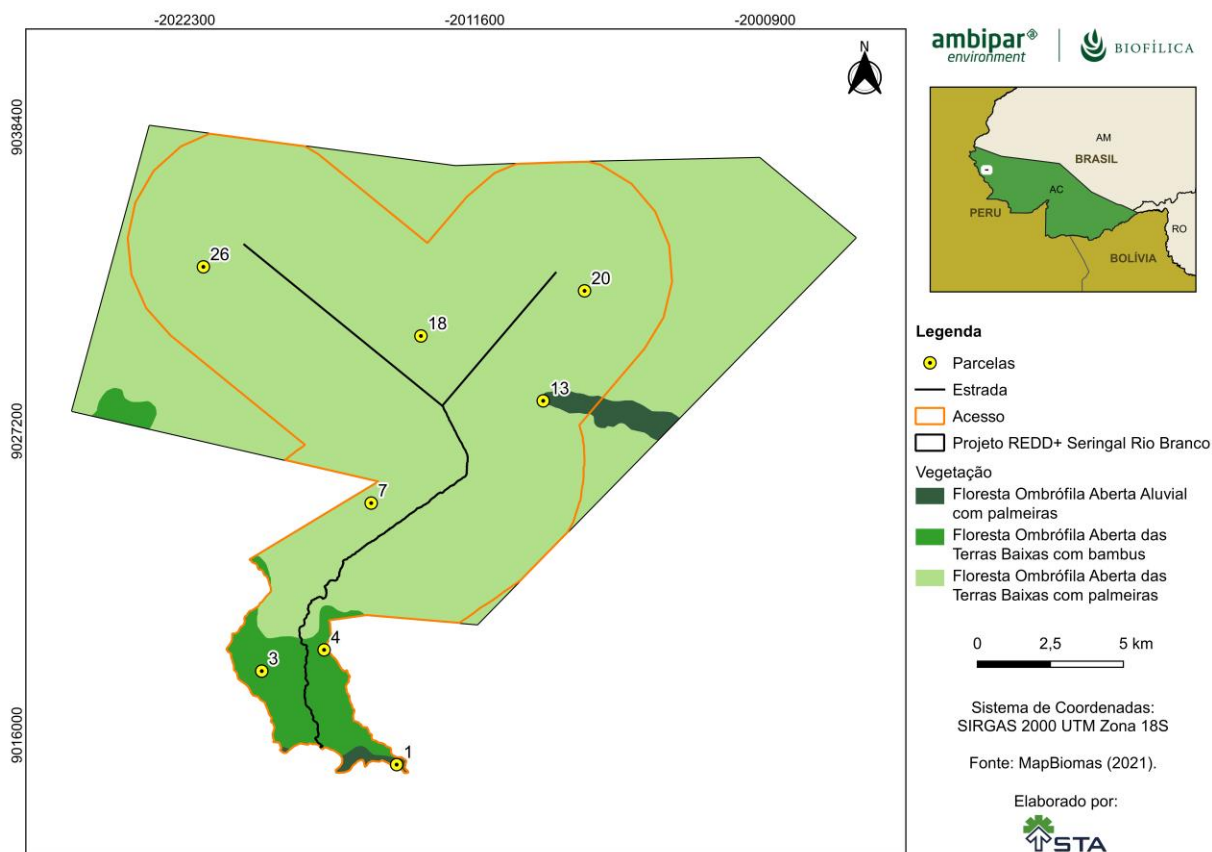


Figure 39: Map showing the distribution of the plots used to carry out the Floristic Inventory in the project area.

The data was analyzed separately for each forest type. The diametric structure and phytosociological parameters of the populations were analyzed, namely: absolute density (D_{ab} , in number of individuals per hectare), relative density (D_r , in %), absolute frequency (F_{ab} , in number of transects with the presence of the species), relative frequency (F_r , in %), absolute dominance (DO_{ab} = basal area (AB , in m^2), relative dominance (DO_r) and importance value index (IVI , in %). The phytosociological parameters were calculated according to Brower et al (1998).

The results for the three forest typologies studied showed high species diversity and very similar abundance patterns. The density of individuals and basal area (Table 44) reveals very high values for open forests, especially the FABD (bamboo forest) which showed the highest values of individuals density and basal area.

Table 42: Density of individuals and basal area

Plots	Density (individuals ha ⁻¹)	Basal area (m ² ha ⁻¹)	S	H'	J	Comparative Estimated Richness*		
						q=0	q=1	q=2
1. FAP-Alluvial – Plot 1	415	18,074	64					
2. FAP-Alluvial – Plot 13	608	24,994	83					
FAP-Alluvial (general)	511,5	21,534	107	4,67	0,84	119,8 3	90,2 8	73,05
3. FABD – Plot 3	549	30,735	85					

4. FABD – Plot 4	595	27,956	82					
FABD (general)	572	29,346	112	4,03	0,85	124,0 0	94,0 3	78,58
5. FAP – Plot 7	510	26,794	72					
6. FAP – Plot 18	546	25,193	95					
7. FAP – Plot 20	519	29,789	82					
8. FAP – Plot 26	580	26,789	143					
FAP (general)	538,75	27,141	143	3,95	0,80	156,8 4	98,5 8	74,96
* Maximum sampling coverage = 98,8%								

The results of the Shannon-Weaver indices and Pielou's evenness can be well visualized in the abundance model, as proposed by Hubbell (2013) (Figure x). In this figure we can see that the curves of the three forest types are very similar, especially among the very abundant species (also called hyperdominant). In the three forest types, the number of hyperdominant species (between 11 and 14 species held 50% of the individuals) was very similar. Hubbell (2013) also showed that tropical forests have a long list of rare species, as well observed in the long tail formed mainly by the FAP (Figure x). The longer tail in the FAP is formed by the high number of rare species, which was also seen in the differences in the richness estimates at $q=0$ and which also justifies the lower value of Pielou's uniformity.

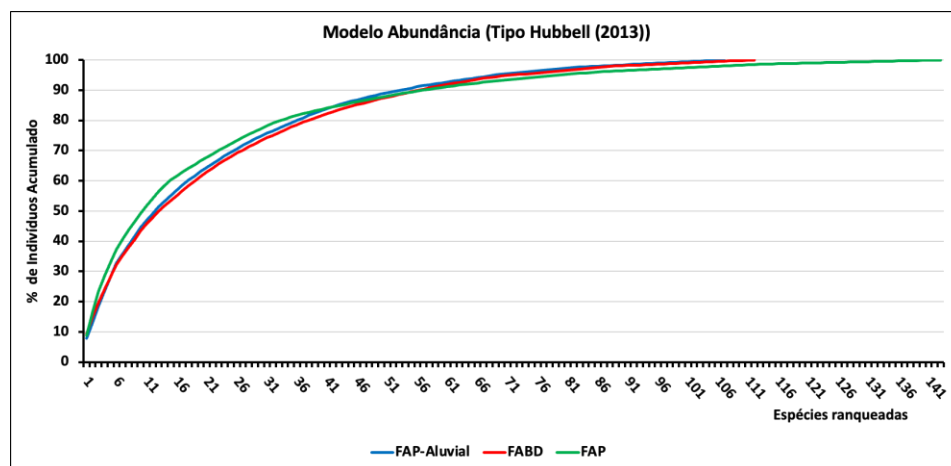


Figure 40: Hubbell's abundance model (2013).

The floristic similarity analysis corroborates the results shown above: the three typologies are highly similar in their floristic composition. Species sharing between the typologies is high (Figure 17), the Sorensen coefficients were greater than 0.5; between Alluvial FAP and FABD it was 0.64; between Alluvial FAP and FAP it was 0.62; and between FABD and FAP it was 0.74. These results reinforce the idea that these three typologies can be united as a single typology, that is, at the level of subgroups of formations, as proposed by the IBGE (2012). Considering the diametric distribution of the inventoried area, it was possible to observe great similarity in the three forest types, showing a negative exponential behavior, commonly called J-inverted (Figure x). The negative exponential diametric distribution is found in natural, undisturbed, self-regenerating environments with trees of different ages and species. This pattern usually reveals stable environments, found in undisturbed Amazonian forests (Souza et al 2006; Santos et al 2016; Lima et al 2019). In this distribution we can see that

the majority of individuals are in the first DBH class (around 60% of individuals) and as the DBH class increases the number of individuals decreases exponentially.

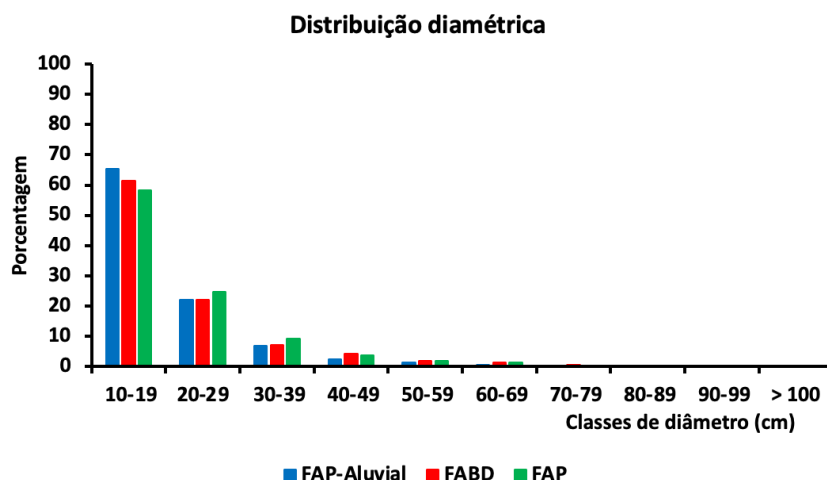


Figure 41: Diametric distribution of the inventoried area classified by the different forest typologies.

In the eight hectares, 4322 individuals belonging to 185 species were inventoried; of these, 108 individuals of 19 species were identified only by common name and 117 species were identified by specific epithet. The floristic composition found in the inventoried plots is typical of the region and is in line with the most dominant species in the Amazon, according to the results of Ter Steege et al. (2013, see especially the appendix presented by this author).

According to Flora e Funga do Brasil (2023), the state of Acre contains 4,331 species of Angiosperms, that is, the list of species in the project area makes up 44.6% of the state list. Tree species alone (956 species) account for 19% of the list of nominally known tree species in the Amazon (Ter Steege et al., 2020). This confirms the enormous regional diversity found in these three analyzed municipalities and adds to the pattern of high diversity found in western Amazonia.

Considering the lists of probable occurrence and confirmed occurrence in the project area, 1,931 species belonging to 138 botanical families were identified, with only 10 families being the richest in species (containing 40% of the species): Fabaceae, Rubiaceae, Araceae, Melastomataceae, Annonaceae, Euphorbiaceae, Piperaceae, Moraceae, Arecaceae and Malvaceae.

• Threatened Species

Considering the IUCN Red List (2023b), MMA (2022), CNCFLORA (2023), 17 species were found in the surrounding project area, four of which are near threatened and 13 of which are threatened to some degree; eleven species are of potential occurrence and six species of confirmed occurrence. Among the species of potential occurrence, five are on the IUCN list, six are on the MMA list and eleven are on the CNCFLORA list; among the species of confirmed occurrence, five are on the IUCN list, three are on the MMA list and six are on the CNCFLORA list.

The species *Bertholletia excelsa*, *Virola surinamensis* and *Swietenia macrophylla* are species of high commercial value, which have already suffered strong reductions in their populations; *Cedrela odorata*, *Handroanthus impetiginosus* and *Handroanthus serratifolius* are also species of high commercial value, not yet widely exploited, but with a projected reduction in their populations to occur in the future if they continue to be exploited; *Pleurothyrium prancei* has an area of occurrence < 20. 000 km², it is a species that only occurs in the state of Acre, and its area of occurrence is under threat from deforestation; *Cathedra paraenses*, which is on the list because its area of occupation is < 500 km², shows continued decline in: (i) extent of occurrence; (ii) area of occupation; (iii) area, extent and/or quality of habitat. The situation of *Qualea brevipedicellata* is similar to that of *Cathedra paraenses*, however its decline is linked to the fact that its locations are under threat. The reduction in the size of populations is directly related to their exploitation, while the problem of distribution area is related to habitat loss, due to deforestation and climate change. Forest communities may experience sharp declines in species in the near future (Gomes et al 2019); drought-adapted species may be expanding their geographical range and increasing in abundance (Esquivel-Muelbert et al 2019).

Table 43: List of extinction threatened species with potential or confirmed occurrence in the project area and area of influence. Degree of threat: VU = vulnerable; EM = endangered; NT = near threatened.

Nº	Species (Common name)	Global	National (Brazil)	
		IUCN (2023b)	MMA (2022)	CNC FLORA (2023)
Potential occurrence				
1	<i>Apuleia leiocarpa</i> (Vogel) J.F. Macbr. (cumaru-ferro; Cumaru-cetim; Amarelão)	-	VU	VU
2	<i>Bertholletia excelsa</i> Bonpl. (Castanha-do-Brasil)	VU	VU	VU
3	<i>Cathedra paraensis</i> Sleumer (Cajuzinho)	EN	EN	EN
4	<i>Cissus acreensis</i> Lombardi	-	-	NT
5	<i>Heteropsis flexuosa</i> (Kunth) G.S.Bunting (cipó-titica)	-	VU	VU
6	<i>Heteropsis spruceana</i> Schott (cipó-titica)	-	-	NT
7	<i>Hymenaea parvifolia</i> Huber (Jutaí)	-	VU	VU
8	<i>Pleurothyrium prancei</i> van der Werff	VU	VU	VU
9	<i>Qualea brevipedicellata</i> Stafleu (Mandioqueira)	NT	-	NT
10	<i>Tapura acreana</i> (Ule) Rizzini (pau-ferrugem)	-	-	EN
11	<i>Virola surinamensis</i> (Rol. ex Rottb.) Warb. (Ucuúba)	EN	VU	VU
Confirmed occurrence				
12	<i>Cedrela odorata</i> L. (Cedro)	VU	VU	VU
13	<i>Cordia hadrantha</i> (Standl.) C.H.Perss. & Delprete (Apurui)	-	EN	EN
14	<i>Handroanthus impetiginosus</i> (Mart. ex DC.) Mattos (Sinônimo de <i>Tabebuia impetiginosa</i> (Mart. ex DC.) Standl.)(Ipê rosa)	NT	-	NT
16	<i>Handroanthus serratifolius</i> (Vahl) S.Grose (Ipê amarelo)	EN	-	NT

17	<i>Swietenia macrophylla</i> King (Mogno)	VU	VU	VU
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Fauna

The vertebrate inventory carried out in the Seringal Rio Branco REDD+ Project area was based on a 12-day field expedition, with 7 to 9 actual days of data collection. All vertebrate groups were sampled, including terrestrial mammals, birds, fish, reptiles and amphibians. For all groups, 10 to 18 sampling points were defined, distributed along a main footpath of around 14 km, opened up in the study area (Figure 6).

It should be noted that the survey's main objective was to confirm the occurrence of as many species as possible in the area, in order to quantitatively and qualitatively characterize the region's fauna. In this way, the sampling design was designed with the distribution of sampling points in mind, combining the logistics available in the field with the maximum spatialization of the points, in order to sample the maximum environmental variability in the study area (Figure 6, Figure 7 and Figure 8).

The data collection was based on bibliographic consultation of all lists of threatened species, including ICMBio's national list and the IUCN's international list.

• Herpetofauna

In order to collect the herpetofauna data, a systematic search methodology was applied. For each of the 9 sampling points, a stretch of about 1,000 meters was covered on the main footpath, and every 2,000 meters a stretch of about 300 meters of footpath perpendicular to the main footpath was delimited, where the search was carried out on both sides. The effort used for each point was 3 hours during the day and 3 hours at night on each footpath, totaling 36 hours of searching over 8 days.

Another active search method used was looking for breeding sites, in which the expert herpetologist located and walked around the perimeter, along the banks and/or inside marshes, streams, ponds and puddles in the study area, during the twilight and night periods. In all the searches, the species found were recorded by visualization or vocalization. In the case of sightings, as the transect was traveled in only one direction, each sighting was considered a record, and the observer never returned to the footpath to avoid counting the same animal more than once. In the case of vocalizations, the direction and distance of the vocalization were evaluated in order to consider different records of the same species. Samplings on the main footpath were also efficient for detecting snakes in particular (Fitch, 1987).

As a complementary method, an exhaustive secondary survey was carried out, using data from bibliographies, databases and scientific collections, in an attempt to record as many species of herpetofauna as possible with confirmed occurrence in the Seringal Rio Branco REDD+ Project region.

Based on the methods described, a total of 29 species were confirmed in 48 herpetofauna records in the Seringal Rio Branco REDD+ Project area. With the exception of one species of exotic lizard (*Hemidactylus mabouia*), all the other species of reptiles and amphibians recorded are native to the Amazon. All the species utilize the area for use, reproduction and nesting, and no migratory species were recorded in the primary survey. Reptiles were the richest, with 17 species, while amphibians accounted for 12 species. The anuran group had a higher species richness than any reptile group, and also had a higher abundance of total records. Among the amphibians, the most commonly sampled species were *Alobates femoralis* (Brilliant Tighed Poison Frog), and *Rhinella marina* (South American Cane Toad).

All the anuran and squamate species inventoried, which are considered endemic to the Amazon region, are listed as "of little concern" on the national and international threatened species list, probably because they are abundant in the area they occupy and have a wide distribution in the Biome (Rodrigues et al., 2010). Among the reptiles, the most common species was the lizard *Anolis fuscoauratus* (figure x). This species occurs on trunks, branches and lianas, and can be found in lightly disturbed forests, but can also be found in degraded areas or forest edges.

The two species of chelonian reptiles with confirmed occurrence in the area, *Chelonoidis denticulata* (Yellow-footed Tortoise) and *Podocnemis unifilis* (Yellow-spotted River Turtle), are classified on the international list of threatened species as vulnerable (VU) (figure x). The species *Podocnemis unifilis* is endemic to the Amazon region but is quite common in the places where it occurs. However, these two species of chelonians are highly prized for consumption by human populations in the Amazon, both for their meat and eggs, which could be a threat to the species in some parts of this Biome (Salera et al., 2009).



Figura 42: Reptiles lizards and chelonians recorded in the area of the REDD+ Project Seringal Rio Branco, Municipality of Cruzeiro do Sul, AC. A) *Gonatodes humeralis* (Trinidad Gecko); (B) *Anolis fuscoauratus* (Brown-eared Anole); (C) *Podocnemis unifilis* (Yellow-spotted River Turtle); (D) *Chelonoideis denticulata* (Yellow-footed Tortoise).

The survey carried out in the project area recorded the amphibian species *Rhinella marina* (South American Cane Toad), a native species that can be quite common in the Amazon. Elsewhere on the planet, this species

can be considered a voracious invader (Global Invasive Species, 2005). The amphibian species *Allobates femoralis* and *Oreobates quixensis* (figure x) can be considered bioindicator species of degraded habitats, since their natural habitats are low-altitude tropical forests, freshwater swamps and highly degraded secondary forests.

Only one exotic species of lizard was recorded, *Hemidactylus mabouia*, which is the common lizard, an exotic species from Africa, but quite common in Brazil (Anjos & Rocha, 2008) and appears on the international list of threatened species. It is worth highlighting the species *Alopoglossus buchkleyi*, a species of lizard recorded in the primary survey that is considered rare and has few records (figure x).

For the snake group we can highlight the species *Bothrops atrox* (Common Lancehead) and *Bothrops bilineatus* (Green Jararaca), which are very venomous species and occur in forest environments. All the amphibian and squamate species recorded in the inventory of the Seringal Rio Branco REDD+ Project area, which appear on the national or international threatened species list, are considered to be in the "Least Concern" category. Only the chelonian species recorded are considered to be in the "Vulnerable" category on the international list of threatened species.

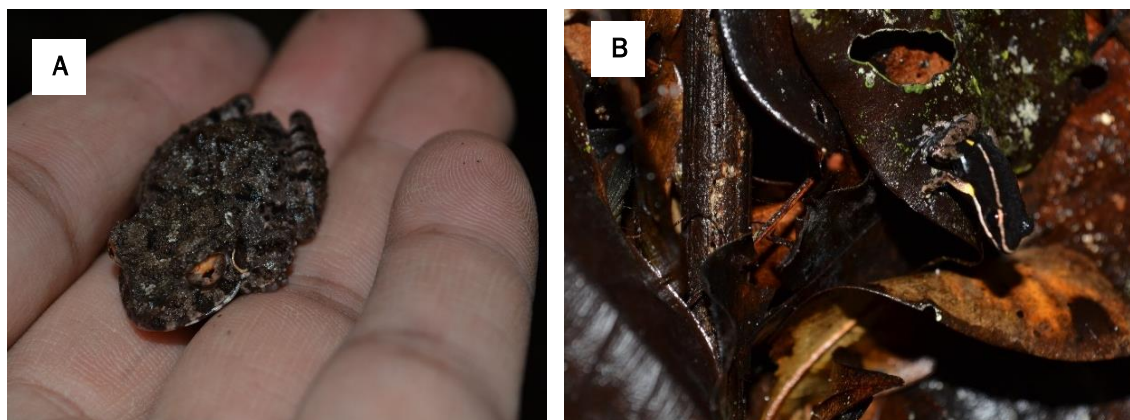


Figure 43: Anuran amphibians recorded in the Seringal Rio Branco REDD+ Project area, Cruzeiro do Sul municipality, AC. (A) *Allobates femoralis* (Brilliant Tighed Poison Frog) and (B) *Oreobates quixensis* (Common-Bigheaded Frog).



Figure 44: Reptile lizards recorded in the area of the REDD+ Project Seringal Rio Branco, Municipality of Cruzeiro do Sul, AC. (A) *Alopoglossus buchkleyi* and (B) *Hemidactylus mabouia* (Lizard).

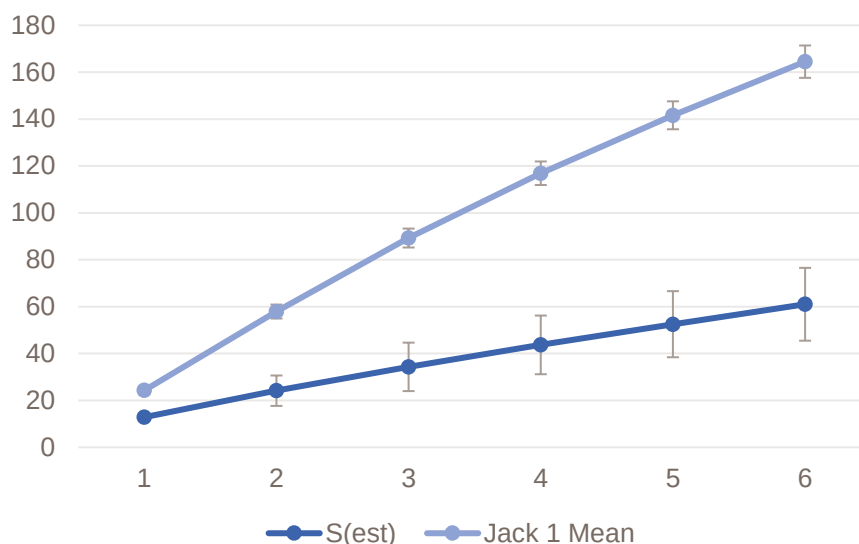
- **Avifauna**

In the bird inventory of the Seringal Rio Branco REDD+ Project area, the listening point census method was used to sample the birds. Using the main footpath, which is approximately 14 km long, stretches of 2,000 meters were defined to be walked daily, during the day and night. The samplings were carried out along this transect, where an observer and an assistant walked the entire transect, stopping for 10 minutes every 200 meters. A vocalization recorder and camera were used to facilitate species identification. The census was carried out at dawn from 5am to 8am and at night from 4pm to 8pm. All the individuals observed and/or vocalizing were recorded on each stretch. Occasional sightings outside the area delimited for the census were also taken into account, especially for species richness and composition. In addition, to complement the sampling already described, active searching between listening points was used. At this point, all sounds and visualizations between the listening points were observed. In the end, the entire methodology lasted around 3-4 hours per section. The methodology applied in the twilight period allows for the survey of nocturnal species of the Strigidae (Owls), Caprimulgidae (Nightjar) and Nyctibiidae (Poor-me-ones) families.

In the case of vocalizations, the quadrant system was used to check the direction of the vocalization, in order to avoid duplicate vocalization records for the same species. If vocalizations of the same species came from the same quadrant, only one record of the species was counted (Vielliard et al., 2010).

As a complementary method, an exhaustive secondary survey was carried out, using data from bibliographies, databases and scientific collections, in an attempt to record as many bird species as possible with confirmed occurrence in the Seringal Rio Branco REDD+ Project region. For the bibliographic search, keywords related to the taxonomic group were used, adding names of nearby regions, such as the Conservation Unit closest to the Project area, cities and municipalities, considering the Project's reference area.

In all, 61 species were confirmed in 93 bird records in the Seringal Rio Branco REDD+ Project area. All the bird species surveyed are native to the Amazon, and no exotic species were recorded. In addition, they all use the area for breeding and nesting, and no migratory species were recorded in the primary survey. The species accumulation curve carried out with the aim of verifying sampling sufficiency showed that the avifauna assembly was not fully accessed through the survey carried out (Figure x). The Jackknife I estimator estimated around 170 bird species for the sampling area.



Graph 4: Species accumulation curve based on medium and large bird sampling, considering the observed values (S - Sobs Mao Tau) and values estimated by the non-parametric first-order Jackknife estimator (Jack 1 Mean), recorded in the area of the REDD+ Project Seringal Rio Branco, Municipality of Cruzeiro do Sul, AC.
In both curves, the vertical bars represent the Confidence Intervals (95%).

Most of the species recorded in the field survey are considered to be of little concern or are not on the national or international threatened species lists. However, we can highlight four species that are considered "vulnerable" on at least one of these lists, such as the Piciformes order, *Ramphastos tucanus* (Red-billed Toucan) and *Ramphastus vitellinus* (Channel-billed Toucan); the Columbiformes order, the *Patagioenas subvinacea* species (Ruddy Pigeon); and the Tinamiformes order, the *Tinamus tao* species (Grey Tinamou). In addition, we highlight two species considered "Endangered" by the IUCN, which are from the Coraciiformes order, *Momotus momota* (Amazonian Motmot) and from the Galbuliformes order, *Monasa morphoeus* (White-fronted Numbird). These species were confirmed in the Seringal Rio Branco REDD+ Project area through the primary survey.

From the secondary survey of bird species, it was possible to catalog 500 species expected to occur in the area of the REDD + Seringal Rio Branco Project. The most represented order was the Passariformes, with 281 species recorded. In terms of threat categories according to national and international lists, in addition to the species already mentioned, recorded by the primary survey, the species *Celeus flavus* and *Pulsatrix perspicillata*, considered "Critically Endangered" by the national list and recorded by the secondary survey, still stand out.

Finally, we highlight the *Micromonacha lanceolata* species as it was the first record of this rare species for the state of Acre. The record was made during the primary survey in the Seringal Rio Branco REDD+ Project area. The known distribution of these species occurs in Costa Rica, Panama, Colombia, Ecuador, Bolivia, Peru and in Brazil, until now, it had only been recorded in the state of Amazonas (Rasmussen et al. 2020).

- **Mastofauna**

The method used to survey the mastofauna in the Seringal Rio Branco REDD+ Project area consisted of walking along the previously built and cleared footpaths, at a constant speed of approximately 1 km/h, in total silence, observing and noting all sightings or traces of mammals on the footpath (Pardini et al., 2003; Peres & Cunha, 2011). Traces include carcasses, feces, vocalizations, burrows and other signs that allow us to reliably identify the species recorded. Occasional sightings outside the census can also be considered in inventories, especially for species richness and composition. This method makes it possible to record species with different habits (i.e. semi-arboreal, arboreal, terrestrial). It is a very versatile and generalist process for detecting mammals in the field (Peres & Cunha, 2011), and is carried out during the day and night. This method was applied in the inventory of the project area, and to systematize the search, using the main footpath of approximately 14 km in length, stretches of 2,000 m were defined to be walked daily, during the day and night. Sampling was carried out along this transect. The census was taken at dawn from 6am to 9am and at night from 4pm to 8pm.

Another complementary method was used, which was photographic trapping with the installation of automatic cameras with heat sensors and monitoring (Srbek - Araujo and Chiarello, 2007). This methodology covers the detection of terrestrial mammals and a few arboreal mammals that descend into the lower vegetation (such as semi-arboreal mammals), although it favors the detection of nocturnal animals, which are difficult to sample using the linear transect method. The recordings are made as the animals pass in front of the equipment, which is installed in trees at a height of around 40 cm from the ground, and programmed to work uninterruptedly, with intervals of 30 seconds per record, throughout the sampling period, marking the date and time of each record. Eleven camera traps were installed, distributed in pairs along the 14 km footpath.

In this inventory, medium and large terrestrial and arboreal mammals were sampled and recorded. Sightings were only counted on the outward journey along the footpath. In the case of camera traps, if the same species was recorded on the same camera, photos with an interval of at least 60 minutes were counted as different records. As a complementary method, an exhaustive secondary survey was carried out, using data from bibliographies, databases and scientific collections, in an attempt to record as many species of mastofauna as possible with confirmed occurrence in the project region.

A total of 30 species were confirmed in 182 independent records of medium and large terrestrial and arboreal mammals in the project area. All the mammal species surveyed are native to the Amazon, and no exotic species were recorded. In addition, they all use the area for breeding and nesting, and no migratory species were recorded in the primary survey. The species accumulation curve carried out with the aim of verifying sampling sufficiency showed that the assemblage of mastofauna was not fully accessed through the survey carried out. It estimated around 42 species of terrestrial and arboreal mastofauna for the sampling area.

The most represented order was Primates, with 10 species recorded in the primary survey. The most commonly recorded species were four species of primate: *Alouatta juara* (Juruá Red Howler Monkey), *Cacajao calvus* (Bald-headed Uacari), *Callicebus cupreus* (Coppery Tia) and *Saimiri macrodon* (Squirrel Monkey), as well as the White-lipped peccari (*Tayassu pecari*). Most of the species recorded in the field survey are considered to be of little concern or are not on the national or international endangered species lists. However, we can highlight species considered "Vulnerable" on at least one or both of these lists, such as *Tayassu pecari* (White-lipped peccari), *Puma yagouaroundi* (Jaguarundi), *Priodontes maximus* (Giant Armadillo), *Tapirus terrestris* (Lowland Tapir), and three primate species, *Cacajao calvus*

(Bald-headed Uacari), *Callimico goeldii* (Goeldi's Monkey) and *Cebuella niveiventris* (Eastern Pygmy Marmoset) (figure x, y).

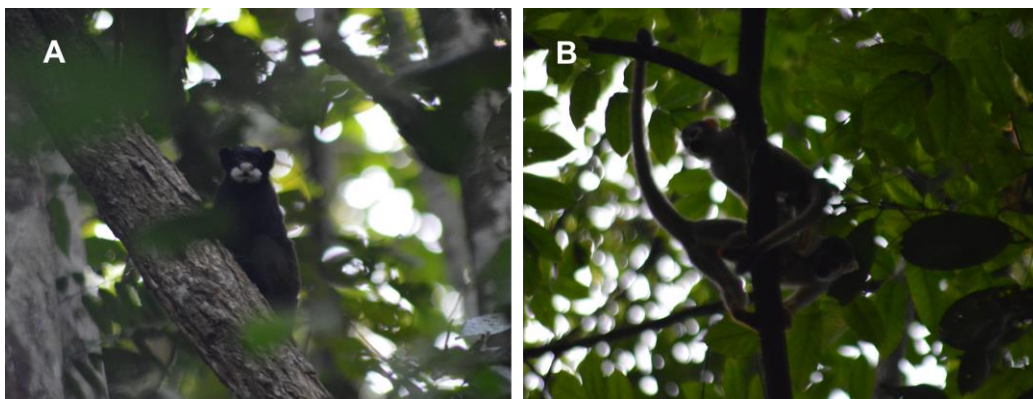


Figure 45: Mammals recorded through direct sightings in the area of the REDD+ Project Seringal Rio Branco, Municipality of Cruzeiro do Sul, AC. (A) *Saguinus mystax* (Moustached Tamarin); (B) *Saimiri macrodon* (Squirrel Monkey).



Figure 46: Mammals recorded using camera traps in the Seringal Rio Branco REDD+ Project area, Cruzeiro do Sul municipality, AC. (A) *Dasyprocta fuliginosa* (Black Agouti); (B) *Cuniculus paca* (Paca); (C) *Mazama americana* *Dasyprocta novemcinctus* (Nine-banded Armadillo); (D) *Atelocynus microtis* (Short-eared Dog); (E) *Tamandua tetradactyla* (Southern Tamandua); (F) *Leopardus pardalis* (Ocelot).

Most species of medium and large mammals are widely distributed in the Amazon and other biomes in Brazil. However, some species are endemic to this biome, and we can highlight the canid species *Atelocynus microtis* (Short-eared Dog), and all the primate species recorded. Some species can be considered bioindicators because they occur preferentially in areas of preserved primary forest in the Amazon, although they also occur in other habitats, including in other more open biomes. This is the case of the Jaguar (*Panthera onca*) and the White-lipped peccari (*Tayassu pecari*) (Teixeira-Santos et al., 2020). The species *Atelocynus microtis* (Short-eared Dog) is also restricted to more conserved primary forest areas (Rocha et al. 2020), as well as all primate species, since Neotropical primates are strictly arboreal and extremely sensitive to forest loss (da Silva et al. 2022).

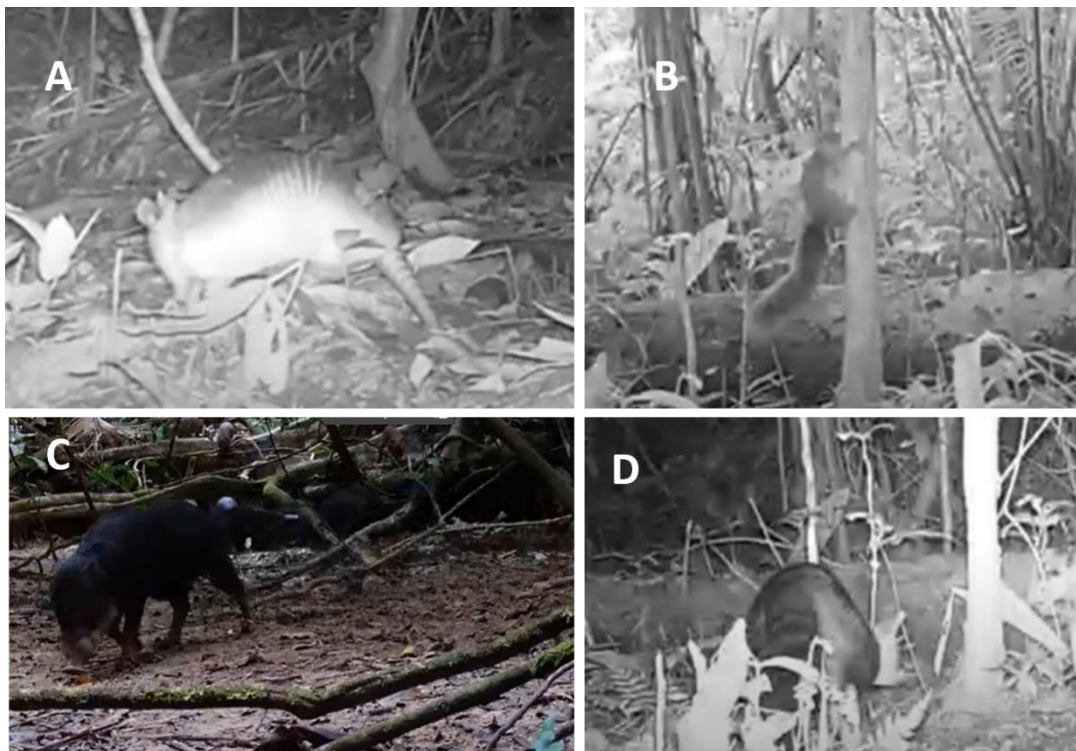


Figure 47: Mammals recorded using camera traps, in the area of the REDD+ Project Seringal Rio Branco, Municipality of Cruzeiro do Sul, AC. (A) *Dasypus novemcinctus* (Nine-banded Armadillo); (B) *Hadrosiurus spadiceus* (Southern Amazon Red Squirrel); (C) *Tayassu pecari* (White-lipped peccari); (D) *Procyon cancrivorus* (Crab-eating Raccoon).

- **Ictiofauna**

To diagnose the project's ichthyofauna, a fish inventory was carried out without the use of invasive specimen collection methodologies. Initially, the streams closest to the sampling points defined for the other terrestrial groups were visited. In addition to actively searching the streams to spot and recognize fish species, a general assessment was made of the condition of the bodies of water evaluated, as an indication of the conservation of the fish fauna.

As a complementary method, an exhaustive secondary survey was carried out, using data from bibliographies, databases and scientific collections, in an attempt to record as many species of ichthyofauna as possible with confirmed occurrence in the project region.

A total of 30 fish species were confirmed in 68 records. All the fish species surveyed are native to the Amazon, and no exotic species were recorded. In addition, they all use the area for breeding and nesting, and no migratory species were recorded in the primary survey. The most represented order was Characiformes with 8 species. This group includes piranhas, douradas and lambaris, and is exclusively freshwater. The most commonly recorded species were *Astyanax lacustris* (Lambari-do-Rabo-Amareno), and *Gasteropelecus sternicla* (Hetchetfish), both of which have a wide distribution (figure x).

One species that deserves to be highlighted is *Astyanax lacustris*, which belongs to the Characidae family. It is a small species native to the upper Paraná basin, with an omnivorous feeding habit, easy environmental adaptation, rapid growth and spawning more than once a year, which is why it is used as a biological model (Carneiro-Leite et al. 2020). This species can be considered a bioindicator of degraded areas. The species *Hemigrammus ocellifer*, *Ituglanis amazonicus* and *Megalechis picta* are endemic to the Amazon Biome, but are considered to be of low concern on the national and international threatened species lists, as are all the other species observed in the Project area (figure x).

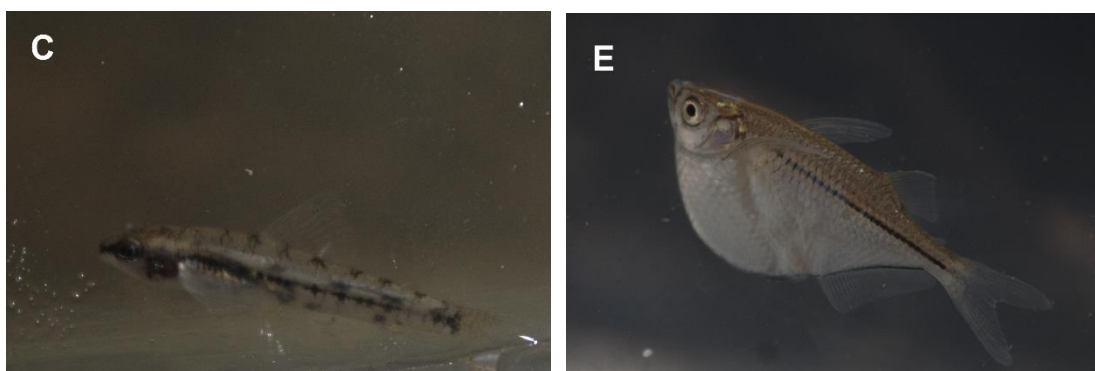


Figure 48: Fish with the highest frequency of records in the Seringal Rio Branco REDD+ Project area, Municipality of Cruzeiro do Sul, AC. *Characidium zebra* (Canivete); *Gasteropelecus sternicla* (Hatchetfish).





Figure 49: Fish recorded in the area of the REDD+ Project Seringal Rio Branco, Municipality of Cruzeiro do Sul, AC. *Hemigrammus ocellifer* (Head-and-tail light tetra), *Ituglanis amazonicus* (Cat fish) and *Megalechis picta* (Tamboatá).

5.1.2 High Conservation Values (CCB, B1.2)

The High Conservation Values (HCV) are a set of criteria for identifying and evaluating areas and elements that have significant importance for biodiversity conservation and human well-being. These criteria were developed by the HCV Resource Network, a global network of organizations working to promote the conservation and responsible use of natural resources.

The HCVs are based on six main categories of conservation values, such as significant concentration of biodiversity, seasonal concentration of species, threatened and rare ecosystems, presence of endangered species, provision of essential ecosystem services, social, historical and cultural values, among others.

High conservation value	AAVC 1 - Areas containing significant concentrations of biodiversity values on a global, national or regional scale: sites that are home to significant concentrations of endangered, endemic or rare species, or where the site contains an extraordinary concentration in terms of abundance of some ecologically relevant species. This attribute includes protected areas and critical sites for the life stage or cycle of plant and animal species, even if only seasonally.
Qualifying attribute	According to the uniqueness criteria applied to species of confirmed occurrence, threatened with extinction and endemic species (we only consider as endemic those species with a restricted occurrence in the state of Acre, according to Flora e Funga do Brasil (2023)), the results showed that two species fit the uniqueness criteria with globally significant populations, considering the estimates of Ter Steege et al. (2020), <i>Aphandra natalia</i> (Fiber Palm) and <i>Attalea tessmannii</i> (Cocão). These two species have 7.3 and 24.6% of their populations in the Seringal Rio Branco REDD+ project area. In addition, the project area has significant concentrations of

	threatened species that meet the vulnerability criterion, such as <i>Cedrela odorata</i> and <i>Swietenia macrophylla</i> .
Focal area	Total project area

5.1.3 Without-project Scenario: Biodiversity (CCB, B1.3)

From the description of the initial conditions of the Project Area, it is known that the diversity of fauna and flora is considerably high, being one of the main characteristics of the region, mainly due to the location of the project area in the Inanbari Endemism Center. Thus, based on the current conditions and the trends observed in the surrounding area, it was possible to outline some likely scenarios in the absence of the Seringal Rio Branco REDD+ Project.

The expectation for the area's scenario in the absence of the project is that it will remain susceptible to the threats that can normally occur, such as illegal logging, the advance of agriculture and livestock, and especially deforestation, which is directly related to the first two factors. It is therefore worth highlighting specific characteristics of these factors.

- Illegal logging, invasions and land grabbing are important threats to the Seringal Rio Branco REDD+ Project area; these threats cause forest degradation and put the preservation of the area at risk. Illegal logging increases the risk of fires, infestation by lianas and pioneer trees, and there is no guarantee that rare species, which are sometimes the main targets of illegal logging, will be conserved.
- Expansion of Agriculture and Livestock: The loss and eventual conversion of habitats to pasture and agriculture represents a major threat to all forest species in the Seringal Rio Branco REDD+ Project region. As fragmentation increases, the loss of connectivity between fragments occurs more slowly, as does the loss of the forest seed bank in the soil; the increased distance of these environments from native forest environments leads to a gradual increase in biodiversity loss. Therefore, these changes in habitat represent a much more significant threat in the medium and long term, increasing the level of threat. This threat is justified by the connection that the area has directly with the Francisco Pimentel PDS and with the riverside communities, especially on the Juruá-Mirim river.
- With deforestation, the Project area is being pressured by external factors such as the scarcity of natural resources outside its boundaries, which creates a scenario of pressure on existing resources within the Project area, where a greater number of extractive activities can be observed, both for timber and non-timber products, such as açaí, copaíba, cipó-titica, etc.
- Deforestation also exacerbates local and regional climate change, making the remaining forests drier, which also increases the risk of forest fires, as well as modifying the habitat of many forest species, reducing their areas of occupancy (AOO) and extent of occurrence (EOO). Here we highlight mainly threatened and endemic species.

Future scenarios for the Amazon have been the focus of several studies in recent decades, including the study by Ter Steege et al (2015), which presented population losses of vascular plants, especially woody plants, for two deforestation scenarios for the entire Amazon.:

1) The BAU scenario, where around 40% of forests would be lost by 2050 - in this scenario, 51% of common species would probably be threatened under IUCN criterion A4 and 43% of rare species would lose $\geq 30\%$ of their population; and, the Improved Governance Scenario (IGS), where 21% of forests would be lost by 2050 - in this scenario, 16% of common species would be threatened and 25% of rare species would lose $\geq 30\%$ of their population. In both scenarios, forest loss in the Eastern Amazon would be more severe (in this region forest loss would be between 42 and 76%, respectively in the BAU and GA scenarios); in this

region, in the BAU scenario, rare species would be more severely affected, with more than 86% of rare species showing population losses of more than 80% of their individuals.

The idea presented by Ter Steege et al. (2015) is interesting, but these authors did not evaluate the configuration of the landscape resulting from deforestation. It is true that habitat loss due to deforestation reduces the number of species and the size of populations, but depending on the configuration of the landscape, these results may be different. Sampaio et al. (2010) and Laques et al. (2012) worked with a landscape complexity index based on the conservation of forest structure and predominant ecological groups. These authors showed that the configuration of the landscape influences the index analyzed.

Several authors have pointed out that the factors influencing diversity vary at the different geographical scales analyzed, with the main scale that strongly influences diversity being the landscape scale, which is why the configuration of the landscape is an important factor. The degree of environmental homogenization within the same land use type and the significant differences between land use types are key factors in assessing the loss of diversity in rural areas, where deforestation leads to forest fragments mixed with agricultural areas, the so-called 'rural mosaics' (Do Vale et al., 2015, 2017, 2018, 2020); Arroyo-Rodriguez et al., 2020).

Forest fragmentation contributes to species loss (Taubert et al., 2018), causing changes in tree community composition (Laurance et al., 2006). Small forest fragments will also rapidly lose species and biomass due to overhunting, causing a decrease in large animal populations (Bicknell and Peres, 2010) and further reducing the species richness of trees that depend on these species for seed dispersal (Bello et al., 2015). Large tree species are also widely affected by fragmentation due to the influence of wind turbulence, desiccation and liana infestation, common effects observed near forest edges (Laurance et al 2000). These species have a strong influence on the structure, composition and hydrology of the forest, as well as contributing to carbon storage (Feldpausch et al., 2005).

At the landscape scale, land use types are a consequence of regional governance, often stimulated by public policies, as seen in other areas of Pará (Laques et al., 2012; Lavelle et al., 2016; Almeida et al., 2020). In areas of Nova Ipixuna, Parauapebas and Pacajá, in the state of Pará, Do Vale (2018, 2020) showed that 40% forest cover in agricultural settlement areas would be the minimum to conserve most species; this same value has been discussed by other authors (Arroyo-Rodriguez et al., 2017, 2020). In this sense, the importance of friendly land uses, such as agroforestry systems and tree plantations, help to increase the quality of the matrix, offering more resources, greater connectivity between forest fragments and buffering negative edge effects (Arroyo-Rodriguez et al., 2020).

These ideas could be useful to the Seringal Rio Branco REDD+ Project when evaluating monitoring data and making decisions for the future. Encouraging the implementation of friendly systems, such as agroforestry systems and tree plantations, can help improve environmental quality in the area and consequently ensure habitat for forest species, which are usually the first to be lost in areas disturbed by humans (Do Vale et al., 2020; Jakovac et al., 2021).

5.2 Net Positive Biodiversity Impacts

5.2.1 Expected Biodiversity Changes (VCS, 3.19; CCB, B2.1)

Expected changes to biodiversity as a result of the Project were estimated using the theory of change method, better defined in Section 2.1.1 in which all the activities were described. From this definition, we can better envision the cause and effect relation between the Project activities defined so far, the actions involved, their expected results and impacts in the short, medium and long term.

That said, all the activities have been defined with a view to promoting the long-term preservation of forest cover in the Project Area and, consequently, promoting changes in the region's future expectations. For biodiversity, it is clear that the biggest changes, based on the scenario without the Project, are associated with the preservation of natural habitats, providing for the maintenance and improvement of flora and fauna species, which would be threatened by planned and unplanned deforestation, and which, without the intervention of a REDD+ mechanism, would tend to increase for the Amazon region as a whole.

In this way, it is clear that ensuring forest protection through REDD+ activities and mechanisms will make it possible for the expected changes in biodiversity to be positive. Furthermore, as an essential tool for this, the biodiversity monitoring plan will make it possible to verify and ascertain whether any changes are having a negative impact on biological diversity, portraying a photograph of the local biota throughout the life cycle of the Project, providing greater clarity on the population dynamics of species, including endemic and vulnerable ones, as well as the conflicts generated by the coexistence of man and nature.

As a result, the changes expected for the Project's biodiversity are:

- Maintenance and addition of species identified in the without-project scenario;
- Ensure the conservation of habitats and species in the Brazilian Amazon;
- Reduce illegal hunting and fishing activities through Project activities;
- Increase awareness of environmental issues in local communities, reducing conflicts between communities and biodiversity;
- Ensure the conservation of endangered and endemic animal and plant species;
- Promote an ecologically balanced environment;
- Maintain landscape connectivity for biological diversity

Biodiversity element	Activities of the Seringal Rio Branco REDD+ Project
Estimated change	Biodiversity Scope
Justification of change	Maintenance of biodiversity

5.2.2 Mitigation Measures (VCS, 3.19; CCB, B2.3)

The data collected for the biodiversity studies was satisfactory in terms of assessing the current context of biodiversity conservation in the Project Zone, its surroundings and with a focus on the Project Area. However, longer-term studies are needed to elucidate the variations that occur in the biotic community over the course of forest modifications, whether arising from the reduction in forest area in the Project Zone or outside the Project Area, climate change or management activities, in order to better understand its dynamics (HENRIQUES et al., 2008).

In addition, according to the Brazilian Diagnosis of Biodiversity & Ecosystem Services, the main vectors threatening biodiversity are: a) climate changes, which alter the configuration and functioning of ecosystems, and changes in land use - in other words: deforestation, or any activity that involves converting areas of native vegetation to other uses, such as agriculture, livestock or mining; and b) the introduction and spread of invasive exotic species that lead to the loss of native species and alterations in interspecific relationships, ecological processes and the provision of ecosystem services.

In order to improve the population conditions of the species and mitigate the impacts caused by the internal and external factors described above, all of the project's activities, especially "biodiversity monitoring", were

designed and structured to act as mitigating measures against the main threats to biodiversity, as well as mitigating negative adverse factors in the conservation and maintenance of HCVAs.

With this in mind, although no negative impacts on biodiversity are expected, a constant evaluation of the results obtained through the monitoring of activities ensures that the project will be able to take swifter action if negative impacts are identified.

5.2.3 Net Positive Biodiversity Impacts (CCB, B2.2, GL1.4)

REDD+ projects have additional benefits (or positive impacts) for biodiversity conservation, especially considering the various ecosystem services that come from biodiversity (Karousakis, 2009). In addition to the main objective of reducing the impacts of climate change, effective actions to protect against deforestation and forest degradation (REDD+) bring benefits in terms of ecosystem services, such as biodiversity conservation and benefits on a local and regional scale (MILES & DICKSON, 2010; PITMAN, 2011).

The Seringal Rio Branco REDD+ Project area can be considered a 'key area' for biodiversity conservation (Langhammer et al., 2007), which reinforces the positive impact expected from the installation of the project. In addition, the project will contribute to maintaining the local climate, minimizing global climate change, with the maintenance of habitat connectivity, especially considering a larger spatial scale, such as in the Reference Region; in addition to protecting ecosystem services, such as pollination, seed dispersal, soil protection, hydrographic basins, etc.

As a positive effect of the implementation of the project, we can cite the preservation of forest areas in the project area. Most of the species with records for the study area are dependent on conserved forest formations, as well as having extensive areas in which to live. That said, it is essential to maintain these large areas of forest, as well as ecological corridors to ensure connections between forested areas, thus helping with the flow of individuals and regional gene flow.

Another positive impact is the monitoring of fauna that is directly associated with the implementation of the REDD+ project, which will allow the population trends of species that occur in and around the area to be monitored. In this way, it is possible to select species that respond quickly to changes in their habitat and are useful for measuring the efficiency of the project in maintaining forested areas and "High Conservation Value" attributes for fauna.

5.2.4 High Conservation Values Protected (CCB, B2.4)

The Project Area has a High Conservation Value attribute related to biodiversity, which has already been described in Section 5.1.2 and relates to i) areas that contain significant concentrations of biodiversity values on a global, regional or national scale, in which there are viable populations of most, if not all, naturally occurring species in natural patterns of distribution and abundance.

The measures proposed to guarantee the integrity of the HCVAs and thus maintain and improve these attributes, ensuring that they are not negatively affected by the Project, have been considered and incorporated into the defined Project activities. Therefore, no negative impact has been predicted for these areas and, furthermore, the proposed activities and their implementation throughout the life of the Project will generate positive impacts on these attributes.

As such, it is not expected that the attributes of the HCVAs will be negatively affected by the Project. On the contrary, by reducing deforestation and forest degradation in the Project Area, what is expected is the preservation of intact and appropriate habitats for the entire biotic community, including the recovery of ecological niches for endemic, rare or endangered species.

5.2.5 Species Used (VCS, 3.19; CCB, B2.5, B2.6)

The Project Area has a High Conservation Value attribute related to biodiversity, which has already been described in Section 5.1.2 and relates to i) areas that contain significant concentrations of biodiversity values on a global, regional or national scale, in which viable populations of most if not all naturally occurring species exist in natural patterns of distribution and abundance.

The measures proposed to guarantee the integrity of the HCVAs and thus maintain and improve these attributes, ensuring that they are not negatively affected by the Project, have been considered and incorporated into the defined Project activities. Therefore, no negative impact has been predicted for these areas and, furthermore, the proposed activities and their implementation throughout the life of the Project will generate positive impacts on these attributes.

As such, it is not expected that the attributes of the HCVAs will be negatively affected by the Project. On the contrary, by reducing deforestation and forest degradation in the Project Area, what is expected is the preservation of intact and appropriate habitats for the entire biotic community, including the recovery of ecological niches for endemic, rare or endangered species.

5.2.6 Invasive Species (VCS, 3.19; CCB, B2.5)

Based on the data collected in the field diagnosis of the project area, it can be stated that there are no species of flora or fauna considered invasive in the project zone. Should the occurrence of such species be observed, the Seringal Rio Branco REDD+ Project guarantees that no activities will be promoted that aid in the development and propagation of such species.

5.2.7 GMO Exclusion (CCB, B2.7)

Through the Seringal Rio Branco REDD+ Project it is guaranteed that no genetically modified organisms (GMOs) will be used. The reduction or removal of greenhouse gas emissions will be achieved by reducing deforestation and forest degradation.

5.2.8 Inputs Justification (VCS, 3.19; CCB, B2.8)

In the Seringal Rio Branco REDD+ Project Zone, where the activities are intended, there is no intention to use any chemical pesticides, biological control agents or other types of chemical inputs in the activities to be implemented.

In the event of the application of any chemical compound, or the use of biological control agents or any other type of input by the responsible parties, these will be reported in the monitoring report.

5.2.9 Waste Products (VCS, 3.19; CCB, B2.9)

The project's activities will not generate waste, so it is not necessary for the project to draw up procedures for identifying, classifying and properly handling waste. If waste is generated, the project will draw up the procedures and present them at the next validation/verification audit.

5.3 Offsite Biodiversity Impacts

5.3.1 Negative Offsite Biodiversity Impacts (CCB, B3.1) and Mitigation Measures (CCB, B3.2)

Generally speaking, a REDD+ Project, based on the premises of the VM0007 methodology, has the central objective of reducing emissions from planned and unplanned deforestation and forest degradation. That said, it is undeniable that the conservation provided by this mechanism is considered the greatest mitigating measure, benefiting the Project Zone as a whole.

Specifically for the Seringal Rio Branco REDD+ Project, the promotion of greater control over the occurrence of anthropogenic disturbances that would negatively impact biodiversity, such as habitat loss due to deforestation for the practice of predominant economic activities in the region such as agriculture, livestock and timber extraction, would act even more representatively in mitigating negative adverse effects. However, once measures aimed at conserving the forest and its resources are strengthened, these disturbances and activities may naturally move to areas outside the Project Zone, where they are more vulnerable to such events.

With regard to the mitigation measures resulting from the Project, mention should be made of the permanence and strengthening of alternative economic activities in the Project Area that generate income and employment, such as the promotion of sustainable practices, and education and awareness of the importance of biodiversity. In other words, the Project's social activities were designed to mitigate possible negative impacts, with an emphasis on environmental education, valuing the forest and the sustainable use of forest resources for local communities.

It's worth saying that all the project activities described in section 2.1.17 were discussed and refined in discussions with community members for their validation and understanding of conservation efforts.

Furthermore, although negative impacts on biodiversity outside the site are unlikely, the Project will seek to mitigate them through the articulation of the proponents, who will have to practice adaptive management and collectively address any additional negative impact on biodiversity outside the site that is identified later, articulating and promoting partnerships that increase such efforts.

5.3.2 Net Offsite Biodiversity Benefits (VCS, 3.19; CCB, B3.3)

The main positive impact expected outside the Project Zone is the favoring of biodiversity by maintaining large forest remnants that can function as ecological corridors, which will serve as a refuge and protection for threatened species and ecosystems and ecological processes can occur without any human intervention or only through sustainable use. Thus, despite the possible displacement of activities and disturbances outside the Project Zone, which could cause specific negative impacts, the connectivity of the landscape and the favoring of gene flow between forest areas, the maintenance of habitat and the in situ monitoring of threatened species are benefits that justify the presence of the Seringal Rio Branco REDD+ Project in the region.

5.4 Biodiversity Impact Monitoring

5.4.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

Biological diversity, or biodiversity, is characterized by the variety of life on Earth, encompassing all species of plants, animals and microorganisms, all genetic variability among species and also all the diversity of terrestrial and aquatic ecosystems - continental and marine - and the ecological complexes of which they are part (Secretariat of the Convention on Biological Diversity 2010).

Although Brazil, especially the Amazon biome, has a rich biodiversity, the exact knowledge and identification of all species is still quite scarce. That said, monitoring all the biodiversity in a given region or location is impossible in terms of logistics, finances and available and adequate knowledge. According to the "Proposal for a Brazilian Biodiversity Monitoring System", a reasonable way to solve this problem, making biodiversity monitoring more practical and feasible, is to monitor specific groups of animals and plants that respond in a calculable way to environmental changes. These groups of animals and plants that have such potential are called biological indicators and, in order for them to be considered good indicators in fact, three characteristics are taken into account: high rationality, high performance and high possibility of implementation.

The methodological roadmap considers four reasonable groups of biological indicators to be monitored: tree plants, selected groups of birds, medium and large mammals and frugivorous butterflies. It is important to remember that in defining these groups, their sensitivity to climate change issues was taken into account.

a) Technical description of monitoring tasks

Taking the aforementioned roadmap into account, the biodiversity monitoring plan for the Seringal Rio Branco REDD+ Project will focus on biological indicators that should be monitored and evaluated in the face of forest modifications and anthropogenic actions, whether arising from the decrease in forest area in the Project Zone or climate change, in order to better understand their dynamics and track the repercussions of these elements on biological diversity.

In the scenario without the Project, described in Section 5.1, the diagnosis carried out considered different sampling areas. This plan will therefore seek, whenever possible, to carry out expeditions to monitor biodiversity under the same conditions and methodologies used to diagnose the without-Project scenario, so that it is possible to compare, observe and gauge the effects of the Project's activities on biodiversity over time.

That said, in order to monitor the fauna, the project will carry out expeditions (to be established), taking into account the periods of low and high rainfall, so that the presence of migratory species and reproductive

periods can be taken into account. As for flora monitoring, given that the dynamics and floristic changes occur more slowly, the expeditions will be carried out in periods close to each verification of the Project in order to assess any changes that may arise in the structure and composition of floristic species.

In addition, it is expected that species of relevance, whether endemic, threatened and rare, be prioritized in the expeditions and collected during the monitoring campaigns, especially the key species, considered to be of great biological importance and high degree of threat, according to IUCN criteria. In addition, we emphasize that the exotic species recorded in the biodiversity diagnosis, such as Javali and domestic animals, will be duly considered in the project's monitoring process. Although so far we have not identified any negative impacts resulting from their presence, we recognize that these species represent a potential risk to local diversity. It is therefore essential to closely monitor their interaction with native biodiversity.

Finally, the plan will seek to protect the High Conservation Value (HCV) attributes by stimulating and improving knowledge about local biodiversity through continuous in situ monitoring. The evaluation of the effectiveness of the measures adopted to maintain and improve the HCVs is covered by the tasks described, especially in the monitoring of indicators related to the richness of both fauna and flora species. In this way, the monitoring plan makes it possible to track the permanence of relevant species, whether they are threatened, rare or endemic in these areas defined as priorities for conservation.

b) Data to be collected

Data / Parameter	Number of documents focused on biodiversity scope
Data unit	Number
Description	This parameter will be responsible for accounting for the quantity of all material produced, from presentations to teaching tools, designed to implement and monitor the Project's biodiversity scope.
Data source	Calculated by means of reports, meeting minutes, monitoring guides and memos, focused on issues related to the Project's biodiversity scope (biodiversity monitoring)
Description of the methods and measurement procedures to be Applied	All documents produced for the project will be stored in digital files throughout the project's accreditation period. In this way, these reports from the "Biodiversity monitoring" activity will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the registration of the Project
Monitoring equipment	Does not apply
QA/QC procedures to be Applied	The information systematized in these documents will be validated among the proponents, allowing for greater reliability and data quality. In addition, the project will undergo continuous evaluation of the information generated, allowing for the identification of improvements in the collection and recording processes, and the incorporation of these into the project's strategic planning when they are identified.
Purpose of the data	Does not apply

Calculation method	Does not apply
Comments	-

Data / Parameter	Morphospecies richness of the monitored plant Community
Data unit	Number
Description	Number of morphospecies found during monitoring being the fundamental unit for assessing the homogeneity of the monitored environment. Due to the great difficulty in identifying an individual at a specific level, the parameter is focused on morphospecies, which are defined by morphological similarity.
Data source	Field sheets, forest inventories, monitoring reports, additional records
Description of the methods and measurement procedures to be Applied	To be established (ideally similar to the one carried out in the initial diagnoses)
Frequency of monitoring/recording	One year before a verification event
Value applied	To be accounted for after Project registration
Monitoring equipment	Does not apply
QA/QC procedures to be Applied	To be established
Purpose of the data	Data sheet
Calculation method	-
Comments	Field sheets, forest inventories, monitoring reports, additional records

Data / Parameter	Presence of relevant species
Data unit	Percentage
Description	Monitoring of fauna and flora species in the Project Area and tracking their status on the IUCN Red List of Threatened Species, with an emphasis on species cited as critically endangered (CR), endangered (EN), vulnerable (VU) and endemic species. Quantification of the % of species monitored in the Project Area with high relevance for protection.
Data source	Field sheets, forest inventories, monitoring reports, additional records
Description of the methods and measurement procedures to be Applied	Systematisation and comparison of the data and information collected during the fauna expeditions with the IUCN Official List, available at: https://www.iucnredlist.org
Frequency of monitoring/recording	At least once before each verification event
Value applied	Not applicable

Monitoring equipment	Not applicable
QA/QC procedures to be Applied	To be established
Purpose of the data	Parameter for evaluating the effectiveness of the Project in protecting and maintaining highly threatened and endemic species within the Project Area, especially key species, ensuring the improvement and maintenance of HCVA 1
Calculation method	Data sheet
Comments	-

5.4.2 Biodiversity Monitoring Plan Dissemination (CCB, B4.3)

The monitoring plan, as well as the results obtained from monitoring the Seringal Rio Branco REDD+ Project, will be made available to the public via a page on the official website of Biofílica Ambipar Environment Investments. The summarized documents referring to the monitoring plan and results, as well as relevant information, will be made available to communities and stakeholders through meetings, lectures and physically on the premises of Amazônia Agroindústria EIRELI.

5.5 Optional Criterion: Exceptional Biodiversity Benefits

5.5.1 High Biodiversity Conservation Priority Status (CCB, GL3.1)

As described in detail in section 5.1.1, the Seringal Rio Branco REDD+ Project area is considered a Priority Area for Biodiversity, classified as extremely high by the Ministry of the Environment⁵⁶, with a high rate of biological diversity and species listed with some degree of endemism and threat. The location of the area in the so-called Inanbari Endemism Centre and its proximity to areas with a high degree of conservation, such as the Serra do Divisor National Park, justifies the high biodiversity found in the area.

Based on this, in compliance with the criterion of demonstrating the presence in the project area of at least a single individual of a species that is critically endangered (CR) or endangered (EN), according to the classification of the IUCN Red List of Threatened Species, the Seringal Rio Branco REDD+ Project observed in the field inventory the presence of at least six individuals of the floral species *Handroanthus serratifolius* (Vahl) S. Grose (Yellow Ipe), classified as an 'endangered' species by the IUCN. In addition, other species of fauna and flora classified as 'vulnerable' were listed, but which represent some kind of importance for the project area due to their degree of endemism or even their use by riverside communities (table 46).

It should be noted that the flora species listed in table below are those confirmed to occur in the project area and identified through the phytosociological study carried out on eight hectares distributed over the three forest types in the Seringal Rio Branco REDD+ project area. As well as the fauna species, which were sighted in a single observation event, directly influencing the number of individuals of each species, not representing the real size of the local population.

In this way, it can be said that the presence of these species in the project area demonstrates the area's high priority for generating exceptional benefits for biodiversity, and that it is of great importance for the project area to monitor and protect these species.

Table 44: List of fauna and flora species in the project area that are classified as VU or EN according to the IUCN.

Threatened species with confirmed occurrence - Flora		
Scientific name	IUCN Classification	Number of individuals
<i>Handroanthus serratifolius</i> (Vahl) S.Grose	EN	6
<i>Annona calcarata</i> (R.E.Fr.) H. Rainer	VU	13
<i>Cedrela odorata</i> L.	VU	4
<i>Swietenia macrophylla</i> King	VU	1
Threatened species with confirmed occurrence - Fauna		
<i>Podocnemis unifilis</i>	VU	1
<i>Chelonoidis denticulata</i>	VU	1

⁵⁶ BRASIL. Ministério do Meio Ambiente e Mudança do Clima. Áreas prioritárias para Biodiversidade. 2024. Disponível em: <https://www.gov.br/mma/pt-br/assuntos/biodiversidade-e-ecossistemas/ecossistemas/areas-prioritarias-para-biodiversidade>. Acesso em: 23 maio 2024.

<i>Patagioenas subvinacea</i>	VU	1
<i>Ramphastos tucanus</i>	VU	6
<i>Ramphastos vitellinus</i>	VU	1
<i>Tinamus tao</i>	VU	1
<i>Tayassu pecari</i>	VU	21
<i>Priodontes maximus</i>	VU	2
<i>Tapirus terrestres</i>	VU	1
<i>Callimico goeldii</i>	VU	2
<i>Cebuella niveiventris</i>	VU	4

5.5.2 Trigger Species Population Trends (CCB, GL3.2, GL3.3)

Flora:

Trigger species	<i>Handroanthus serratifolius</i> (Vahl) S.Grose
Population trend at start of project	The species is considered Endangered (EN) according to the IUCN ⁵⁷ international classification. Despite its wide geographical distribution in Brazil and other countries in South America, the species has shown a downward population trend in recent years due to its high added commercial value.
Without-project scenario	In a no-project scenario, deforestation is expected to increase with a view to conversion into non-forest areas and logging. The species is highly commercialised in Brazil and is the target of illegal logging in non-permitted areas ⁵⁸ , which jeopardises long-term population recovery, given its difficulty in regenerating ⁵⁹ .
With-project scenario	Through its activities, the Seringal Rio Branco REDD+ Project aims to reduce deforestation and forest degradation, promoting the maintenance of the population of this species in the project area. It will also carry out surveillance activities to help prevent illegal logging in the project area. In addition, the project's biodiversity monitoring plan will consider the monitoring of this species, which has been chosen to reflect the project's results. In this way, it is hoped that in the scenario

⁵⁷ IUCN 2024. Lista Vermelha de Espécies Ameaçadas da IUCN. Versão 2023-1. <<https://www.iucnredlist.org>>

⁵⁸ UICN/TRÁFEGO. 2019. Inclusão de Trombetas *Handroanthus* spp., *Tabebuia* spp. e *Roseodendron* spp. no Apêndice II com anotação #6. Análises de propostas para CoP18 da UICN/TRAFFIC.

⁵⁹ Schulze, M., Grogan, J., Uhl, C., Lentini, M. e Vidal, E. 2008. Avaliando a exploração madeireira de ipê (*Tabebuia*, Bignoniaceae) na Amazônia: Manejo sustentável ou catalisador para a degradação florestal? *Conservação Biológica* 141(8): 2071-2085.

	with the project, there will be a favourable improvement in the population trend of the <i>Handroanthus serratifolius</i> (Yellow Ipe) species.
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Trigger species	<i>Annona calcarata</i> (R.E.Fr.) H. Rainer
Population trend at start of project	The species is considered Vulnerable (VU) according to the IUCN international classification ⁶⁰ . This species is endemic to the state of Acre and its area of occurrence is threatened by deforestation and forest degradation. The loss of this species' habitat due to intense deforestation has led to a downward population trend in recent years. ⁶¹
Without-project scenario	In a non-project scenario, deforestation is expected to increase with a view to conversion into non-forest areas and logging, leading to degradation of the area, loss of forest cover and consequently loss of habitat for numerous species with limited ranges, such as the <i>Annona calcarata</i> and other species endemic to the region ⁶² .
With-project scenario	Through its activities, the Seringal Rio Branco REDD+ Project aims to reduce deforestation and forest degradation, promoting the increase or maintenance of the population of this species in the project area. Satellite monitoring and surveillance activities will also help prevent deforestation in the project area, preventing habitat loss for the region's endemic species and favouring the conservation of local biodiversity. In addition, this project's biodiversity monitoring plan will consider the monitoring of this species being chosen to reflect the project's results. In this way, it is hoped that in the scenario with the project, there will be a favourable improvement in the population trend of the <i>Annona calcarata</i> (Biribá-Bravo) species.

Trigger species	<i>Cedrela odorata</i> L.
Population trend at start of project	The species is considered Vulnerable (VU) according to the IUCN international classification ⁶³ . Despite its wide geographical distribution in Brazil and the other countries

⁶⁰ IUCN 2024. Lista Vermelha de Espécies Ameaçadas da IUCN. Versão 2023-1. <<https://www.iucnredlist.org>>

⁶¹ CNCFlora. *Annona calcarata* in Lista Vermelha da flora brasileira versão 2012.2 Centro Nacional de Conservação da Flora. Disponível em <[http://cncflora.jbrj.gov.br/portal/pt-br/profile/Annona calcarata](http://cncflora.jbrj.gov.br/portal/pt-br/profile/Annona%20calcarata)>. Acesso em 23 maio 2024.

⁶² Da Silva, JMC, Rylands, AB e Da Fonseca, GAB 2005. O destino das áreas de endemismo amazônico. 19: 689-694.

⁶³ UCN 2024. Lista Vermelha de Espécies Ameaçadas da IUCN. Versão 2023-1. <<https://www.iucnredlist.org>>

	where it occurs, the species has shown a downward population trend in recent years due to its high added commercial value, overexploitation and intense deforestation.
Without-project scenario	In a no-project scenario, deforestation is expected to increase with a view to conversion into non-forest areas and logging. The species is highly commercialised in Brazil and is the target of intense exploitation for both domestic use and export ⁶⁴ , leading to a decline in the species' development. In addition, according to the socio-economic diagnosis carried out specifically in the project area, it was possible to observe that all the communities approached use <i>Cedrela odorata</i> wood to build furniture and handicrafts, which intensifies its exploitation.
With-project scenario	Through its activities, the Seringal Rio Branco REDD+ Project aims to reduce deforestation and forest degradation, promoting an increase or maintenance of the population of this species in the project area. As well as surveillance and monitoring activities that will help prevent illegal logging in the project area. As well as environmental education activities, aimed at raising awareness of the appropriate use of forest products in the project area. In addition, this project's biodiversity monitoring plan will consider the monitoring of this species, which was chosen to reflect the project's results. In this way, it is hoped that in the scenario with the project, there will be a favourable improvement in the population trend of the <i>Cedrela odorata</i> (Cedar) species.

Fauna:

Trigger species	<i>Tayassu pecari</i>
Population trend at start of project	The species is considered Vulnerable (VU) according to the IUCN international classification ⁶⁵ . Despite its wide geographical distribution in Brazil and other countries where it occurs, the species has shown a downward population trend in recent years due to habitat loss from deforestation and illegal poaching. The species forms large herds and needs extensive areas of habitat to obtain resources throughout the year, which has been highly threatened by the advance of widespread deforestation, exposing the species more and

⁶⁴ Convenção sobre o Comércio Internacional de Espécies da Fauna e da Flora Selvagens Ameaçadas de Extinção (CITES). 2007. Proposta 33 Inclusão no Apêndice II de *Cedrela odorata* e todas as outras espécies de *Cedrela* por motivos de semelhança. Décima Quarta Reunião da Conferência das Partes Haia (Holanda).

⁶⁵ IUCN 2024. Lista Vermelha de Espécies Ameaçadas da IUCN. Versão 2023-1. <<https://www.iucnredlist.org>>

	more and making it more susceptible to pressure from illegal hunting ⁶⁶ .
Without-project scenario	In a no-project scenario, deforestation is expected to increase with a view to conversion into non-forest areas. Considering the species' characteristics, the loss of forest cover jeopardises the permanence of individuals that live in flocks, causing a decrease in their population. In addition, the species is hunted in Brazil, which tends to increase with the deforestation of the area due to the exposure of individuals. In addition, according to the socio-economic diagnosis carried out specifically in the project area, some communities around the project area hunt this animal, which tends to be intensified over time in a scenario without the project.
With-project scenario	Through its activities, the Seringal Rio Branco REDD+ Project aims to reduce deforestation and forest degradation, promoting the maintenance of the habitat and population of this species in the project area. As well as asset surveillance and monitoring activities that will help prevent deforestation in the area, with the aim of protecting the habitat of the species that live there. As well as environmental education activities, aimed at raising awareness about destructive hunting of animals. In addition, this project's biodiversity monitoring plan will consider the monitoring of this species being chosen to reflect the project's results. In this way, it is hoped that in the scenario with the project, there will be a favourable improvement in the population trend of the <i>Tayassu pecari</i> (Queixada) species.
Trigger species	<i>Tapirus terrestres</i>
Population trend at start of project	The species is considered Vulnerable (VU) according to the IUCN international classification ⁶⁷ . The species has a wide geographical distribution in Brazil and other countries where it occurs, but has shown a downward population trend in recent years due to pressure from illegal poaching and its difficulty in adapting to constant large-scale habitat changes and resource availability due to deforestation of forest areas ⁶⁸

⁶⁶ Azevedo, FCC e Conforti, VC 2008. Declínio de queixadas em uma floresta subtropical protegida do Brasil: em direção a questões de conservação. Mamíferos 72: 82-88.

⁶⁷ IUCN 2024. Lista Vermelha de Espécies Ameaçadas da IUCN. Versão 2023-1. <<https://www.iucnredlist.org>>

⁶⁸ IUCN 2024. Lista Vermelha de Espécies Ameaçadas da IUCN. Versão 2023-1. <<https://www.iucnredlist.org>>

Without-project scenario	In a no-project scenario, deforestation is expected to increase with a view to conversion into non-forest areas. Considering the species' characteristics, the loss of forest cover will jeopardise the permanence of individuals, since Brazilian tapirs tend to be more dependent on the forest, leading to a decrease in their population. In addition, the species is hunted in Brazil, which tends to increase with the deforestation of the area due to the exposure of individuals. In addition, according to the socio-economic diagnosis carried out specifically in the project area, some communities around the project area hunt this animal, which tends to be intensified over time in a scenario without the project.
With-project scenario	Through its activities, the Seringal Rio Branco REDD+ Project aims to reduce deforestation and forest degradation, promoting the maintenance of the habitat and population of this species in the project area. As well as asset surveillance and monitoring activities that will help prevent deforestation in the area, with the aim of protecting the habitat of the species that live there. As well as environmental education activities, aimed at raising awareness about destructive hunting of animals. In addition, this project's biodiversity monitoring plan will consider the monitoring of this species being chosen to reflect the project's results. In this way, it is hoped that in the scenario with the project, there will be a favourable improvement in the population trend of the <i>Tapirus terrestres</i> (Tapir) species.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION.

Section	Information	Justification
NPRT	Economic Model	Commercially strategic information
2.1.9	Ownership and benefit sharing contract	Personal data and business strategic information
3	Emissions spreadsheet	Commercially strategic information
2.1.8	Contracts with Other entities	Personal data and business strategic information
NPRT	Documents generated in the risk analysis, such as property and owner documents	Personal data and business strategic information